MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project Customer: **Mansourah Massarah Gold Project**
MA'ADEN GOLD AND BASE METALS (MA'ADEN)

Title: **DESIGN CALCULATION**
3610 POX CONDITIONING, LEACHING AND ADSORPTION
DESIGN OF MISCELLANEOUS FOUNDATION

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TABLE OF CONTENTS

DEFINITIONS AND ABBREVIATIONS ……..……4

1. INTENT……..……...5

2. EXECUTIVE SUMMARY……..……5

3. SCOPE…….....6

4. DESIGN BASIS

4.1 Materials……..……...6

4.2 Loading……..……6

4.3 Load Combinations……..……7

4.4 Minimum Reinforcing Steel (Rebar) Size……..……8

4.5 Minimum Concrete Cover……..……..8

4.6 Allowable Soil Bearing Capacity……..……...8

4.7 Vertical Modulus of Subgrade Reaction……....8

5. DESIGN CALCULATION

5.1 Design of Footing Type F1……..……..……..…….....9

5.2 Design of Footing Type F2……..……..……..…….....20

5.3 Design of Footing Type F5……..……..……..…….....26

5.4 Design of Footing Type F6……..……..……..…….....32

5.5 Design of Footing Type F7……..……..……..…….....43

5.6 Design of Footing Type F8/F9……..……..……..…….....49

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5.7 Design of Footing Type F10.....55

5.8 Design of Footing Type F11.....58

5.9 Design of Footing Type F13.....69

5.10 Design of Footing Type F14 & F15.....80

5.11 Design of Footing Type F16.....83

5.13 Design of column C1.....89

5.14 Design of column C2.....90

5.15 Design of column C5.....91

5.16 Design of column C6.....92

5.17 Design of column C7.....93

5.18 Design of column C8 & C9.....94

5.19 Design of column C10.....95

5.20 Design of column C11.....96

References.....97

ATTACHMENT 1 STAAD INPUT FILE FOR F1 DESIGN.....98

ATTACHMENT 2 STAAD INPUT FILE FOR F2 & F5 DESIGN 109

ATTACHMENT 3 STAAD INPUT FILE FOR F6 DESIGN 122

ATTACHMENT 4 STAAD INPUT FILE FOR F7 DESIGN.....129

ATTACHMENT 5 STAAD INPUT FILE FOR F8 & F9 DESIGN.....182

ATTACHMENT 6 STAAD INPUTFILE FOR F16 DESIGN 214

ATTACHMENT 7 FOUNDATION DESIGN LOAD DATA.....220

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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	'EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities

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1. INTENT

The purpose of this document is to analysis and design of **POX CONDITIONING, LEACHING AND ADSORPTION MISCELLANEOUS FOUNDATION**. In this document design & analysis of foundation have been covered. The analysis & design is carried out in accordance with project specifications and codes & standards. The design calculation ensures the safety and serviceability of the foundation system.

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week

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3. SCOPE

The analysis has been carried out in Staad pro & designed by Excel Sheet. Strength design of column and tie beams are performed in STAAD Pro. Load data for foundation design taken from document no. **MD-216-3610-EG-ST-CAL-2001, MD-216-3610-EG-ST-CAL-2280**, from respective superstructure Staad model and foundation reaction summary shared for individual structures.

4. DESIGN BASIS

4.1 Material

A. Concrete Strength of 28 days

Foundation and Structural Members: 35 MPa

Blinding Concrete below foundation bases: 15 MPa

B. Reinforcement Grade

Deformed Bar : ASTM A 615 / A 615M Grade 60 (fy = 420MPa)

Welded Wire Fabric : ASTM A 185(Plain) / A 497(Deformed)

C. Grouting

Grout shall be 50 mm cementitious of Strength 40 MPa.

4.2 Loading

A. Primary Load

LOAD 1 = DEAD LOAD (DL)

LOAD 2 = LIVE LOAD (LL)

LOAD 3 = WL1 (+X)

LOAD 4 = WL2 (-X)

LOAD 5 = WL3 (+Z)

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LOAD 6 = WL4 (-Z)

LOAD 7 =TL1_TEMP (+25)

4.3 Load combination

A. The following load combination are considered for member strength calculation.

101 1.4 DL

102 1.2(DL+TL1) +1.6LL

103 1.2(DL+TL2) +1.6 LL

104 1.2 DL+1.6 WL (+X) + 0.5 LL

105 1.2 DL+1.6 WL (-X) + 0.5 LL

106 1.2 DL+1.6 WL (+Z) + 0.5 LL

107 1.2 DL+1.6 WL (-Z) + 0.5 LL

108 0.9 DL+1.6 WL (+X)

109 0.9 DL+1.6 WL (-X)

110 0.9 DL+1.6 WL (+Z)

111 0.9 DL+1.6 WL (-Z)

112 1.2 DL+0.8 WL (+X)

113 1.2 DL+0.8 WL (-X)

114 1.2DL+0.8 WL (+Z)

115 1.2DL+0.8 WL (-Z)

B) The following load combination are considered for base pressure, sliding & overturning check.

201 DL

202 (DL+ TL1) + LL

203 (DL+ TL2) + LL

204 DL+ WL (+X)

205 DL+ WL (-X)

206 DL+ WL (+Z)

207 DL+ WL (-Z)

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208 0.6DL+WL (+X)

209 0.6DL+WL (-X)

210 0.6DL+WL (+Z)

211 0.6DL+WL (-Z)

212 DL+0.75WL (+X) +0.75LL

213 DL+0.75WL (-X) +0.75LL

214 DL+0.75WL (+Z) +0.75LL

215 DL+0.75WL (-Z)+0.75LL

4.4 Minimum Reinforcing Steel (Rebar) Size

The minimum rebar size for use as tension, compression reinforcement shall be 12 mm except where welded wire fabric is used for ties and spirals the minimum rebar size shall be 10 mm.

4.5 Minimum concrete cover

	Member	Minimum Cover(mm)
Exposed to weather or contact with ground	Foundation & Columns	50

4.6 Allowable Soil Bearing Capacity

The net safe bearing capacity of soil for foundations design has been considered as 450 kN/m2 at 1.2m depth below EGL (RL 912.200) refer Table 5 - BH-11 of Geotechnical Report (MD-216-3000-EG-CV-RPT-1008).

4.7 Vertical Modulus of Subgrade Reaction

The Vertical Modulus of Subgrade Reaction has been considered as 40500 kN/m^2/m at 1.2m depth below EGL (RL 912.500) refer Table 5 - BH-11 of Geotechnical Report (MD-216-3000-EG-CV-RPT-1008).

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5.1 DESIGN OF FOOTING TYPE F1

	Along-X	Along-Z
Size of column :	0.60m	0.70m
Size of fdn. raft :	2.00m	2.00m
Thickness of raft :	0.40	m
Original ground level:	0.00	m
Finished floor level :	0.00	m
U/S level of foundation :	-1.20	m
U/S level Base Plate :	0.50	m
Unit weight of conc :	24	kN/m^3
Unit weight of soil :	18	kN/m^3

SYMBOLS AND NOTATIONS

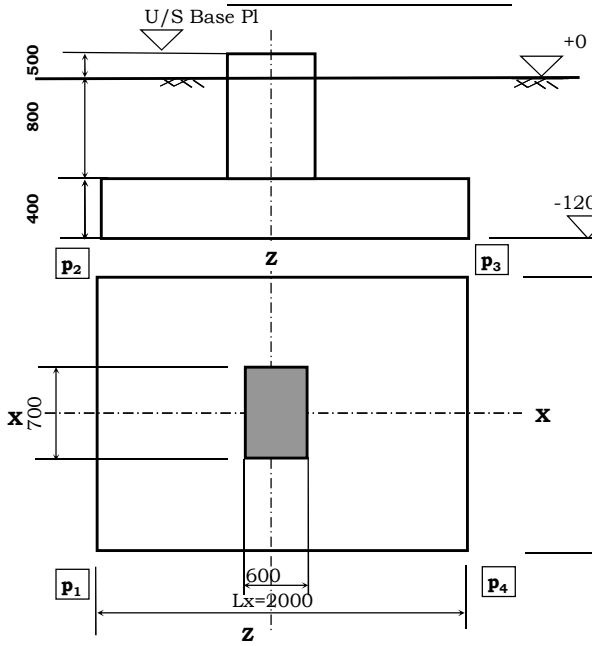
Dead Load	DL
Normal Operational Fluid Load	FL1
Max. Fluid Load	FL2
Wind Load	WL
Seismic Load	SL

Calculation of Weight of Foundation :

ght of raft & Col.=	(2*2*0.4*24)+(0.6*0.7*1.3*24) =	51.50KN
Weight of soil =	((2*2)-(0.6*0.7))*0.8*18 =	51.55KN
	Total =	103.06KN

Design Input:

Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	50mm
Dia. of bars provd. along X direction at bottom =	12mm
Dia. of bars provd. along X direction at top =	12mm
Dia. of bars provided along Z direction at bottom=	12mm
Dia. of bars provided along Z direction at top=	12mm
Mininum percentage of reinforcement at bottom=	0.18%

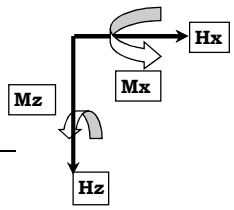


Effective depth of slab along X direction = 344mm = 0.344m

Effective depth of slab along Z direction = 332mm = 0.332m

Clear projection along Z direction = 0.65m

Clear projection along X direction = 0.7m



H_X=Force along X-X direction

Hz=Force along Z-Z direction

M_X=Moment about X-X direction

Mz=Moment about Z-Z direction

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5.1 DESIGN OF FOOTING TYPE F1

Combined Load Cases:	**Load at bottom of foundation**					**Load at bottom of foundation + self weight**				
Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
Allowable Stress Design										
201 1. DL	231.3	0.0	-18.7	2.5	-11.8	334.35	0.03	-18.74	2.47	-11.78
202 1 (DL + TL1) + 1 LL	354.7	3.3	-25.3	-6.1	-0.3	457.81	3.25	-25.28	-6.13	-0.29
203 1 (DL + TL2) + 1 LL	314.0	-0.5	-43.5	9.9	-37.1	417.01	-0.51	-43.54	9.95	-37.11
204 1 DL + 0.75 TL1 + 0.75 LL	323.9	2.4	-23.6	-4.0	-3.2	426.94	2.45	-23.64	-3.98	-3.16
205 1 DL + 0.75 TL2 + 0.75 LL	293.3	-0.4	-37.3	8.1	-30.8	396.35	-0.37	-37.34	8.08	-30.78
206 1 DL+ 1 WL(+X)	129.4	-30.3	47.5	1.4	-5.8	232.49	-30.30	47.52	1.44	-5.77
207 1 DL+ 1 WL(-X)	333.1	30.4	-85.0	3.5	-17.8	436.20	30.35	-85.01	3.51	-17.78
208 1 DL+ 1 WL(+Z)	310.8	2.2	-29.0	-25.2	-67.4	413.91	2.15	-28.99	-25.20	-67.38
209 1 DL+ 1 WL(-Z)	151.7	-2.1	-8.5	30.1	43.8	254.79	-2.10	-8.49	30.15	43.83
210 1 DL + 0.7 EL(+X)	218.1	-2.9	-12.5	1.9	-11.8	321.14	-2.88	-12.53	1.87	-11.82
211 1 DL + 0.7 EL(-X)	244.5	2.9	-25.0	3.1	-11.7	347.55	2.93	-24.95	3.07	-11.73
212 1 DL + 0.7 EL(+Z)	239.5	0.1	-19.6	0.5	-15.6	342.52	0.10	-19.55	0.51	-15.62
213 1 DL + 0.7 EL(-Z)	223.1	0.0	-17.9	4.4	-7.9	326.17	-0.04	-17.93	4.44	-7.93
214 0.6 DL + 1 WL(+X)	36.9	-30.3	55.0	0.5	-1.1	139.98	-30.31	55.02	0.45	-1.06
215 0.6 DL + 1 WL(-X)	240.6	30.3	-77.5	2.5	-13.1	343.69	30.34	-77.51	2.52	-13.07
216 0.6 DL + 1 WL(+Z)	218.3	2.1	-21.5	-26.2	-62.7	321.39	2.14	-21.49	-26.19	-62.67
217 0.6 DL + 1 WL(-Z)	59.2	-2.1	-1.0	29.2	48.5	162.27	-2.11	-1.00	29.16	48.54
218 0.6 DL + 0.7 EL(+X)	125.6	-2.9	-5.0	0.9	-7.1	228.63	-2.89	-5.04	0.88	-7.11
219 0.6 DL + 0.7 EL(-X)	152.0	2.9	-17.5	2.1	-7.0	255.04	2.92	-17.45	2.08	-7.02
220 0.6 DL + 0.7 EL(+Z)	147.0	0.1	-12.1	-0.5	-10.9	250.01	0.09	-12.06	-0.48	-10.91
221 0.6 DL + 0.7 EL(-Z)	130.6	-0.1	-10.4	3.4	-3.2	233.65	-0.05	-10.44	3.45	-3.22
222 1 DL+ 0.75 WL(+X)+ 0.75 LL	232.2	-21.7	19.2	1.3	-12.5	335.25	-21.71	19.21	1.28	-12.47
223 1 DL+ 0.75 WL(-X)+ 0.75 LL	385.0	23.8	-80.2	2.8	-21.5	488.03	23.78	-80.19	2.82	-21.47
224 1 DL+ 0.75 WL(+Z)+ 0.75 LL	368.3	2.6	-38.2	-18.7	-58.7	471.31	2.63	-38.18	-18.71	-58.68
225 1 DL+ 0.75 WL(-Z)+ 0.75 LL	248.9	-0.6	-22.8	22.8	24.7	351.97	-0.56	-22.80	22.81	24.74
226 1 DL + 0.525 EL(+X)+ 0.75 LL	298.7	-1.1	-25.8	1.6	-17.0	401.74	-1.14	-25.83	1.60	-17.00
227 1 DL + 0.525 EL(-X)+ 0.75 LL	318.5	3.2	-35.1	2.5	-16.9	421.55	3.22	-35.15	2.50	-16.94
228 1 DL + 0.525 EL(+Z)+ 0.75 LL	314.7	1.1	-31.1	0.6	-19.9	417.78	1.09	-31.10	0.58	-19.86
229 1 DL + 0.525 EL(-Z)+ 0.75 LL	300.4	1.0	-29.7	4.0	-13.1	403.47	0.97	-29.68	4.01	-13.12

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Combined Load Cases:	**Load at bottom of foundation**					**Load at bottom of foundation + self weight**				
Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
Allowable Strength Design										
101 1.4 DL	323.8	0.0	-26.2	3.5	-16.5	426.86	0.04	-26.24	3.46	-16.49
102 1.2 (DL + TL1) + 1.6 LL	466.9	4.4	-36.6	-7.6	-3.1	569.98	4.44	-36.60	-7.58	-3.12
103 1.2 (DL + TL2) + 1.6 LL	418.0	-0.1	-58.5	11.7	-47.3	521.03	-0.07	-58.51	11.71	-47.30
104 1.2 DL+ 1.6 WL(+X) + 0.5 LL	166.1	-47.8	75.7	1.0	-8.0	269.17	-47.82	75.70	1.03	-7.99
105 1.2 DL+ 1.6 WL(-X) + 0.5 LL	492.0	49.2	-136.3	4.3	-27.2	595.10	49.23	-136.35	4.34	-27.20
106 1.2 DL+ 1.6 WL(+Z) + 0.5 LL	456.4	4.1	-46.7	-41.6	-106.6	559.43	4.10	-46.72	-41.60	-106.57
107 1.2 DL+ 1.6 WL(-Z) + 0.5 LL	201.8	-2.7	-13.9	47.0	71.4	304.84	-2.69	-13.93	46.96	71.38
108 1.2 DL + 1.0 EL(+X) + 0.5 LL	310.2	-3.4	-21.5	1.8	-17.7	413.27	-3.44	-21.45	1.83	-17.66
109 1.2 DL + 1.0 EL(-X) + 0.5 LL	347.9	4.9	-39.2	3.5	-17.5	451.00	4.86	-39.19	3.54	-17.53
110 1.2 DL + 1.0 EL(+Z) + 0.5 LL	340.8	0.8	-31.5	-0.1	-23.1	443.82	0.80	-31.48	-0.12	-23.09
111 1.2 DL + 1.0 EL(-Z) + 0.5 LL	317.4	0.6	-29.2	5.5	-12.1	420.45	0.61	-29.17	5.49	-12.10
112 0.9 DL + 1.6 WL(+X)	45.2	-48.5	89.2	0.6	-1.0	148.25	-48.50	89.16	0.57	-1.00
113 0.9 DL + 1.6 WL(-X)	371.1	48.5	-122.9	3.9	-20.2	474.18	48.55	-122.89	3.88	-20.20
114 0.9 DL + 1.6 WL(+Z)	335.5	3.4	-33.3	-42.1	-99.6	438.51	3.42	-33.27	-42.05	-99.57
115 0.9 DL + 1.6 WL(-Z)	80.9	-3.4	-0.5	46.5	78.4	183.92	-3.37	-0.47	46.50	78.37
116 0.9 DL + 1.0 EL(+X)	189.3	-4.1	-8.0	1.4	-10.7	292.35	-4.12	-8.00	1.37	-10.66
117 0.9 DL + 1.0 EL(-X)	227.0	4.2	-25.7	3.1	-10.5	330.08	4.18	-25.74	3.08	-10.54
118 0.9 DL + 1.0 EL(+Z)	219.8	0.1	-18.0	-0.6	-16.1	322.90	0.12	-18.02	-0.58	-16.09
119 0.9 DL + 1.0 EL(-Z)	196.5	-0.1	-15.7	5.0	-5.1	299.53	-0.07	-15.71	5.03	-5.10
120 1.2 DL + 0.8 WL(+X)	196.1	-24.2	30.5	2.1	-9.3	299.12	-24.23	30.52	2.14	-9.33
121 1.2 DL + 0.8 WL(-X)	359.0	24.3	-75.5	3.8	-18.9	462.09	24.29	-75.50	3.79	-18.93
122 1.2 DL + 0.8 WL(+Z)	341.2	1.7	-30.7	-19.2	-58.6	444.25	1.73	-30.69	-19.17	-58.62
123 1.2 DL + 0.8 WL(-Z)	341.2	1.7	-30.7	-19.2	-58.6	444.25	1.73	-30.69	-19.17	-58.62

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Check for overturning and sliding:

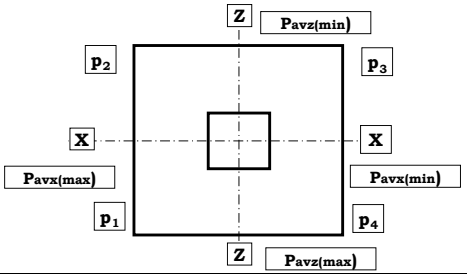
Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	Overturning		Sliding	
						F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
201 1. DL	334.3	0.0	-18.7	2.5	-11.8	17.8	28.4	4776.4	54.1
202 1 (DL + TL1) + 1 LL	457.8	3.3	-25.3	-6.1	-0.3	18.1	1557.2	56.3	29.9
203 1 (DL + TL2) + 1 LL	417.0	-0.5	-43.5	9.9	-37.1	9.6	11.2	329.7	16.8
204 1 DL + 0.75 TL1 + 0.75 LL	426.9	2.4	-23.6	-4.0	-3.2	18.1	134.9	69.8	42.9
205 1 DL + 0.75 TL2 + 0.75 LL	396.3	-0.4	-37.3	8.1	-30.8	10.6	12.9	426.2	19.6
206 1 DL+ 1 WL(+X)	232.5	-30.3	47.5	1.4	-5.8	4.9	40.3	3.1	64.6
207 1 DL+ 1 WL(-X)	436.2	30.4	-85.0	3.5	-17.8	5.1	24.5	5.7	49.8
208 1 DL+ 1 WL(+Z)	413.9	2.2	-29.0	-25.2	-67.4	14.3	6.1	77.0	6.6
209 1 DL+ 1 WL(-Z)	254.8	-2.1	-8.5	30.1	43.8	30.0	5.8	48.6	3.4
210 1 DL + 0.7 EL(+X)	321.1	-2.9	-12.5	1.9	-11.8	25.6	27.2	44.7	68.7
211 1 DL + 0.7 EL(-X)	347.6	2.9	-25.0	3.1	-11.7	13.9	29.6	47.4	45.2
212 1 DL + 0.7 EL(+Z)	342.5	0.1	-19.6	0.5	-15.6	17.5	21.9	1427.2	269.2
213 1 DL + 0.7 EL(-Z)	326.2	0.0	-17.9	4.4	-7.9	18.2	41.1	3345.3	29.4
214 0.6 DL + 1 WL(+X)	140.0	-30.3	55.0	0.5	-1.1	2.5	131.7	1.8	124.4
215 0.6 DL + 1 WL(-X)	343.7	30.3	-77.5	2.5	-13.1	4.4	26.3	4.5	54.6
216 0.6 DL + 1 WL(+Z)	321.4	2.1	-21.5	-26.2	-62.7	15.0	5.1	60.1	4.9
217 0.6 DL + 1 WL(-Z)	162.3	-2.1	-1.0	29.2	48.5	162.8	3.3	30.8	2.2
218 0.6 DL + 0.7 EL(+X)	228.6	-2.9	-5.0	0.9	-7.1	45.4	32.2	31.7	103.7
219 0.6 DL + 0.7 EL(-X)	255.0	2.9	-17.5	2.1	-7.0	14.6	36.3	34.9	49.0
229 1 DL + 0.525 EL(-Z)+ 0.75 LL	403.5	1.0	-29.7	4.0	-13.1	13.6	30.7	166.5	40.2
					Min.=	2.5	3.3	1.8	2.2
						OK	OK	OK	OK

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5.1 DESIGN OF FOOTING TYPE F1



Load Combination & Calculation of Base pressure

Max. Base pressure = p_{max} = (V/(Lx.Lz)) + (6.Mz/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)

Min. Base pressure = p_{min} = (V/(Lx.Lz)) - (6.Mz/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)

Surcharge load = 10 kN/m^2

Net SBC of soil for DL, LL= 450 kN/m^2 For WL/SL case = 562.5 kN/m^2

Gross SBC of soil for DL, LL= 471.6 kN/m^2 For WL/SL case = 584.1 kN/m^2

Allowable Stress Design

Load Combination	Design Load/moments			Base Pressure (kN/m^2)													
	V (kN)	Mz (kN-m)	Mx (kN-m)	V/ (Lx.Lz)	6.Mz/ (Lx)^2.Lz	6.Mx/ Lx.(Lz)^2	p_1 (p_{max})	p_2	p_4	p_3 (p_{min})	(p_{max}-γh)	$p_{avx(max)}$	$p_{avx(min)}$	$p_{avz(max)}$	$p_{avz(min)}$	Remarks
201 1. DL	334.35	-18.74	-11.78	83.59	14.06	8.83	106.47	88.81	78.36	60.70	84.87	97.64	69.53	92.42	74.76	**O.K**
202 1 (DL + TL1) + 1 LL	457.81	-25.28	-0.29	114.45	18.96	0.22	133.63	133.19	95.72	95.27	112.03	133.41	95.50	114.68	114.23	**O.K**
203 1 (DL + TL2) + 1 LL	417.01	-43.54	-37.11	104.25	32.65	27.83	164.74	109.08	99.43	43.77	143.14	136.91	71.60	132.09	76.43	**O.K**
204 1 DL + 0.75 TL1 + 0.75 LL	426.94	-23.64	-3.16	106.74	17.73	2.37	126.84	122.09	91.38	86.63	105.24	124.47	89.01	109.11	104.36	**O.K**
205 1 DL + 0.75 TL2 + 0.75 LL	396.35	-37.34	-30.78	99.09	28.00	23.08	150.17	104.01	94.16	48.00	128.57	127.09	71.08	122.17	76.01	**O.K**
206 1 DL+ 1 WL(+X)	232.49	47.52	-5.77	58.12	35.64	4.33	98.09	89.43	26.81	18.15	76.49	93.76	22.48	62.45	53.79	**O.K**
207 1 DL+ 1 WL(-X)	436.20	-85.01	-17.78	109.05	63.75	13.33	186.14	159.47	58.63	31.96	164.54	172.81	45.30	122.39	95.72	**O.K**
208 1 DL+ 1 WL(+Z)	413.91	-28.99	-67.38	103.48	21.74	50.54	175.76	74.68	132.27	31.20	154.16	125.22	81.74	154.02	52.94	**O.K**
209 1 DL+ 1 WL(-Z)	254.79	-8.49	43.83	63.70	6.37	32.87	102.94	37.19	90.20	24.45	81.34	70.07	57.33	96.57	30.82	**O.K**
210 1 DL + 0.7 EL(+X)	321.14	-12.53	-11.82	80.28	9.40	8.87	98.55	80.82	79.75	62.02	76.95	89.69	70.89	89.15	71.42	**O.K**
211 1 DL + 0.7 EL(-X)	347.55	-24.95	-11.73	86.89	18.71	8.80	114.40	96.80	76.97	59.38	92.80	105.60	68.18	95.69	78.09	**O.K**
212 1 DL + 0.7 EL(+Z)	342.52	-19.55	-15.62	85.63	14.66	11.72	112.01	88.58	82.68	59.25	90.41	100.30	70.97	97.35	73.92	**O.K**
213 1 DL + 0.7 EL(-Z)	326.17	-17.93	-7.93	81.54	13.45	5.95	100.94	89.05	74.04	62.15	79.34	95.00	68.10	87.49	75.60	**O.K**
214 0.6 DL + 1 WL(+X)	139.98	55.02	-1.06	34.99	41.26	0.80	85.00	85.00	0.00	0.00	63.40	85.00	0.00	42.50	42.50	**O.K**
215 0.6 DL + 1 WL(-X)	343.69	-77.51	-13.07	85.92	58.13	9.80	153.85	134.25	37.59	17.99	132.25	144.05	27.79	95.72	76.12	**O.K**
216 0.6 DL + 1 WL(+Z)	321.39	-21.49	-62.67	80.35	16.12	47.00	143.47	49.46	111.23	17.22	121.87	96.47	64.23	127.35	33.34	**O.K**
217 0.6 DL + 1 WL(-Z)	162.27	-1.00	48.54	40.57	0.75	36.41	77.72	4.91	76.23	3.41	56.12	41.32	39.82	76.98	4.16	**O.K**
218 0.6 DL + 0.7 EL(+X)	228.63	-5.04	-7.11	57.16	3.78	5.33	66.27	55.60	58.71	48.05	44.67	60.94	53.38	62.49	51.83	**O.K**
219 0.6 DL + 0.7 EL(-X)	255.04	-17.45	-7.02	63.76	13.09	5.27	82.12	71.58	55.94	45.40	60.52	76.85	50.67	69.03	58.49	**O.K**
229 1 DL + 0.525 EL(-Z)+ 0.75 LL	403.47	-2										123.13	78.61	110.71	91.03	**O.K**

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Strength Design

101 1.4 DL	426.86	-26.24	-16.49	106.72	19.68	12.36	138.76	114.03	99.40	74.67	117.16	126.40	87.04	119.08	94.35	
102 1.2 (DL + TL1) + 1.6 LL	569.98	-36.60	-3.12	142.50	27.45	2.34	172.28	167.60	117.39	112.71	150.68	169.94	115.05	144.84	140.16	
103 1.2 (DL + TL2) + 1.6 LL	521.03	-58.51	-47.30	130.26	43.88	35.48	209.62	138.67	121.85	50.90	188.02	174.15	86.38	165.74	94.79	
104 1.2 DL+ 1.6 WL(+X) + 0.5 LL	269.17	75.70	-7.99	67.29	56.78	5.99	130.06	118.07	16.51	4.52	108.46	124.07	10.52	73.29	61.30	
105 1.2 DL+ 1.6 WL(-X) + 0.5 LL	595.10	-136.35	-27.20	148.78	102.26	20.40	271.43	230.64	66.91	26.12	249.83	251.04	46.52	169.17	128.38	
106 1.2 DL+ 1.6 WL(+Z) + 0.5 LL	559.43	-46.72	-106.57	139.86	35.04	79.92	254.82	94.97	184.74	24.89	233.22	174.90	104.82	219.78	59.93	
107 1.2 DL+ 1.6 WL(-Z) + 0.5 LL	304.84	-13.93	71.38	76.21	10.44	53.53	140.19	33.12	119.30	12.23	118.59	86.66	65.77	129.75	22.68	
108 1.2 DL + 1.0 EL(+X) + 0.5 LL	413.27	-21.45	-17.66	103.32	16.09	13.24	132.65	106.17	100.47	73.98	111.05	119.41	87.23	116.56	90.08	
109 1.2 DL + 1.0 EL(-X) + 0.5 LL	451.00	-39.19	-17.53	112.75	29.39	13.15	155.29	129.00	96.50	70.21	133.69	142.15	83.36	125.90	99.61	
110 1.2 DL + 1.0 EL(+Z) + 0.5 LL	443.82	-31.48	-23.09	110.95	23.61	17.32	151.88	117.25	104.66	70.03	130.28	134.57	87.35	128.27	93.64	
111 1.2 DL + 1.0 EL(-Z) + 0.5 LL	420.45	-29.17	-12.10	105.11	21.87	9.07	136.06	117.91	92.31	74.17	114.46	126.99	83.24	114.19	96.04	
112 0.9 DL + 1.6 WL(+X)	148.25	89.16	-1.00	37.06	66.87	0.75	104.68	103.18	-29.06	-30.55	83.08	103.93	-29.81	37.81	36.32	
113 0.9 DL + 1.6 WL(-X)	474.18	-122.89	-20.20	118.55	92.17	15.15	225.86	195.56	41.53	11.23	204.26	210.71	26.38	133.70	103.40	
114 0.9 DL + 1.6 WL(+Z)	438.51	-33.27	-99.57	109.63	24.95	74.68	209.25	59.90	159.36	10.00	187.65	134.58	84.68	184.31	34.95	
115 0.9 DL + 1.6 WL(-Z)	183.92	-0.47	78.37	45.98	0.35	58.78	105.11	-12.45	104.41	-13.15	83.51	46.33	45.63	104.76	-12.80	
116 0.9 DL + 1.0 EL(+X)	292.35	-8.00	-10.66	73.09	6.00	8.00	87.08	71.09	75.08	59.09	65.48	79.09	67.09	81.08	65.09	
117 0.9 DL + 1.0 EL(-X)	330.08	-25.74	-10.54	82.52	19.30	7.90	109.72	93.92	71.12	55.32	88.12	101.82	63.22	90.42	74.62	
123 1.2 DL + 0.8 WL(-Z)	444.25	-30.69	-58.62	111.06	23.02	43.96	178.04	90.12	132.01	44.08	156.44	134.08	88.05	155.03	67.10	
Max :	595	89	78													
Min :	140	-136	-107													

Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL cases) = 143.14 kN/m^2

Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL cases) = 153.14 kN/m^2

Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL+SL/WL cases) = 164.54 kN/m^2

Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL+WL/SL cases) = 174.54 kN/m^2

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Sample calculation of values of base pressure:

Load case : ↄ2 1 (DL + TL1) + 1 I

Maximum pressure, p_1 (pmax) = 114.45+18.96+0.22= 133.63 kN/m^2

p_2= 114.45+18.96-0.22= 133.19 kN/m^2

p_3= 114.45-(18.96+0.22)= 95.27 kN/m^2

p_4= 114.45-18.96+0.22= 95.72 kN/m^2

Maxm pressure excluding weight of soil (p_{max}-γh)= 133.63-1.2*18= 112.03 kN/m^2

Maximum average Pressure along X, $p_{avx(max)}$ = (133.63+133.19)/2= 133.41 kN/m^2

Minimum average Pressure along X, $p_{avx(minx)}$ = (95.27+95.72)/2= 95.50 kN/m^2

Maximum average Pressure along Y, $p_{avy(max)}$ = (133.63+95.72)/2= 114.67 kN/m^2

Minimum average Pressure along Y, $p_{avy(minx)}$ = (95.27+133.19)/2= 114.23 kN/m^2

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5.1 DESIGN OF FOOTING TYPE F1

Calculation of Bending Moment and Shear Force in X direction :

p_{1x} =pressure at the face of column at section 1x-1x in kN/m^2.

p_{2x} =pressure at the face of column at section 2x-2x in kN/m^2.

p_{3x} =pressure at a distance 'd' from the face of column at section 3x-3x in kN/m^2.

BM_{1x} =Moment at the face of column at section 1x-1x in kN-m/m.

BM_{2x} =Moment at the face of column 2x-2x in kN-m/m.

SF_{3x} = Oneway shear force at a distance 'd' from the face of column at section 3x-3x in kN/m.

SF (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Punching perimeter= ((0.6+2*0.5*0.332)+(0.7+2*0.5*0.332))*2

= 3.928 m

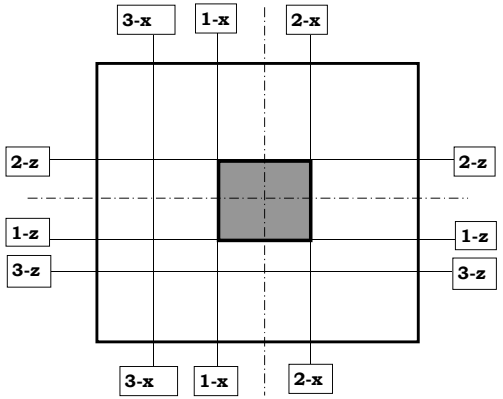
Punching Area= ((0.6+2*0.5*0.332)*(0.7+2*0.5*0.332))

= 0.96 m^2 Punching shear stress to be checked

Downward pressure due to self weight of raft and soil = (0.4*24+0.8*18)

= 24.00 kN/m^2

Area of footing, A_f= 4.00 m^2



Load Combination	p_{1x} kN/m2	p_{2x} kN/m2	p_{3x} kN/m2	$B.M_{1x}$ kNm	$B.M_{2x}$ kNm	$S.F_{3x}$ kN	$S.F_{(punching)}$ kN
Ultimate Conditions							
101 1.4 DL	112.62	100.82	119.39	23.96	16.57	35.21	251.45
102 1.2 (DL + TL1) + 1.6 LL	150.73	134.26	160.17	34.19	23.88	50.22	360.23
103 1.2 (DL + TL2) + 1.6 LL	143.43	117.10	158.53	34.28	17.79	50.67	323.02
104 1.2 DL+ 1.6 WL(+X) + 0.5 LL	84.33	50.26	103.86	21.27	-0.06	32.03	131.61
105 1.2 DL+ 1.6 WL(-X) + 0.5 LL	179.46	118.10	214.64	49.78	11.36	74.35	379.32
106 1.2 DL+ 1.6 WL(+Z) + 0.5 LL	150.37	129.35	162.43	34.97	21.80	51.50	352.21
107 1.2 DL+ 1.6 WL(-Z) + 0.5 LL	79.35	73.08	82.94	14.75	10.83	21.65	158.72
108 1.2 DL + 1.0 EL(+X) + 0.5 LL	108.15	98.49	113.68	22.46	16.41	32.95	241.12
109 1.2 DL + 1.0 EL(-X) + 0.5 LL	121.57	103.94	131.69	27.27	16.22	40.20	269.80
110 1.2 DL + 1.0 EL(+Z) + 0.5 LL	118.04	103.88	126.16	25.74	16.87	37.87	264.34
111 1.2 DL + 1.0 EL(-Z) + 0.5 LL	111.68	98.55	119.20	23.98	15.76	35.28	246.58
112 0.9 DL + 1.6 WL(+X)	57.12	17.00	80.12	15.76	-9.36	24.22	39.71
113 0.9 DL + 1.6 WL(-X)	146.19	90.90	177.90	40.48	5.85	60.63	287.42
114 0.9 DL + 1.6 WL(+Z)	117.12	102.15	125.70	25.67	16.29	37.79	260.31
115 0.9 DL + 1.6 WL(-Z)	46.09	45.88	46.21	5.45	5.32	7.93	66.82
116 0.9 DL + 1.0 EL(+X)	74.89	71.29	76.95	13.15	10.90	19.23	149.23
117 0.9 DL + 1.0 EL(-X)	88.31	76.73	94.95	17.96	10.71	26.48	177.90
123 1.2 DL + 0.8 WL(-Z)	117.97	104.16	125.89	25.65	17.01	37.73	264.67

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Sample calculation of BM & SF

Load case : 101 1.4 DL

p_{1x} = 126.4-(126.4-87.04)*0.7/2 = 112.62 kN/m^2 Overhang = 0.7 m

p_{2x} = 87.04+(126.4-87.04)*0.7/2 = 100.82 kN/m^2 Effective depth = 0.344 m

p_{3x} = 126.4-(126.4-87.04)*(0.7-0.344)/2 = 119.39 kN/m^2

BM_{1x} = (112.62-24)*0.7*0.7/2+0.5*(126.4-112.62)*0.7*0.7*0.67 = 23.96 kN-m/m

BM_{2x} = (100.82-24)*0.7*0.7/2-0.5*(-87.04+100.82)*0.7*0.7*0.67 = 16.57 kN-m/m

SF_{3x} = (119.39-24)*(0.7-0.344)+0.5*(126.4-119.39)*(0.7-0.344) = 35.21 kN/m

$S.F_{(punching)}$ = (4-0.96)*(106.72-24) = 251.45 kN

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	49.78	0.42	0.56	0.001	463.80	309.60	12ϕT	243.85	300	150	753.98
At Top	-9.36	-0.08	0.11	0.000	86.54	309.60	12ϕT	365.30	300	250	452.39

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc)/bd = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 74.35 kN/m

Shear stress , τ_v= (74.35*1000)/(1000*344)

= 0.22 N/mm^2 **O.K.**

Check against Two-way shear force at a distance d/2 from face of column:

Design punching shear at a distance d/2 from face of column, Vu= 379.32 kN

Strength of Concrete in shear Vc1 = 4441.11 kN **Table 22.6.5.2_ (a)**

Strength of Concrete in shear Vc2 = 2499.14 kN **Table 22.6.5.2_ (b)**

Strength of Concrete in shear Vc3 = 2664.67 kN **Table 22.6.5.2_ (c)**

Design Vc= 2499.14 kN

Allowable shear stress, (ϕ Vc)/bd = 1.39 N/mm^2

Design punching shear stress, τ = (379.32*1000)/(3928*344)

= 0.28 N/mm^2 **No shear reinforcement reqd.**

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Calculation of Bending Moment and Shear Force in Z direction :

p1z =pressure at the face of column at section 1z-1z in kN/m^2.
p2z =pressure at the face of column at section 2z-2z in kN/m^2.
p3z =pressure at a distance d from the face of column at section 3z-3z in kN/m^2.
B.M1z =Moment at the face of column at section 1z-1z in kN-m/m.
B.M2z =Moment at the face of column at section 2z-2z in kN-m/m.
S.F 3z = Oneway Shear Force at a distance d from the face of column at section 3z-3z in kN/m.
S.F (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Load Combination	p_{1z} kN/m2	p_{2z} kN/m2	p_{3z} kN/m2	$B.M_{1z}$ kNm	$B.M_{2z}$ kNm	$S.F_{3z}$ kN
Ultimate Conditions						
101 1.4 DL	111.04	102.39	115.15	19.53	15.42	29.61
102 1.2 (DL + TL1) + 1.6 LL	143.32	141.68	144.10	25.42	24.64	38.31
103 1.2 (DL + TL2) + 1.6 LL	142.68	117.85	154.46	28.34	16.56	43.28
104 1.2 DL+ 1.6 WL(+X) + 0.5 LL	69.39	65.20	71.38	10.14	8.15	15.37
105 1.2 DL+ 1.6 WL(-X) + 0.5 LL	155.91	141.64	162.68	29.74	22.97	45.13
106 1.2 DL+ 1.6 WL(+Z) + 0.5 LL	167.83	111.88	194.36	37.74	11.21	58.22
107 1.2 DL+ 1.6 WL(-Z) + 0.5 LL	94.95	57.48	112.73	19.91	2.15	30.92
108 1.2 DL + 1.0 EL(+X) + 0.5 LL	107.95	98.69	112.35	18.95	14.56	28.76
109 1.2 DL + 1.0 EL(-X) + 0.5 LL	117.36	108.15	121.72	20.93	16.57	31.74
110 1.2 DL + 1.0 EL(+Z) + 0.5 LL	117.02	104.89	122.76	21.24	15.50	32.28
111 1.2 DL + 1.0 EL(-Z) + 0.5 LL	108.29	101.94	111.30	18.64	15.63	28.22
112 0.9 DL + 1.6 WL(+X)	37.33	36.80	37.57	2.88	2.64	4.35
113 0.9 DL + 1.6 WL(-X)	123.85	113.25	128.88	22.49	17.46	34.12
114 0.9 DL + 1.6 WL(+Z)	135.77	83.49	160.56	30.48	5.70	47.20
115 0.9 DL + 1.6 WL(-Z)	66.55	25.41	86.07	14.40	-5.11	22.71
116 0.9 DL + 1.0 EL(+X)	75.88	70.29	78.54	11.70	9.04	17.75
117 0.9 DL + 1.0 EL(-X)	85.29	79.76	87.91	13.67	11.05	20.72
123 1.2 DL + 0.8 WL(-Z)	126.45	95.68	141.05	25.69	11.10	39.44

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Sample calculation of BM & SF

Load case : 101 1.4 DL

p_{1z} = 119.08-(119.08-94.35)*0.65/2= 111.04 kN/m^2

p_{2z} = 94.35+(119.08-94.35)*0.65/2= 102.39 kN/m^2

p_{3z} = 119.08-(119.08-94.35)*(0.65-0.332)/2= 115.15 kN/m^2

Overhang = 0.65 m

Effective depth = 0.332 m

BM_{1z} = (111.04-24)*0.65*0.65/2+0.5*(119.08-111.04)*0.65*0.65*0.67 = 19.53 kN-m/m

BM_{2z} = (102.39-24)*0.65*0.65/2-0.5*(-94.35+102.39)*0.65*0.65*0.67 = 15.42 kN-m/m

SF_{3z} = (115.15-24)*(0.65-0.332)+0.5*(119.08-115.15)*(0.65-0.332) = 29.61 kN/m

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	37.74	0.342	0.456	0.0011	376.80	298.80	12ϕT	300.15	300	150	754.0
Top	-5.11	-0.046	-0.062	-0.0001	-0.49	298.80	12ϕT	378.51	300	250	452.4

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc) = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 58.22 kN/m

Shear stress , τ_v= (58.22*1000)/(1000*332)

= 0.18 N/mm^2 **O.K.**

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :			
At Bottom	**12ϕT**	**150**	c/c
At Top	**12ϕT**	**250**	c/c
Along Z Direction :			
At Bottom	**12ϕT**	**150**	c/c
At Top	**12ϕT**	**250**	c/c

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5.2 DESIGN OF FOOTING TYPE F2

Size of Columns at node	Along-X	Along-Z
1	0.60m	0.60m
2	0.60m	0.60m
Size of fdn. raft:	7.10m	1.80m
Thickness of raft:	0.60	m
Original Gr. Level:	0.00	m
Finished Gr. Level:	0.00	m
U/S of B/Pl Above GL:	0.84	m
U/S level of fdn:	-1.20	m
Unit weight of conc:	25	kN/m^3
Unit weight of soil:	18	kN/m^3

Design Input:

Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	50mm

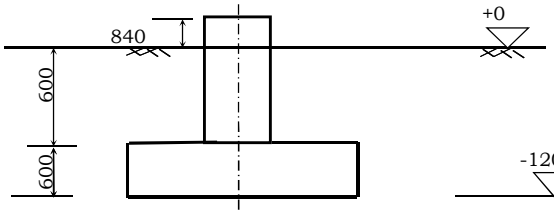
Calculation of Weight of Foundation :

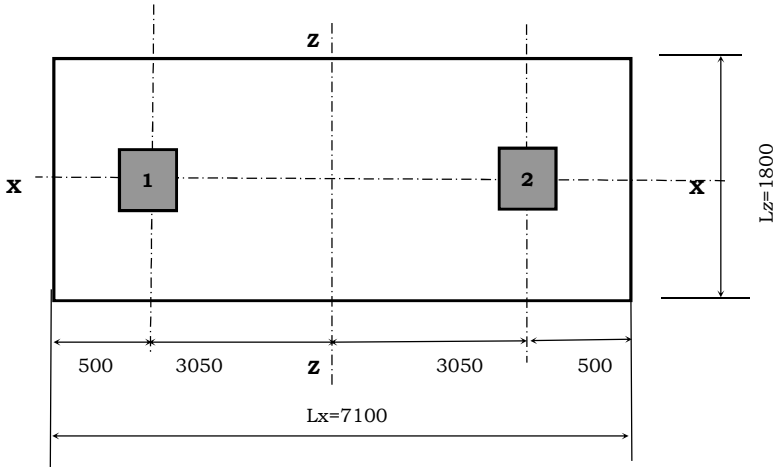
Weight of raft = (7.1*1.8*0.6*25) = 191.70KN

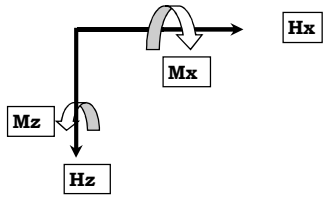
Weight of soil = ((7.1*1.8)-2*(0.6*0.6))*0.6*18 = 130.25KN

Total = 321.95KN

Submerged wt. = 321.95-7.1*1.8*1.2*10 = 168.59KN







H_X=Force along X-X direction

Hz=Force along Z-Z direction

M_X=Moment about X-X direction

Mz=Moment about Z-Z direction

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Check for overturning and sliding of footing F1

Load at bottom of fooring	Node 1					Node 2				
L/C No.	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
201 1. DL	172.2	7.8	-16.0	-2.0	-4.0	148.7	-10.8	22.0	-5.9	-12.0
202 1 (DL) + 1 LL	379.2	23.5	-48.0	-2.9	-6.0	275.2	-30.4	62.0	-13.7	-28.0
203 1 (DL) + 1 LL	379.2	23.5	-48.0	-2.9	-6.0	275.2	-30.4	62.0	-13.7	-28.0
204 1 DL+ 1 WL(+X)	44.8	-12.7	26.0	-3.9	-8.0	276.2	-98.1	200.1	-3.9	-8.0
205 1 DL+ 1 WL(-X)	299.7	28.4	-58.0	-3.9	-8.0	21.2	76.5	-156.0	-3.9	-8.0
206 1 DL+ 1 WL(+Z)	248.7	7.8	-16.0	-40.2	-82.0	208.5	-10.8	22.0	-31.4	-64.0
207 1 DL+ 1 WL(-Z)	95.8	7.8	-16.0	36.3	74.0	88.9	-10.8	22.0	19.6	40.0
208 0.6 DL + 1 WL(+X)	-24.1	-15.9	32.4	-3.1	-6.4	216.7	-93.8	191.3	-1.6	-3.2
209 0.6 DL + 1 WL(-X)	230.8	25.3	-51.6	-3.1	-6.4	-38.3	80.8	-164.8	-1.6	-3.2
210 0.6 DL + 1 WL(+Z)	179.8	4.7	-9.6	-39.4	-80.4	149.0	-6.5	13.2	-29.0	-59.2
211 0.6 DL + 1 WL(-Z)	26.9	4.7	-9.6	37.1	75.6	29.4	-6.5	13.2	22.0	44.8
212 1 DL+ 0.75 WL(+X)+ 0.75 LL	231.8	4.2	-8.5	-4.2	-8.5	339.2	-91.0	185.6	-10.3	-21.0
213 1 DL+ 0.75 WL(-X)+ 0.75 LL	423.1	35.1	-71.5	-4.2	-8.5	148.0	40.0	-81.5	-10.3	-21.0
214 1 DL+ 0.75 WL(+Z)+ 0.75 LL	384.8	19.6	-40.0	-31.4	-64.0	288.5	-25.5	52.0	-30.9	-63.0
215 1 DL+ 0.75 WL(-Z)+ 0.75 LL	270.1	19.6	-40.0	26.0	53.0	198.7	-25.5	52.0	7.4	15.0

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Check for overturning and sliding:

L/C No.	Resultant Load at bottom of footing								
	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	Overturning		Sliding	
						F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
201 1. DL	642.9	-2.94	-65.8	-7.8	-16.0	34.7	36.2	87.4	32.8
202 1 (DL) + 1 LL	976.3	-6.87	-303.0	-16.7	-34.0	11.4	25.8	56.9	23.4
203 1 (DL) + 1 LL	976.3	-6.87	-303.0	-16.7	-34.0	11.4	25.8	56.9	23.4
204 1 DL+ 1 WL(+X)	642.9	-110.82	931.9	-7.8	-16.0	2.4	36.2	2.3	32.8
205 1 DL+ 1 WL(-X)	642.9	104.93	-1063.5	-7.8	-16.0	2.1	36.2	2.5	32.8
206 1 DL+ 1 WL(+Z)	779.2	-2.94	-116.6	-71.6	-146.0	23.7	4.8	105.9	4.4
207 1 DL+ 1 WL(-Z)	506.6	-2.94	-14.9	55.9	114.0	120.4	4.0	68.9	3.6
208 0.6 DL + 1 WL(+X)	514.5	-109.64	958.3	-4.7	-9.6	1.9	48.2	1.9	43.7
209 0.6 DL + 1 WL(-X)	514.5	106.11	-1037.2	-4.7	-9.6	1.8	48.2	1.9	43.7
210 0.6 DL + 1 WL(+Z)	650.8	-1.77	-90.3	-68.5	-139.6	25.6	4.2	147.5	3.8
211 0.6 DL + 1 WL(-Z)	378.2	-1.77	11.4	59.0	120.4	118.0	2.8	85.7	2.6
212 1 DL+ 0.75 WL(+X)+ 0.75 LL	893.0	-86.79	504.6	-14.5	-29.5	6.3	27.2	4.1	24.7
213 1 DL+ 0.75 WL(-X)+ 0.75 LL	893.0	75.02	-992.0	-14.5	-29.5	3.2	27.2	4.8	24.7
214 1 DL+ 0.75 WL(+Z)+ 0.75 LL	995.2	-5.88	-281.9	-62.3	-127.0	12.5	7.1	67.7	6.4
215 1 DL+ 0.75 WL(-Z)+ 0.75 LL	790.7	-5.88	-205.6	33.3	68.0	13.7	10.5	53.8	9.5
					Min.=	**1.8**	**2.8**	**1.9**	**2.6**

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Base pressure check of footing F1

	Node	L/C	Horizontal Fx N/mm2	Vertical Fy N/mm2	Horizontal Fz N/mm2
Max Px	164	201	0	0.08	0
Min Px	164	201	0	0.08	0
Max Py	483	214	0	0.21	0
Min Py	777	209	0	-0.05	0
Max Pz	164	201	0	0.08	0
Min Pz	164	201	0	0.08	0

Gross Maximum Base pressure, p_{max} = 209.0 KN/m^2

Net Allowable Safe Bearing Capacity = 450 KN/m^2
Depth of soil above footing = 0.7 m
Unit weight of soil = 18 KN/m^3

Gross Safe Bearing Capacity = 462.6 KN/m^2
> 209.0 KN/m^2
Hence OK

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Design of foundation raft F1

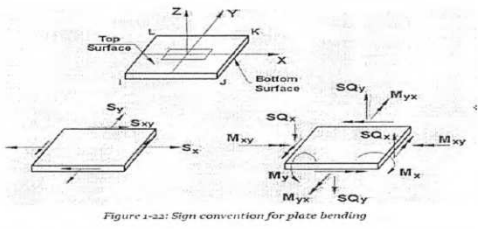
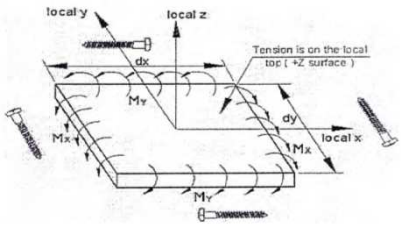


Figure 1-22: Sign convention for plate bending



Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:
PLATE CENTER STRESS : FDN_F1_MOMENT At the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	859	0.405	-0.117	0	0	0	-82.4	17.935	16.279	-98.677	34.214	104 1.2 DL+ 1.6 WL(+X) + 0.5 LL
Min Qx	809	-0.354	-0.012	0	0	0	-59.87	39.752	0	-59.867	39.752	106 1.2 DL+ 1.6 WL(+Z) + 0.5 LL
Max Qy	777	0.029	0.564	0	0	0	-7.247	49.653	-26.219	-33.466	75.872	104 1.2 DL+ 1.6 WL(+X) + 0.5 LL
Min Qy	861	0.012	-0.57	0	0	0	-6.194	54.321	25.977	-32.171	80.298	104 1.2 DL+ 1.6 WL(+X) + 0.5 LL
Max Sx	554	-0.025	0.045	0	0	0	-0.95	2.795	0.298	-1.248	3.093	101 1.4 DL
Min Sx	554	-0.025	0.045	0	0	0	-0.95	2.795	0.298	-1.248	3.093	101 1.4 DL
Max Sy	554	-0.025	0.045	0	0	0	-0.95	2.795	0.298	-1.248	3.093	101 1.4 DL
Min Sy	554	-0.025	0.045	0	0	0	-0.95	2.795	0.298	-1.248	3.093	101 1.4 DL
Max Sxy	554	-0.025	0.045	0	0	0	-0.95	2.795	0.298	-1.248	3.093	101 1.4 DL
Min Sxy	554	-0.025	0.045	0	0	0	-0.95	2.795	0.298	-1.248	3.093	101 1.4 DL
Max Mx	830	0.218	-0.142	0	0	0	170.23	4.357	-20.893	191.126	25.25	109 0.9 DL + 1.6 WL(-X)
Min Mx	802	0.02	-0.124	0	0	0	-115	29.779	-34.784	-149.784	64.563	104 1.2 DL+ 1.6 WL(+X) + 0.5 LL
Max My	833	-0.304	-0.357	0	0	0	28.099	91.118	24.26	52.359	115.378	104 1.2 DL+ 1.6 WL(+X) + 0.5 LL
Min My	805	0.187	-0.242	0	0	0	-20.91	-32.876	22.483	-43.39	-55.359	109 0.9 DL + 1.6 WL(-X)
Max Mxy	753	-0.105	0.014	0	0	0	-67.39	-1.858	42.059	-109.45	-43.917	106 1.2 DL+ 1.6 WL(+Z) + 0.5 LL
Min Mxy	669	-0.062	0.031	0	0	0	-78.49	-7.395	-47.417	-125.908	-54.812	107 1.2 DL+ 1.6 WL(-Z) + 0.5 LL

PLATE CENTER STRESS : FDN_F1_SHEAR At a distance "d" from the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	859	0.405	-0.117	0	0	0	-82.4	17.935	16.279	-98.677	34.214	104 1.2 DL+ 1.6 WL(+X) + 0.5 LL
Min Qx	809	-0.354	-0.012	0	0	0	-59.87	39.752	0	-59.867	39.752	106 1.2 DL+ 1.6 WL(+Z) + 0.5 LL
Max Qy	777	0.029	0.564	0	0	0	-7.247	49.653	-26.219	-33.466	75.872	104 1.2 DL+ 1.6 WL(+X) + 0.5 LL
Min Qy	861	0.012	-0.57	0	0	0	-6.194	54.321	25.977	-32.171	80.298	104 1.2 DL+ 1.6 WL(+X) + 0.5 LL
Max Sx	554	-0.025	0.045	0	0	0	-0.95	2.795	0.298	-1.248	3.093	101 1.4 DL
Min Sx	554	-0.025	0.045	0	0	0	-0.95	2.795	0.298	-1.248	3.093	101 1.4 DL
Max Sy	554	-0.025	0.045	0	0	0	-0.95	2.795	0.298	-1.248	3.093	101 1.4 DL
Min Sy	554	-0.025	0.045	0	0	0	-0.95	2.795	0.298	-1.248	3.093	101 1.4 DL
Max Sxy	554	-0.025	0.045	0	0	0	-0.95	2.795	0.298	-1.248	3.093	101 1.4 DL
Min Sxy	554	-0.025	0.045	0	0	0	-0.95	2.795	0.298	-1.248	3.093	101 1.4 DL
Max Mx	830	0.218	-0.142	0	0	0	170.23	4.357	-20.893	191.126	25.25	109 0.9 DL + 1.6 WL(-X)
Min Mx	802	0.02	-0.124	0	0	0	-254.4	29.779	-34.784	-289.196	64.563	104 1.2 DL+ 1.6 WL(+X) + 0.5 LL
Max My	833	-0.304	-0.357	0	0	0	28.099	91.118	24.26	52.359	115.378	104 1.2 DL+ 1.6 WL(+X) + 0.5 LL
Min My	805	0.187	-0.242	0	0	0	-20.91	-32.876	22.483	-43.39	-55.359	109 0.9 DL + 1.6 WL(-X)
Max Mxy	753	-0.105	0.014	0	0	0	-67.39	-1.858	42.059	-109.45	-43.917	106 1.2 DL+ 1.6 WL(+Z) + 0.5 LL
Min Mxy	669	-0.062	0.031	0	0	0	-78.49	-7.395	-47.417	-125.908	-54.812	107 1.2 DL+ 1.6 WL(-Z) + 0.5 LL

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	50	mm
	top face	50	mm
Thicknes of slab =		600	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

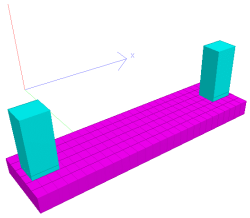
Force Values: Load condition	Factored Mt, Mx	Tension face due to bending	Depth: clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
109 0.9 D	191.13	top face	50	542	191.13	0.87	0.0021	1136.3	487.8	16	177.02	150	1341	18.91	23.64	0.066	tension control
104 1.2 D	-149.78	bottom face	50	542	149.78	0.68	0.0016	887.6		16	226.62	200	1005.8	14.18	17.73	0.089	tension control
106 1.2 D	-109.45	bottom face	50	542	109.45	0.50	0.0012	646.5		16	311.12	150	1341	18.91	23.64	0.066	tension control
107 1.2 D	-125.91	bottom face	50	542	125.91	0.57	0.0014	744.7		16	270.1	150	1341	18.91	23.64	0.066	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	φ(V_c +0.66√ f_c'bd)≥V_u		Rema rks	φV_c	Remar ks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qx	0.405	1	243	98.677	0.002	0.542	543.47	535.839	929.889	535.84	1989	> 243	ok	401.9	φVc>Vu	not required
	Min Qx	0.354	1	212.4	59.867			556.88			535.84	1989	> 212.4	ok	401.9	φVc>Vu	not required

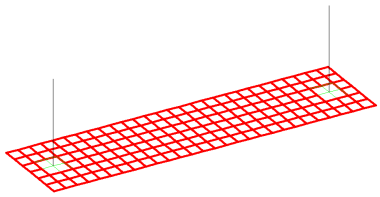
Reinforcement Along Z:

Force Values: Load condition	Factored Mt, My	Tension face due to bending	Depth: clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
104 1.2 D	115.38	top face	50	544	115.38	0.52	0.0012	679.3	489.6	12	166.56	150	754.3	10.64	13.29	0.12	tension control
109 0.9 D	-55.36	bottom face	50	544	55.359	0.25	0.0006	324.4		12	231.95	200	565.8	7.978	9.972	0.161	tension control
106 1.2 D	-43.92	bottom face	50	542	43.917	0.20	0.0005	258.1		16	412.35	150	1341	18.91	23.64	0.066	tension control
107 1.2 D	-54.81	top face	50	542	54.812	0.25	0.0006	322.4		16	412.35	150	1341	18.91	23.64	0.066	tension control

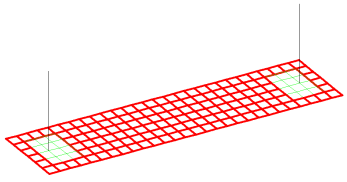
Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	φ(V_c +0.66√ f_c'bd)≥V_u		Rema rks	φV_c	Remar ks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	104 1.2 D	0.564	1	338.4	75.872	0.001	0.544	568.152	535.839	fc'929.9	535.84	561	> 338.4	ok	401.9	φVc>Vu	not required
	104 1.2 D	0.57	1	342	80.298			565.668			535.84	561	> 342	ok	401.9	φVc>Vu	not required



3D rendered view



Highlighted plate for Moment at face of column



Highlighted plate for Shear at 'd' distance from face of column

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :			
At Bottom	16ϕT	200	c/c
At Top	16ϕT	150	c/c
Along Z Direction :			
At Bottom	12ϕT	200	c/c
At Top	12ϕT	150	c/c

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5.3 DESIGN OF FOOTING TYPE F5

Size of Columns at node	Along-X	Along-Z
1	0.45m	0.55m
2	0.45m	0.55m
Size of fdn. raft:	5.00m	1.80m
Thickness of raft:	0.40	m
Original Gr. Level:	0.00	m
Finished Gr. Level:	0.00	m
U/S of B/Pl Above GL:	0.84	m
U/S level of fdn:	-1.20	m
Unit weight of conc:	25	kN/m^3
Unit weight of soil:	18	kN/m^3

Design Input:

Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	50mm

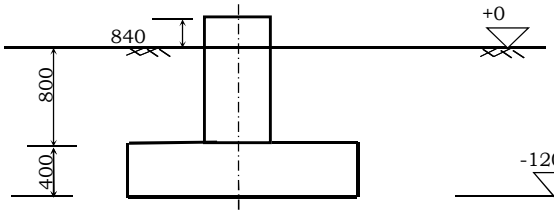
Calculation of Weight of Foundation :

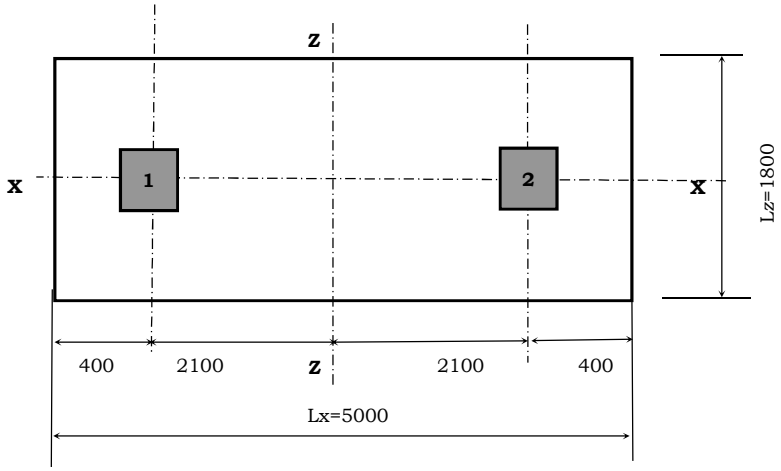
Weight of raft = (5*1.8*0.4*25) = 90.00KN

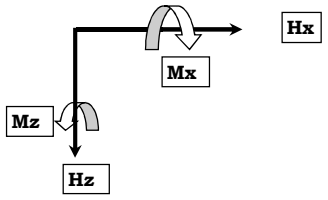
Weight of soil = ((5*1.8)-2*(0.45*0.55))*0.8*18 = 122.47KN

Total = 212.47KN

Submerged wt. = 212.47-5*1.8*1.2*10 = 104.47KN







H_X=Force along X-X direction

Hz=Force along Z-Z direction

M_X=Moment about X-X direction

Mz=Moment about Z-Z direction

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Check for overturning and sliding of footing F1

Load at bottom of fooring	Node 1					Node 2				
L/C No.	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
201 1. DL	84.0	0.0	0.0	0.0	0.0	84.0	0.0	0.0	0.0	0.0
202 1 (DL) + 1 LL	131.0	0.0	0.0	0.0	0.0	130.1	0.0	0.0	0.0	0.0
203 1 (DL) + 1 LL	131.0	0.0	0.0	0.0	0.0	130.1	0.0	0.0	0.0	0.0
204 1 DL+ 1 WL(+X)	62.4	-11.8	19.1	0.0	0.0	105.6	-11.8	19.1	0.0	0.0
205 1 DL+ 1 WL(-X)	105.6	11.8	-19.1	0.0	0.0	62.4	11.8	-19.1	0.0	0.0
206 1 DL+ 1 WL(+Z)	84.0	0.0	0.0	0.0	0.0	84.0	0.0	0.0	0.0	0.0
207 1 DL+ 1 WL(-Z)	84.0	0.0	0.0	0.0	0.0	84.0	0.0	0.0	0.0	0.0
208 0.6 DL + 1 WL(+X)	28.8	-11.8	19.1	0.0	0.0	72.0	-11.8	19.1	0.0	0.0
209 0.6 DL + 1 WL(-X)	72.0	11.8	-19.1	0.0	0.0	28.8	11.8	-19.1	0.0	0.0
210 0.6 DL + 1 WL(+Z)	50.4	0.0	0.0	0.0	0.0	50.4	0.0	0.0	0.0	0.0
211 0.6 DL + 1 WL(-Z)	50.4	0.0	0.0	0.0	0.0	50.4	0.0	0.0	0.0	0.0
212 1 DL+ 0.75 WL(+X)+ 0.75 LL	103.1	-8.8	14.3	0.0	0.0	134.7	-8.8	14.3	0.0	0.0
213 1 DL+ 0.75 WL(-X)+ 0.75 LL	135.5	8.8	-14.3	0.0	0.0	102.4	8.8	-14.3	0.0	0.0
214 1 DL+ 0.75 WL(+Z)+ 0.75 LL	119.3	0.0	0.0	0.0	0.0	118.5	0.0	0.0	0.0	0.0
215 1 DL+ 0.75 WL(-Z)+ 0.75 LL	119.3	0.0	0.0	0.0	0.0	118.5	0.0	0.0	0.0	0.0

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Check for overturning and sliding:

L/C No.	Resultant Load at bottom of footing								
	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	Overturning		Sliding	
						F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
201 1. DL	380.4	0.00	0.0	0.0	0.0	NA	NA	NA	NA
202 1 (DL) + 1 LL	473.6	0.00	-2.1	0.0	0.0	575.3	NA	NA	NA
203 1 (DL) + 1 LL	473.6	0.00	-2.1	0.0	0.0	575.3	NA	NA	NA
204 1 DL+ 1 WL(+X)	380.4	-23.54	128.7	0.0	0.0	7.4	NA	6.5	NA
205 1 DL+ 1 WL(-X)	380.4	23.54	-128.7	0.0	0.0	7.4	NA	6.5	NA
206 1 DL+ 1 WL(+Z)	380.4	0.00	0.0	0.0	0.0	NA	NA	NA	NA
207 1 DL+ 1 WL(-Z)	380.4	0.00	0.0	0.0	0.0	NA	NA	NA	NA
208 0.6 DL + 1 WL(+X)	313.2	-23.54	128.7	0.0	0.0	6.1	NA	5.3	NA
209 0.6 DL + 1 WL(-X)	313.2	23.54	-128.7	0.0	0.0	6.1	NA	5.3	NA
210 0.6 DL + 1 WL(+Z)	313.2	0.00	0.0	0.0	0.0	NA	NA	NA	NA
211 0.6 DL + 1 WL(-Z)	313.2	0.00	0.0	0.0	0.0	NA	NA	NA	NA
212 1 DL+ 0.75 WL(+X)+ 0.75 LL	450.3	-17.65	95.0	0.0	0.0	11.8	NA	10.2	NA
213 1 DL+ 0.75 WL(-X)+ 0.75 LL	450.3	17.65	-98.1	0.0	0.0	11.5	NA	10.2	NA
214 1 DL+ 0.75 WL(+Z)+ 0.75 LL	450.3	0.00	-1.5	0.0	0.0	729.3	NA	NA	NA
215 1 DL+ 0.75 WL(-Z)+ 0.75 LL	450.3	0.00	-1.5	0.0	0.0	729.3	NA	NA	NA
					Min.=	**6.1**	**0.0**	**5.3**	**0.0**

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Base pressure check of footing F1

	Node	L/C	Horizontal Fx N/mm2	Vertical Fy N/mm2	Horizontal Fz N/mm2
Max Px	475	201	0	0.05	0
Min Px	475	201	0	0.05	0
Max Py	952	213	0	0.08	0
Min Py	486	208	0	0.02	0
Max Pz	475	201	0	0.05	0
Min Pz	475	201	0	0.05	0

Gross Maximum Base pressure, p_{max} = 83.0 KN/m^2

Net Allowable Safe Bearing Capacity = 450 KN/m^2
Depth of soil above footing = 0.7 m
Unit weight of soil = 18 KN/m^3

Gross Safe Bearing Capacity = 462.6 KN/m^2
> 83.0 KN/m^2
Hence OK

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Design of foundation raft F1

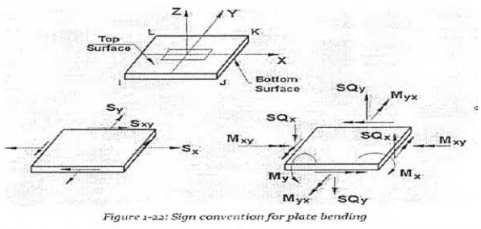
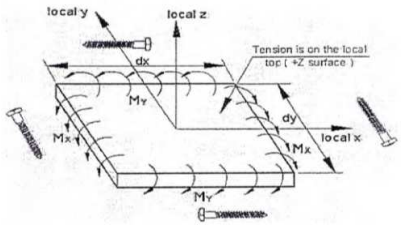


Figure 1-22: Sign convention for plate bending



Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:
PLATE CENTER STRESS : FDN_F1_MOMENT At the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	947	0.348	0.308	0	0	0	6.887	24.19	-1.709	8.596	25.899	102 1.2 (DL) + 1.6 LL
Min Qx	932	-0.351	0.311	0	0	0	7	24.396	0	7	24.396	102 1.2 (DL) + 1.6 LL
Max Qy	931	0.327	0.394	0	0	0	24.922	29.101	10.427	35.349	39.528	105 1.2 DL+ 1.6 WL(-X) + 0.5 LL
Min Qy	951	0.327	-0.394	0	0	0	24.922	29.101	-10.427	35.349	39.528	105 1.2 DL+ 1.6 WL(-X) + 0.5 LL
Max Sx	890	-0.002	0.019	0	0	0	0.399	1.012	0.022	0.421	1.034	101 1.4 DL
Min Sx	890	-0.002	0.019	0	0	0	0.399	1.012	0.022	0.421	1.034	101 1.4 DL
Max Sy	890	-0.002	0.019	0	0	0	0.399	1.012	0.022	0.421	1.034	101 1.4 DL
Min Sy	890	-0.002	0.019	0	0	0	0.399	1.012	0.022	0.421	1.034	101 1.4 DL
Max Sxy	890	-0.002	0.019	0	0	0	0.399	1.012	0.022	0.421	1.034	101 1.4 DL
Min Sxy	890	-0.002	0.019	0	0	0	0.399	1.012	0.022	0.421	1.034	101 1.4 DL
Max Mx	932	-0.302	0.269	0	0	0	31.646	16.149	-9.655	41.301	25.804	104 1.2 DL+ 1.6 WL(+X) + 0.5 LL
Min Mx	939	-0.007	0	0	0	0	-49.65	0.613	0.04	-49.685	0.653	102 1.2 (DL) + 1.6 LL
Max My	931	0.327	0.394	0	0	0	24.922	29.101	10.427	35.349	39.528	105 1.2 DL+ 1.6 WL(-X) + 0.5 LL
Min My	906	0.02	0.01	0	0	0	5.603	0.016	0.271	5.874	0.287	109 0.9 DL + 1.6 WL(-X)
Max Mxy	932	-0.193	0.17	0	0	0	-21.78	18.3	12.092	-33.875	30.392	105 1.2 DL+ 1.6 WL(-X) + 0.5 LL
Min Mxy	952	-0.193	-0.17	0	0	0	-21.78	18.3	-12.092	-33.875	30.392	105 1.2 DL+ 1.6 WL(-X) + 0.5 LL

PLATE CENTER STRESS : FDN_F1_SHEAR At a distance "d" from the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	947	0.348	0.308	0	0	0	6.887	24.19	-1.709	8.596	25.899	102 1.2 (DL) + 1.6 LL
Min Qx	932	-0.351	0.311	0	0	0	7	24.396	0	7	24.396	102 1.2 (DL) + 1.6 LL
Max Qy	931	0.327	0.394	0	0	0	24.922	29.101	10.427	35.349	39.528	105 1.2 DL+ 1.6 WL(-X) + 0.5 LL
Min Qy	951	0.327	-0.394	0	0	0	24.922	29.101	-10.427	35.349	39.528	105 1.2 DL+ 1.6 WL(-X) + 0.5 LL
Max Sx	890	-0.002	0.019	0	0	0	0.399	1.012	0.022	0.421	1.034	101 1.4 DL
Min Sx	890	-0.002	0.019	0	0	0	0.399	1.012	0.022	0.421	1.034	101 1.4 DL
Max Sy	890	-0.002	0.019	0	0	0	0.399	1.012	0.022	0.421	1.034	101 1.4 DL
Min Sy	890	-0.002	0.019	0	0	0	0.399	1.012	0.022	0.421	1.034	101 1.4 DL
Max Sxy	890	-0.002	0.019	0	0	0	0.399	1.012	0.022	0.421	1.034	101 1.4 DL
Min Sxy	890	-0.002	0.019	0	0	0	0.399	1.012	0.022	0.421	1.034	101 1.4 DL
Max Mx	932	-0.302	0.269	0	0	0	31.646	16.149	-9.655	41.301	25.804	104 1.2 DL+ 1.6 WL(+X) + 0.5 LL
Min Mx	939	-0.007	0	0	0	0	-49.65	0.613	0.04	-49.685	0.653	102 1.2 (DL) + 1.6 LL
Max My	931	0.327	0.394	0	0	0	24.922	29.101	10.427	35.349	39.528	105 1.2 DL+ 1.6 WL(-X) + 0.5 LL
Min My	906	0.02	0.01	0	0	0	5.603	0.016	0.271	5.874	0.287	109 0.9 DL + 1.6 WL(-X)
Max Mxy	932	-0.193	0.17	0	0	0	-21.78	18.3	12.092	-33.875	30.392	105 1.2 DL+ 1.6 WL(-X) + 0.5 LL
Min Mxy	952	-0.193	-0.17	0	0	0	-21.78	18.3	-12.092	-33.875	30.392	105 1.2 DL+ 1.6 WL(-X) + 0.5 LL

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	50	mm
	top face	50	mm
Thicknes of slab =		400	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

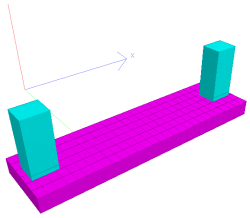
Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
104 1.2 D	41.30	top face	50	344	41.301	0.47	0.0011	384.2	309.6	12	294.51	150	754.3	10.64	13.29	0.075	tension control
102 1.2 (	-49.69	bottom face	50	344	49.685	0.56	0.0013	462.9		12	244.41	200	565.8	7.978	9.972	0.1	tension control
105 1.2 D	-33.88	top face	50	344	33.875	0.38	0.0009	314.6		12	359.59	150	754.3	10.64	13.29	0.075	tension control
105 1.2 D	-33.88	top face	50	344	33.875	0.38	0.0009	314.6		12	359.59	150	754.3	10.64	13.29	0.075	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c + 0.66\sqrt{f_c'}bd) \geq V_u$		Rema	φV_c	Remar	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN		rks	KN	ks	
Load condition	Max Qx	0.348	1	139.2	8.596	0.002	0.344	397.053	338.444	590.188	338.44	1261	> 139.2	ok	253.8	φVc>Vu	not required
	Min Qx	0.351	1	140.4	7			414.096			338.44	1261	> 140.4	ok	253.8	φVc>Vu	not required

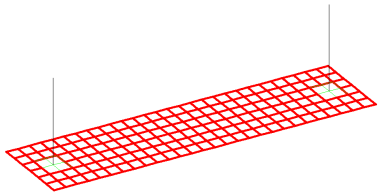
Reinforcement Along Z:

Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, My	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
105 1.2 D	39.53	top face	50	344	39.528	0.45	0.0011	367.6	309.6	12	307.82	250	452.6	6.382	7.977	0.126	tension control
109 0.9 D	0.29	bottom face	50	344	0.287	0.00	0.0000	2.6		12	365.45	200	565.8	7.978	9.972	0.1	tension control
105 1.2 D	30.39	bottom face	50	344	30.392	0.34	0.0008	282.1		12	365.45	150	754.3	10.64	13.29	0.075	tension control
105 1.2 D	30.39	top face	50	344	30.392	0.34	0.0008	282.1		12	365.45	150	754.3	10.64	13.29	0.075	tension control

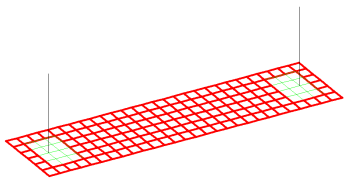
Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c + 0.66\sqrt{f_c'}bd) \geq V_u$		Rema	φV_c	Remar	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN		rks	KN	ks	
Load condition	105 1.2 D	0.394	1	157.6	39.528	0.001	0.344	343.208	338.444	fc'590.2	338.44	355	> 157.6	ok	253.8	φVc>Vu	not required
	105 1.2 D	0.394	1	157.6	39.528			343.208			338.44	355	> 157.6	ok	253.8	φVc>Vu	not required



3D rendered view



Highlighted plate for Moment at face of column



Highlighted plate for Shear at 'd' distance from face of column

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :			
At Bottom	12ϕT	200	c/c
At Top	12ϕT	150	c/c
Along Z Direction :			
At Bottom	12ϕT	200	c/c
At Top	12ϕT	250	c/c

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5.4 DESIGN OF FOOTING TYPE F6

	Along-X	Along-Z
Size of column :	0.50m	0.50m
Size of fdn. raft :	1.80m	1.80m
Thickness of raft :	0.35	m
Original ground level:	0.00	m
Finished floor level :	0.00	m
U/S level of foundation :	-1.50	m
U/S level Base Plate :	0.48	m
Unit weight of conc :	24	kN/m^3
Unit weight of soil :	18	kN/m^3

SYMBOLS AND NOTATIONS

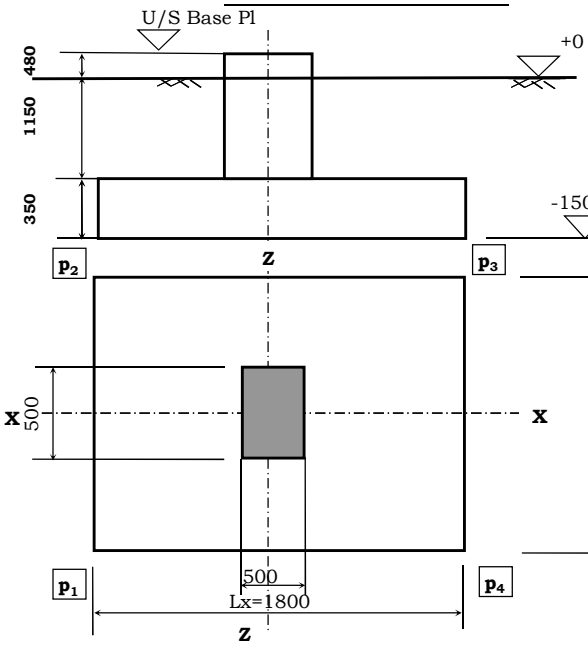
Dead Load	DL
Normal Operational Fluid Load	FL1
Max. Fluid Load	FL2
Wind Load	WL
Seismic Load	SL

Calculation of Weight of Foundation :

ght of raft & Col.=	(1.8*1.8*0.35*24)+(0.5*0.5*1.63*24) =	37.00KN
Weight of soil =	((1.8*1.8)-(0.5*0.5))*1.15*18 =	61.89KN
	Total =	98.89KN

Design Input:

Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	50mm
Dia. of bars provd. along X direction at bottom =	12mm
Dia. of bars provd. along X direction at top =	12mm
Dia. of bars provided along Z direction at bottom=	12mm
Dia. of bars provided along Z direction at top=	12mm
Mininum percentage of reinforcement at bottom=	0.18%

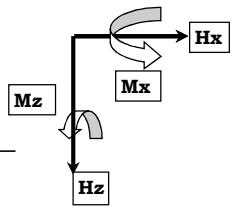


Effective depth of slab along X direction = 294mm = 0.294m

Effective depth of slab along Z direction = 282mm = 0.282m

Clear projection along Z direction = 0.65m

Clear projection along X direction = 0.65m



H_x=Force along X-X direction

Hz=Force along Z-Z direction

M_x=Moment about X-X direction

Mz=Moment about Z-Z direction

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5.4 DESIGN OF FOOTING TYPE F6

Combined Load Cases:	**Load at bottom of foundation**					**Load at bottom of foundation + self weight**				
Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
Allowable Stress Design										
201 1. DL	138.9	23.3	21.1	-0.9	-1.0	237.80	23.28	21.15	-0.87	-0.98
202 1 (DL+TL1) + 1 LL	224.4	34.6	33.5	-3.3	-3.8	323.29	34.60	33.55	-3.34	-3.77
203 1 (DL+TL2) + 1 LL	224.3	33.7	31.5	-2.3	-2.6	323.17	33.65	31.49	-2.31	-2.62
204 1 DL+ 1 WL(+X)	148.3	12.9	41.7	-2.6	-3.1	247.21	12.88	41.66	-2.56	-3.11
205 1 DL+ 1 WL(-X)	129.5	33.7	0.6	0.8	1.1	228.38	33.72	0.58	0.82	1.15
206 1 DL+ 1 WL(+Z)	121.8	20.1	23.6	-24.7	-28.8	220.68	20.11	23.60	-24.69	-28.82
207 1 DL+ 1 WL(-Z)	156.0	26.5	18.7	22.8	26.4	254.92	26.47	18.68	22.77	26.44
208 0.6 DL + 1 WL(+X)	92.8	3.6	33.2	-2.2	-2.7	191.65	3.57	33.20	-2.22	-2.71
209 0.6 DL + 1 WL(-X)	73.9	24.4	-7.9	1.2	1.5	172.82	24.40	-7.88	1.17	1.54
210 0.6 DL + 1 WL(+Z)	66.2	10.8	15.1	-24.3	-28.4	165.12	10.80	15.14	-24.34	-28.43
211 0.6 DL + 1 WL(-Z)	100.5	17.2	10.2	23.1	26.8	199.36	17.16	10.22	23.12	26.83
212 1 DL+ 0.75 WL(+X)+ 0.75 LL	210.0	23.6	45.1	-3.6	-4.2	308.93	23.61	45.06	-3.60	-4.23
213 1 DL+ 0.75 WL(-X)+ 0.75 LL	195.9	39.2	14.3	-1.1	-1.0	294.81	39.24	14.25	-1.07	-1.04
214 1 DL+ 0.75 WL(+Z)+ 0.75 LL	190.1	29.0	31.5	-20.2	-23.5	289.04	29.04	31.52	-20.20	-23.52
215 1 DL+ 0.75 WL(-Z)+ 0.75 LL	215.8	33.8	27.8	15.4	17.9	314.72	33.80	27.82	15.40	17.93
0	0.0	0.0	0.0	0.0	0.0	98.89	0.00	0.00	0.00	0.00
0	0.0	0.0	0.0	0.0	0.0	98.89	0.00	0.00	0.00	0.00
0	0.0	0.0	0.0	0.0	0.0	98.89	0.00	0.00	0.00	0.00
0	0.0	0.0	0.0	0.0	0.0	98.89	0.00	0.00	0.00	0.00
0	0.0	0.0	0.0	0.0	0.0	98.89	0.00	0.00	0.00	0.00
0	0.0	0.0	0.0	0.0	0.0	98.89	0.00	0.00	0.00	0.00

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Combined Load Cases:	**Load at bottom of foundation**					**Load at bottom of foundation + self weight**				
Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
Allowable Strength Design										
101 1.4 DL	194.5	32.6	29.6	-1.2	-1.4	293.37	32.60	29.61	-1.22	-1.37
102 1.2 (DL+TL1) + 1.6 LL	303.5	45.9	44.8	-4.8	-5.4	402.35	45.86	44.81	-4.79	-5.40
103 1.2 (DL+TL2) + 1.6 LL	303.3	44.7	42.3	-3.6	-4.0	402.20	44.72	42.34	-3.55	-4.03
104 1.2 DL+ 1.6 WL(+X) + 0.5 LL	224.5	16.7	63.9	-4.7	-5.7	323.36	16.71	63.88	-4.73	-5.68
105 1.2 DL+ 1.6 WL(-X) + 0.5 LL	194.3	50.1	-1.8	0.7	1.1	293.23	50.05	-1.84	0.69	1.12
106 1.2 DL+ 1.6 WL(+Z) + 0.5 LL	182.0	28.3	35.0	-40.1	-46.8	280.91	28.29	34.99	-40.13	-46.83
107 1.2 DL+ 1.6 WL(-Z) + 0.5 LL	236.8	38.5	27.1	35.8	41.6	335.69	38.46	27.11	35.81	41.59
108 0.9 DL + 1.6 WL(+X)	140.1	4.3	51.9	-3.5	-4.3	238.97	4.31	51.85	-3.49	-4.28
109 0.9 DL + 1.6 WL(-X)	110.0	37.6	-13.9	1.9	2.5	208.84	37.65	-13.88	1.92	2.52
110 0.9 DL + 1.6 WL(+Z)	97.6	15.9	23.0	-38.9	-45.4	196.52	15.88	22.96	-38.89	-45.43
111 0.9 DL + 1.6 WL(-Z)	152.4	26.1	15.1	37.0	43.0	251.30	26.05	15.08	37.05	42.99
112 1.2 DL + 0.8 WL(+X)	174.2	19.6	41.8	-2.4	-2.9	273.11	19.62	41.79	-2.40	-2.88
113 1.2 DL + 0.8 WL(-X)	159.2	36.3	8.9	0.3	0.5	258.05	36.29	8.92	0.31	0.53
114 1.2 DL + 0.8 WL(+Z)	153.0	25.4	27.3	-20.1	-23.4	251.89	25.40	27.34	-20.10	-23.45
115 1.2 DL + 0.8 WL(-Z)	153.0	25.4	27.3	-20.1	-23.4	251.89	25.40	27.34	-20.10	-23.45

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Check for overturning and sliding:

						Overturning		Sliding	
Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
201 1. DL	237.8	23.3	21.1	-0.9	-1.0	10.1	218.6	4.1	109.1
202 1 (DL+TL1) + 1 LL	323.3	34.6	33.5	-3.3	-3.8	8.7	77.3	3.7	38.8
203 1 (DL+TL2) + 1 LL	323.2	33.7	31.5	-2.3	-2.6	9.2	111.1	3.8	56.0
204 1 DL+ 1 WL(+X)	247.2	12.9	41.7	-2.6	-3.1	5.3	71.7	7.7	38.6
205 1 DL+ 1 WL(-X)	228.4	33.7	0.6	0.8	1.1	353.8	179.0	2.7	111.4
206 1 DL+ 1 WL(+Z)	220.7	20.1	23.6	-24.7	-28.8	8.4	6.9	4.4	3.6
207 1 DL+ 1 WL(-Z)	254.9	26.5	18.7	22.8	26.4	12.3	8.7	3.9	4.5
208 0.6 DL + 1 WL(+X)	191.6	3.6	33.2	-2.2	-2.7	5.2	63.6	21.5	34.6
209 0.6 DL + 1 WL(-X)	172.8	24.4	-7.9	1.2	1.5	19.7	101.1	2.8	59.1
210 0.6 DL + 1 WL(+Z)	165.1	10.8	15.1	-24.3	-28.4	9.8	5.2	6.1	2.7
211 0.6 DL + 1 WL(-Z)	199.4	17.2	10.2	23.1	26.8	17.6	6.7	4.6	3.4
212 1 DL+ 0.75 WL(+X)+ 0.75 LL	308.9	23.6	45.1	-3.6	-4.2	6.2	65.7	5.2	34.3
213 1 DL+ 0.75 WL(-X)+ 0.75 LL	294.8	39.2	14.3	-1.1	-1.0	18.6	254.4	3.0	110.6
214 1 DL+ 0.75 WL(+Z)+ 0.75 LL	289.0	29.0	31.5	-20.2	-23.5	8.3	11.1	4.0	5.7
215 1 DL+ 0.75 WL(-Z)+ 0.75 LL	314.7	33.8	27.8	15.4	17.9	10.2	15.8	3.7	8.2
0	98.9	0.0	0.0	0.0	0.0	NA	NA	NA	NA
0	98.9	0.0	0.0	0.0	0.0	NA	NA	NA	NA
0	98.9	0.0	0.0	0.0	0.0	NA	NA	NA	NA
0	98.9	0.0	0.0	0.0	0.0	NA	NA	NA	NA
					Min.=	5.2	5.2	2.7	2.7
						OK	OK	OK	OK

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5.4 DESIGN OF FOOTING TYPE F6

Load Combination & Calculation of Base pressure

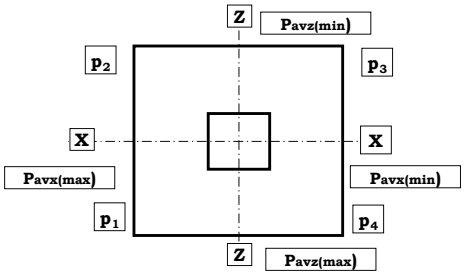
Max. Base pressure = p_{max} = (V/(Lx.Lz)) + (6.Mz/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)

Min. Base pressure = p_{min} = (V/(Lx.Lz)) - (6.Mz/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)

Surcharge load = 10 kN/m^2

Net SBC of soil for DL, LL= 350 kN/m^2 For WL/SL case = 437.5 kN/m^2

Gross SBC of soil for DL, LL= 377 kN/m^2 For WL/SL case = 464.5 kN/m^2



Allowable Stress Design																	
Load Combination	Design Load/moments						Base Pressure (kN/m^2)										Remarks
	V (kN)	Mz (kN-m)	Mx (kN-m)	V/ (Lx.Lz)	6.Mz/ (Lx)^2.Lz	6.Mx/ Lx.(Lz)^2	p_1 (p_{max})	p_2	p_4	p_3 (p_{min})	(p_{max}-γh)	$p_{avx(max)}$	$p_{avx(min)}$	$p_{avz(max)}$	$p_{avz(min)}$		
201 1. DL	237.80	21.15	-0.98	73.40	21.76	1.01	96.16	94.15	52.64	50.63	69.16	95.16	51.64	74.40	72.39		O.K
202 1 (DL+TL1) + 1 LL	323.29	33.55	-3.77	99.78	34.51	3.87	138.17	130.42	69.14	61.39	111.17	134.30	65.27	103.66	95.91		O.K
203 1 (DL+TL2) + 1 LL	323.17	31.49	-2.62	99.74	32.40	2.69	134.84	129.45	70.04	64.65	107.84	132.15	67.35	102.44	97.05		O.K
204 1 DL+ 1 WL(+X)	247.21	41.66	-3.11	76.30	42.86	3.19	122.35	115.96	36.64	30.25	95.35	119.16	33.45	79.50	73.11		O.K
205 1 DL+ 1 WL(-X)	228.38	0.58	1.15	70.49	0.60	1.18	72.27	69.91	71.07	68.71	45.27	71.09	69.89	71.67	69.31		O.K
206 1 DL+ 1 WL(+Z)	220.68	23.60	-28.82	68.11	24.28	29.65	122.05	62.75	73.48	14.18	95.05	92.40	43.83	97.77	38.47		O.K
207 1 DL+ 1 WL(-Z)	254.92	18.68	26.44	78.68	19.21	27.20	125.09	70.69	86.67	32.27	98.09	97.89	59.47	105.88	51.48		O.K
208 0.6 DL + 1 WL(+X)	191.65	33.20	-2.71	59.15	34.16	2.79	96.10	90.51	27.79	22.20	69.10	93.31	25.00	61.95	56.36		O.K
209 0.6 DL + 1 WL(-X)	172.82	-7.88	1.54	53.34	8.11	1.58	63.03	59.86	46.82	43.65	36.03	61.45	45.24	54.93	51.76		O.K
210 0.6 DL + 1 WL(+Z)	165.12	15.14	-28.43	50.96	15.58	29.25	95.79	37.30	64.63	6.14	68.79	66.55	35.39	80.21	21.72		O.K
211 0.6 DL + 1 WL(-Z)	199.36	10.22	26.83	61.53	10.51	27.60	99.64	44.43	78.63	23.42	72.64	72.04	51.03	89.14	33.93		O.K
212 1 DL+ 0.75 WL(+X)+ 0.75 LL	308.93	45.06	-4.23	95.35	46.36	4.35	146.06	137.35	53.35	44.64	119.06	141.71	49.00	99.71	91.00		O.K
213 1 DL+ 0.75 WL(-X)+ 0.75 LL	294.81	14.25	-1.04	90.99	14.66	1.07	106.73	104.58	77.40	75.26	79.73	105.66	76.33	92.07	89.92		O.K
214 1 DL+ 0.75 WL(+Z)+ 0.75 LL	289.04	31.52	-23.52	89.21	32.43	24.20	145.83	97.44	80.98	32.59	118.83	121.64	56.79	113.41	65.02		O.K
215 1 DL+ 0.75 WL(-Z)+ 0.75 LL	314.72	27.82	17.93	97.13	28.62	18.44	144.20	107.31	86.95	50.07	117.20	125.76	68.51	115.58	78.69		O.K

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Strength Design

101 1.4 DL	293.37	29.61	-1.37	90.55	30.46	1.41	122.42	119.60	61.49	58.67	95.42	121.01	60.08	91.96	89.14	
102 1.2 (DL+TL1) + 1.6 LL	402.35	44.81	-5.40	124.18	46.10	5.56	175.84	164.72	83.64	72.52	148.84	170.28	78.08	129.74	118.62	
103 1.2 (DL+TL2) + 1.6 LL	402.20	42.34	-4.03	124.14	43.56	4.14	171.84	163.55	84.72	76.43	144.84	167.70	80.58	128.28	119.99	
104 1.2 DL+ 1.6 WL(+X) + 0.5 LL	323.36	63.88	-5.68	99.80	65.72	5.85	171.37	159.68	39.93	28.23	144.37	165.53	34.08	105.65	93.96	
105 1.2 DL+ 1.6 WL(-X) + 0.5 LL	293.23	-1.84	1.12	90.50	1.90	1.15	93.55	91.25	89.76	87.45	66.55	92.40	88.61	91.66	89.35	
106 1.2 DL+ 1.6 WL(+Z) + 0.5 LL	280.91	34.99	-46.83	86.70	36.00	48.17	170.88	74.53	98.88	2.53	143.88	122.71	50.71	134.88	38.53	
107 1.2 DL+ 1.6 WL(-Z) + 0.5 LL	335.69	27.11	41.59	103.61	27.89	42.79	174.28	88.71	118.51	32.94	147.28	131.50	75.73	146.40	60.83	
108 0.9 DL + 1.6 WL(+X)	238.97	51.85	-4.28	73.76	53.34	4.41	131.51	122.69	24.82	16.01	104.51	127.10	20.42	78.17	69.35	
109 0.9 DL + 1.6 WL(-X)	208.84	-13.88	2.52	64.46	14.27	2.59	81.33	76.14	52.78	47.59	54.33	78.74	50.19	67.06	61.87	
110 0.9 DL + 1.6 WL(+Z)	196.52	22.96	-45.43	60.65	23.62	46.73	131.01	37.54	83.77	-9.70	104.01	84.28	37.04	107.39	13.92	
111 0.9 DL + 1.6 WL(-Z)	251.30	15.08	42.99	77.56	15.51	44.23	137.30	48.84	106.28	17.83	110.30	93.07	62.06	121.79	33.34	
112 1.2 DL + 0.8 WL(+X)	273.11	41.79	-2.88	84.29	42.99	2.96	130.24	124.33	44.26	38.34	103.24	127.29	41.30	87.25	81.34	
113 1.2 DL + 0.8 WL(-X)	258.05	8.92	0.53	79.65	9.18	0.54	89.37	88.28	71.01	69.92	62.37	88.83	70.47	80.19	79.10	
114 1.2 DL + 0.8 WL(+Z)	251.89	27.34	-23.45	77.74	28.13	24.12	130.00	81.75	73.74	25.49	103.00	105.88	49.62	101.87	53.62	
115 1.2 DL + 0.8 WL(-Z)	251.89	27.34	-23.45	77.74	28.13	24.12	130.00	81.75	73.74	25.49	103.00	105.88	49.62	101.87	53.62	
Max :	402	64	43													
Min :	165	-14	-47													

Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL cases) = 107.84 kN/m^2

Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL cases) = 117.84 kN/m^2

Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL+SL/WL cases) = 119.06 kN/m^2

Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL+WL/SL cases) = 129.06 kN/m^2

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Sample calculation of values of base pressure:

Load case : 02 1 (DL+TL1) + 1 L

Maximum pressure, p_1 (pmax) = 99.78+34.51+3.87= 138.17 kN/m^2

p_2= 99.78+34.51-3.87= 130.42 kN/m^2

p_3= 99.78-(34.51+3.87)= 61.39 kN/m^2

p_4= 99.78-34.51+3.87= 69.14 kN/m^2

Maxm pressure excluding weight of soil (p_{max}-γh)= 138.17-1.5*18= 111.17 kN/m^2

Maximum average Pressure along X, $p_{avx(max)}$ = (138.17+130.42)/2= 134.30 kN/m^2

Minimum average Pressure along X, $p_{avx(minx)}$ = (61.39+69.14)/2= 65.27 kN/m^2

Maximum average Pressure along Y, $p_{avy(max)}$ = (138.17+69.14)/2= 103.66 kN/m^2

Minimum average Pressure along Y, $p_{avy(minx)}$ = (61.39+130.42)/2= 95.91 kN/m^2

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5.4 DESIGN OF FOOTING TYPE F6

Calculation of Bending Moment and Shear Force in X direction :

p_{1x} =pressure at the face of column at section 1x-1x in kN/m^2.

p_{2x} =pressure at the face of column at section 2x-2x in kN/m^2.

p_{3x} =pressure at a distance 'd' from the face of column at section 3x-3x in kN/m^2.

BM_{1x} =Moment at the face of column at section 1x-1x in kN-m/m.

BM_{2x} =Moment at the face of column 2x-2x in kN-m/m.

SF_{3x} = Oneway shear force at a distance 'd' from the face of column at section 3x-3x in kN/m.

SF (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Punching perimeter= ((0.5+2*0.5*0.282)+(0.5+2*0.5*0.282))*2

= 3.128 m

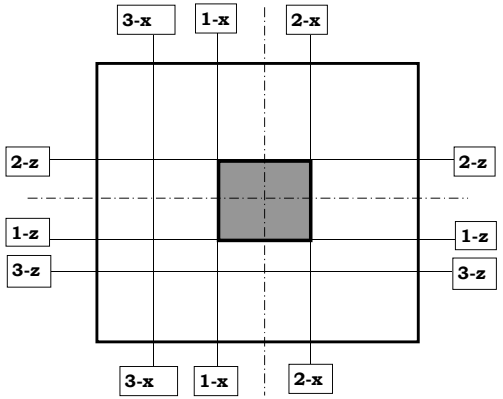
Punching Area= ((0.5+2*0.5*0.282)*(0.5+2*0.5*0.282))

= 0.61 m^2 Punching shear stress to be checked

Downward pressure due to self weight of raft and soil = (0.35*24+1.15*18)

= 29.10 kN/m^2

Area of footing, A_f= 3.24 m^2



Load Combination	p_{1x} kN/m2	p_{2x} kN/m2	p_{3x} kN/m2	$B.M_{1x}$ kNm	$B.M_{2x}$ kNm	$S.F_{3x}$ kN	$S.F_{(punching)}$ kN
Ultimate Conditions							
101 1.4 DL	99.01	82.08	108.96	17.87	8.09	30.57	161.60
102 1.2 (DL+TL1) + 1.6 LL	136.99	111.37	152.04	27.48	12.69	47.01	250.06
103 1.2 (DL+TL2) + 1.6 LL	136.24	112.04	150.47	27.06	13.09	46.27	249.95
104 1.2 DL+ 1.6 WL(+X) + 0.5 LL	118.06	81.55	139.53	25.48	4.39	43.94	185.95
105 1.2 DL+ 1.6 WL(-X) + 0.5 LL	91.03	89.98	91.65	13.28	12.67	22.40	161.49
106 1.2 DL+ 1.6 WL(+Z) + 0.5 LL	96.71	76.71	108.47	17.94	6.40	30.79	151.49
107 1.2 DL+ 1.6 WL(-Z) + 0.5 LL	111.36	95.87	120.47	20.21	11.27	34.49	195.96
108 0.9 DL + 1.6 WL(+X)	88.58	58.94	106.00	17.99	0.88	31.13	117.44
109 0.9 DL + 1.6 WL(-X)	68.43	60.50	73.09	9.76	5.18	16.67	92.99
110 0.9 DL + 1.6 WL(+Z)	67.22	54.10	74.94	10.46	2.88	17.98	82.99
111 0.9 DL + 1.6 WL(-Z)	81.87	73.26	86.94	12.73	7.75	21.68	127.46
112 1.2 DL + 0.8 WL(+X)	96.24	72.35	110.28	18.56	4.76	31.93	145.16
113 1.2 DL + 0.8 WL(-X)	82.20	77.10	85.20	12.15	9.21	20.62	132.93
114 1.2 DL + 0.8 WL(+Z)	85.56	69.94	94.75	14.79	5.77	25.35	127.93
115 1.2 DL + 0.8 WL(-Z)	85.56	69.94	94.75	14.79	5.77	25.35	127.93

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Sample calculation of BM & SF

Load case : 101 1.4 DL

p_{1x} = 121.01-(121.01-60.08)*0.65/1.8 = 99.01 kN/m^2 Overhang = 0.65 m

p_{2x} = 60.08+(121.01-60.08)*0.65/1.8 = 82.08 kN/m^2 Effective depth = 0.294 m

p_{3x} = 121.01-(121.01-60.08)*(0.65-0.294)/1.8 = 108.96 kN/m^2

BM_{1x} = (99.01-29.1)*0.65*0.65/2+0.5*(121.01-99.01)*0.65*0.65*0.67 = 17.87 kN-m/m

BM_{2x} = (82.08-29.1)*0.65*0.65/2-0.5*(-60.08+82.08)*0.65*0.65*0.67 = 8.09 kN-m/m

SF_{3x} = (108.96-29.1)*(0.65-0.294)+0.5*(121.01-108.96)*(0.65-0.294) = 30.57 kN/m

$S.F_{(punching)}$ = (3.24-0.61)*(90.55-29.1) = 161.60 kN

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	27.48	0.32	0.42	0.001	298.87	529.20	12ϕT	213.71	300	150	753.98
At Top	0.00	0.00	0.00	0.000	0.00	264.60	12ϕT	427.43	300	250	452.39

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc)/bd = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 47.01 kN/m

Shear stress , τ_v= (47.01*1000)/(1000*294)

= 0.16 N/mm^2 **O.K.**

Check against Two-way shear force at a distance d/2 from face of column:

Design punching shear at a distance d/2 from face of column, Vu= 250.06 kN

Strength of Concrete in shear Vc1 = 2720.31 kN **Table 22.6.5.2_ (a)**

Strength of Concrete in shear Vc2 = 1759.04 kN **Table 22.6.5.2_ (b)**

Strength of Concrete in shear Vc3 = 1813.54 kN **Table 22.6.5.2_ (c)**

Design Vc= 1759.04 kN

Allowable shear stress, (ϕ Vc)/bd = 1.43 N/mm^2

Design punching shear stress, τ = (250.06*1000)/(3128*294)

= 0.27 N/mm^2 **No shear reinforcement reqd.**

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Calculation of Bending Moment and Shear Force in Z direction :

p1z =pressure at the face of column at section 1z-1z in kN/m^2.
p2z =pressure at the face of column at section 2z-2z in kN/m^2.
p3z =pressure at a distance d from the face of column at section 3z-3z in kN/m^2.
B.M1z =Moment at the face of column at section 1z-1z in kN-m/m.
B.M2z =Moment at the face of column at section 2z-2z in kN-m/m.
S.F 3z = Oneway Shear Force at a distance d from the face of column at section 3z-3z in kN/m.
S.F (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Load Combination	p_{1z} kN/m2	p_{2z} kN/m2	p_{3z} kN/m2	$B.M_{1z}$ kNm	$B.M_{2z}$ kNm	$S.F_{3z}$ kN
Ultimate Conditions						
101 1.4 DL	90.94	90.16	91.38	13.21	12.75	23.03
102 1.2 (DL+TL1) + 1.6 LL	125.72	122.64	127.47	20.98	19.19	36.62
103 1.2 (DL+TL2) + 1.6 LL	125.29	122.98	126.59	20.74	19.41	36.19
104 1.2 DL+ 1.6 WL(+X) + 0.5 LL	101.43	98.18	103.26	15.88	14.00	27.73
105 1.2 DL+ 1.6 WL(-X) + 0.5 LL	90.83	90.18	91.19	13.16	12.79	22.94
106 1.2 DL+ 1.6 WL(+Z) + 0.5 LL	100.09	73.32	115.18	19.92	4.42	35.30
107 1.2 DL+ 1.6 WL(-Z) + 0.5 LL	115.50	91.73	128.91	22.63	8.86	39.95
108 0.9 DL + 1.6 WL(+X)	74.99	72.54	76.37	10.14	8.72	17.73
109 0.9 DL + 1.6 WL(-X)	65.19	63.74	66.00	7.89	7.05	13.77
110 0.9 DL + 1.6 WL(+Z)	73.64	47.67	88.28	14.19	-0.85	25.29
111 0.9 DL + 1.6 WL(-Z)	89.85	65.28	103.71	17.35	3.12	30.78
112 1.2 DL + 0.8 WL(+X)	85.12	83.47	86.04	12.14	11.18	21.18
113 1.2 DL + 0.8 WL(-X)	79.80	79.49	79.97	10.77	10.59	18.76
114 1.2 DL + 0.8 WL(+Z)	84.45	71.04	92.01	14.16	6.39	24.96
115 1.2 DL + 0.8 WL(-Z)	84.45	71.04	92.01	14.16	6.39	24.96

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Sample calculation of BM & SF

Load case : 101 1.4 DL

p_{1z} = 91.96-(91.96-89.14)*0.65/1.8= 90.94 kN/m^2

p_{2z} = 89.14+(91.96-89.14)*0.65/1.8= 90.16 kN/m^2

p_{3z} = 91.96-(91.96-89.14)*(0.65-0.282)/1.8= 91.38 kN/m^2

Overhang = 0.65 m

Effective depth = 0.282 m

BM_{1z} = (90.94-29.1)*0.65*0.65/2+0.5*(91.96-90.94)*0.65*0.65*0.67 = 13.21 kN-m/m

BM_{2z} = (90.16-29.1)*0.65*0.65/2-0.5*(-89.14+90.16)*0.65*0.65*0.67 = 12.75 kN-m/m

SF_{3z} = (91.38-29.1)*(0.65-0.282)+0.5*(91.96-91.38)*(0.65-0.282) = 23.03 kN/m

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	22.63	0.285	0.379	0.0009	267.26	507.60	12ϕT	222.81	300	150	754.0
Top	-0.85	-0.011	-0.014	0.0000	-0.10	253.80	12ϕT	445.62	300	250	452.4

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc) = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 39.95 kN/m

Shear stress , τ_v= (39.95*1000)/(1000*282)

= 0.14 N/mm^2 **O.K.**

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :

At Bottom **12ϕT** **150** c/c

At Top **12ϕT** **250** c/c

Along Z Direction :

At Bottom **12ϕT** **150** c/c

At Top **12ϕT** **250** c/c

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5.5 DESIGN OF FOOTING TYPE F7

Size of Columns at node	Along-X	Along-Z
1	0.50m	0.50m
2	0.50m	0.50m
Size of fdn. raft:	4.00m	2.00m
Thickness of raft:	0.50	m
Original Gr. Level:	0.00	m
Finished Gr. Level:	0.00	m
U/S of B/Pl Above GL:	0.65	m
U/S level of fdn:	-1.20	m
Unit weight of conc:	25	kN/m^3
Unit weight of soil:	18	kN/m^3

Design Input:

Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	50mm

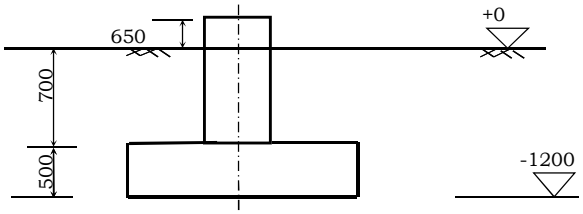
Calculation of Weight of Foundation :

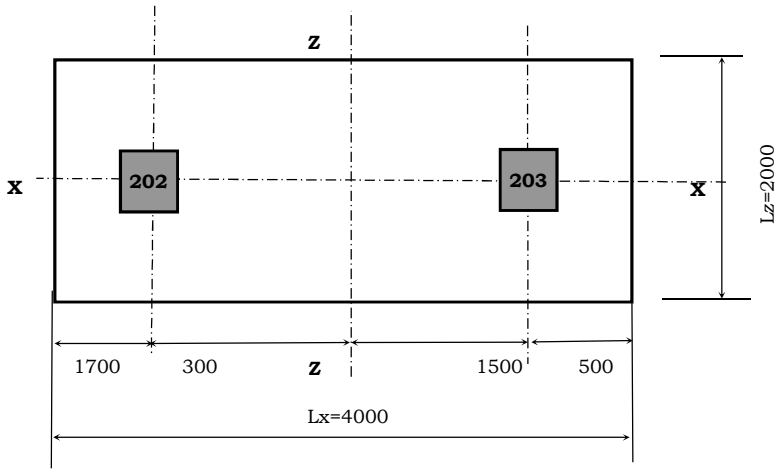
Weight of raft = (4*2*0.5*25) = 100.00KN

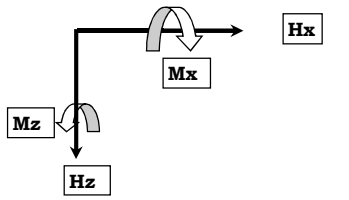
Weight of soil = ((4*2)-2*(0.5*0.5))*0.7*18 = 94.50KN

Total = 194.50KN

Submerged wt. = 194.5-4*2*1.2*10 = 98.50KN







H_X=Force along X-X direction

Hz=Force along Z-Z direction

M_X=Moment about X-X direction

Mz=Moment about Z-Z direction

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Check for overturning and sliding of footing F1

Load at bottom of fooring	Node 202					Node 203				
L/C No.	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
201 1. DL	29.2	-5.4	10.0	-9.4	-17.5	100.0	5.6	-10.4	0.0	0.0
202 1 (DL+TL1) + 1 LL	43.9	-16.3	30.1	-29.5	-54.5	286.2	17.4	-32.2	0.0	0.0
203 1 (DL+TL2) + 1 LL	35.4	-18.3	33.8	-23.1	-42.6	289.1	18.9	-34.9	0.0	0.0
204 1 DL+ 1 WL(+X)	-76.7	-43.8	81.0	-9.9	-18.3	207.2	2.8	-5.1	0.0	0.0
205 1 DL+ 1 WL(-X)	135.1	33.0	-61.0	-9.0	-16.7	-7.3	2.8	-5.1	0.0	0.0
206 1 DL+ 1 WL(+Z)	38.8	-5.8	10.6	-15.9	-29.4	121.9	7.3	-13.6	-0.8	-1.5
207 1 DL+ 1 WL(-Z)	19.6	-5.1	9.4	-3.0	-5.6	78.0	3.9	-7.2	0.8	1.5
208 0.6 DL + 1 WL(+X)	-88.4	-41.6	77.0	-6.1	-11.3	167.3	0.5	-1.0	0.0	0.0
209 0.6 DL + 1 WL(-X)	123.5	35.1	-65.0	-5.2	-9.7	-47.3	0.5	-1.0	0.0	0.0
210 0.6 DL + 1 WL(+Z)	27.2	-3.6	6.6	-12.1	-22.4	82.0	5.1	-9.4	-0.8	-1.5
211 0.6 DL + 1 WL(-Z)	7.9	-2.9	5.4	0.8	1.4	38.0	1.6	-3.0	0.8	1.5
212 1 DL+ 0.75 WL(+X)+ 0.75 LL	-42.4	-43.1	79.7	-22.4	-41.4	321.2	12.9	-23.8	0.0	0.0
213 1 DL+ 0.75 WL(-X)+ 0.75 LL	116.5	14.5	-26.8	-21.7	-40.2	160.2	12.9	-23.8	0.0	0.0
214 1 DL+ 0.75 WL(+Z)+ 0.75 LL	44.3	-14.6	26.9	-26.9	-49.8	257.2	16.3	-30.1	-0.6	-1.1
215 1 DL+ 0.75 WL(-Z)+ 0.75 LL	29.8	-14.1	26.0	-17.2	-31.9	224.2	13.7	-25.3	0.6	1.1

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Check for overturning and sliding:

L/C No.	Resultant Load at bottom of footing								
	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	Overturning		Sliding	
						F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
201 1. DL	323.7	0.18	105.8	-9.4	-17.5	6.1	18.5	719.2	13.7
202 1 (DL+TL1) + 1 LL	524.6	1.11	361.3	-29.5	-54.5	2.9	9.6	189.1	7.1
203 1 (DL+TL2) + 1 LL	518.9	0.61	379.4	-23.1	-42.6	2.7	12.2	340.3	9.0
204 1 DL+ 1 WL(+X)	325.0	-41.05	252.4	-9.9	-18.3	2.0	17.8	3.2	13.2
205 1 DL+ 1 WL(-X)	322.3	35.73	-279.8	-9.0	-16.7	2.3	19.4	3.6	14.3
206 1 DL+ 1 WL(+Z)	355.3	1.58	121.7	-16.7	-30.9	5.8	11.5	89.9	8.5
207 1 DL+ 1 WL(-Z)	292.0	-1.22	89.9	-2.2	-4.1	6.5	72.1	95.8	53.3
208 0.6 DL + 1 WL(+X)	273.4	-41.12	279.4	-6.1	-11.3	1.7	24.2	2.7	17.9
209 0.6 DL + 1 WL(-X)	270.6	35.66	-322.2	-5.2	-9.7	1.7	28.0	3.0	20.7
210 0.6 DL + 1 WL(+Z)	303.6	1.51	79.4	-12.9	-23.9	7.6	12.7	80.5	9.4
211 0.6 DL + 1 WL(-Z)	240.4	-1.29	47.6	1.6	2.9	10.1	81.9	74.4	60.6
212 1 DL+ 0.75 WL(+X)+ 0.75 LL	473.3	-30.23	601.3	-22.4	-41.4	1.6	11.4	6.3	8.5
213 1 DL+ 0.75 WL(-X)+ 0.75 LL	471.2	27.35	15.0	-21.7	-40.2	62.9	11.7	6.9	8.7
214 1 DL+ 0.75 WL(+Z)+ 0.75 LL	496.0	1.74	316.2	-27.5	-50.9	3.1	9.7	114.0	7.2
215 1 DL+ 0.75 WL(-Z)+ 0.75 LL	448.5	-0.36	292.3	-16.6	-30.7	3.1	14.6	498.4	10.8
					Min.=	**1.6**	**9.6**	**2.7**	**7.1**

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Design of foundation raft F8 & F9

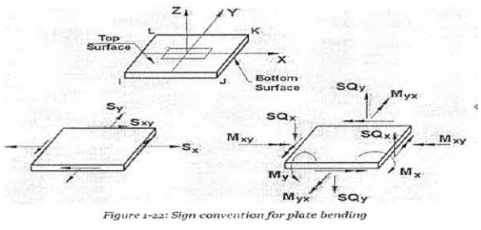
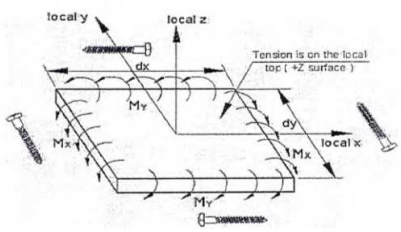


Figure 1-22: Sign convention for plate bending



Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:
PLATE CENTER STRESS : FDN_F8&F9_MOMENT At the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	1376	0.634	0.051	0	0	0	41.513	62.51	3.409	44.922	65.919	103 1.2 (DL+TL2) + 1.6 LL
Min Qx	1340	-0.605	-0.033	0	0	0	-130.8	-20.232	0	-130.804	-20.232	104 1.2 DL+ 1.6 WL(+X) + 0.5 LL
Max Qy	1100	0.123	0.473	0	0	0	41.252	40.396	0.473	41.725	40.869	103 1.2 (DL+TL2) + 1.6 LL
Min Qy	1716	0.123	-0.487	0	0	0	39.781	39.615	5.97	45.751	45.585	103 1.2 (DL+TL2) + 1.6 LL
Max Sx	552	-0.017	-0.004	0	0	0	-5.289	-0.3	-1.762	-7.051	-2.062	101 1.4 DL
Min Sx	552	-0.017	-0.004	0	0	0	-5.289	-0.3	-1.762	-7.051	-2.062	101 1.4 DL
Max Sy	552	-0.017	-0.004	0	0	0	-5.289	-0.3	-1.762	-7.051	-2.062	101 1.4 DL
Min Sy	552	-0.017	-0.004	0	0	0	-5.289	-0.3	-1.762	-7.051	-2.062	101 1.4 DL
Max Sxy	552	-0.017	-0.004	0	0	0	-5.289	-0.3	-1.762	-7.051	-2.062	101 1.4 DL
Min Sxy	552	-0.017	-0.004	0	0	0	-5.289	-0.3	-1.762	-7.051	-2.062	101 1.4 DL
Max Mx	1396	0.548	-0.044	0	0	0	99.974	29.015	-4.622	104.596	33.637	109 0.9 DL + 1.6 WL(-X)
Min Mx	1340	-0.605	-0.033	0	0	0	-130.8	-20.232	-8.861	-139.665	-29.093	104 1.2 DL+ 1.6 WL(+X) + 0.5 LL
Max My	1387	-0.186	0.012	0	0	0	-8.478	68.346	6.394	-14.872	74.74	103 1.2 (DL+TL2) + 1.6 LL
Min My	1065	-0.004	-0.193	0	0	0	-23	-39.209	-8.616	-31.616	-47.825	102 1.2 (DL+TL1) + 1.6 LL
Max Mxy	1575	-0.164	-0.156	0	0	0	-2.645	11.002	30.303	-32.948	41.305	104 1.2 DL+ 1.6 WL(+X) + 0.5 LL
Min Mxy	1228	-0.504	-0.228	0	0	0	-113.9	-27.911	-21.169	-135.043	-49.08	104 1.2 DL+ 1.6 WL(+X) + 0.5 LL

PLATE CENTER STRESS : FDN_F8&F9_SHEAR At a distance "d" from the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	1376	0.634	0.051	0	0	0	41.513	62.51	3.409	44.922	65.919	103 1.2 (DL+TL2) + 1.6 LL
Min Qx	1340	-0.605	-0.033	0	0	0	-130.8	-20.232	0	-130.804	-20.232	104 1.2 DL+ 1.6 WL(+X) + 0.5 LL
Max Qy	1100	0.123	0.473	0	0	0	41.252	40.396	0.473	41.725	40.869	103 1.2 (DL+TL2) + 1.6 LL
Min Qy	1716	0.123	-0.487	0	0	0	39.781	39.615	5.97	45.751	45.585	103 1.2 (DL+TL2) + 1.6 LL
Max Sx	552	-0.017	-0.004	0	0	0	-5.289	-0.3	-1.762	-7.051	-2.062	101 1.4 DL
Min Sx	552	-0.017	-0.004	0	0	0	-5.289	-0.3	-1.762	-7.051	-2.062	101 1.4 DL
Max Sy	552	-0.017	-0.004	0	0	0	-5.289	-0.3	-1.762	-7.051	-2.062	101 1.4 DL
Min Sy	552	-0.017	-0.004	0	0	0	-5.289	-0.3	-1.762	-7.051	-2.062	101 1.4 DL
Max Sxy	552	-0.017	-0.004	0	0	0	-5.289	-0.3	-1.762	-7.051	-2.062	101 1.4 DL
Min Sxy	552	-0.017	-0.004	0	0	0	-5.289	-0.3	-1.762	-7.051	-2.062	101 1.4 DL
Max Mx	1396	0.548	-0.044	0	0	0	99.974	29.015	-4.622	104.596	33.637	109 0.9 DL + 1.6 WL(-X)
Min Mx	1340	-0.605	-0.033	0	0	0	-130.8	-20.232	-8.861	-139.665	-29.093	104 1.2 DL+ 1.6 WL(+X) + 0.5 LL
Max My	1387	-0.186	0.012	0	0	0	-8.478	68.346	6.394	-14.872	74.74	103 1.2 (DL+TL2) + 1.6 LL
Min My	1065	-0.004	-0.193	0	0	0	-23	-39.209	-8.616	-31.616	-47.825	102 1.2 (DL+TL1) + 1.6 LL
Max Mxy	1575	-0.164	-0.156	0	0	0	-2.645	11.002	30.303	-32.948	41.305	104 1.2 DL+ 1.6 WL(+X) + 0.5 LL
Min Mxy	1228	-0.504	-0.228	0	0	0	-113.9	-27.911	-21.169	-135.043	-49.08	104 1.2 DL+ 1.6 WL(+X) + 0.5 LL

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	50	mm
	top face	50	mm
Thicknes of slab =		500	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

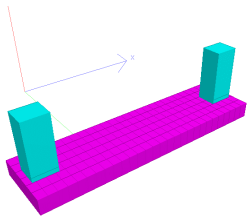
Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,provided}$	a	$c = a/\beta_1$	$\varepsilon_t = \varepsilon_{cu}/c *(d-c)$	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
109 0.9 D	104.60	top face	50	442	104.6	0.71	0.0017	760.5	397.8	16	264.49	150	1341	18.91	23.64	0.053	tension control
104 1.2 D	-139.67	bottom face	50	442	139.67	0.95	0.0023	1019.7		16	197.25	150	1341	18.91	23.64	0.053	tension control
104 1.2 D	-32.95	top face	50	442	32.948	0.22	0.0005	237.5		16	505.64	200	1005.8	14.18	17.73	0.072	tension control
104 1.2 D	-135.04	top face	50	442	135.04	0.92	0.0022	985.4		16	204.12	200	1005.8	14.18	17.73	0.072	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c + 0.66\sqrt{f_c'}bd) \geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qx	0.634	1	317	44.922	0.003	0.442	489.49	441.182	758.323	441.18	1625	> 317	ok	330.9	φVc>Vu	not required
	Min Qx	0.605	1	302.5	130.8			441.688			441.18	1625	> 302.5	ok	330.9	φVc>Vu	not required

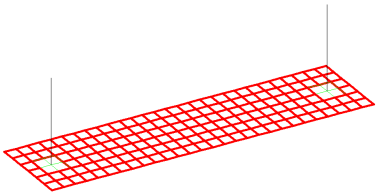
Reinforcement Along Z:

Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, My	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,provided}$	a	$c = a/\beta_1$	$\varepsilon_t = \varepsilon_{cu}/c *(d-c)$	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
103 1.2 (	74.74	top face	50	444	74.74	0.51	0.0012	539.0	399.6	12	209.91	250	452.6	6.382	7.977	0.164	tension control
102 1.2 (	-47.83	bottom face	50	444	47.825	0.32	0.0008	343.8		12	284.42	150	754.3	10.64	13.29	0.097	tension control
104 1.2 D	41.31	bottom face	50	444	41.305	0.28	0.0007	296.7		12	284.42	200	565.8	7.978	9.972	0.131	tension control
104 1.2 D	-49.08	top face	50	444	49.08	0.33	0.0008	352.9		12	284.42	200	565.8	7.978	9.972	0.131	tension control

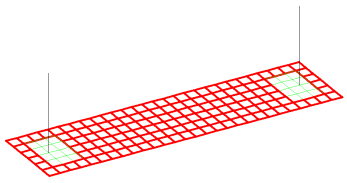
Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c + 0.66\sqrt{f_c'}bd) \geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	103 1.2 (	0.473	1	236.5	40.869	0.001	0.444	476.694	441.182	fc'758.3	441.18	460	> 236.5	ok	330.9	φVc>Vu	not required
	103 1.2 (	0.487	1	243.5	45.585			472.209			441.18	460	> 243.5	ok	330.9	φVc>Vu	not required



3D rendered view



Highlighted plate for Moment at face of column



Highlighted plate for Shear at 'd' distance from face of column

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :			
At Bottom	**16ϕT**	**150**	c/c
At Top	**16ϕT**	**150**	c/c
Along Z Direction :			
At Bottom	**12ϕT**	**150**	c/c
At Top	**12ϕT**	**250**	c/c

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5.6 DESIGN OF FOOTING F8 & F9

Size of Columns at node	Along-X	Along-Z
1	0.45m	0.45m
2	0.45m	0.45m
Size of fdn. raft:	2.80m	1.50m
Thickness of raft:	0.40	m
Original Gr. Level:	0.00	m
Finished Gr. Level:	0.00	m
U/S of B/Pl Above GL:	0.15	m
U/S level of fdn:	-1.20	m
Unit weight of conc:	25	kN/m^3
Unit weight of soil:	18	kN/m^3

Design Input:

Characteristic strength of concrete, f'_c=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	50mm

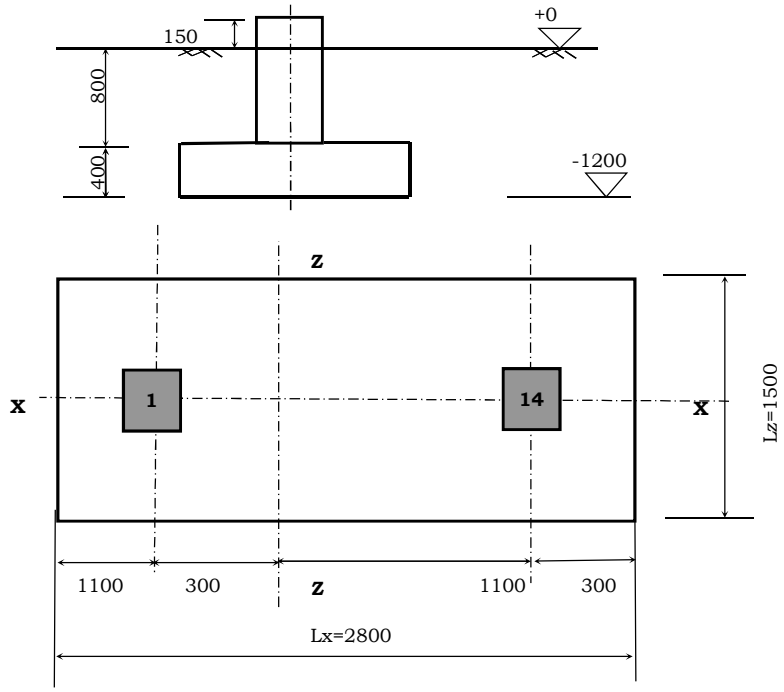
Calculation of Weight of Foundation :

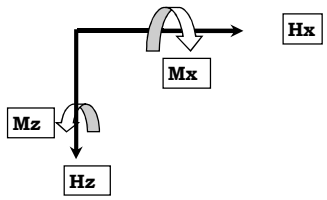
Weight of raft = (2.8*1.5*0.4*25) = 42.00KN

Weight of soil = ((2.8*1.5)-2*(0.45*0.45))*0.8*18 = 54.65KN

Total = 96.65KN

Submerged wt. = 96.65-2.8*1.5*1.2*10 = 46.25KN





H_x=Force along X-X direction

Hz=Force along Z-Z direction

M_x=Moment about X-X direction

Mz=Moment about Z-Z direction

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Check for overturning and sliding of footing

Load at bottom of fooring	Node 1					Node 14				
L/C No.	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
201 1. DL	32.1	0.9	-1.2	-0.6	-0.8	21.1	-0.8	1.1	0.6	0.8
202 1 (DL+TL1) + 1 LL	90.6	41.0	-55.4	0.5	0.6	54.8	-38.6	52.1	1.9	2.6
203 1 (DL+TL2) + 1 LL	85.5	-35.4	47.8	-5.0	-6.7	59.7	34.3	-46.3	2.4	3.2
204 1 DL+ 1 WL(+X)	16.0	-13.9	18.8	-2.0	-2.7	36.8	-13.6	18.3	0.6	0.8
205 1 DL+ 1 WL(-X)	48.1	15.5	-20.9	0.8	1.1	5.5	12.2	-16.5	0.7	0.9
206 1 DL+ 1 WL(+Z)	32.1	0.9	-1.2	-13.2	-17.8	21.2	-0.8	1.0	-10.6	-14.3
207 1 DL+ 1 WL(-Z)	32.1	0.9	-1.2	12.1	16.3	21.1	-0.8	1.1	11.9	16.1
208 0.6 DL + 1 WL(+X)	3.2	-14.3	19.3	-1.7	-2.4	28.4	-13.3	17.9	0.3	0.4
209 0.6 DL + 1 WL(-X)	35.3	15.2	-20.5	1.1	1.4	-3.0	12.5	-16.9	0.4	0.6
210 0.6 DL + 1 WL(+Z)	19.2	0.5	-0.7	-13.0	-17.5	12.7	-0.4	0.6	-10.8	-14.6
211 0.6 DL + 1 WL(-Z)	19.3	0.5	-0.7	12.3	16.6	12.7	-0.5	0.7	11.6	15.7
212 1 DL+ 0.75 WL(+X)+ 0.75 LL	62.0	-8.8	11.8	-2.9	-3.9	60.0	-11.4	15.4	1.7	2.3
213 1 DL+ 0.75 WL(-X)+ 0.75 LL	86.1	13.3	-17.9	-0.8	-1.0	36.4	7.9	-10.7	1.8	2.5
214 1 DL+ 0.75 WL(+Z)+ 0.75 LL	74.0	2.3	-3.1	-11.3	-15.2	48.2	-1.8	2.4	-6.6	-9.0
215 1 DL+ 0.75 WL(-Z)+ 0.75 LL	74.1	2.3	-3.1	7.7	10.3	48.2	-1.8	2.5	10.2	13.8

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Check for overturning and sliding:

L/C No.	Resultant Load at bottom of footing								
						Overturning		Sliding	
	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
201 1. DL	149.9	0.12	-12.2	0.1	0.1	17.2	1677.7	499.6	1199.0
202 1 (DL+TL1) + 1 LL	242.0	2.42	-42.6	2.4	3.3	7.9	55.5	40.0	40.0
203 1 (DL+TL2) + 1 LL	241.8	-1.12	-26.9	-2.6	-3.5	12.6	52.5	86.4	37.8
204 1 DL+ 1 WL(+X)	149.5	-27.50	60.0	-1.4	-1.9	3.5	58.9	2.2	42.4
205 1 DL+ 1 WL(-X)	150.2	27.74	-84.4	1.5	2.0	2.5	55.3	2.2	39.8
206 1 DL+ 1 WL(+Z)	149.9	0.15	-12.2	-23.8	-32.1	17.2	3.5	399.7	2.5
207 1 DL+ 1 WL(-Z)	149.9	0.09	-12.2	24.0	32.3	17.2	3.5	666.1	2.5
208 0.6 DL + 1 WL(+X)	128.2	-27.55	64.9	-1.4	-1.9	2.8	49.8	1.9	35.9
209 0.6 DL + 1 WL(-X)	129.0	27.69	-79.5	1.5	2.0	2.3	48.1	1.9	34.6
210 0.6 DL + 1 WL(+Z)	128.6	0.10	-7.3	-23.8	-32.2	24.7	3.0	504.3	2.2
211 0.6 DL + 1 WL(-Z)	128.6	0.04	-7.3	23.9	32.3	24.5	3.0	1224.5	2.1
212 1 DL+ 0.75 WL(+X)+ 0.75 LL	218.6	-20.20	25.0	-1.1	-1.5	12.2	107.0	4.3	77.0
213 1 DL+ 0.75 WL(-X)+ 0.75 LL	219.2	21.23	-83.3	1.1	1.4	3.7	115.4	4.1	83.1
214 1 DL+ 0.75 WL(+Z)+ 0.75 LL	218.9	0.54	-29.1	-17.9	-24.2	10.5	6.8	162.2	4.9
215 1 DL+ 0.75 WL(-Z)+ 0.75 LL	218.9	0.50	-29.1	17.9	24.1	10.5	6.8	176.9	4.9
					Min.=	**2.3**	**3.0**	**1.9**	**2.1**

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Base pressure check of footing

	Node	L/C	Horizontal Fx N/mm2	Vertical Fy N/mm2	Horizontal Fz N/mm2
Max Px	167	201	0	0.04	0
Min Px	167	201	0	0.04	0
Max Py	1576	212	0	0.09	0
Min Py	497	208	0	-0.01	0
Max Pz	167	201	0	0.04	0
Min Pz	167	201	0	0.04	0

Gross Maximum Base pressure, p_{max} = 93.0 KN/m^2

Net Allowable Safe Bearing Capacity = 450 KN/m^2
Depth of soil above footing = 0.8 m
Unit weight of soil = 18 KN/m^3

Gross Safe Bearing Capacity = 464.4 KN/m^2
> 93.0 KN/m^2
Hence OK

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Design of foundation raft F8 & F9

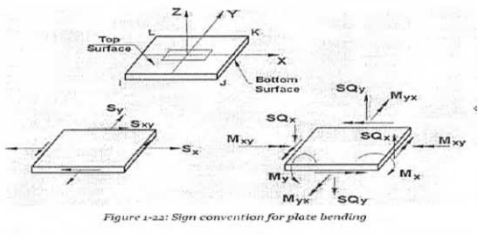
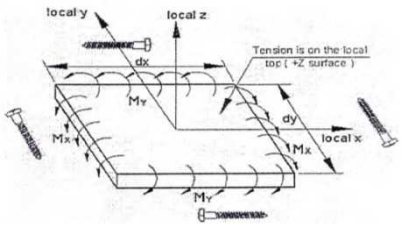


Figure 1-22: Sign convention for plate bending



Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:
PLATE CENTER STRESS : FDN_F8&F9_MOMENT At the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	1409	0.538	-0.073	0	0	0	46.319	19.836	-2.336	48.655	22.172	102 1.2 (DL+TL1) + 1.6 LL
Min Qx	1418	-0.492	-0.053	0	0	0	71.324	20.505	0	71.324	20.505	103 1.2 (DL+TL2) + 1.6 LL
Max Qy	1167	-0.004	0.551	0	0	0	-0.376	22.387	-4.926	-5.302	27.313	102 1.2 (DL+TL1) + 1.6 LL
Min Qy	1671	0.001	-0.57	0	0	0	-0.406	20.288	5.007	-5.413	25.295	102 1.2 (DL+TL1) + 1.6 LL
Max Sx	552	0	0	0	0	0	0.006	0.008	-0.001	0.007	0.009	101 1.4 DL
Min Sx	552	0	0	0	0	0	0.006	0.008	-0.001	0.007	0.009	101 1.4 DL
Max Sy	552	0	0	0	0	0	0.006	0.008	-0.001	0.007	0.009	101 1.4 DL
Min Sy	552	0	0	0	0	0	0.006	0.008	-0.001	0.007	0.009	101 1.4 DL
Max Sxy	552	0	0	0	0	0	0.006	0.008	-0.001	0.007	0.009	101 1.4 DL
Min Sxy	552	0	0	0	0	0	0.006	0.008	-0.001	0.007	0.009	101 1.4 DL
Max Mx	1418	-0.492	-0.053	0	0	0	71.324	20.505	4.918	76.242	25.423	103 1.2 (DL+TL2) + 1.6 LL
Min Mx	1437	-0.18	0.032	0	0	0	-76.66	10.597	5.708	-82.365	16.305	102 1.2 (DL+TL1) + 1.6 LL
Max My	1278	-0.179	0.363	0	0	0	4.135	27.102	-13.921	18.056	41.023	102 1.2 (DL+TL1) + 1.6 LL
Min My	1134	0.011	-0.129	0	0	0	2.715	-16.662	-0.606	3.321	-17.268	110 0.9 DL + 1.6 WL(+Z)
Max Mxy	1663	0.119	0.098	0	0	0	-36.86	-1.515	16.717	-53.581	-18.232	102 1.2 (DL+TL1) + 1.6 LL
Min Mxy	1164	0.043	0.335	0	0	0	-7.85	23.947	-15.476	-23.326	39.423	102 1.2 (DL+TL1) + 1.6 LL

PLATE CENTER STRESS : FDN_F8&F9_SHEAR At a distance "d" from the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	1409	0.538	-0.073	0	0	0	46.319	19.836	-2.336	48.655	22.172	102 1.2 (DL+TL1) + 1.6 LL
Min Qx	1418	-0.492	-0.053	0	0	0	71.324	20.505	0	71.324	20.505	103 1.2 (DL+TL2) + 1.6 LL
Max Qy	1167	-0.004	0.551	0	0	0	-0.376	22.387	-4.926	-5.302	27.313	102 1.2 (DL+TL1) + 1.6 LL
Min Qy	1671	0.001	-0.57	0	0	0	-0.406	20.288	5.007	-5.413	25.295	102 1.2 (DL+TL1) + 1.6 LL
Max Sx	552	0	0	0	0	0	0.006	0.008	-0.001	0.007	0.009	101 1.4 DL
Min Sx	552	0	0	0	0	0	0.006	0.008	-0.001	0.007	0.009	101 1.4 DL
Max Sy	552	0	0	0	0	0	0.006	0.008	-0.001	0.007	0.009	101 1.4 DL
Min Sy	552	0	0	0	0	0	0.006	0.008	-0.001	0.007	0.009	101 1.4 DL
Max Sxy	552	0	0	0	0	0	0.006	0.008	-0.001	0.007	0.009	101 1.4 DL
Min Sxy	552	0	0	0	0	0	0.006	0.008	-0.001	0.007	0.009	101 1.4 DL
Max Mx	1418	-0.492	-0.053	0	0	0	71.324	20.505	4.918	76.242	25.423	103 1.2 (DL+TL2) + 1.6 LL
Min Mx	1437	-0.18	0.032	0	0	0	-76.66	10.597	5.708	-82.365	16.305	102 1.2 (DL+TL1) + 1.6 LL
Max My	1278	-0.179	0.363	0	0	0	4.135	27.102	-13.921	18.056	41.023	102 1.2 (DL+TL1) + 1.6 LL
Min My	1134	0.011	-0.129	0	0	0	2.715	-16.662	-0.606	3.321	-17.268	110 0.9 DL + 1.6 WL(+Z)
Max Mxy	1663	0.119	0.098	0	0	0	-36.86	-1.515	16.717	-53.581	-18.232	102 1.2 (DL+TL1) + 1.6 LL
Min Mxy	1164	0.043	0.335	0	0	0	-7.85	23.947	-15.476	-23.326	39.423	102 1.2 (DL+TL1) + 1.6 LL

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	50	mm
	top face	50	mm
Thicknes of slab =		400	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

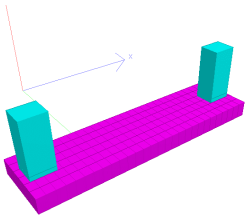
Force Values			Depth		Reinforcement Calculation													
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
103 1.2 (	76.24	top face	50	342	76.242	0.87	0.0021	718.4	307.8	16	280	200	1005.8	14.18	17.73	0.055	tension control
102 1.2 (	-82.37	bottom face	50	342	82.365	0.94	0.0023	777.0		16	258.87	200	1005.8	14.18	17.73	0.055	tension control
102 1.2 (	-53.58	top face	50	342	53.581	0.61	0.0015	502.6		16	400.22	200	1005.8	14.18	17.73	0.055	tension control
102 1.2 (	-23.33	top face	50	342	23.326	0.27	0.0006	217.5		16	653.49	200	1005.8	14.18	17.73	0.055	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c + 0.66\sqrt{f_c'}bd) \geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qx	0.538	1	215.2	48.655	0.003	0.342	349.592	340.826	586.757	340.83	1257	> 215.2	ok	255.6	φVc>Vu	not required
	Min Qx	0.492	1	196.8	71.324			339.863			339.86	1256	> 196.8	ok	254.9	φVc>Vu	not required

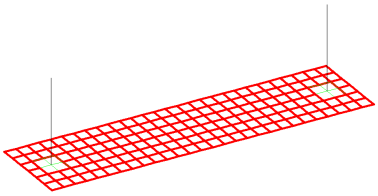
Reinforcement Along Z:

Force Values			Depth		Reinforcement Calculation													
Load condition	Factored Mt, My	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
102 1.2 (	41.02	top face	50	344	41.023	0.46	0.0011	381.6	309.6	12	296.52	200	565.8	7.978	9.972	0.1	tension control
110 0.9 D	-17.27	bottom face	50	344	17.268	0.19	0.0005	159.9		12	367.59	200	565.8	7.978	9.972	0.1	tension control
102 1.2 (	-18.23	bottom face	50	344	18.232	0.21	0.0005	168.8		12	367.59	200	565.8	7.978	9.972	0.1	tension control
102 1.2 (	39.42	top face	50	344	39.423	0.44	0.0011	366.6		12	308.65	200	565.8	7.978	9.972	0.1	tension control

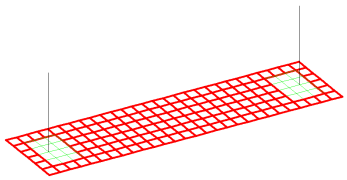
Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c + 0.66\sqrt{f_c'}bd) \geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	102 1.2 (	0.551	1	220.4	27.313	0.002	0.344	370.916	340.826	fc'586.8	340.83	356	> 220.4	ok	255.6	φVc>Vu	not required
	102 1.2 (	0.57	1	228	25.295			376.437			340.83	356	> 228	ok	255.6	φVc>Vu	not required



3D rendered view



Highlighted plate for Moment at face of column



Highlighted plate for Shear at 'd' distance from face of column

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :			
At Bottom	**16ϕT**	**200**	c/c
At Top	**16ϕT**	**200**	c/c
Along Z Direction :			
At Bottom	**12ϕT**	**200**	c/c
At Top	**12ϕT**	**200**	c/c

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5.7 DESIGN OF FOOTING TYPE F10

Size of Columns at node	Along-X	Along-Z
47	0.60m	0.60m
22	0.50m	0.50m
Size of fdn. raft:	2.50m	1.50m
Thickness of raft:	0.35 m	
Original Gr. Level:	0.00 m	
Finished Gr. Level:	0.00 m	
U/S of B/Pl Above GL:	0.84 m	
U/S level of fdn:	-1.20 m	
Unit weight of conc:	25 kN/m^3	
Unit weight of soil:	18 kN/m^3	

Design Input:

Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	50mm

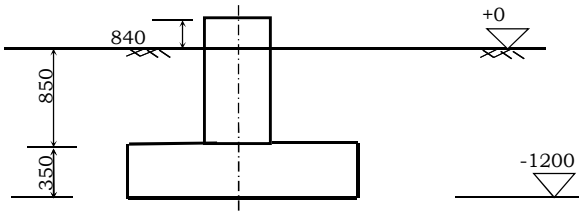
Calculation of Weight of Foundation :

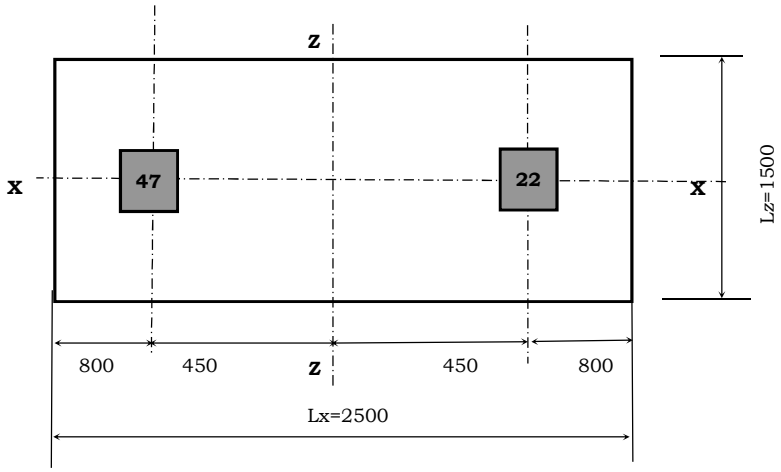
Weight of raft = (2.5*1.5*0.35*25) = 32.81KN

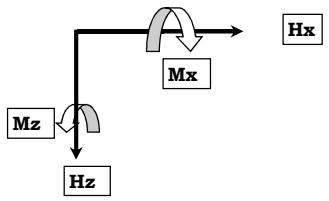
Weight of soil = ((2.5*1.5)-2*(0.6*0.6))*0.85*18 = 46.36KN

Total = 79.17KN

Submerged wt. = 79.17-2.5*1.5*1.2*10 = 34.17KN







H_x=Force along X-X direction

Hz=Force along Z-Z direction

M_x=Moment about X-X direction

Mz=Moment about Z-Z direction

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Check for overturning and sliding of footing F10

Load at bottom of fooring	Node 1					Node 2				
L/C No.	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
201 1. DL	44.6	-0.1	0.1	0.0	0.0	16.1	0.4	-0.7	0.1	0.2
202 1 (DL+TL1) + 1 LL	79.6	-0.1	-0.4	0.0	0.0	22.0	1.8	-3.7	0.4	0.7
203 1 (DL+TL2) + 1 LL	80.0	-0.1	1.0	0.0	0.0	20.6	0.6	-1.2	0.1	0.2
204 1 DL+ 1 WL(+X)	42.9	-0.1	0.2	0.0	0.0	16.5	-0.5	1.0	0.2	0.5
205 1 DL+ 1 WL(-X)	46.2	-0.1	0.1	0.0	0.0	15.6	1.2	-2.5	-0.1	-0.2
206 1 DL+ 1 WL(+Z)	44.7	-0.1	0.1	0.0	0.0	5.4	0.2	-0.5	-14.3	-29.6
207 1 DL+ 1 WL(-Z)	44.4	-0.1	0.1	0.0	0.0	26.8	0.5	-1.0	14.5	29.9
208 0.6 DL + 1 WL(+X)	25.1	0.0	0.1	0.0	0.0	10.0	-0.6	1.3	0.2	0.4
209 0.6 DL + 1 WL(-X)	28.4	0.0	0.0	0.0	0.0	9.2	1.1	-2.2	-0.1	-0.3
210 0.6 DL + 1 WL(+Z)	26.9	0.0	0.1	0.0	0.0	-1.0	0.1	-0.2	-14.3	-29.7
211 0.6 DL + 1 WL(-Z)	26.6	0.0	0.1	0.0	0.0	20.3	0.3	-0.7	14.4	29.9
212 1 DL+ 0.75 WL(+X)+ 0.75 LL	69.8	-0.1	0.3	0.0	0.0	20.3	0.3	-0.7	0.3	0.6
213 1 DL+ 0.75 WL(-X)+ 0.75 LL	72.2	-0.1	0.2	0.0	0.0	19.6	1.6	-3.4	0.1	0.1
214 1 DL+ 0.75 WL(+Z)+ 0.75 LL	71.1	-0.1	0.2	0.0	0.0	12.0	0.9	-1.8	-10.6	-21.9
215 1 DL+ 0.75 WL(-Z)+ 0.75 LL	70.9	-0.1	0.2	0.0	0.0	28.0	1.1	-2.2	11.0	22.7

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Check for overturning and sliding:

L/C No.	Resultant Load at bottom of footing					Overturning		Sliding	
	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
201 1. DL	139.8	0.30	-13.4	0.1	0.2	13.0	631.7	186.4	699.1
202 1 (DL+TL1) + 1 LL	180.8	1.67	-30.1	0.4	0.7	7.5	182.0	43.3	200.9
203 1 (DL+TL2) + 1 LL	179.8	0.43	-26.9	0.1	0.2	8.4	543.6	167.2	599.2
204 1 DL+ 1 WL(+X)	138.6	-0.54	-10.7	0.2	0.5	16.1	228.4	102.6	251.9
205 1 DL+ 1 WL(-X)	141.0	1.16	-16.2	-0.1	-0.2	10.9	510.8	48.6	564.0
206 1 DL+ 1 WL(+Z)	129.3	0.18	-18.1	-14.3	-29.6	8.9	3.3	287.4	3.6
207 1 DL+ 1 WL(-Z)	150.3	0.42	-8.8	14.5	29.9	21.4	3.8	143.2	4.2
208 0.6 DL + 1 WL(+X)	114.3	-0.66	-5.4	0.2	0.4	26.6	220.4	69.3	243.2
209 0.6 DL + 1 WL(-X)	116.7	1.04	-10.8	-0.1	-0.3	13.5	320.7	44.9	353.7
210 0.6 DL + 1 WL(+Z)	105.1	0.06	-12.7	-14.3	-29.7	10.3	2.7	700.3	2.9
211 0.6 DL + 1 WL(-Z)	126.1	0.30	-3.4	14.4	29.9	46.0	3.2	168.1	3.5
212 1 DL+ 0.75 WL(+X)+ 0.75 LL	169.2	0.23	-22.7	0.3	0.6	9.3	201.1	291.7	221.9
213 1 DL+ 0.75 WL(-X)+ 0.75 LL	171.0	1.51	-26.8	0.1	0.1	8.0	950.2	45.4	1052.5
214 1 DL+ 0.75 WL(+Z)+ 0.75 LL	162.3	0.77	-28.2	-10.6	-21.9	7.2	5.6	84.1	6.1
215 1 DL+ 0.75 WL(-Z)+ 0.75 LL	178.0	0.95	-21.3	11.0	22.7	10.5	5.9	74.8	6.5
					Min.=	**7.2**	**2.7**	**43.3**	**2.9**

Note: Foundation is critical in Stability criteria. Hence, detail reinforcement calc not performed.

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5.8 DESIGN OF FOOTING TYPE F11

	Along-X	Along-Z
Size of column :	0.60m	0.60m
Size of fdn. raft :	0.90m	0.90m
Thickness of raft :	0.35	m
Original ground level:	0.00	m
Finished floor level :	0.00	m
U/S level of foundation :	-1.20	m
U/S level Base Plate :	0.87	m
Unit weight of conc :	24	kN/m^3
Unit weight of soil :	18	kN/m^3

SYMBOLS AND NOTATIONS

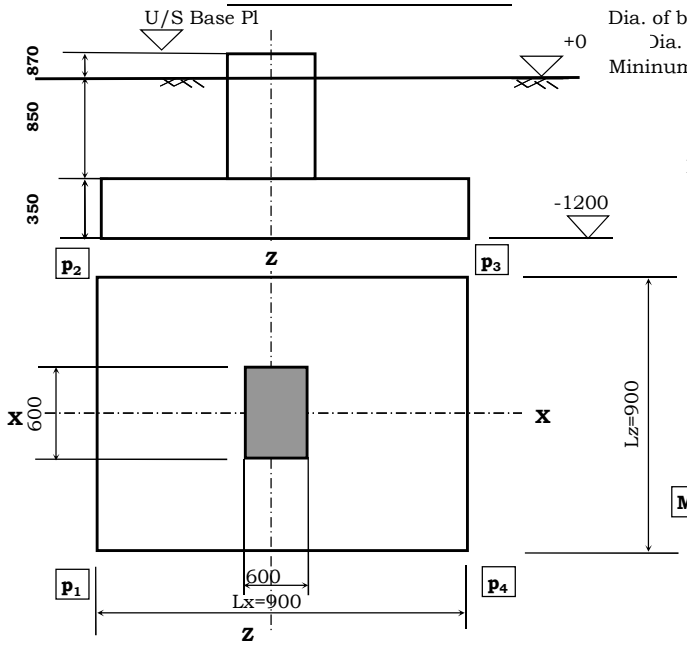
Dead Load	DL
Normal Operational Fluid Load	FL1
Max. Fluid Load	FL2
Wind Load	WL
Seismic Load	SL

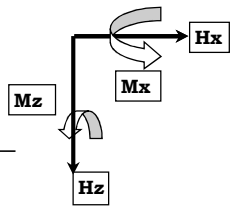
Calculation of Weight of Foundation :

ght of raft & Col.=	(0.9*0.9*0.35*24)+(0.6*0.6*1.72*24) =	21.66KN
Weight of soil =	((0.9*0.9)-(0.6*0.6))*0.85*18 =	6.89KN
	Total =	28.55KN

Design Input:

Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	50mm
Dia. of bars provd. along X direction at bottom =	12mm
Dia. of bars provd. along X direction at top =	12mm
Dia. of bars provided along Z direction at bottom=	12mm
Dia. of bars provided along Z direction at top=	12mm
Mininum percentage of reinforcement at bottom=	0.18%





Effective depth of slab along X direction = 294mm = 0.294m

Effective depth of slab along Z direction = 282mm = 0.282m

Clear projection along Z direction = 0.15m

Clear projection along X direction = 0.15m

H_x=Force along X-X direction

Hz=Force along Z-Z direction

M_x=Moment about X-X direction

Mz=Moment about Z-Z direction

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5.8 DESIGN OF FOOTING TYPE F11

Combined Load Cases:	**Load at bottom of foundation**					**Load at bottom of foundation + self weight**				
Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
Allowable Stress Design										
201 1. DL	44.6	-0.1	0.1	0.0	0.0	73.11	-0.06	0.12	0.00	0.00
202 1 (DL+TL1) + 1 LL	79.6	-0.1	-0.4	0.0	0.0	108.20	-0.13	-0.44	0.00	0.00
203 1 (DL+TL2) + 1 LL	80.0	-0.1	1.0	0.0	0.0	108.52	-0.13	0.98	0.00	0.00
204 1 DL+ 1 WL(+X)	42.9	-0.1	0.2	0.0	0.0	71.46	-0.06	0.16	0.00	0.00
205 1 DL+ 1 WL(-X)	46.2	-0.1	0.1	0.0	0.0	74.76	-0.06	0.08	0.00	0.00
206 1 DL+ 1 WL(+Z)	44.7	-0.1	0.1	0.0	0.0	73.28	-0.06	0.12	0.00	0.00
207 1 DL+ 1 WL(-Z)	44.4	-0.1	0.1	0.0	0.0	72.94	-0.06	0.12	0.00	0.00
208 0.6 DL + 1 WL(+X)	25.1	0.0	0.1	0.0	0.0	53.64	-0.04	0.12	0.00	0.00
209 0.6 DL + 1 WL(-X)	28.4	0.0	0.0	0.0	0.0	56.94	-0.04	0.04	0.00	0.00
210 0.6 DL + 1 WL(+Z)	26.9	0.0	0.1	0.0	0.0	55.46	-0.04	0.08	0.00	0.00
211 0.6 DL + 1 WL(-Z)	26.6	0.0	0.1	0.0	0.0	55.12	-0.04	0.08	0.00	0.00
212 1 DL+ 0.75 WL(+X)+ 0.75 LL	69.8	-0.1	0.3	0.0	0.0	98.31	-0.11	0.26	0.00	0.00
213 1 DL+ 0.75 WL(-X)+ 0.75 LL	72.2	-0.1	0.2	0.0	0.0	100.78	-0.11	0.20	0.00	0.00
214 1 DL+ 0.75 WL(+Z)+ 0.75 LL	71.1	-0.1	0.2	0.0	0.0	99.67	-0.11	0.23	0.00	0.00
215 1 DL+ 0.75 WL(-Z)+ 0.75 LL	70.9	-0.1	0.2	0.0	0.0	99.42	-0.11	0.23	0.00	0.00

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Combined Load Cases:	**Load at bottom of foundation**					**Load at bottom of foundation + self weight**				
Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
Allowable Strength Design										
101 1.4 DL	62.4	-0.1	0.2	0.0	0.0	90.93	-0.08	0.17	0.00	0.00
102 1.2 (DL+TL1) + 1.6 LL	109.7	-0.2	-0.5	0.0	0.0	138.23	-0.18	-0.47	0.00	0.00
103 1.2 (DL+TL2) + 1.6 LL	110.1	-0.2	1.2	0.0	0.0	138.61	-0.18	1.23	0.00	0.00
104 1.2 DL+ 1.6 WL(+X) + 0.5 LL	68.5	-0.1	0.3	0.0	0.0	97.01	-0.11	0.29	0.00	0.00
105 1.2 DL+ 1.6 WL(-X) + 0.5 LL	73.7	-0.1	0.2	0.0	0.0	102.29	-0.11	0.16	0.00	0.00
106 1.2 DL+ 1.6 WL(+Z) + 0.5 LL	71.4	-0.1	0.2	0.0	0.0	99.92	-0.11	0.22	0.00	0.00
107 1.2 DL+ 1.6 WL(-Z) + 0.5 LL	70.8	-0.1	0.2	0.0	0.0	99.37	-0.11	0.22	0.00	0.00
108 0.9 DL + 1.6 WL(+X)	37.5	-0.1	0.2	0.0	0.0	66.01	-0.05	0.18	0.00	0.00
109 0.9 DL + 1.6 WL(-X)	42.7	-0.1	0.0	0.0	0.0	71.29	-0.05	0.05	0.00	0.00
110 0.9 DL + 1.6 WL(+Z)	40.4	-0.1	0.1	0.0	0.0	68.92	-0.05	0.11	0.00	0.00
111 0.9 DL + 1.6 WL(-Z)	39.8	-0.1	0.1	0.0	0.0	68.38	-0.05	0.11	0.00	0.00
112 1.2 DL + 0.8 WL(+X)	52.2	-0.1	0.2	0.0	0.0	80.70	-0.07	0.18	0.00	0.00
113 1.2 DL + 0.8 WL(-X)	54.8	-0.1	0.1	0.0	0.0	83.34	-0.07	0.12	0.00	0.00
114 1.2 DL + 0.8 WL(+Z)	53.6	-0.1	0.1	0.0	0.0	82.16	-0.07	0.15	0.00	0.00
115 1.2 DL + 0.8 WL(-Z)	53.6	-0.1	0.1	0.0	0.0	82.16	-0.07	0.15	0.00	0.00

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Check for overturning and sliding:

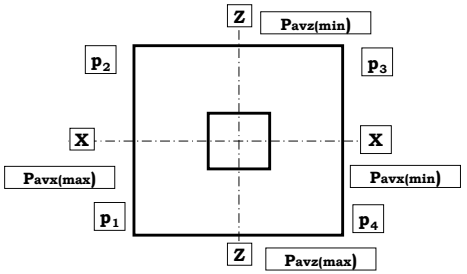
						Overturning		Sliding	
Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
201 1. DL	73.1	-0.1	0.1	0.0	0.0	265.3	NA	487.4	NA
202 1 (DL+TL1) + 1 LL	108.2	-0.1	-0.4	0.0	0.0	110.4	NA	332.9	NA
203 1 (DL+TL2) + 1 LL	108.5	-0.1	1.0	0.0	0.0	49.9	NA	333.9	NA
204 1 DL+ 1 WL(+X)	71.5	-0.1	0.2	0.0	0.0	196.1	NA	476.4	NA
205 1 DL+ 1 WL(-X)	74.8	-0.1	0.1	0.0	0.0	400.5	NA	498.4	NA
206 1 DL+ 1 WL(+Z)	73.3	-0.1	0.1	0.0	0.0	265.9	NA	488.5	NA
207 1 DL+ 1 WL(-Z)	72.9	-0.1	0.1	0.0	0.0	264.7	NA	486.3	NA
208 0.6 DL + 1 WL(+X)	53.6	0.0	0.1	0.0	0.0	209.9	NA	595.9	NA
209 0.6 DL + 1 WL(-X)	56.9	0.0	0.0	0.0	0.0	732.0	NA	632.6	NA
210 0.6 DL + 1 WL(+Z)	55.5	0.0	0.1	0.0	0.0	332.7	NA	616.2	NA
211 0.6 DL + 1 WL(-Z)	55.1	0.0	0.1	0.0	0.0	330.7	NA	612.4	NA
212 1 DL+ 0.75 WL(+X)+ 0.75 LL	98.3	-0.1	0.3	0.0	0.0	168.2	NA	348.0	NA
213 1 DL+ 0.75 WL(-X)+ 0.75 LL	100.8	-0.1	0.2	0.0	0.0	223.4	NA	356.8	NA
214 1 DL+ 0.75 WL(+Z)+ 0.75 LL	99.7	-0.1	0.2	0.0	0.0	192.5	NA	352.8	NA
215 1 DL+ 0.75 WL(-Z)+ 0.75 LL	99.4	-0.1	0.2	0.0	0.0	192.0	NA	351.9	NA
					Min.=	49.9	0.0	332.9	0.0
						OK	-	OK	-

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5.8 DESIGN OF FOOTING TYPE F11



Load Combination & Calculation of Base pressure

Max. Base pressure = p_{max} = (V/(Lx.Lz)) + (6.Mz/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)

Min. Base pressure = p_{min} = (V/(Lx.Lz)) - (6.Mz/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)

Surcharge load = 10 kN/m^2

Net SBC of soil for DL, LL= 350 kN/m^2 For WL/SL case = 437.5 kN/m^2

Gross SBC of soil for DL, LL= 371.6 kN/m^2 For WL/SL case = 459.1 kN/m^2

Allowable Stress Design

Load Combination	Design Load/moments			Base Pressure (kN/m^2)													
	V (kN)	Mz (kN-m)	Mx (kN-m)	V/ (Lx.Lz)	6.Mz/ (Lx)^2.Lz	6.Mx/ Lx.(Lz)^2	p_1 (p_{max})	p_2	p_4	p_3 (p_{min})	(p_{max}-γh)	$p_{avx(max)}$	$p_{avx(min)}$	$p_{avz(max)}$	$p_{avz(min)}$	Remarks
201 1. DL	73.11	0.12	0.00	90.26	1.02	0.00	91.28	91.28	89.24	89.24	69.68	91.28	89.24	90.26	90.26	O.K
202 1 (DL+TL1) + 1 LL	108.20	-0.44	0.00	133.58	3.63	0.00	137.21	137.21	129.95	129.95	115.61	137.21	129.95	133.58	133.58	O.K
203 1 (DL+TL2) + 1 LL	108.52	0.98	0.00	133.97	8.06	0.00	142.03	142.03	125.92	125.92	120.43	142.03	125.92	133.98	133.98	O.K
204 1 DL+ 1 WL(+X)	71.46	0.16	0.00	88.22	1.35	0.00	89.57	89.57	86.87	86.87	67.97	89.57	86.87	88.22	88.22	O.K
205 1 DL+ 1 WL(-X)	74.76	0.08	0.00	92.29	0.69	0.00	92.99	92.99	91.60	91.60	71.39	92.99	91.60	92.30	92.30	O.K
206 1 DL+ 1 WL(+Z)	73.28	0.12	0.00	90.47	1.02	0.00	91.49	91.49	89.45	89.45	69.89	91.49	89.45	90.47	90.47	O.K
207 1 DL+ 1 WL(-Z)	72.94	0.12	0.00	90.05	1.02	0.00	91.07	91.07	89.03	89.03	69.47	91.07	89.03	90.05	90.05	O.K
208 0.6 DL + 1 WL(+X)	53.64	0.12	0.00	66.22	0.95	0.00	67.16	67.16	65.27	65.27	45.56	67.16	65.27	66.22	66.22	O.K
209 0.6 DL + 1 WL(-X)	56.94	0.04	0.00	70.29	0.29	0.00	70.58	70.58	70.00	70.00	48.98	70.58	70.00	70.29	70.29	O.K
210 0.6 DL + 1 WL(+Z)	55.46	0.08	0.00	68.46	0.62	0.00	69.08	69.08	67.85	67.85	47.48	69.08	67.85	68.47	68.47	O.K
211 0.6 DL + 1 WL(-Z)	55.12	0.08	0.00	68.04	0.62	0.00	68.66	68.66	67.43	67.43	47.06	68.66	67.43	68.05	68.05	O.K
212 1 DL+ 0.75 WL(+X)+ 0.75 LL	98.31	0.26	0.00	121.37	2.16	0.00	123.53	123.53	119.20	119.20	101.93	123.53	119.20	121.37	121.37	O.K
213 1 DL+ 0.75 WL(-X)+ 0.75 LL	100.78	0.20	0.00	124.42	1.67	0.00	126.09	126.09	122.75	122.75	104.49	126.09	122.75	124.42	124.42	O.K
214 1 DL+ 0.75 WL(+Z)+ 0.75 LL	99.67	0.23	0.00	123.05	1.92	0.00	124.97	124.97	121.14	121.14	103.37	124.97	121.14	123.06	123.06	O.K
215 1 DL+ 0.75 WL(-Z)+ 0.75 LL	99.42	0.23	0.00	122.74	1.92	0.00	124.66	124.66	120.82	120.82	103.06	124.66	120.82	122.74	122.74	O.K

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Strength Design

101 1.4 DL	90.93	0.17	0.00	112.26	1.43	0.00	113.69	113.69	110.83	110.83	92.09	113.69	110.83	112.26	112.26	
102 1.2 (DL+TL1) + 1.6 LL	138.23	-0.47	0.00	170.65	3.88	0.00	174.53	174.53	166.78	166.78	152.93	174.53	166.78	170.66	170.66	
103 1.2 (DL+TL2) + 1.6 LL	138.61	1.23	0.00	171.13	10.15	0.00	181.27	181.27	160.98	160.98	159.67	181.27	160.98	171.13	171.13	
104 1.2 DL+ 1.6 WL(+X) + 0.5 LL	97.01	0.29	0.00	119.76	2.35	0.00	122.10	122.10	117.41	117.41	100.50	122.10	117.41	119.76	119.76	
105 1.2 DL+ 1.6 WL(-X) + 0.5 LL	102.29	0.16	0.00	126.28	1.29	0.00	127.57	127.57	124.99	124.99	105.97	127.57	124.99	126.28	126.28	
106 1.2 DL+ 1.6 WL(+Z) + 0.5 LL	99.92	0.22	0.00	123.35	1.82	0.00	125.17	125.17	121.54	121.54	103.57	125.17	121.54	123.36	123.36	
107 1.2 DL+ 1.6 WL(-Z) + 0.5 LL	99.37	0.22	0.00	122.68	1.82	0.00	124.50	124.50	120.86	120.86	102.90	124.50	120.86	122.68	122.68	
108 0.9 DL + 1.6 WL(+X)	66.01	0.18	0.00	81.50	1.45	0.00	82.94	82.94	80.05	80.05	61.34	82.94	80.05	81.50	81.50	
109 0.9 DL + 1.6 WL(-X)	71.29	0.05	0.00	88.01	0.40	0.00	88.41	88.41	87.62	87.62	66.81	88.41	87.62	88.02	88.02	
110 0.9 DL + 1.6 WL(+Z)	68.92	0.11	0.00	85.09	0.92	0.00	86.01	86.01	84.17	84.17	64.41	86.01	84.17	85.09	85.09	
111 0.9 DL + 1.6 WL(-Z)	68.38	0.11	0.00	84.42	0.92	0.00	85.34	85.34	83.50	83.50	63.74	85.34	83.50	84.42	84.42	
112 1.2 DL + 0.8 WL(+X)	80.70	0.18	0.00	99.63	1.49	0.00	101.12	101.12	98.14	98.14	79.52	101.12	98.14	99.63	99.63	
113 1.2 DL + 0.8 WL(-X)	83.34	0.12	0.00	102.89	0.96	0.00	103.85	103.85	101.93	101.93	82.25	103.85	101.93	102.89	102.89	
114 1.2 DL + 0.8 WL(+Z)	82.16	0.15	0.00	101.43	1.23	0.00	102.65	102.65	100.20	100.20	81.05	102.65	100.20	101.43	101.43	
115 1.2 DL + 0.8 WL(-Z)	82.16	0.15	0.00	101.43	1.23	0.00	102.65	102.65	100.20	100.20	81.05	102.65	100.20	101.43	101.43	
Max :	139	1	0													
Min :	54	0	0													

Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL cases) = 120.43 kN/m²

Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL cases) = 130.43 kN/m²

Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL+SL/WL cases) = 120.43 kN/m²

Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL+WL/SL cases) = 130.43 kN/m²

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Sample calculation of values of base pressure:

Load case : 02 1 (DL+TL1) + 1 L

Maximum pressure, p_1 (pmax) = 133.58+3.63+0.00= 137.21 kN/m^2

p_2= 133.58+3.63-0.00= 137.21 kN/m^2

p_3= 133.58-(3.63+0.00)= 129.95 kN/m^2

p_4= 133.58-3.63+0.00= 129.95 kN/m^2

Maxm pressure excluding weight of soil (p_{max}-γh)= 137.21-1.2*18= 115.61 kN/m^2

Maximum average Pressure along X, $p_{avx(max)}$ = (137.21+137.21)/2= 137.21 kN/m^2

Minimum average Pressure along X, $p_{avx(minx)}$ = (129.95+129.95)/2= 129.95 kN/m^2

Maximum average Pressure along Y, $p_{avy(max)}$ = (137.21+129.95)/2= 133.58 kN/m^2

Minimum average Pressure along Y, $p_{avy(minx)}$ = (129.95+137.21)/2= 133.58 kN/m^2

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5.8 DESIGN OF FOOTING TYPE F11

Calculation of Bending Moment and Shear Force in X direction :

p_{1x} =pressure at the face of column at section 1x-1x in kN/m^2.

p_{2x} =pressure at the face of column at section 2x-2x in kN/m^2.

p_{3x} =pressure at a distance 'd' from the face of column at section 3x-3x in kN/m^2.

BM_{1x} =Moment at the face of column at section 1x-1x in kN-m/m.

BM_{2x} =Moment at the face of column 2x-2x in kN-m/m.

SF_{3x} = Oneway shear force at a distance 'd' from the face of column at section 3x-3x in kN/m.

SF (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Punching perimeter= ((0.6+2*0.5*0.282)+(0.6+2*0.5*0.282))*2

= 3.528 m

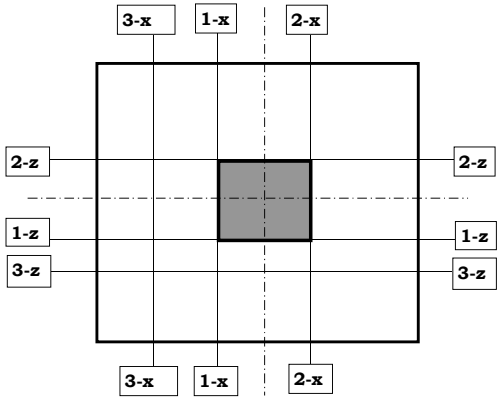
Punching Area= ((0.6+2*0.5*0.282)*(0.6+2*0.5*0.282))

= 0.78 m^2 Punching shear stress to be checked

Downward pressure due to self weight of raft and soil = (0.35*24+0.85*18)

= 23.70 kN/m^2

Area of footing, A_f= 0.81 m^2



Load Combination	p_{1x} kN/m2	p_{2x} kN/m2	p_{3x} kN/m2	$B.M_{1x}$ kNm	$B.M_{2x}$ kNm	$S.F_{3x}$ kN	$S.F_{(punching)}$ kN
Ultimate Conditions							
101 1.4 DL	113.21	111.31	114.15	1.01	0.98	-12.99	2.66
102 1.2 (DL+TL1) + 1.6 LL	173.24	168.07	175.77	1.69	1.61	-21.81	4.41
103 1.2 (DL+TL2) + 1.6 LL	177.89	164.36	184.52	1.76	1.56	-22.92	4.42
104 1.2 DL+ 1.6 WL(+X) + 0.5 LL	121.32	118.19	122.85	1.10	1.06	-14.22	2.88
105 1.2 DL+ 1.6 WL(-X) + 0.5 LL	127.14	125.42	127.98	1.17	1.14	-14.99	3.08
106 1.2 DL+ 1.6 WL(+Z) + 0.5 LL	124.57	122.15	125.75	1.14	1.10	-14.65	2.99
107 1.2 DL+ 1.6 WL(-Z) + 0.5 LL	123.89	121.47	125.08	1.13	1.10	-14.56	2.97
108 0.9 DL + 1.6 WL(+X)	82.46	80.53	83.40	0.66	0.64	-8.56	1.73
109 0.9 DL + 1.6 WL(-X)	88.28	87.75	88.54	0.73	0.72	-9.33	1.93
110 0.9 DL + 1.6 WL(+Z)	85.70	84.48	86.30	0.70	0.68	-8.99	1.84
111 0.9 DL + 1.6 WL(-Z)	85.03	83.81	85.63	0.69	0.67	-8.90	1.82
112 1.2 DL + 0.8 WL(+X)	100.62	98.64	101.60	0.87	0.84	-11.18	2.28
113 1.2 DL + 0.8 WL(-X)	103.53	102.25	104.16	0.90	0.88	-11.56	2.38
114 1.2 DL + 0.8 WL(+Z)	102.24	100.61	103.04	0.89	0.86	-11.40	2.33
115 1.2 DL + 0.8 WL(-Z)	102.24	100.61	103.04	0.89	0.86	-11.40	2.33

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Sample calculation of BM & SF

Load case : 101 1.4 DL

p_{1x} = 113.69-(113.69-110.83)*0.15/0.9 = 113.21 kN/m^2 Overhang = 0.15 m

p_{2x} = 110.83+(113.69-110.83)*0.15/0.9 = 111.31 kN/m^2 Effective depth = 0.294 m

p_{3x} = 113.69-(113.69-110.83)*(0.15-0.294)/0.9 114.15 kN/m^2

BM_{1x} = (113.21-23.7)*0.15*0.15/2+0.5*(113.69-113.21)*0.15*0.15*0.67 = 1.01 kN-m/m

BM_{2x} = (111.31-23.7)*0.15*0.15/2-0.5*(-110.83+111.31)*0.15*0.15*0.67 = 0.98 kN-m/m

SF_{3x} = (114.15-23.7)*(0.15-0.294)+0.5*(113.69-114.15)*(0.15-0.294) = -12.99 kN/m

$S.F_{(punching)}$ = (0.81-0.78)*(112.26-23.7) = 2.66 kN

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	1.76	0.02	0.03	0.000	19.01	529.20	12ϕT	213.71	300	150	753.98
At Top	0.00	0.00	0.00	0.000	0.00	264.60	12ϕT	427.43	300	250	452.39

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc)/bd = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= -8.56 kN/m

Shear stress , τ_v= (-8.56*1000)/(1000*294)

= -0.03 N/mm^2 **O.K.**

Check against Two-way shear force at a distance d/2 from face of column:

Design punching shear at a distance d/2 from face of column, Vu= 4.42 kN

Strength of Concrete in shear Vc1 = 3068.17 kN **Table 22.6.5.2_ (a)**

Strength of Concrete in shear Vc2 = 1874.99 kN **Table 22.6.5.2_ (b)**

Strength of Concrete in shear Vc3 = 2045.45 kN **Table 22.6.5.2_ (c)**

Design Vc= 1874.99 kN

Allowable shear stress, (ϕ Vc)/bd = 1.36 N/mm^2

Design punching shear stress, τ = (4.42*1000)/(3528*294)

= 0.00 N/mm^2 **No shear reinforcement reqd.**

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Calculation of Bending Moment and Shear Force in Z direction :

p1z =pressure at the face of column at section 1z-1z in kN/m^2.
p2z =pressure at the face of column at section 2z-2z in kN/m^2.
p3z =pressure at a distance d from the face of column at section 3z-3z in kN/m^2.
B.M1z =Moment at the face of column at section 1z-1z in kN-m/m.
B.M2z =Moment at the face of column at section 2z-2z in kN-m/m.
S.F 3z = Oneway Shear Force at a distance d from the face of column at section 3z-3z in kN/m.
S.F (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Load Combination	p_{1z} kN/m2	p_{2z} kN/m2	p_{3z} kN/m2	$B.M_{1z}$ kNm	$B.M_{2z}$ kNm	$S.F_{3z}$ kN
Ultimate Conditions						
101 1.4 DL	112.26	112.26	112.26	1.00	1.00	-11.69
102 1.2 (DL+TL1) + 1.6 LL	170.66	170.66	170.66	1.65	1.65	-19.40
103 1.2 (DL+TL2) + 1.6 LL	171.13	171.13	171.13	1.66	1.66	-19.46
104 1.2 DL+ 1.6 WL(+X) + 0.5 LL	119.76	119.76	119.76	1.08	1.08	-12.68
105 1.2 DL+ 1.6 WL(-X) + 0.5 LL	126.28	126.28	126.28	1.15	1.15	-13.54
106 1.2 DL+ 1.6 WL(+Z) + 0.5 LL	123.36	123.36	123.36	1.12	1.12	-13.16
107 1.2 DL+ 1.6 WL(-Z) + 0.5 LL	122.68	122.68	122.68	1.11	1.11	-13.07
108 0.9 DL + 1.6 WL(+X)	81.50	81.50	81.50	0.65	0.65	-7.63
109 0.9 DL + 1.6 WL(-X)	88.02	88.02	88.02	0.72	0.72	-8.49
110 0.9 DL + 1.6 WL(+Z)	85.09	85.09	85.09	0.69	0.69	-8.10
111 0.9 DL + 1.6 WL(-Z)	84.42	84.42	84.42	0.68	0.68	-8.02
112 1.2 DL + 0.8 WL(+X)	99.63	99.63	99.63	0.85	0.85	-10.02
113 1.2 DL + 0.8 WL(-X)	102.89	102.89	102.89	0.89	0.89	-10.45
114 1.2 DL + 0.8 WL(+Z)	101.43	101.43	101.43	0.87	0.87	-10.26
115 1.2 DL + 0.8 WL(-Z)	101.43	101.43	101.43	0.87	0.87	-10.26

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Sample calculation of BM & SF

Load case : 101 1.4 DL

p_{1z} = 112.26-(112.26-112.26)*0.15/0.9= 112.26 kN/m^2

p_{2z} = 112.26+(112.26-112.26)*0.15/0.9= 112.26 kN/m^2

p_{3z} = 112.26-(112.26-112.26)*(0.15-0.282)/0.9 112.26 kN/m^2

Overhang = 0.15 m

Effective depth = 0.282 m

BM_{1z} = (112.26-23.7)*0.15*0.15/2+0.5*(112.26-112.26)*0.15*0.15*0.67 = 1.00 kN-m/m

BM_{2z} = (112.26-23.7)*0.15*0.15/2-0.5*(-112.26+112.26)*0.15*0.15*0.67 = 1.00 kN-m/m

SF_{3z} = (112.26-23.7)*(0.15-0.282)+0.5*(112.26-112.26)*(0.15-0.282) = -11.69 kN/m

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	1.66	0.021	0.028	0.0001	19.48	507.60	12ϕT	222.81	300	150	754.0
Top	0.00	0.000	0.000	0.0000	0.00	253.80	12ϕT	445.62	300	250	452.4

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc) = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= -7.63 kN/m

Shear stress , τ_v= (-7.63*1000)/(1000*282)

= -0.03 N/mm^2 **O.K.**

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :

At Bottom **12ϕT** **150** c/c

At Top **12ϕT** **250** c/c

Along Z Direction :

At Bottom **12ϕT** **150** c/c

At Top **12ϕT** **250** c/c

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5.9 DESIGN OF FOOTING TYPE F13

	Along-X	Along-Z
Size of column :	0.50m	0.50m
Size of fdn. raft :	1.50m	1.50m
Thickness of raft :	0.35	m
Original ground level:	0.00	m
Finished floor level :	0.00	m
U/S level of foundation :	-1.20	m
U/S level Base Plate :	0.87	m
Unit weight of conc :	24	kN/m^3
Unit weight of soil :	18	kN/m^3

SYMBOLS AND NOTATIONS

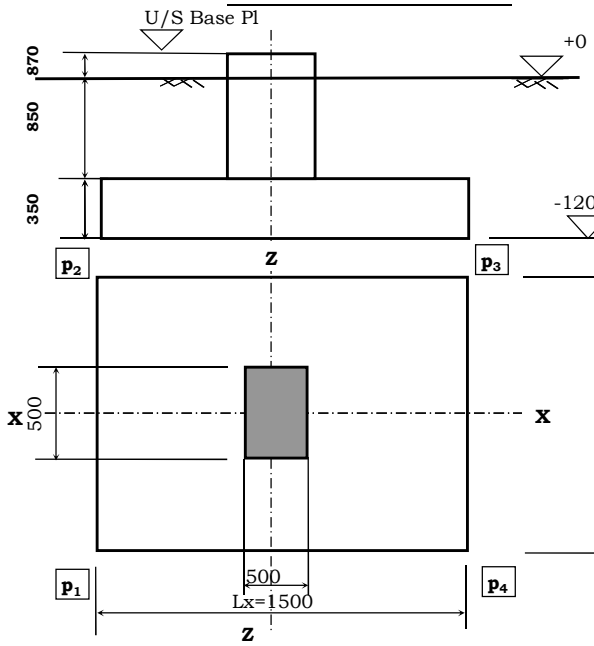
Dead Load	DL
Normal Operational Fluid Load	FL1
Max. Fluid Load	FL2
Wind Load	WL
Seismic Load	SL

Calculation of Weight of Foundation :

ght of raft & Col.=	(1.5*1.5*0.35*24)+(0.5*0.5*1.72*24) =	29.22KN
Weight of soil =	((1.5*1.5)-(0.5*0.5))*0.85*18 =	30.60KN
	Total =	59.82KN

Design Input:

Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	50mm
Dia. of bars provd. along X direction at bottom =	12mm
Dia. of bars provd. along X direction at top =	10mm
Dia. of bars provided along Z direction at bottom=	12mm
Dia. of bars provided along Z direction at top=	10mm
Mininum percentage of reinforcement at bottom=	0.18%

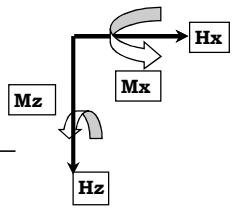


Effective depth of slab along X direction = 294mm = 0.294m

Effective depth of slab along Z direction = 282mm = 0.282m

Clear projection along Z direction = 0.5m

Clear projection along X direction = 0.5m



H_x=Force along X-X direction

Hz=Force along Z-Z direction

M_x=Moment about X-X direction

Mz=Moment about Z-Z direction

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5.9 DESIGN OF FOOTING TYPE F13

Combined Load Cases:	**Load at bottom of foundation**					**Load at bottom of foundation + self weight**				
Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
Allowable Stress Design										
201 1. DL	24.0	0.0	0.0	0.0	0.0	83.83	0.02	-0.04	0.00	0.00
202 1 (DL+TL1) + 1 LL	37.8	0.0	0.0	0.0	0.0	97.63	0.02	-0.04	0.00	0.00
203 1 (DL+TL2) + 1 LL	38.3	0.1	-0.2	0.0	0.0	98.09	0.12	-0.25	0.00	0.00
204 1 DL+ 1 WL(+X)	31.5	-12.4	25.6	0.0	0.0	91.29	-12.36	25.59	0.00	0.00
205 1 DL+ 1 WL(-X)	16.5	12.4	-25.7	0.0	0.0	76.37	12.40	-25.67	0.00	0.00
206 1 DL+ 1 WL(+Z)	19.3	0.0	-0.1	0.0	0.0	79.13	0.04	-0.08	0.00	0.00
207 1 DL+ 1 WL(-Z)	28.7	0.0	0.0	0.0	0.0	88.53	0.00	0.00	0.00	0.00
208 0.6 DL + 1 WL(+X)	21.9	-12.4	25.6	0.0	0.0	81.69	-12.37	25.60	0.00	0.00
209 0.6 DL + 1 WL(-X)	6.9	12.4	-25.7	0.0	0.0	66.77	12.39	-25.65	0.00	0.00
210 0.6 DL + 1 WL(+Z)	9.7	0.0	-0.1	0.0	0.0	69.53	0.03	-0.07	0.00	0.00
211 0.6 DL + 1 WL(-Z)	19.1	0.0	0.0	0.0	0.0	78.93	-0.01	0.02	0.00	0.00
212 1 DL+ 0.75 WL(+X)+ 0.75 LL	40.1	-9.2	19.1	0.0	0.0	99.95	-9.23	19.10	0.00	0.00
213 1 DL+ 0.75 WL(-X)+ 0.75 LL	28.9	9.3	-19.3	0.0	0.0	88.76	9.34	-19.34	0.00	0.00
214 1 DL+ 0.75 WL(+Z)+ 0.75 LL	31.0	0.1	-0.2	0.0	0.0	90.83	0.07	-0.15	0.00	0.00
215 1 DL+ 0.75 WL(-Z)+ 0.75 LL	38.1	0.0	-0.1	0.0	0.0	97.88	0.04	-0.09	0.00	0.00

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Combined Load Cases:	**Load at bottom of foundation**					**Load at bottom of foundation + self weight**				
Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
Allowable Strength Design										
101 1.4 DL	33.6	0.0	-0.1	0.0	0.0	93.43	0.03	-0.06	0.00	0.00
102 1.2 (DL+TL1) + 1.6 LL	51.0	0.0	-0.1	0.0	0.0	110.80	0.04	-0.09	0.00	0.00
103 1.2 (DL+TL2) + 1.6 LL	51.5	0.2	-0.3	0.0	0.0	111.35	0.16	-0.34	0.00	0.00
104 1.2 DL+ 1.6 WL(+X) + 0.5 LL	47.8	-19.8	40.9	0.0	0.0	107.58	-19.76	40.90	0.00	0.00
105 1.2 DL+ 1.6 WL(-X) + 0.5 LL	23.9	19.9	-41.1	0.0	0.0	83.71	19.86	-41.10	0.00	0.00
106 1.2 DL+ 1.6 WL(+Z) + 0.5 LL	28.3	0.1	-0.2	0.0	0.0	88.13	0.08	-0.17	0.00	0.00
107 1.2 DL+ 1.6 WL(-Z) + 0.5 LL	43.3	0.0	0.0	0.0	0.0	103.17	0.02	-0.04	0.00	0.00
108 0.9 DL + 1.6 WL(+X)	33.5	-19.8	41.0	0.0	0.0	93.36	-19.79	40.97	0.00	0.00
109 0.9 DL + 1.6 WL(-X)	9.7	19.8	-41.0	0.0	0.0	69.49	19.83	-41.04	0.00	0.00
110 0.9 DL + 1.6 WL(+Z)	14.1	0.1	-0.1	0.0	0.0	73.91	0.05	-0.10	0.00	0.00
111 0.9 DL + 1.6 WL(-Z)	29.1	0.0	0.0	0.0	0.0	88.95	-0.01	0.03	0.00	0.00
112 1.2 DL + 0.8 WL(+X)	34.8	-9.9	20.5	0.0	0.0	94.60	-9.88	20.45	0.00	0.00
113 1.2 DL + 0.8 WL(-X)	22.8	9.9	-20.6	0.0	0.0	82.66	9.93	-20.55	0.00	0.00
114 1.2 DL + 0.8 WL(+Z)	25.1	0.0	-0.1	0.0	0.0	84.87	0.04	-0.08	0.00	0.00
115 1.2 DL + 0.8 WL(-Z)	25.1	0.0	-0.1	0.0	0.0	84.87	0.04	-0.08	0.00	0.00

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Check for overturning and sliding:

						Overturning		Sliding	
Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
201 1. DL	83.8	0.0	0.0	0.0	0.0	1533.4	NA	1676.6	NA
202 1 (DL+TL1) + 1 LL	97.6	0.0	0.0	0.0	0.0	1785.9	NA	1952.6	NA
203 1 (DL+TL2) + 1 LL	98.1	0.1	-0.2	0.0	0.0	296.6	NA	327.0	NA
204 1 DL+ 1 WL(+X)	91.3	-12.4	25.6	0.0	0.0	2.7	NA	3.0	NA
205 1 DL+ 1 WL(-X)	76.4	12.4	-25.7	0.0	0.0	2.2	NA	2.5	NA
206 1 DL+ 1 WL(+Z)	79.1	0.0	-0.1	0.0	0.0	715.0	NA	791.3	NA
207 1 DL+ 1 WL(-Z)	88.5	0.0	0.0	0.0	0.0	NA	NA	NA	NA
208 0.6 DL + 1 WL(+X)	81.7	-12.4	25.6	0.0	0.0	2.4	NA	2.6	NA
209 0.6 DL + 1 WL(-X)	66.8	12.4	-25.7	0.0	0.0	2.0	NA	2.2	NA
210 0.6 DL + 1 WL(+Z)	69.5	0.0	-0.1	0.0	0.0	790.1	NA	869.1	NA
211 0.6 DL + 1 WL(-Z)	78.9	0.0	0.0	0.0	0.0	3482.0	NA	3946.3	NA
212 1 DL+ 0.75 WL(+X)+ 0.75 LL	99.9	-9.2	19.1	0.0	0.0	3.9	NA	4.3	NA
213 1 DL+ 0.75 WL(-X)+ 0.75 LL	88.8	9.3	-19.3	0.0	0.0	3.4	NA	3.8	NA
214 1 DL+ 0.75 WL(+Z)+ 0.75 LL	90.8	0.1	-0.2	0.0	0.0	454.1	NA	497.7	NA
215 1 DL+ 0.75 WL(-Z)+ 0.75 LL	97.9	0.0	-0.1	0.0	0.0	834.2	NA	910.5	NA
					Min.=	2.0	0.0	2.2	0.0
						OK	-	OK	-

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5.9 DESIGN OF FOOTING TYPE F13

Load Combination & Calculation of Base pressure

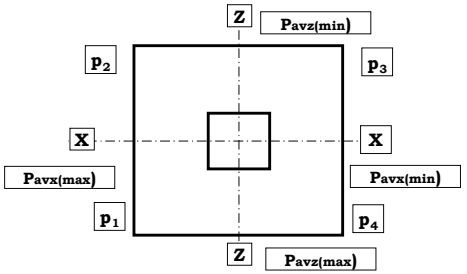
Max. Base pressure = p_{max} = (V/(Lx.Lz)) + (6.Mz/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)

Min. Base pressure = p_{min} = (V/(Lx.Lz)) - (6.Mz/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)

Surcharge load = 10 kN/m^2

Net SBC of soil for DL, LL= 350 kN/m^2 For WL/SL case = 437.5 kN/m^2

Gross SBC of soil for DL, LL= 371.6 kN/m^2 For WL/SL case = 459.1 kN/m^2



Allowable Stress Design

Load Combination	Design Load/moments			Base Pressure (kN/m^2)													
	V (kN)	Mz (kN-m)	Mx (kN-m)	V/ (Lx.Lz)	6.Mz/ (Lx)^2.Lz	6.Mx/ Lx.(Lz)^2	p_1 (p_{max})	p_2	p_4	p_3 (p_{min})	(p_{max}-γh)	$p_{avx(max)}$	$p_{avx(min)}$	$p_{avz(max)}$	$p_{avz(min)}$	Remarks
201 1. DL	83.83	-0.04	0.00	37.26	0.07	0.00	37.33	37.33	37.18	37.18	15.73	37.33	37.18	37.26	37.26	**O.K**
202 1 (DL+TL1) + 1 LL	97.63	-0.04	0.00	43.39	0.07	0.00	43.46	43.46	43.32	43.32	21.86	43.46	43.32	43.39	43.39	**O.K**
203 1 (DL+TL2) + 1 LL	98.09	-0.25	0.00	43.59	0.44	0.00	44.04	44.04	43.15	43.15	22.44	44.04	43.15	43.60	43.60	**O.K**
204 1 DL+ 1 WL(+X)	91.29	25.59	0.00	40.57	45.48	0.00	86.06	86.06	0.00	0.00	64.46	86.06	0.00	43.03	43.03	**O.K**
205 1 DL+ 1 WL(-X)	76.37	-25.67	0.00	33.94	45.63	0.00	79.57	79.57	0.00	0.00	57.97	79.57	0.00	39.79	39.79	**O.K**
206 1 DL+ 1 WL(+Z)	79.13	-0.08	0.00	35.17	0.15	0.00	35.32	35.32	35.02	35.02	13.72	35.32	35.02	35.17	35.17	**O.K**
207 1 DL+ 1 WL(-Z)	88.53	0.00	0.00	39.35	0.00	0.00	39.35	39.35	39.35	39.35	17.75	39.35	39.35	39.35	39.35	**O.K**
208 0.6 DL + 1 WL(+X)	81.69	25.60	0.00	36.30	45.51	0.00	81.82	81.82	0.00	0.00	60.22	81.82	0.00	40.91	40.91	**O.K**
209 0.6 DL + 1 WL(-X)	66.77	-25.65	0.00	29.67	45.60	0.00	100.00	100.00	0.00	0.00	78.40	100.00	0.00	50.00	50.00	**O.K**
210 0.6 DL + 1 WL(+Z)	69.53	-0.07	0.00	30.90	0.12	0.00	31.02	31.02	30.78	30.78	9.42	31.02	30.78	30.90	30.90	**O.K**
211 0.6 DL + 1 WL(-Z)	78.93	0.02	0.00	35.08	0.03	0.00	35.11	35.11	35.05	35.05	13.51	35.11	35.05	35.08	35.08	**O.K**
212 1 DL+ 0.75 WL(+X)+ 0.75 LL	99.95	19.10	0.00	44.42	33.96	0.00	78.38	78.38	10.46	10.46	56.78	78.38	10.46	44.42	44.42	**O.K**
213 1 DL+ 0.75 WL(-X)+ 0.75 LL	88.76	-19.34	0.00	39.45	34.38	0.00	73.83	73.83	5.07	5.07	52.23	73.83	5.07	39.45	39.45	**O.K**
214 1 DL+ 0.75 WL(+Z)+ 0.75 LL	90.83	-0.15	0.00	40.37	0.27	0.00	40.63	40.63	40.10	40.10	19.03	40.63	40.10	40.37	40.37	**O.K**
215 1 DL+ 0.75 WL(-Z)+ 0.75 LL	97.88	-0.09	0.00	43.50	0.16	0.00	43.66	43.66	43.34	43.34	22.06	43.66	43.34	43.50	43.50	**O.K**

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Strength Design

101 1.4 DL	93.43	-0.06	0.00	41.52	0.10	0.00	41.63	41.63	41.42	41.42	20.03	41.63	41.42	41.53	41.53	
102 1.2 (DL+TL1) + 1.6 LL	110.80	-0.09	0.00	49.25	0.16	0.00	49.41	49.41	49.08	49.08	27.81	49.41	49.08	49.25	49.25	
103 1.2 (DL+TL2) + 1.6 LL	111.35	-0.34	0.00	49.49	0.60	0.00	50.09	50.09	48.89	48.89	28.49	50.09	48.89	49.49	49.49	
104 1.2 DL+ 1.6 WL(+X) + 0.5 LL	107.58	40.90	0.00	47.81	72.71	0.00	120.53	120.53	-24.90	-24.90	98.93	120.53	-24.90	47.82	47.82	
105 1.2 DL+ 1.6 WL(-X) + 0.5 LL	83.71	-41.10	0.00	37.20	73.07	0.00	110.28	110.28	-35.87	-35.87	88.68	110.28	-35.87	37.21	37.21	
106 1.2 DL+ 1.6 WL(+Z) + 0.5 LL	88.13	-0.17	0.00	39.17	0.30	0.00	39.47	39.47	38.87	38.87	17.87	39.47	38.87	39.17	39.17	
107 1.2 DL+ 1.6 WL(-Z) + 0.5 LL	103.17	-0.04	0.00	45.85	0.06	0.00	45.91	45.91	45.79	45.79	24.31	45.91	45.79	45.85	45.85	
108 0.9 DL + 1.6 WL(+X)	93.36	40.97	0.00	41.49	72.83	0.00	114.32	114.32	-31.33	-31.33	92.72	114.32	-31.33	41.50	41.50	
109 0.9 DL + 1.6 WL(-X)	69.49	-41.04	0.00	30.88	72.96	0.00	103.84	103.84	-42.08	-42.08	82.24	103.84	-42.08	30.88	30.88	
110 0.9 DL + 1.6 WL(+Z)	73.91	-0.10	0.00	32.85	0.18	0.00	33.03	33.03	32.66	32.66	11.43	33.03	32.66	32.85	32.85	
111 0.9 DL + 1.6 WL(-Z)	88.95	0.03	0.00	39.53	0.05	0.00	39.58	39.58	39.48	39.48	17.98	39.58	39.48	39.53	39.53	
112 1.2 DL + 0.8 WL(+X)	94.60	20.45	0.00	42.04	36.36	0.00	78.40	78.40	5.68	5.68	56.80	78.40	5.68	42.04	42.04	
113 1.2 DL + 0.8 WL(-X)	82.66	-20.55	0.00	36.74	36.54	0.00	73.27	73.27	0.20	0.20	51.67	73.27	0.20	36.74	36.74	
114 1.2 DL + 0.8 WL(+Z)	84.87	-0.08	0.00	37.72	0.15	0.00	37.87	37.87	37.57	37.57	16.27	37.87	37.57	37.72	37.72	
115 1.2 DL + 0.8 WL(-Z)	84.87	-0.08	0.00	37.72	0.15	0.00	37.87	37.87	37.57	37.57	16.27	37.87	37.57	37.72	37.72	
Max :	111	41	0													
Min :	67	-41	0													

Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL cases) = 22.44 kN/m^2

Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL cases) = 32.44 kN/m^2

Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL+SL/WL cases) = 78.40 kN/m^2

Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL+WL/SL cases) = 88.40 kN/m^2

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Sample calculation of values of base pressure:

Load case : ;02 1 (DL+TL1) + 1 L

Maximum pressure, p_1 (pmax) =	43.39+0.07+0.00=	43.46 kN/m^2	Maximum average Pressure along X, $p_{avx(max)}$ =	(43.46+43.46)/2=	43.46 kN/m^2
p_2=	43.39+0.07-0.00=	43.46 kN/m^2	Minimum average Pressure along X, $p_{avx(minx)}$ =	(43.32+43.32)/2=	43.32 kN/m^2
p_3=	43.39-(0.07+0.00)=	43.32 kN/m^2	Maximum average Pressure along Y, $p_{avy(max)}$ =	(43.46+43.32)/2=	43.39 kN/m^2
p_4=	43.39-0.07+0.00=	43.32 kN/m^2	Minimum average Pressure along Y, $p_{avy(minx)}$ =	(43.32+43.46)/2=	43.39 kN/m^2
Maxm pressure excluding weight of soil (p_{max}-γh)=	43.46-1.2*18=	21.86 kN/m^2			

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5.9 DESIGN OF FOOTING TYPE F13

Calculation of Bending Moment and Shear Force in X direction :

p_{1x} =pressure at the face of column at section 1x-1x in kN/m^2.

p_{2x} =pressure at the face of column at section 2x-2x in kN/m^2.

p_{3x} =pressure at a distance 'd' from the face of column at section 3x-3x in kN/m^2.

BM_{1x} =Moment at the face of column at section 1x-1x in kN-m/m.

BM_{2x} =Moment at the face of column 2x-2x in kN-m/m.

SF_{3x} = Oneway shear force at a distance 'd' from the face of column at section 3x-3x in kN/m.

SF (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Punching perimeter= ((0.5+2*0.5*0.282)+(0.5+2*0.5*0.282))*2

= 3.128 m

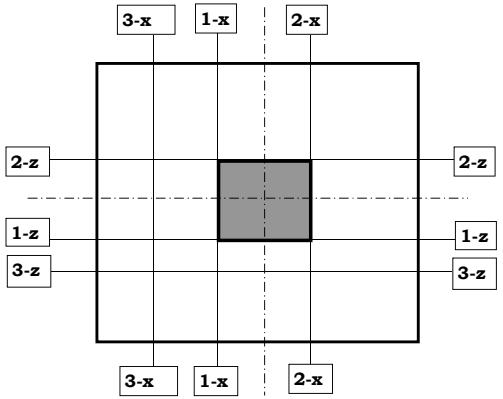
Punching Area= ((0.5+2*0.5*0.282)*(0.5+2*0.5*0.282))

= 0.61 m^2 Punching shear stress to be checked

Downward pressure due to self weight of raft and soil = (0.35*24+0.85*18)

= 23.70 kN/m^2

Area of footing, A_f= 2.25 m^2



Load Combination	p_{1x} kN/m2	p_{2x} kN/m2	p_{3x} kN/m2	$B.M_{1x}$ kNm	$B.M_{2x}$ kNm	$S.F_{3x}$ kN	$S.F_{(punching)}$ kN
Ultimate Conditions							
101 1.4 DL	41.56	41.49	41.60	2.24	2.22	3.69	29.23
102 1.2 (DL+TL1) + 1.6 LL	49.30	49.19	49.36	3.21	3.18	5.29	41.89
103 1.2 (DL+TL2) + 1.6 LL	49.69	49.29	49.93	3.28	3.17	5.42	42.30
104 1.2 DL+ 1.6 WL(+X) + 0.5 LL	72.05	23.58	100.56	10.08	-4.06	17.89	39.55
105 1.2 DL+ 1.6 WL(-X) + 0.5 LL	61.56	12.85	90.21	8.79	-5.42	15.77	22.15
106 1.2 DL+ 1.6 WL(+Z) + 0.5 LL	39.27	39.07	39.39	1.96	1.90	3.24	25.37
107 1.2 DL+ 1.6 WL(-Z) + 0.5 LL	45.87	45.83	45.89	2.77	2.76	4.57	36.33
108 0.9 DL + 1.6 WL(+X)	65.77	17.22	94.32	9.30	-4.86	16.61	29.18
109 0.9 DL + 1.6 WL(-X)	55.20	6.56	83.80	7.99	-6.20	14.44	11.78
110 0.9 DL + 1.6 WL(+Z)	32.91	32.78	32.98	1.16	1.13	1.92	15.00
111 0.9 DL + 1.6 WL(-Z)	39.55	39.51	39.57	1.98	1.97	3.27	25.96
112 1.2 DL + 0.8 WL(+X)	54.16	29.92	68.41	5.83	-1.24	10.24	30.08
113 1.2 DL + 0.8 WL(-X)	48.91	24.56	63.24	5.18	-1.92	9.18	21.38
114 1.2 DL + 0.8 WL(+Z)	37.77	37.67	37.83	1.77	1.74	2.91	22.99
115 1.2 DL + 0.8 WL(-Z)	37.77	37.67	37.83	1.77	1.74	2.91	22.99

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Sample calculation of BM & SF

Load case : 101 1.4 DL

p_{1x} = 41.63-(41.63-41.42)*0.5/1.5 = 41.56 kN/m^2 Overhang = 0.5 m

p_{2x} = 41.42+(41.63-41.42)*0.5/1.5 = 41.49 kN/m^2 Effective depth = 0.294 m

p_{3x} = 41.63-(41.63-41.42)*(0.5-0.294)/1.5 = 41.60 kN/m^2

BM_{1x} = (41.56-23.7)*0.5*0.5/2+0.5*(41.63-41.56)*0.5*0.5*0.67 = 2.24 kN-m/m

BM_{2x} = (41.49-23.7)*0.5*0.5/2-0.5*(-41.42+41.49)*0.5*0.5*0.67 = 2.22 kN-m/m

SF_{3x} = (41.60-23.7)*(0.5-0.294)+0.5*(41.63-41.60)*(0.5-0.294) = 3.69 kN/m

$S.F_{(punching)}$ = (2.25-0.61)*(41.52-23.7) = 29.23 kN

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	10.08	0.12	0.16	0.000	109.17	529.20	12ϕT	213.71	300	150	753.98
At Top	-6.20	-0.07	0.10	0.000	67.01	264.60	10ϕT	296.82	300	200	392.70

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc)/bd = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 17.89 kN/m

Shear stress , τ_v= (17.89*1000)/(1000*294)

= 0.06 N/mm^2 **O.K.**

Check against Two-way shear force at a distance d/2 from face of column:

Design punching shear at a distance d/2 from face of column, Vu= 42.30 kN

Strength of Concrete in shear Vc1 = 2720.31 kN **Table 22.6.5.2_ (a)**

Strength of Concrete in shear Vc2 = 1759.04 kN **Table 22.6.5.2_ (b)**

Strength of Concrete in shear Vc3 = 1813.54 kN **Table 22.6.5.2_ (c)**

Design Vc= 1759.04 kN

Allowable shear stress, (ϕ Vc)/bd = 1.43 N/mm^2

Design punching shear stress, τ = (42.3*1000)/(3128*294)

= 0.05 N/mm^2 **No shear reinforcement reqd.**

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Calculation of Bending Moment and Shear Force in Z direction :

p1z =pressure at the face of column at section 1z-1z in kN/m^2.
p2z =pressure at the face of column at section 2z-2z in kN/m^2.
p3z =pressure at a distance d from the face of column at section 3z-3z in kN/m^2.
B.M1z =Moment at the face of column at section 1z-1z in kN-m/m.
B.M2z =Moment at the face of column at section 2z-2z in kN-m/m.
S.F 3z = Oneway Shear Force at a distance d from the face of column at section 3z-3z in kN/m.
S.F (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Load Combination	p_{1z} kN/m2	p_{2z} kN/m2	p_{3z} kN/m2	$B.M_{1z}$ kNm	$B.M_{2z}$ kNm	$S.F_{3z}$ kN
Ultimate Conditions						
101 1.4 DL	41.53	41.53	41.53	2.23	2.23	3.89
102 1.2 (DL+TL1) + 1.6 LL	49.25	49.25	49.25	3.19	3.19	5.57
103 1.2 (DL+TL2) + 1.6 LL	49.49	49.49	49.49	3.22	3.22	5.62
104 1.2 DL+ 1.6 WL(+X) + 0.5 LL	47.82	47.82	47.82	3.02	3.02	5.26
105 1.2 DL+ 1.6 WL(-X) + 0.5 LL	37.21	37.21	37.21	1.69	1.69	2.95
106 1.2 DL+ 1.6 WL(+Z) + 0.5 LL	39.17	39.17	39.17	1.93	1.93	3.37
107 1.2 DL+ 1.6 WL(-Z) + 0.5 LL	45.85	45.85	45.85	2.77	2.77	4.83
108 0.9 DL + 1.6 WL(+X)	41.50	41.50	41.50	2.23	2.23	3.88
109 0.9 DL + 1.6 WL(-X)	30.88	30.88	30.88	0.90	0.90	1.57
110 0.9 DL + 1.6 WL(+Z)	32.85	32.85	32.85	1.14	1.14	1.99
111 0.9 DL + 1.6 WL(-Z)	39.53	39.53	39.53	1.98	1.98	3.45
112 1.2 DL + 0.8 WL(+X)	42.04	42.04	42.04	2.29	2.29	4.00
113 1.2 DL + 0.8 WL(-X)	36.74	36.74	36.74	1.63	1.63	2.84
114 1.2 DL + 0.8 WL(+Z)	37.72	37.72	37.72	1.75	1.75	3.06
115 1.2 DL + 0.8 WL(-Z)	37.72	37.72	37.72	1.75	1.75	3.06

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Sample calculation of BM & SF

Load case : 101 1.4 DL

p_{1z} = 41.53-(41.53-41.53)*0.5/1.5= 41.53 kN/m^2

p_{2z} = 41.53+(41.53-41.53)*0.5/1.5= 41.53 kN/m^2

p_{3z} = 41.53-(41.53-41.53)*(0.5-0.282)/1.5= 41.53 kN/m^2

Overhang = 0.5 m

Effective depth = 0.282 m

BM_{1z} = (41.53-23.7)*0.5*0.5/2+0.5*(41.53-41.53)*0.5*0.5*0.67 = 2.23 kN-m/m

BM_{2z} = (41.53-23.7)*0.5*0.5/2-0.5*(-41.53+41.53)*0.5*0.5*0.67 = 2.23 kN-m/m

SF_{3z} = (41.53-23.7)*(0.5-0.282)+0.5*(41.53-41.53)*(0.5-0.282) = 3.89 kN/m

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	3.22	0.041	0.054	0.0001	37.87	507.60	12ϕT	222.81	300	150	754.0
Top	0.00	0.000	0.000	0.0000	0.00	253.80	10ϕT	309.46	300	200	392.7

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc) = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 5.62 kN/m

Shear stress , τ_v= (5.62*1000)/(1000*282)

= 0.02 N/mm^2 **O.K.**

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :

At Bottom **12ϕT** **150** c/c

At Top **10ϕT** **200** c/c

Along Z Direction :

At Bottom **12ϕT** **150** c/c

At Top **10ϕT** **200** c/c

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5.10 DESIGN OF FOOTING TYPE F14 & F15

Size of Columns at node	Along-X	Along-Z
47	0.50m	0.50m
22	0.50m	0.50m
Size of fdn. raft:	2.00m	1.50m
Thickness of raft:	0.35	m
Original Gr. Level:	0.00	m
Finished Gr. Level:	0.00	m
U/S of B/Pl Above GL:	0.30	m
U/S level of fdn:	-1.20	m
Unit weight of conc:	25	kN/m^3
Unit weight of soil:	18	kN/m^3

Design Input:

Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f$_y$=	420 MPa
Clear cover for main reinforcement bars =	50mm

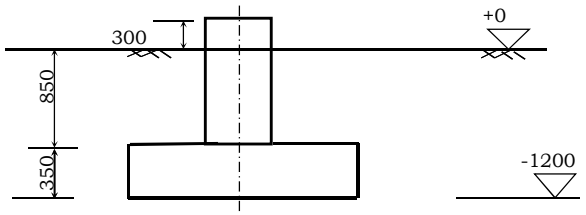
Calculation of Weight of Foundation :

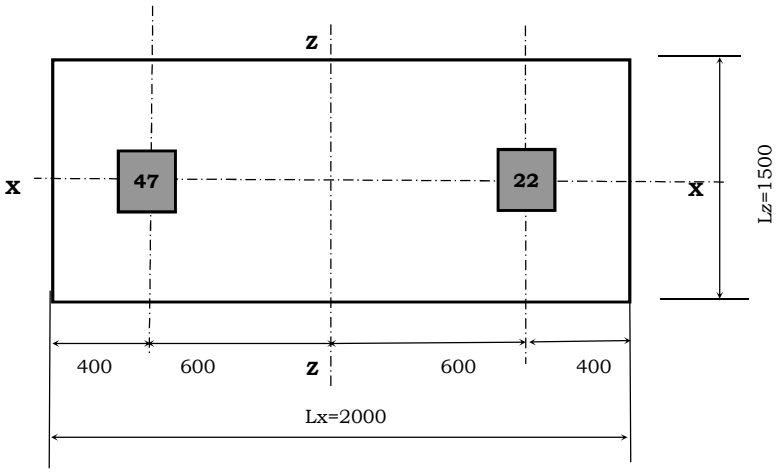
Weight of raft = (2*1.5*0.35*25) = 26.25KN

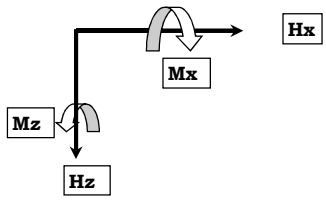
Weight of soil = ((2*1.5)-2*(0.5*0.5))*0.85*18 = 38.25KN

Total = 64.50KN

Submerged wt. = 64.5-2*1.5*1.2*10 = 28.50KN







H_x=Force along X-X direction

Hz=Force along Z-Z direction

M_x=Moment about X-X direction

Mz=Moment about Z-Z direction

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Check for overturning and sliding

Load at bottom of fooring	Node 1					Node 2				
L/C No.	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
201 1. DL	16.1	0.0	0.0	-1.3	-2.0	15.4	0.0	0.0	0.2	0.3
202 1 (DL+TL1) + 1 LL	25.1	0.0	0.0	-5.1	-7.6	22.5	0.0	0.0	0.9	1.4
203 1 (DL+TL2) + 1 LL	24.3	0.0	0.0	-3.4	-5.1	22.6	0.0	0.0	0.6	0.9
204 1 DL+ 1 WL(+X)	17.3	-1.3	1.9	-3.7	-5.6	13.2	-1.3	2.0	4.8	7.1
205 1 DL+ 1 WL(-X)	14.9	1.3	-1.9	1.1	1.6	17.6	1.3	-2.0	-4.3	-6.4
206 1 DL+ 1 WL(+Z)	20.8	0.0	0.0	-13.3	-20.0	17.2	0.0	0.0	-6.3	-9.4
207 1 DL+ 1 WL(-Z)	11.3	0.0	0.0	10.7	16.0	13.6	0.0	0.0	6.7	10.1
208 0.6 DL + 1 WL(+X)	10.8	-1.3	1.9	-3.2	-4.8	7.0	-1.3	2.0	4.7	7.0
209 0.6 DL + 1 WL(-X)	8.4	1.3	-1.9	1.6	2.4	11.4	1.3	-2.0	-4.4	-6.6
210 0.6 DL + 1 WL(+Z)	14.4	0.0	0.0	-12.8	-19.2	11.0	0.0	0.0	-6.4	-9.6
211 0.6 DL + 1 WL(-Z)	4.9	0.0	0.0	11.2	16.8	7.4	0.0	0.0	6.6	10.0
212 1 DL+ 0.75 WL(+X)+ 0.75 LL	23.5	-0.9	1.4	-5.3	-8.0	19.1	-1.0	1.5	4.0	6.0
213 1 DL+ 0.75 WL(-X)+ 0.75 LL	21.7	0.9	-1.4	-1.7	-2.6	22.4	1.0	-1.5	-2.8	-4.1
214 1 DL+ 0.75 WL(+Z)+ 0.75 LL	26.1	0.0	0.0	-12.5	-18.8	22.1	0.0	0.0	-4.3	-6.4
215 1 DL+ 0.75 WL(-Z)+ 0.75 LL	19.0	0.0	0.0	5.5	8.2	19.4	0.0	0.0	5.5	8.3

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Check for overturning and sliding:

L/C No.	Resultant Load at bottom of footing								
						Overturning		Sliding	
	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
201 1. DL	95.9	0.00	-0.4	-1.1	-1.6	231.7	44.0	NA	35.2
202 1 (DL+TL1) + 1 LL	112.1	0.00	-1.6	-4.2	-6.3	70.0	13.4	NA	10.7
203 1 (DL+TL2) + 1 LL	111.4	0.00	-1.0	-2.8	-4.2	111.2	19.8	NA	15.8
204 1 DL+ 1 WL(+X)	94.9	-2.57	1.4	1.0	1.6	67.5	45.6	14.8	36.5
205 1 DL+ 1 WL(-X)	96.9	2.57	-2.2	-3.2	-4.8	43.4	15.0	15.1	12.0
206 1 DL+ 1 WL(+Z)	102.5	-0.04	-2.1	-19.6	-29.4	48.0	2.6	1024.9	2.1
207 1 DL+ 1 WL(-Z)	89.3	0.04	1.3	17.4	26.1	68.3	2.6	893.5	2.1
208 0.6 DL + 1 WL(+X)	82.4	-2.57	1.6	1.5	2.2	52.4	27.9	12.8	22.3
209 0.6 DL + 1 WL(-X)	84.3	2.57	-2.1	-2.8	-4.2	40.8	15.1	13.1	12.1
210 0.6 DL + 1 WL(+Z)	89.9	-0.04	-2.0	-19.2	-28.7	45.6	2.3	899.2	1.9
211 0.6 DL + 1 WL(-Z)	76.8	0.04	1.5	17.9	26.8	52.1	2.1	767.8	1.7
212 1 DL+ 0.75 WL(+X)+ 0.75 LL	107.1	-1.93	0.3	-1.3	-2.0	374.1	41.2	22.2	32.9
213 1 DL+ 0.75 WL(-X)+ 0.75 LL	108.6	1.93	-2.4	-4.5	-6.7	44.4	12.1	22.5	9.7
214 1 DL+ 0.75 WL(+Z)+ 0.75 LL	112.7	-0.03	-2.4	-16.8	-25.2	47.6	3.4	1503.2	2.7
215 1 DL+ 0.75 WL(-Z)+ 0.75 LL	102.9	0.03	0.2	11.0	16.5	488.1	4.7	1371.8	3.7
					Min.=	**40.8**	**2.1**	**12.8**	**1.7**

Note: Foundation is critical in Stability criteria. Hence, detail reinforcement calc not performed.

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5.11 DESIGN OF FOOTING TYPE F16

Size of Columns at node	Along-X	Along-Z
1	0.50m	0.50m
2	0.50m	0.50m
Size of fdn. raft:	2.40m	1.20m
Thickness of raft:	0.40	m
Original Gr. Level:	0.00	m
Finished Gr. Level:	0.00	m
U/S of B/Pl Above GL:	1.00	m
U/S level of fdn:	-1.20	m
Unit weight of conc:	25	kN/m^3
Unit weight of soil:	18	kN/m^3

Design Input:

Characteristic strength of concrete, f'_c=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	50mm

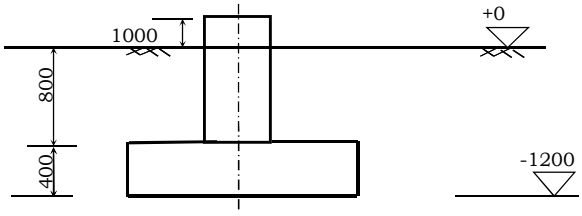
Calculation of Weight of Foundation :

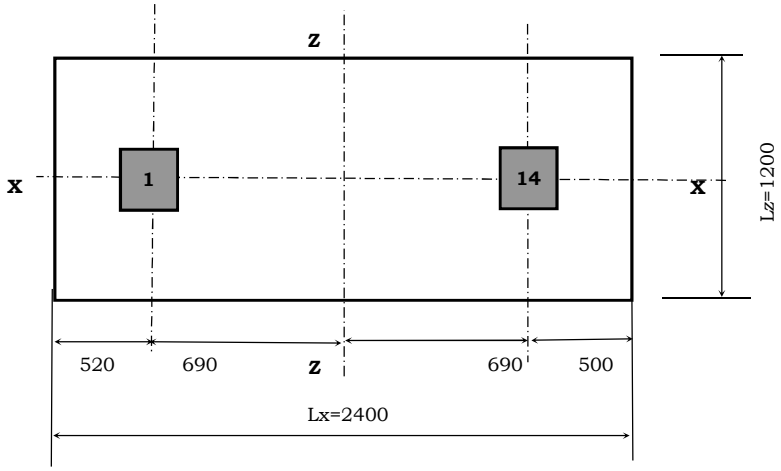
Weight of raft = (2.4*1.2*0.4*25) = 28.80KN

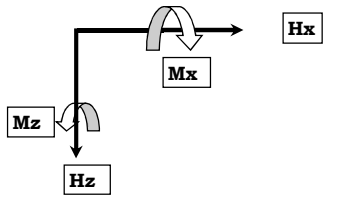
Weight of soil = ((2.4*1.2)-2*(0.5*0.5))*0.8*18 = 34.27KN

Total = 63.07KN

Submerged wt. = 63.07-2.4*1.2*1.2*10 = 28.51KN







H_X=Force along X-X direction

Hz=Force along Z-Z direction

M_X=Moment about X-X direction

Mz=Moment about Z-Z direction

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Check for overturning and sliding

Load at bottom of fooring	Node 1					Node 14				
L/C No.	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
201 1. DL	20.8	-0.3	0.6	-0.1	-0.2	17.3	0.4	-0.9	0.0	0.0
202 1 (DL+TL1) + 1 LL	36.8	-0.8	1.7	-0.7	-1.5	19.0	1.3	-2.9	-0.4	-0.9
203 1 (DL+TL2) + 1 LL	36.5	-1.2	2.5	0.1	0.3	18.6	1.6	-3.4	0.4	0.9
204 1 DL+ 1 WL(+X)	-3.1	1.2	-2.6	0.0	-0.1	41.2	-10.8	23.8	0.1	0.2
205 1 DL+ 1 WL(-X)	44.7	-1.7	3.8	-0.1	-0.3	-6.5	11.6	-25.6	-0.1	-0.2
206 1 DL+ 1 WL(+Z)	19.8	-0.2	0.5	-0.4	-1.0	19.2	-0.4	0.9	-0.5	-1.1
207 1 DL+ 1 WL(-Z)	21.9	-0.3	0.6	0.3	0.6	15.4	1.3	-2.8	0.5	1.1
208 0.6 DL + 1 WL(+X)	-11.4	1.3	-2.9	0.0	0.0	34.2	-11.0	24.1	0.1	0.2
209 0.6 DL + 1 WL(-X)	36.4	-1.6	3.6	-0.1	-0.2	-13.4	11.5	-25.2	-0.1	-0.2
210 0.6 DL + 1 WL(+Z)	11.4	-0.1	0.3	-0.4	-0.9	12.3	-0.6	1.3	-0.5	-1.1
211 0.6 DL + 1 WL(-Z)	13.6	-0.2	0.4	0.3	0.6	8.5	1.1	-2.4	0.5	1.1
212 1 DL+ 0.75 WL(+X)+ 0.75 LL	14.8	0.3	-0.7	-0.2	-0.4	36.3	-7.2	15.9	0.0	0.1
213 1 DL+ 0.75 WL(-X)+ 0.75 LL	50.6	-1.9	4.1	-0.3	-0.6	0.6	9.6	-21.1	-0.1	-0.2
214 1 DL+ 0.75 WL(+Z)+ 0.75 LL	31.9	-0.8	1.7	-0.5	-1.1	19.8	0.6	-1.2	-0.4	-0.9
215 1 DL+ 0.75 WL(-Z)+ 0.75 LL	33.5	-0.8	1.8	0.0	0.0	17.0	1.8	-4.0	0.3	0.8

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Check for overturning and sliding:

L/C No.	Resultant Load at bottom of footing								
	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	Overturning		Sliding	
						F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
201 1. DL	101.3	0.16	-2.8	-0.1	-0.2	43.8	345.2	253.1	506.3
202 1 (DL+TL1) + 1 LL	118.9	0.57	-13.5	-1.1	-2.5	10.5	28.9	83.4	42.5
203 1 (DL+TL2) + 1 LL	118.2	0.41	-13.3	0.5	1.1	10.7	62.0	115.3	90.9
204 1 DL+ 1 WL(+X)	101.2	-9.60	51.6	0.1	0.1	2.4	551.8	4.2	809.3
205 1 DL+ 1 WL(-X)	101.3	9.92	-57.2	-0.2	-0.5	2.1	131.6	4.1	193.0
206 1 DL+ 1 WL(+Z)	102.0	-0.64	1.0	-1.0	-2.1	121.5	29.3	63.8	43.0
207 1 DL+ 1 WL(-Z)	100.4	0.97	-6.6	0.7	1.6	18.1	37.5	41.4	55.0
208 0.6 DL + 1 WL(+X)	85.9	-9.66	52.7	0.1	0.2	2.0	286.3	3.6	419.0
209 0.6 DL + 1 WL(-X)	86.1	9.86	-56.1	-0.2	-0.4	1.8	131.7	3.5	193.4
210 0.6 DL + 1 WL(+Z)	86.8	-0.70	2.1	-0.9	-2.0	49.2	25.8	49.3	37.8
211 0.6 DL + 1 WL(-Z)	85.1	0.91	-5.5	0.8	1.7	18.4	30.5	37.6	44.7
212 1 DL+ 0.75 WL(+X)+ 0.75 LL	114.1	-6.91	30.1	-0.1	-0.3	4.6	210.7	6.6	308.5
213 1 DL+ 0.75 WL(-X)+ 0.75 LL	114.3	7.73	-51.5	-0.3	-0.8	2.7	90.9	5.9	133.3
214 1 DL+ 0.75 WL(+Z)+ 0.75 LL	114.8	-0.19	-7.9	-0.9	-2.0	17.4	34.9	237.9	51.1
215 1 DL+ 0.75 WL(-Z)+ 0.75 LL	113.6	1.02	-13.6	0.4	0.8	10.0	85.4	44.8	125.1
					Min.=	**1.8**	**25.8**	**3.5**	**37.8**

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Base pressure check of footing

	Node	L/C	Horizontal Fx N/mm2	Vertical Fy N/mm2	Horizontal Fz N/mm2
Max Px	495	201	0	0.04	0
Min Px	495	201	0	0.04	0
Max Py	500	213	0	0.09	0
Min Py	503	209	0	-0.03	0
Max Pz	495	201	0	0.04	0
Min Pz	495	201	0	0.04	0

Gross Maximum Base pressure, p_{max} = 88.0 KN/m^2

Net Allowable Safe Bearing Capacity = 450 KN/m^2
Depth of soil above footing = 0.7 m
Unit weight of soil = 18 KN/m^3

Gross Safe Bearing Capacity = 462.6 KN/m^2
> 88.0 KN/m^2
Hence OK

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Design of foundation

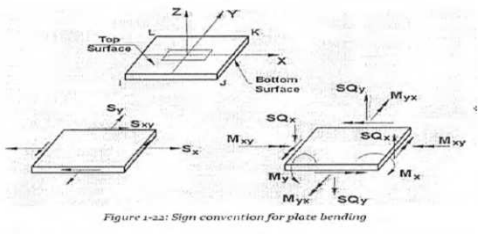
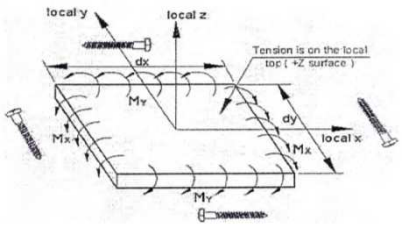


Figure 1-22: Sign convention for plate bending



Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:
PLATE CENTER STRESS : FDN_F8&F9_MOMENT At the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	1127	0.174	-0.024	0	0	0	-11.84	-5.834	1.314	-13.158	-7.148	105 1.2 DL+ 1.6 WL(-X) + 0.5 LL
Min Qx	1127	-0.197	0.027	0	0	0	11.572	9.525	0	11.572	9.525	108 0.9 DL + 1.6 WL(+X)
Max Qy	884	0.027	0.132	0	0	0	-1.8	6.041	-3.016	-4.816	9.057	104 1.2 DL+ 1.6 WL(+X) + 0.5 LL
Min Qy	1413	0.003	-0.131	0	0	0	-0.504	6.231	2.833	-3.337	9.064	104 1.2 DL+ 1.6 WL(+X) + 0.5 LL
Max Sx	556	0.001	0.001	0	0	0	0.015	0.015	-0.003	0.018	0.018	101 1.4 DL
Min Sx	556	0.001	0.001	0	0	0	0.015	0.015	-0.003	0.018	0.018	101 1.4 DL
Max Sy	556	0.001	0.001	0	0	0	0.015	0.015	-0.003	0.018	0.018	101 1.4 DL
Min Sy	556	0.001	0.001	0	0	0	0.015	0.015	-0.003	0.018	0.018	101 1.4 DL
Max Sxy	556	0.001	0.001	0	0	0	0.015	0.015	-0.003	0.018	0.018	101 1.4 DL
Min Sxy	556	0.001	0.001	0	0	0	0.015	0.015	-0.003	0.018	0.018	101 1.4 DL
Max Mx	1164	0.159	-0.02	0	0	0	35.699	1.792	-1.318	37.017	3.11	105 1.2 DL+ 1.6 WL(-X) + 0.5 LL
Min Mx	1164	-0.095	0.014	0	0	0	-34.05	2.009	1.494	-35.545	3.503	108 0.9 DL + 1.6 WL(+X)
Max My	1128	-0.142	0.019	0	0	0	6.517	10.126	-0.492	7.009	10.618	104 1.2 DL+ 1.6 WL(+X) + 0.5 LL
Min My	1129	0.088	-0.014	0	0	0	-3.486	-7.04	0.454	-3.94	-7.494	109 0.9 DL + 1.6 WL(-X)
Max Mxy	1308	-0.02	0.054	0	0	0	-25.11	-0.191	5.3	-30.413	-5.491	104 1.2 DL+ 1.6 WL(+X) + 0.5 LL
Min Mxy	972	-0.027	-0.06	0	0	0	-25.76	-0.688	-5.121	-30.882	-5.809	108 0.9 DL + 1.6 WL(+X)

PLATE CENTER STRESS : FDN_F8&F9_SHEAR At a distance "d" from the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	1127	0.174	-0.024	0	0	0	-11.84	-5.834	1.314	-13.158	-7.148	105 1.2 DL+ 1.6 WL(-X) + 0.5 LL
Min Qx	1127	-0.197	0.027	0	0	0	11.572	9.525	0	11.572	9.525	108 0.9 DL + 1.6 WL(+X)
Max Qy	884	0.027	0.132	0	0	0	-1.8	6.041	-3.016	-4.816	9.057	104 1.2 DL+ 1.6 WL(+X) + 0.5 LL
Min Qy	1413	0.003	-0.131	0	0	0	-0.504	6.231	2.833	-3.337	9.064	104 1.2 DL+ 1.6 WL(+X) + 0.5 LL
Max Sx	556	0.001	0.001	0	0	0	0.015	0.015	-0.003	0.018	0.018	101 1.4 DL
Min Sx	556	0.001	0.001	0	0	0	0.015	0.015	-0.003	0.018	0.018	101 1.4 DL
Max Sy	556	0.001	0.001	0	0	0	0.015	0.015	-0.003	0.018	0.018	101 1.4 DL
Min Sy	556	0.001	0.001	0	0	0	0.015	0.015	-0.003	0.018	0.018	101 1.4 DL
Max Sxy	556	0.001	0.001	0	0	0	0.015	0.015	-0.003	0.018	0.018	101 1.4 DL
Min Sxy	556	0.001	0.001	0	0	0	0.015	0.015	-0.003	0.018	0.018	101 1.4 DL
Max Mx	1164	0.159	-0.02	0	0	0	35.699	1.792	-1.318	37.017	3.11	105 1.2 DL+ 1.6 WL(-X) + 0.5 LL
Min Mx	1164	-0.095	0.014	0	0	0	-34.05	2.009	1.494	-35.545	3.503	108 0.9 DL + 1.6 WL(+X)
Max My	1128	-0.142	0.019	0	0	0	6.517	10.126	-0.492	7.009	10.618	104 1.2 DL+ 1.6 WL(+X) + 0.5 LL
Min My	1129	0.088	-0.014	0	0	0	-3.486	-7.04	0.454	-3.94	-7.494	109 0.9 DL + 1.6 WL(-X)
Max Mxy	1308	-0.02	0.054	0	0	0	-25.11	-0.191	5.3	-30.413	-5.491	104 1.2 DL+ 1.6 WL(+X) + 0.5 LL
Min Mxy	972	-0.027	-0.06	0	0	0	-25.76	-0.688	-5.121	-30.882	-5.809	108 0.9 DL + 1.6 WL(+X)

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	50	mm
	top face	50	mm
Thicknes of slab =		400	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

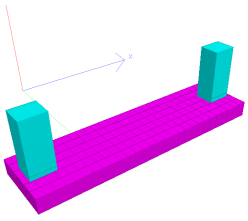
Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
105 1.2 D	37.02	top face	50	344	37.017	0.42	0.0010	344.0	309.6	12	328.87	125	905.2	12.76	15.95	0.062	tension control
108 0.9 D	-35.55	bottom face	50	344	35.545	0.40	0.0010	330.3		12	342.58	175	646.6	9.117	11.4	0.088	tension control
104 1.2 D	-30.41	top face	50	344	30.413	0.34	0.0008	282.3		12	365.45	150	754.3	10.64	13.29	0.075	tension control
108 0.9 D	-30.88	top face	50	344	30.882	0.35	0.0008	286.7		12	365.45	150	754.3	10.64	13.29	0.075	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c + 0.66\sqrt{f_c'}bd) \geq V_u$	Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN		KN		
Load condition	Max Qx	0.174	1	69.6	13.158	0.003	0.344	353.622	341.009	590.188	341.01	1263 > 69.6	ok	255.8	φVc>Vu	not required
	Min Qx	0.197	1	78.8	11.572			361.668			341.01	1263 > 78.8	ok	255.8	φVc>Vu	not required

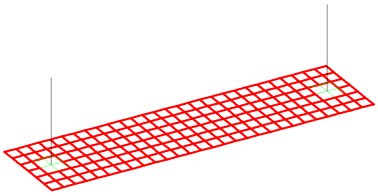
Reinforcement Along Z:

Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, My	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
104 1.2 D	10.62	top face	50	344	10.618	0.12	0.0003	98.2	309.6	12	365.45	250	452.6	6.382	7.977	0.126	tension control
109 0.9 D	-7.49	bottom face	50	344	7.494	0.08	0.0002	69.3		12	365.45	150	754.3	10.64	13.29	0.075	tension control
104 1.2 D	-5.49	bottom face	50	344	5.491	0.06	0.0001	50.7		12	365.45	150	754.3	10.64	13.29	0.075	tension control
108 0.9 D	-5.81	top face	50	344	5.809	0.07	0.0002	53.7		12	365.45	150	754.3	10.64	13.29	0.075	tension control

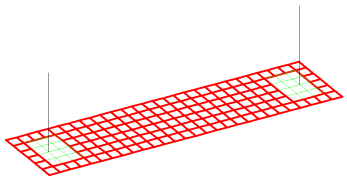
Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c + 0.66\sqrt{f_c'}bd) \geq V_u$	Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN		KN		
Load condition	104 1.2 D	0.132	1	52.8	9.057	0.001	0.344	356.481	341.009	fc'590.2	341.01	356 > 52.8	ok	255.8	φVc>Vu	not required
	104 1.2 D	0.131	1	52.4	9.064			356.224			341.01	356 > 52.4	ok	255.8	φVc>Vu	not required



3D rendered view



Highlighted plate for Moment at face of column



Highlighted plate for Shear at 'd' distance from face of column

SUMMARY OF REINFORCEMENT IN RAFT:

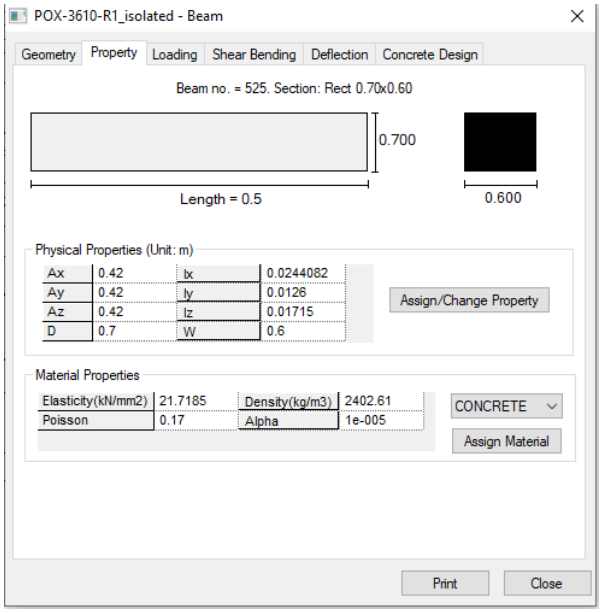
Along X Direction :			
At Bottom	**12ϕT**	**175**	c/c
At Top	**12ϕT**	**125**	c/c
Along Z Direction :			
At Bottom	**12ϕT**	**150**	c/c
At Top	**12ϕT**	**250**	c/c

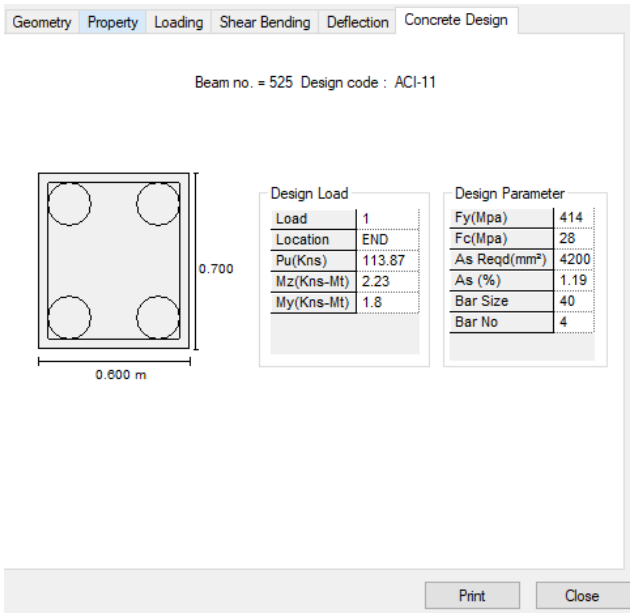
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5.13 Design of Column C1:





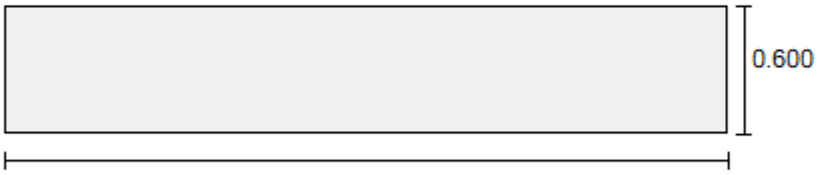
Reinforcement provided 12 nos 25mm dia bar.
Hence, Ast, provided = 5880 mm2 > Ast, required

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5.14 Design of Column C2:

Geometry | Property | Loading | Shear Bending | Deflection | Concrete Design

Beam no. = 548. Section: Rect 0.60x0.60



Node	X-Coord	Y-Coord	Z-Coord
464	0	-2.17	0
490	0	-0.53	0

UNIT: m

Additional Info

Beta Angle: 0 Change Beta

Member

Fire Proofing :

Radius of Curvature :

Gamma Angle : deg

Releases:

Start:

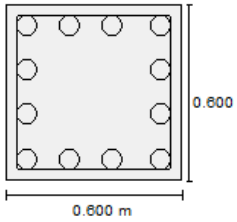
End:

Change Releases At Start....

Change Releases At End ....

Geometry | Property | Loading | Shear Bending | Deflection | Concrete Design

Beam no. = 548 Design code : ACI-11



Design Load

Load	1
Location	END
Pu(Kns)	138.27
Mz(Kns-Mt)	0
My(Kns-Mt)	0

Design Parameter

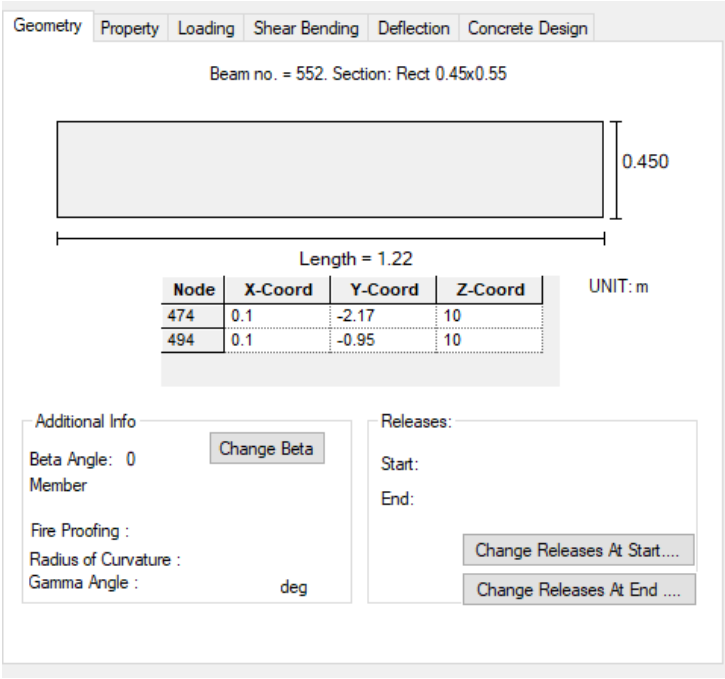
Fy(Mpa)	414
Fc(Mpa)	28
As Reqd(mm²)	3600
As (%)	1.04
Bar Size	20
Bar No	12

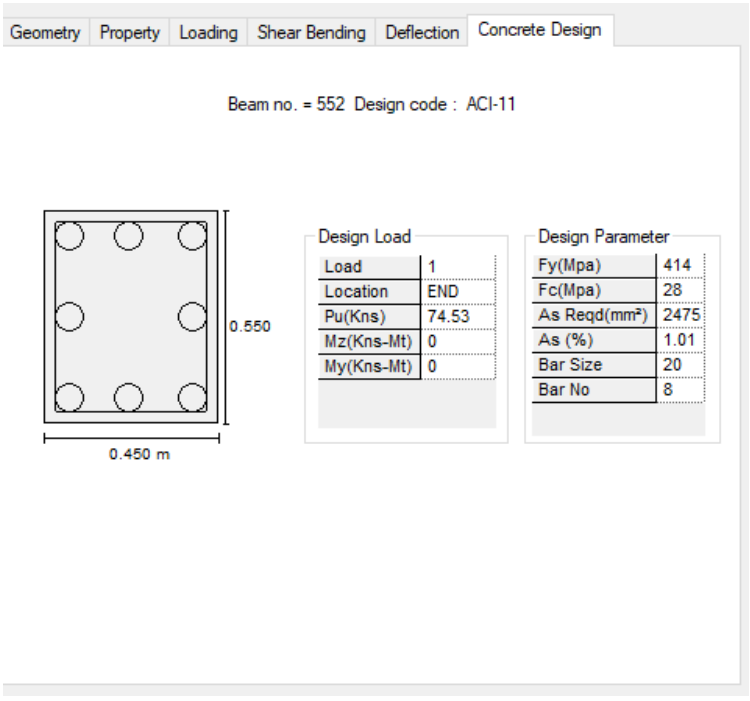
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5.15 Design of Column C5:

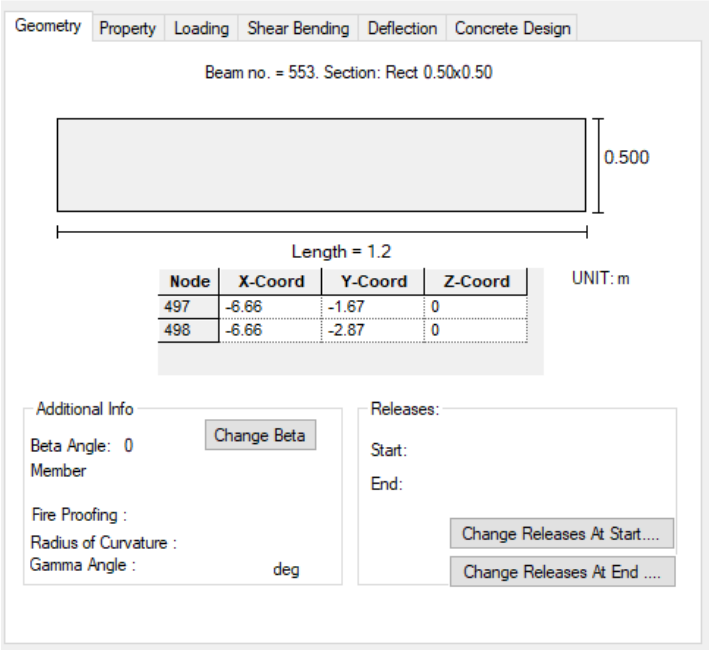


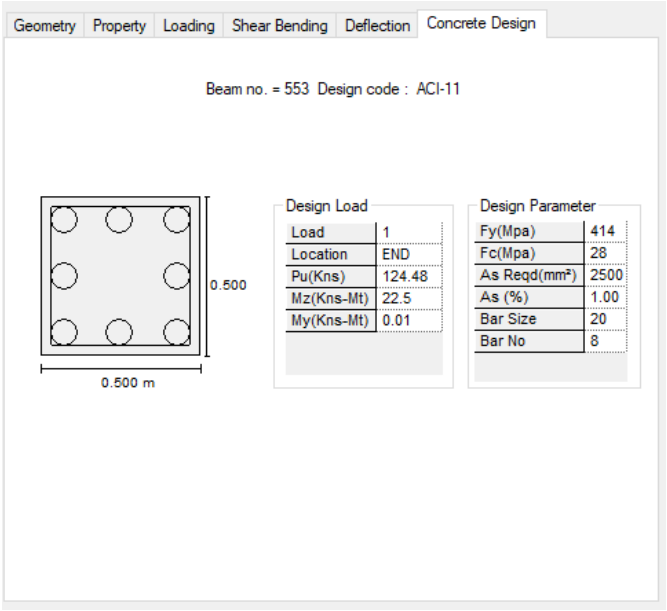


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5.16 Design of Column C6:

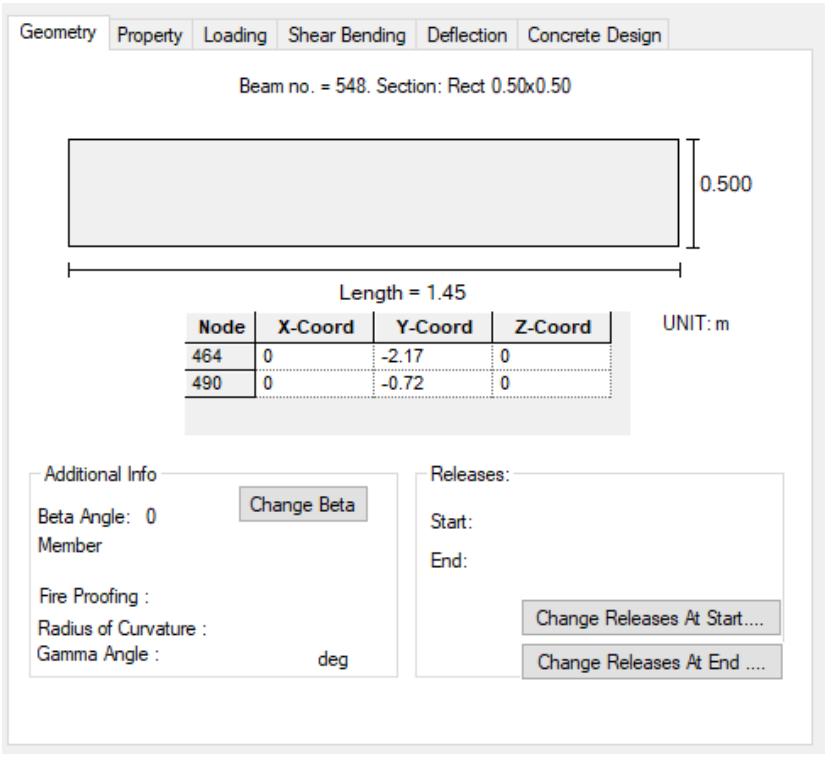


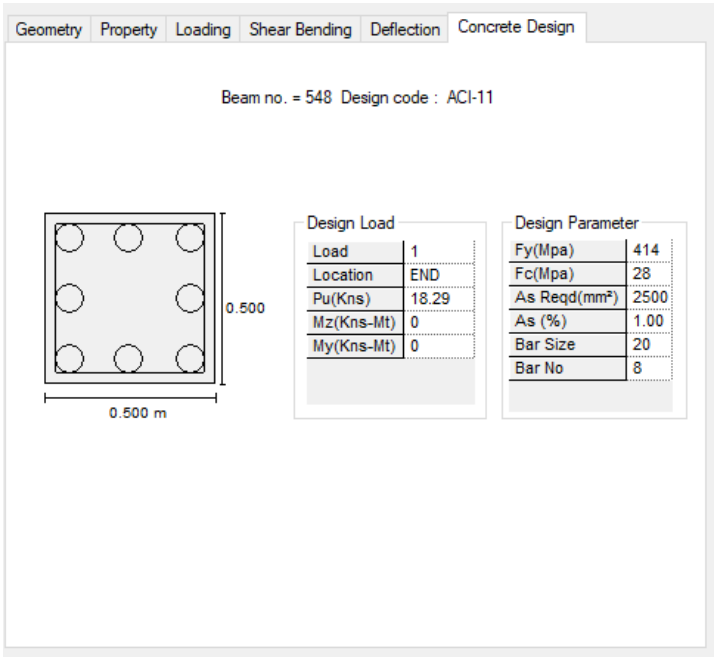


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5.21 Design of Column C7:

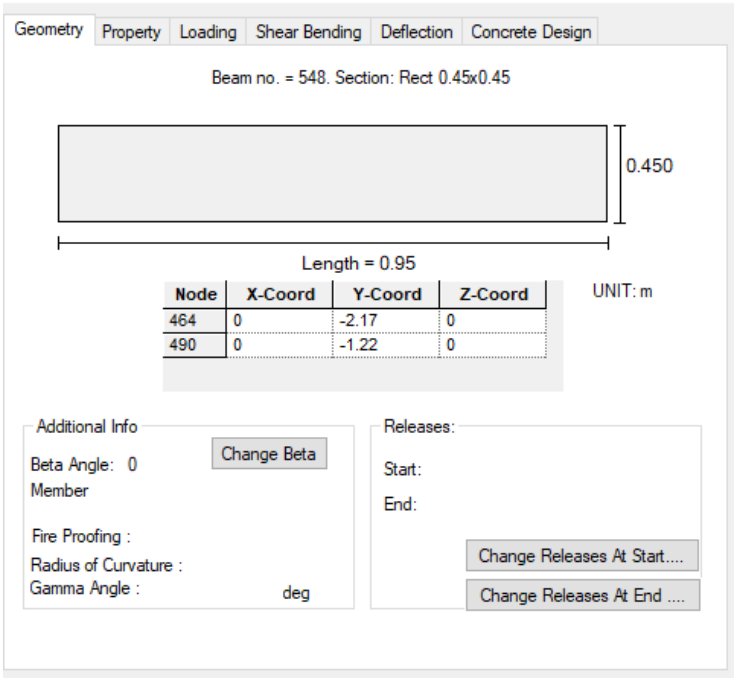


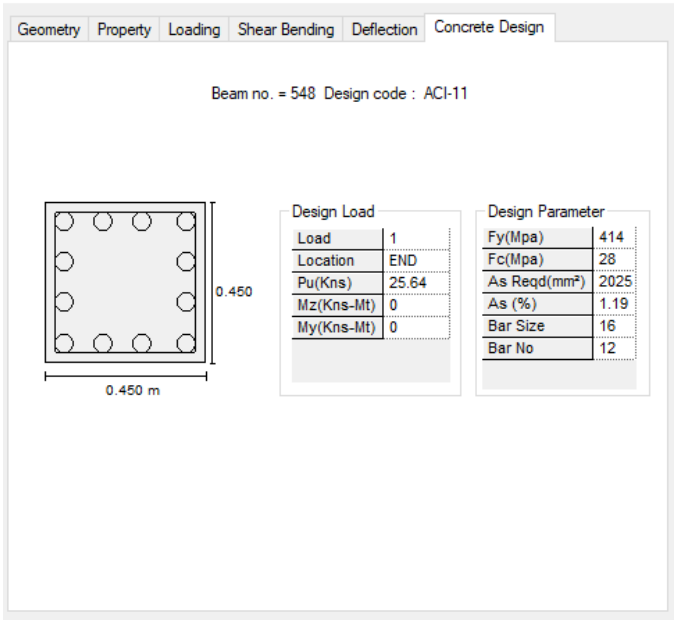


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5.22 Design of Column C8 & C9:

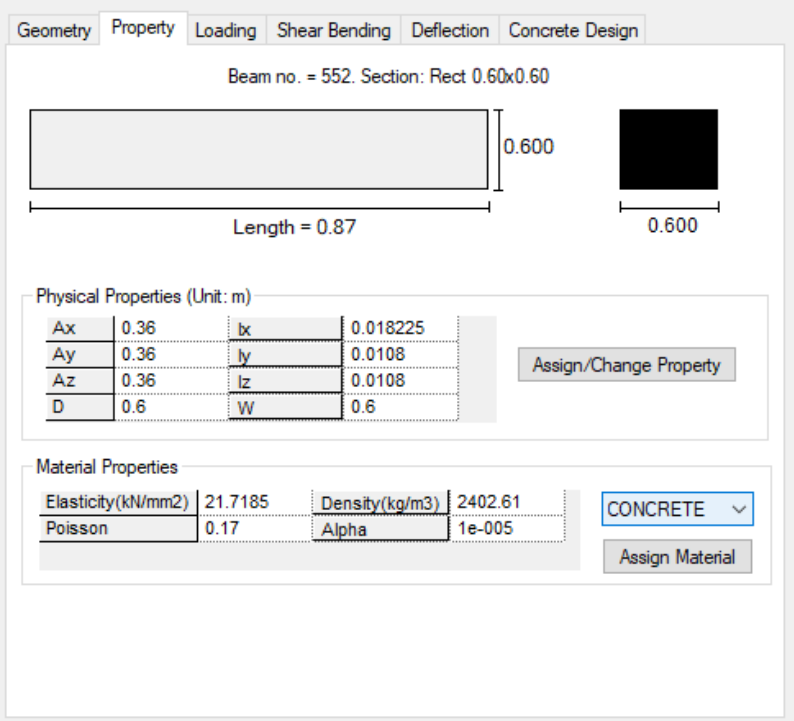


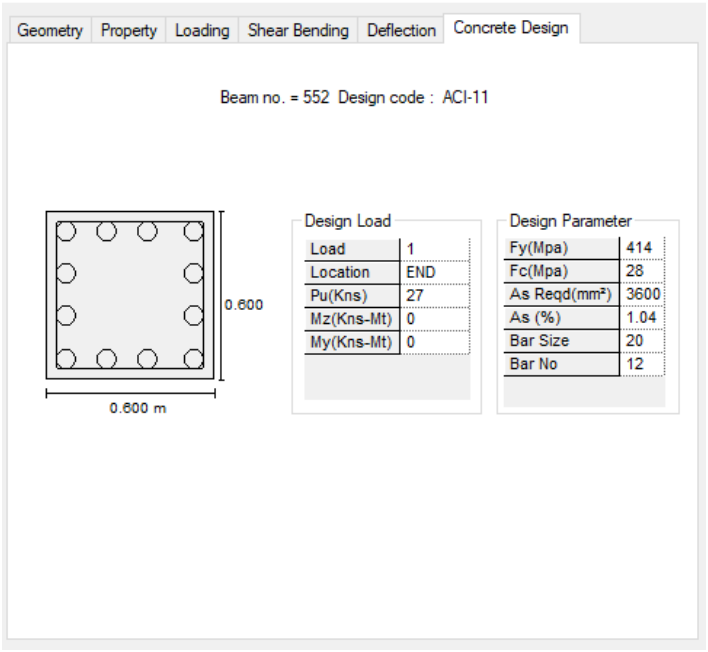


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5.23 Design of Column C10:



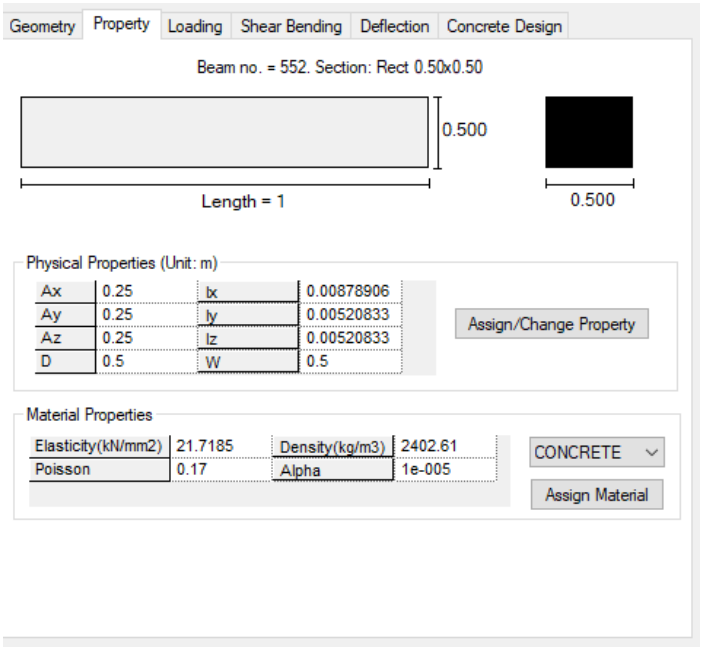


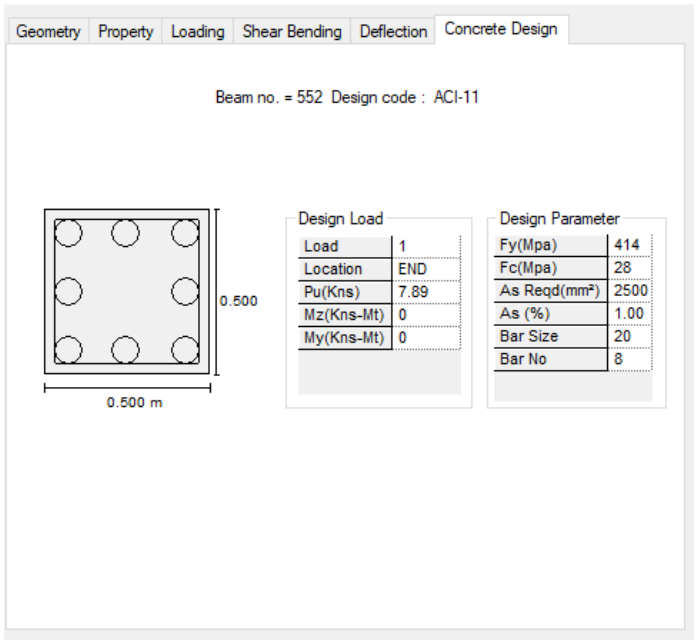
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5.24 Design of Column C11:





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References:

Reference is made to codes, standards and specifications as indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) clause 4.1.

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ATTACHMENT 1 STAAD INPUT FILE FOR F1 DESIGN

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STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 25-May-20
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER MTON
JOINT COORDINATES
1 0 -0.5 0; 2 7.8 -0.5 0; 3 0 -0.5 5; 4 7.8 -0.5 5; 5 0 11.5 0; 6 7.8 11.5 0;
7 0 11.5 5; 8 7.8 11.5 5; 9 4.5 11.5 0; 10 4.5 11.5 5; 11 0 8.9 0;
12 7.8 8.9 0; 13 0 8.9 5; 14 7.8 8.9 5; 19 0 3.7 0; 20 0 3.7 5; 21 7.8 3.7 5;
22 7.8 3.7 0; 23 0 6.3 0; 24 0 6.3 5; 25 7.8 6.3 5; 26 7.8 6.3 0;
27 4.5 -0.5 5; 28 4.5 3.7 5; 29 4.5 -0.5 0; 30 4.5 3.7 0; 31 4.5 8.9 0;
32 4.5 8.9 5; 33 4.5 6.3 0; 34 4.5 6.3 5; 35 0 11.5 6.5; 36 7.8 11.5 6.5;
37 4.5 11.5 6.5; 40 4.5 11.5 1.5; 41 0 11.5 1.5; 42 4.5 11.5 4.1;
43 0 11.5 4.1; 44 1.5 11.5 5; 45 3 11.5 5; 46 3 11.5 0; 47 1.5 11.5 0;
48 3 11.5 1.5; 49 1.5 11.5 1.5; 50 3 11.5 4.1; 51 1.5 11.5 4.1; 52 0 6.3 8;
53 7.8 6.3 8; 54 4.5 6.3 8; 55 0 3.7 7; 56 4.5 3.7 7; 57 7.8 1.2 7; 58 0 2.2 5;
59 7.8 2.2 5; 60 4.5 2.2 5; 61 0 9.2 5; 62 7.8 9.2 5; 63 4.5 9.2 5;
64 1.125 6.3 5; 65 2.25 6.3 5; 66 3.375 6.3 5; 67 1.125 6.3 8; 68 2.25 6.3 8;
69 3.375 6.3 8; 70 1.125 3.7 5; 71 2.25 3.7 5; 72 3.375 3.7 5; 73 1.125 3.7 7;
74 2.25 3.7 7; 75 3.375 3.7 7; 76 5.6 1.2 5.5; 77 6.7 1.2 5.5; 78 5.6 6.3 5;
79 6.7 6.3 5; 80 5.6 6.3 8; 81 6.7 6.3 8; 82 5.6 1.2 7; 83 6.7 1.2 7;
84 1.5 11.5 6.5; 85 3 11.5 6.5; 86 5.6 11.5 5; 87 6.7 11.5 5; 88 6.7 11.5 6.5;
89 5.6 11.5 6.5; 90 3 11.5 1.93333; 91 3 11.5 2.36667; 92 3 11.5 2.6;
93 3 11.5 3.23333; 94 3 11.5 3.46667; 95 1.5 11.5 1.93333; 96 1.5 11.5 2.36667;
97 1.5 11.5 2.6; 98 1.5 11.5 3.23333; 99 1.5 11.5 3.46667; 100 4.5 11.5 2.6;
101 0 11.5 2.6; 102 6.15 8.9 5; 103 6.15 1.2 5.5; 104 6.15 6.3 5;
105 6.15 11.5 5; 106 9.3 1.2 5; 107 9.3 1.2 0; 109 7.8 2.2 0; 110 6.15 11.5 0;
111 6.15 8.9 0; 112 6.15 6.3 0; 113 6.15 3.7 0; 114 0 1.2 5; 115 7.8 1.2 5;
116 0 1.2 0; 117 4.5 1.2 0; 118 7.8 1.2 0; 119 4.5 1.2 5; 120 4.5 1.2 7;
121 8 1.2 2.5; 127 9.3 1.2 2.5; 128 8 1.2 3.75; 129 8 1.2 1.25;
130 9.3 1.2 3.75; 131 9.3 1.2 1.25; 132 6.15 3.7 5; 133 0 1.2 2.5;
134 4.5 1.2 2.5; 135 4.5 1.2 5.5; 136 7.8 1.2 5.5; 139 8 1.2 0; 140 8 1.2 5;
143 7.8 1.2 2.5; 144 0.5 11.5 5; 145 1 11.5 5; 146 1 11.5 6.5;
147 0.5 11.5 6.5; 148 0.5 11.5 6; 149 0.5 11.5 5.5; 150 1 11.5 6;
151 1 11.5 5.5; 152 0 6.3 2.5; 153 4.5 6.3 2.5; 154 1.125 8.9 5;
155 2.25 8.9 5; 156 3.375 8.9 5; 157 3.375 8.9 0; 158 2.25 8.9 0;
159 1.125 8.9 0; 160 3.375 6.3 2.5; 161 2.25 6.3 2.5; 162 1.125 6.3 2.5;
163 2.25 6.3 0; 164 0 -2.57 5; 165 4.5 -2.57 5; 166 7.8 -2.57 5; 167 0 -2.57 0;
168 4.5 -2.57 0; 169 7.8 -2.57 0; 458 0 -1.67 0; 459 4.5 -1.67 0;
460 7.8 -1.67 0; 461 7.8 -1.67 5; 462 4.5 -1.67 5; 463 0 -1.67 5;
464 0 -2.17 0; 465 4.5 -2.17 0; 466 7.8 -2.17 0; 467 7.8 -2.17 5;
468 4.5 -2.17 5; 469 0 -2.17 5;
MEMBER INCIDENCES
1 3 114; 2 1 116; 3 4 115; 4 2 118; 5 5 41; 6 7 144; 7 8 6; 8 6 110; 9 9 46;
10 10 86; 11 9 40; 12 11 5; 13 12 6; 14 13 61; 15 14 62; 16 11 13; 17 13 154;

18 14 12; 19 12 111; 24 23 152; 25 24 64; 26 25 26; 27 26 112; 28 19 23;
29 20 24; 30 19 20; 31 21 25; 32 20 70; 33 22 26; 34 21 22; 35 22 113;
36 23 11; 37 24 13; 39 25 14; 41 26 12; 42 135 76; 43 27 119; 44 30 19;
45 29 117; 46 31 157; 47 31 9; 48 32 102; 49 32 63; 50 33 163; 51 30 33;
52 34 78; 53 28 34; 54 33 31; 55 34 32; 56 30 28; 57 33 153; 58 31 32; 59 8 36;
60 10 37; 61 7 35; 62 36 88; 63 37 85; 67 40 100; 68 41 101; 69 40 48;
70 42 10; 71 43 7; 72 42 50; 73 44 45; 74 45 10; 75 46 47; 76 47 5; 77 48 49;
78 49 41; 79 50 51; 80 51 43; 81 48 90; 82 49 95; 83 46 48; 84 47 49; 85 40 46;
86 46 49; 87 49 5; 88 49 101; 89 7 51; 90 51 44; 91 50 45; 92 143 21; 93 21 26;
94 26 14; 95 14 6; 96 133 20; 97 20 23; 98 23 13; 99 13 5; 100 19 29;
101 33 19; 102 11 33; 103 9 11; 104 27 20; 105 20 34; 106 34 13; 107 13 10;
110 24 52; 111 52 67; 112 54 34; 113 54 80; 114 53 25; 115 20 55; 116 55 73;
117 56 28; 118 120 82; 119 57 136; 120 20 52; 121 28 54; 122 21 53; 123 58 20;
125 60 28; 126 57 4; 127 120 27; 128 55 58; 129 61 7; 130 62 8; 131 63 10;
132 36 62; 133 37 63; 134 35 61; 135 64 65; 136 65 66; 137 66 34; 138 67 68;
139 68 69; 140 69 54; 141 67 64; 142 68 65; 143 69 66; 144 70 71; 145 71 72;
146 72 28; 147 73 74; 148 74 75; 149 75 56; 150 73 70; 151 74 71; 152 75 72;
153 76 103; 154 77 136; 155 78 104; 156 79 25; 157 80 81; 158 81 53; 159 82 83;
160 83 57; 161 80 78; 162 81 79; 163 82 76; 164 83 77; 165 84 146; 166 44 84;
167 85 84; 168 45 85; 169 86 105; 170 87 8; 171 88 89; 172 89 37; 173 89 86;
174 88 87; 175 90 91; 176 91 92; 177 92 93; 178 93 94; 179 94 50; 180 95 96;
181 96 97; 182 97 98; 183 98 99; 184 99 51; 185 100 42; 186 101 43; 187 92 100;
188 97 101; 189 101 51; 190 51 45; 191 45 42; 192 42 92; 193 92 40; 194 102 14;
195 103 77; 196 104 79; 197 105 87; 198 115 140; 199 2 107; 200 106 130;
201 59 21; 202 109 22; 203 107 139; 204 106 4; 205 110 9; 206 111 31;
207 112 33; 208 113 30; 209 114 58; 210 115 59; 211 116 19; 212 117 30;
213 118 109; 214 119 60; 215 140 128; 216 119 135; 217 121 129; 218 4 143;
219 143 22; 220 2 143; 221 127 131; 222 128 121; 223 129 139; 224 130 127;
225 131 107; 226 128 130; 227 129 131; 228 121 127; 229 21 132; 230 132 28;
231 56 60; 232 44 85; 233 70 74; 234 64 68; 235 114 133; 236 133 116;
237 19 133; 238 1 133; 239 3 133; 240 1 30; 241 28 3; 242 119 134; 243 134 117;
244 27 134; 245 29 134; 246 28 134; 247 28 33; 248 33 32; 249 32 9; 250 30 134;
251 135 120; 252 136 115; 253 119 115; 254 139 118; 255 140 106; 256 115 143;
257 143 118; 258 144 145; 259 145 44; 260 146 147; 261 147 35; 262 147 148;
263 146 150; 264 148 149; 265 149 144; 266 150 151; 267 151 145; 268 152 24;
269 153 34; 270 153 160; 271 154 155; 272 155 156; 273 156 32; 274 157 158;
275 158 159; 276 159 11; 277 160 161; 278 161 162; 279 162 152; 280 154 159;
281 155 158; 282 156 157; 283 64 162; 284 65 161; 285 66 160; 286 163 23;
287 152 163; 288 163 153; 289 152 64; 290 64 161; 291 161 66; 292 66 153;
294 3 463; 295 27 462; 296 4 461; 297 1 458; 298 29 459; 299 2 460;
522 458 464; 523 459 465; 524 460 466; 525 461 467; 526 462 468; 527 463 469;
528 458 459; 529 459 460; 530 463 462; 531 462 461; 532 458 463; 533 459 462;
534 460 461; 535 464 167; 536 465 168; 537 466 169; 538 467 166; 539 468 165;
540 469 164;
START GROUP DEFINITION
FLOOR

_FLOOR-1 16 17 25 46 52 58 110 TO 114 135 TO 143 155 TO 158 161 162 196 268 -
269 TO 285
_FLOOR-2 72 177 TO 179 185 187
END GROUP DEFINITION
SUPPORTS
164 TO 169 FIXED
MEMBER RELEASE
5 TO 11 16 TO 19 24 TO 27 30 32 34 35 42 44 46 48 50 52 56 TO 63 69 72 81 -
82 TO 107 110 TO 113 115 116 118 120 TO 122 126 TO 128 132 TO 134 141 TO 143 -
150 TO 152 161 TO 164 166 168 173 174 187 TO 193 198 TO 200 204 215 -
218 TO 220 226 TO 229 231 TO 235 237 TO 242 244 TO 250 253 256 262 263 270 -
280 TO 285 287 TO 292 294 TO 299 528 TO 534 START MY MZ
7 16 18 26 30 34 44 56 58 70 71 74 76 78 80 83 TO 107 112 114 117 120 TO 122 -
126 TO 128 132 TO 134 137 140 TO 143 146 149 TO 152 154 156 158 160 TO 164 -
166 168 170 172 TO 174 179 184 187 TO 194 199 204 TO 208 218 TO 220 223 225 -
226 TO 228 230 TO 234 236 TO 241 243 TO 250 252 TO 254 257 261 265 -
267 TO 269 273 276 279 TO 292 528 TO 534 END MY MZ
DEFINE MATERIAL START
ISOTROPIC STEEL
E 2.09042e+007
POISSON 0.3
DENSITY 7.83341
ALPHA 1.2e-005
DAMP 0.03
TYPE STEEL
STRENGTH RY 1.5 RT 1.2
ISOTROPIC CONCRETE
E 2.21467e+006
POISSON 0.17
DENSITY 2.40261
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
G 946438
TYPE CONCRETE
STRENGTH FCU 2855.21
STRENGTH FCU 2812.28
END DEFINE MATERIAL
MEMBER PROPERTY EUROPEAN
1 TO 4 12 TO 15 28 29 31 33 36 37 39 41 43 45 47 49 51 53 TO 55 123 125 129 -
130 TO 131 201 202 209 TO 214 TABLE ST IPE400
232 TO 234 TABLE ST L75X75X8
17 19 25 27 35 44 46 48 50 135 TO 137 194 206 TO 208 229 230 270 TO 279 -
286 TABLE ST IPE300
92 218 TO 220 244 245 256 257 287 288 TABLE ST L100X100X10
16 18 24 26 30 34 56 TO 58 268 269 TABLE ST IPE300

```
6 TO 10 32 42 52 59 TO 63 73 TO 76 83 84 90 91 110 TO 119 138 TO 174 187 188 -
195 TO 198 200 203 205 215 TO 217 221 TO 228 242 243 251 252 254 255 258 -
259 TO 267 280 TO 285 TABLE ST IPE240
126 TO 128 199 204 231 289 TO 292 TABLE ST L65X65X6
93 96 97 100 104 235 TO 241 246 247 250 TABLE ST L150X150X13
85 TO 89 189 TO 193 TABLE ST L65X65X6
94 95 98 99 101 TO 103 105 TO 107 248 249 TABLE ST L130X130X13
120 TO 122 253 TABLE ST L100X100X10
5 11 67 TO 72 77 TO 80 185 186 TABLE ST HE500A
MEMBER PROPERTY EUROPEAN
81 82 175 TO 184 TAPERED 0.4 0.012 0.4 0.24 0.02 0.24 0.02
132 TO 134 TABLE D UPN140 SP 0.008
MEMBER PROPERTY EUROPEAN
294 TO 299 522 TO 527 535 TO 540 PRIS YD 0.7 ZD 0.6
MEMBER PROPERTY EUROPEAN
528 TO 534 PRIS YD 0.35 ZD 0.25
CONSTANTS
MATERIAL STEEL MEMB 1 TO 19 24 TO 37 39 41 TO 63 67 TO 107 110 TO 123 125 -
126 TO 292
MATERIAL CONCRETE MEMB 294 TO 299 522 TO 540
DEFINE IBC 2003
SDS 0.06 SD1 0.013 S1 0.02 IE 1 RX 3 RZ 4 SCLASS 2 PX 0.332 PZ 0.332
****
*DEFINE IBC 2003
*SDS 0.066 SD1 0.0133 S1 0.02 IE 1 RX 4 RZ 4 SCLASS 2 PX 0.432 PZ 0.432
****
SELFWEIGHT 1.1
****
****
FLOOR WEIGHT
_FLOOR-1 FLOAD 0.02
**LL
_FLOOR-1 FLOAD 0.5
***
*JOINT WEIGHT
*8 10 12 22 25 30 31 34 115 119 WEIGHT 0.5
*103 TO 105 111 113 WEIGHT 1
*48 TO 51 90 TO 99 WEIGHT 0.5
*****LL
*103 TO 105 111 113 WEIGHT 2
*8 10 12 22 25 30 31 34 115 119 WEIGHT 1
*48 TO 51 90 TO 99 WEIGHT 2
JOINT WEIGHT
8 10 12 22 25 30 31 34 115 119 WEIGHT 0.5
103 TO 105 111 113 WEIGHT 1
91 93 96 98 WEIGHT 6
```

```
50 51 94 99 WEIGHT 1
148 TO 151 WEIGHT 1
92 94 97 99 WEIGHT 1.75
90 95 WEIGHT 2
***
***
LOAD 9 LOADTYPE Seismic  TITLE SL1(+X)
IBC LOAD X 1
*JOINT LOAD
*48 TO 51 90 TO 99 FX 0.5
PERFORM ANALYSIS
CHANGE
LOAD 10 LOADTYPE Seismic  TITLE SL2(+Z)
IBC LOAD Z 1
*JOINT LOAD
*48 TO 51 90 TO 99 FZ 0.5
PERFORM ANALYSIS
CHANGE
****
LOAD 1 LOADTYPE Dead  TITLE DL
SELFWEIGHT Y -1.1
FLOOR LOAD
_FLOOR-1 FLOAD -0.02 GY
JOINT LOAD
8 10 12 22 25 30 31 34 115 119 FY -0.5
103 TO 105 111 113 FY -1
91 93 96 98 FY -6
FLOOR LOAD
_FLOOR-2 FLOAD -0.02 GY
JOINT LOAD
50 51 94 99 FY -1
148 TO 151 FY -1
92 94 97 99 FY -1.75
90 95 FY -2
UNIT METER KN
MEMBER LOAD
528 TO 534 UNI GY -5.4
UNIT METER MTON
LOAD 2 LOADTYPE Live  TITLE LL
FLOOR LOAD
_FLOOR-1 FLOAD -0.5 GY
JOINT LOAD
103 TO 105 111 113 FY -2
8 10 12 22 25 30 31 34 115 119 FY -1
FLOOR LOAD
_FLOOR-2 FLOAD -0.5 GY
```

```
LOAD 3 LOADTYPE Wind  TITLE WL1(+X)
JOINT LOAD
48 TO 51 90 TO 99 FX 0.04
MEMBER LOAD
1 TO 4 12 TO 15 28 29 31 33 36 37 39 41 43 45 47 49 51 53 TO 55 123 125 129 -
130 TO 131 201 202 209 TO 214 UNI GX 0.2
LOAD 4 LOADTYPE Wind  TITLE WL2(-X)
JOINT LOAD
48 TO 51 90 TO 99 FX -0.04
MEMBER LOAD
1 TO 4 12 TO 15 28 29 31 33 36 37 39 41 43 45 47 49 51 53 TO 55 123 125 129 -
130 TO 131 201 202 209 TO 214 UNI GX -0.2
LOAD 5 LOADTYPE Wind  TITLE WL3(+Z)
JOINT LOAD
48 TO 51 90 TO 99 FZ 0.04
MEMBER LOAD
1 2 12 14 28 29 36 37 123 129 209 211 UNI GZ 0.2
43 45 47 49 51 53 TO 55 125 131 212 214 UNI GZ 0.3
3 4 13 15 31 33 39 41 130 201 202 210 213 UNI GZ 0.15
LOAD 6 LOADTYPE Wind  TITLE WL4(-Z)
JOINT LOAD
48 TO 51 90 TO 99 FZ -0.04
MEMBER LOAD
1 2 12 14 28 29 36 37 123 129 209 211 UNI GZ -0.2
43 45 47 49 51 53 TO 55 125 131 212 214 UNI GZ -0.3
3 4 13 15 31 33 39 41 130 201 202 210 213 UNI GZ -0.15
LOAD 7 LOADTYPE Temperature  TITLE TMP-1
TEMPERATURE LOAD
1 TO 19 24 TO 37 39 41 TO 63 67 TO 107 110 TO 123 125 TO 184 TEMP 25 0 25
LOAD 8 LOADTYPE Temperature  TITLE TMP-2
TEMPERATURE LOAD
1 TO 19 24 TO 37 39 41 TO 63 67 TO 107 110 TO 123 125 TO 184 TEMP -25 0 -25
***
LOAD 21 LOADTYPE Wind  TITLE TEST(-Z)
MEMBER LOAD
1 2 12 14 28 29 36 37 123 129 209 211 UNI GZ -0.2
43 45 47 49 51 53 TO 55 125 131 212 214 UNI GZ -0.3
3 4 13 15 31 33 39 41 130 201 202 210 213 UNI GZ -0.15
LOAD COMB 101 1.4 DL
1 1.4
LOAD COMB 102 1.2 (DL + TL1) + 1.6 LL
1 1.2 7 1.2 2 1.6
LOAD COMB 103 1.2 (DL + TL2) + 1.6 LL
1 1.2 8 1.2 2 1.6
LOAD COMB 104 1.2 DL+ 1.6 WL(+X) + 0.5 LL
1 1.2 3 1.6 2 0.5
```

LOAD COMB 105 1.2 DL+ 1.6 WL(-X) + 0.5 LL
1 1.2 4 1.6 2 0.5
LOAD COMB 106 1.2 DL+ 1.6 WL(+Z) + 0.5 LL
1 1.2 5 1.6 2 0.5
LOAD COMB 107 1.2 DL+ 1.6 WL(-Z) + 0.5 LL
1 1.2 6 1.6 2 0.5
LOAD COMB 108 1.2 DL + 1.0 EL(+X) + 0.5 LL
1 1.2 9 1.0 2 0.5
LOAD COMB 109 1.2 DL + 1.0 EL(-X) + 0.5 LL
1 1.2 9 -1.0 2 0.5
LOAD COMB 110 1.2 DL + 1.0 EL(+Z) + 0.5 LL
1 1.2 10 1.0 2 0.5
LOAD COMB 111 1.2 DL + 1.0 EL(-Z) + 0.5 LL
1 1.2 10 -1.0 2 0.5
LOAD COMB 112 0.9 DL + 1.6 WL(+X)
1 0.9 3 1.6
LOAD COMB 113 0.9 DL + 1.6 WL(-X)
1 0.9 4 1.6
LOAD COMB 114 0.9 DL + 1.6 WL(+Z)
1 0.9 5 1.6
LOAD COMB 115 0.9 DL + 1.6 WL(-Z)
1 0.9 6 1.6
LOAD COMB 116 0.9 DL + 1.0 EL(+X)
1 0.9 9 1.0
LOAD COMB 117 0.9 DL + 1.0 EL(-X)
1 0.9 9 -1.0
LOAD COMB 118 0.9 DL + 1.0 EL(+Z)
1 0.9 10 1.0
LOAD COMB 119 0.9 DL + 1.0 EL(-Z)
1 0.9 10 -1.0
LOAD COMB 120 1.2 DL + 0.8 WL(+X)
1 1.2 3 0.8
LOAD COMB 121 1.2 DL + 0.8 WL(-X)
1 1.2 4 0.8
LOAD COMB 122 1.2 DL + 0.8 WL(+Z)
1 1.2 5 0.8
LOAD COMB 123 1.2 DL + 0.8 WL(-Z)
1 1.2 5 0.8
****
****Serviceanbility check
LOAD COMB 201 1. DL
1 1.0
LOAD COMB 202 1 (DL + TL1) + 1 LL
1 1.0 7 1.0 2 1.0
LOAD COMB 203 1 (DL + TL2) + 1 LL
1 1.0 8 1.0 2 1.0

LOAD COMB 204 1 DL + 0.75 TL1 + 0.75 LL
1 1.0 7 0.75 2 0.75
LOAD COMB 205 1 DL + 0.75 TL2 + 0.75 LL
1 1.0 8 0.75 2 0.75
LOAD COMB 206 1 DL+ 1 WL(+X)
1 1.0 3 1.0
LOAD COMB 207 1 DL+ 1 WL(-X)
1 1.0 4 1.0
LOAD COMB 208 1 DL+ 1 WL(+Z)
1 1.0 5 1.0
LOAD COMB 209 1 DL+ 1 WL(-Z)
1 1.0 6 1.0
LOAD COMB 210 1 DL + 0.7 EL(+X)
1 1.0 9 0.7
LOAD COMB 211 1 DL + 0.7 EL(-X)
1 1.0 9 -0.7
LOAD COMB 212 1 DL + 0.7 EL(+Z)
1 1.0 10 0.7
LOAD COMB 213 1 DL + 0.7 EL(-Z)
1 1.0 10 -0.7
LOAD COMB 214 0.6 DL + 1 WL(+X)
1 0.6 3 1.0
LOAD COMB 215 0.6 DL + 1 WL(-X)
1 0.6 4 1.0
LOAD COMB 216 0.6 DL + 1 WL(+Z)
1 0.6 5 1.0
LOAD COMB 217 0.6 DL + 1 WL(-Z)
1 0.6 6 1.0
LOAD COMB 218 0.6 DL + 0.7 EL(+X)
1 0.6 9 0.7
LOAD COMB 219 0.6 DL + 0.7 EL(-X)
1 0.6 9 -0.7
LOAD COMB 220 0.6 DL + 0.7 EL(+Z)
1 0.6 10 0.7
LOAD COMB 221 0.6 DL + 0.7 EL(-Z)
1 0.6 10 -0.7
LOAD COMB 222 1 DL+ 0.75 WL(+X)+ 0.75 LL
1 1.0 3 0.75 2 0.75
LOAD COMB 223 1 DL+ 0.75 WL(-X)+ 0.75 LL
1 1.0 4 0.75 2 0.75
LOAD COMB 224 1 DL+ 0.75 WL(+Z)+ 0.75 LL
1 1.0 5 0.75 2 0.75
LOAD COMB 225 1 DL+ 0.75 WL(-Z)+ 0.75 LL
1 1.0 6 0.75 2 0.75
LOAD COMB 226 1 DL + 0.525 EL(+X)+ 0.75 LL
1 1.0 9 0.525 2 0.75

```
LOAD COMB 227 1 DL + 0.525 EL(-X)+ 0.75 LL
1 1.0 9 -0.525 2 0.75
LOAD COMB 228 1 DL + 0.525 EL(+Z)+ 0.75 LL
1 1.0 10 0.525 2 0.75
LOAD COMB 229 1 DL + 0.525 EL(-Z)+ 0.75 LL
1 1.0 10 -0.7 2 0.75
****
PERFORM ANALYSIS
START CONCRETE DESIGN
CODE ACI
UNIT METER KN
CLB 0.04 MEMB 294 TO 299 522 TO 540
CLS 0.04 MEMB 294 TO 299 522 TO 540
CLT 0.04 MEMB 294 TO 299 522 TO 540
FC 28000 MEMB 294 TO 299 522 TO 540
FYMAIN 414000 MEMB 294 TO 299 522 TO 540
MINMAIN 16 MEMB 528 TO 534
MINMAIN 20 MEMB 294 TO 299 522 TO 527 535 TO 540
MINSEC 10 MEMB 294 TO 299 522 TO 540
DESIGN BEAM 528 TO 534
DESIGN COLUMN 294 TO 299 522 TO 527 535 TO 540
END CONCRETE DESIGN
*LOAD LIST 1 TO 10
*PERFORM ANALYSIS
*LOAD LIST 101 TO 123
***
*PARAMETER 1
*CODE LRFD
*FU 41000 MEMB 1 TO 19 24 TO 37 39 41 TO 63 67 TO 107 110 TO 123 125 TO 292
*FYLD 27500 MEMB 1 TO 19 24 TO 37 39 41 TO 63 67 TO 107 110 TO 123 125 TO 292
*KX 1 MEMB 1 TO 19 24 TO 37 39 41 TO 63 67 TO 107 110 TO 123 125 TO 261 268 -
*269 TO 292
*KY 1 MEMB 1 TO 19 24 TO 37 39 41 TO 63 67 TO 107 110 TO 123 125 TO 261 268 -
*269 TO 292
*KZ 0.85 MEMB 1 TO 19 24 TO 37 39 41 TO 63 67 TO 107 110 TO 123 125 TO 292
*LX 1.5 MEMB 81 82 175 TO 184 262 TO 267
*LZ 2 MEMB 262 TO 267
*LY 1.5 MEMB 81 82 175 TO 184 262 TO 267
*LY 5 MEMB 1 TO 4 12 TO 16 18 28 29 31 33 36 37 39 41 47 49 51 53 TO 55 58 -
*123 129 TO 131 201 202 209 TO 211 213 235 236 242 243 256 257 280 TO 282
*LX 3 MEMB 43 45 125 212 214
*LY 3 MEMB 43 45 125 212 214 253
*LZ 3 MEMB 81 82 175 TO 184
*LZ 4 MEMB 8 10 19 27 35 42 48 52 62 113 118 153 TO 160 169 TO 172 194 TO 197 -
*205 TO 208 229 230
*LZ 5 MEMB 1 TO 6 9 11 TO 17 24 25 28 TO 33 36 37 39 41 44 TO 47 49 TO 51 53 -
```

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*54 TO 58 63 67 TO 80 111 116 123 129 TO 131 135 TO 140 144 TO 149 165 167 -
*185 186 201 202 209 TO 213 258 TO 261 268 TO 279 286
*LZ 5 MEMB 43 45 125 212 214
*LZ 4.5 MEMB 17 25 46 50 111 135 TO 140 270 TO 279 286
*MAIN 200 MEMB 1 TO 19 24 TO 37 39 41 TO 63 67 TO 107 110 TO 123 125 TO 292
*RATIO 1 MEMB 1 TO 19 24 TO 37 39 41 TO 63 67 TO 107 110 TO 123 125 TO 292
*TMAIN 300 MEMB 1 TO 19 24 TO 37 39 41 TO 63 67 TO 107 110 TO 123 125 TO 292
*TRACK 2 MEMB 1 TO 19 24 TO 37 39 41 TO 63 67 TO 107 110 TO 123 125 TO 292
*CHECK CODE MEMB 1 TO 19 24 TO 37 39 41 TO 63 67 TO 107 110 TO 123 125 TO 292
*PARAMETER 2
*CODE LRFD
*DJ2 40 MEMB 69 77 78
*DJ1 41 MEMB 69 77 78
*DJ2 10 MEMB 11 67 70 185
*DJ1 9 MEMB 11 67 70 185
*DJ2 7 MEMB 5 68 71 186
*DJ1 5 MEMB 5 68 71 186
*DJ2 50 MEMB 81 175 TO 179
*DJ1 48 MEMB 81 175 TO 179
*DJ2 51 MEMB 82 180 TO 184
*DJ1 49 MEMB 82 180 TO 184
*DJ2 42 MEMB 72 79 80
*DJ1 43 MEMB 72 79 80
*DFF 500 MEMB 5 6 11 67 TO 74 77 TO 82 175 TO 186 258 259
*DFF 360 MEMB 1 TO 4 12 TO 15 28 29 31 33 36 37 39 41 43 45 47 49 51 -
*53 TO 55 123 125 129 TO 131 201 202 209 TO 214
**FREQUENCY CHECK FOR TOP FLOOR DONE BY DEFLECTION VALUES
*DJ1 4 MEMB 3 15 31 39 130 201 210
*DJ2 8 MEMB 3 15 31 39 130 201 210
*DJ2 6 MEMB 4 13 33 41 202 213
*DJ2 10 MEMB 43 49 53 55 125 131 214
*DJ2 9 MEMB 45 47 51 54 212
*DJ2 5 MEMB 2 12 28 36 211
*DJ1 2 MEMB 4 13 33 41 202 213
*DJ1 29 MEMB 45 47 51 54 212
*DJ1 27 MEMB 43 49 53 55 125 131 214
*DJ1 3 MEMB 1 14 29 37 123 129 209
*DJ1 1 MEMB 2 12 28 36 211
*CHECK CODE MEMB 1 TO 19 24 TO 37 39 41 TO 63 67 TO 107 110 TO 123 125 TO 292
*STEEL TAKE OFF LIST 1 TO 19 24 TO 37 39 41 TO 63 67 TO 107 110 TO 123 125 -
*126 TO 292
FINISH
```

ATTACHMENT 2 STAAD INPUT FILE FOR F2 & F5 DESIGN

PROPRIETARY & CONFIDENTIAL

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STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 25-May-20
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER MTON
JOINT COORDINATES
164 0 -2.57 5; 167 0 -2.57 0; 464 0 -2.17 0; 469 0 -2.17 5; 470 6 -2.17 0;
471 6 -2.57 0; 472 6 -2.17 5; 473 6 -2.57 5; 474 0.1 -2.17 10;
475 0.1 -2.57 10; 476 4.1 -2.17 10; 477 4.1 -2.57 10; 478 -0.5 -2.57 -0.9;
479 -0.5 -2.57 0.9; 480 6.5 -2.57 -0.9; 481 6.5 -2.57 0.9; 482 -0.5 -2.57 4.1;
483 -0.5 -2.57 5.9; 484 6.5 -2.57 4.1; 485 6.5 -2.57 5.9; 486 -0.4 -2.57 9.1;
487 -0.4 -2.57 10.9; 488 4.6 -2.57 9.1; 489 4.6 -2.57 10.9; 490 0 -0.53 0;
491 0 -0.53 5; 492 6 -0.53 0; 493 6 -0.53 5; 494 0.1 -0.95 10;
495 4.1 -0.95 10; 496 -0.25 -2.57 -0.9; 497 -0.25 -2.57 -0.6;
498 -0.5 -2.57 -0.6; 499 2.23517e-008 -2.57 -0.9; 500 9.18905e-008 -2.57 -0.6;
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980 1.6 -2.57 10.3; 981 1.85 -2.57 10.3; 982 2.1 -2.57 10.3;
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MEMBER INCIDENCES
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992 995 996 1016 1015; 993 996 997 1017 1016; 994 997 998 1018 1017;
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1007 1010 1011 1031 1030; 1008 1011 1012 1032 1031; 1009 1012 1013 489 1032;
*START GROUP DEFINITION
*FLOOR
*END GROUP DEFINITION
ELEMENT PROPERTY
554 TO 889 THICKNESS 0.6
890 TO 1009 THICKNESS 0.4
SUPPORTS
554 TO 1009 PLATE MAT DIRECT Y SUBGRADE 4129.85 PRINT

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DEFINE MATERIAL START
ISOTROPIC STEEL
E 2.09042e+007
POISSON 0.3
DENSITY 7.83341
ALPHA 1.2e-005
DAMP 0.03
TYPE STEEL
STRENGTH RY 1.5 RT 1.2
ISOTROPIC CONCRETE
E 2.21467e+006
POISSON 0.17
DENSITY 2.40261
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
G 946438
TYPE CONCRETE
STRENGTH FCU 2855.21
STRENGTH FCU 2812.28
END DEFINE MATERIAL
MEMBER PROPERTY EUROPEAN
535 540 TO 542 548 TO 551 PRIS YD 0.6 ZD 0.6
UNIT METER KN
MEMBER PROPERTY
543 544 552 553 PRIS YD 0.45 ZD 0.55
UNIT METER MTON
CONSTANTS
MATERIAL CONCRETE ALL
*****
LOAD 1 LOADTYPE Dead  TITLE DL
UNIT METER KN
SELFWEIGHT Y -1
UNIT METER MTON
JOINT LOAD
492 FX 1.2
492 FY -10.8
492 FZ -0.3
490 FX -1.5 FY -14.1 FZ -0.5
493 FX 1.1 FY -13.4 FZ 0.6
491 FX -0.8 FY -15.8 FZ 0.2
494 495 FY -7.6
UNIT METER KN
ELEMENT LOAD
554 TO 1009 PR GY -12.6
UNIT METER MTON
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LOAD 2 LOADTYPE Live TITLE LL
JOINT LOAD
495 FY -4.7
494 FY -4.8
491 FX -1.6 FY -21.1 FZ 0.1
493 FX 2 FY -12.9 FZ 0.8
490 FX -3 FY -14.9 FZ -0.6
492 FX 2.7 FY -9 FZ -0.5
LOAD 3 LOADTYPE Wind TITLE WL1(+X)
JOINT LOAD
492 FX 4.7 FY -12.3 FZ 0.1
490 FX 4.8 FY 12.3 FZ -0.1
493 FX 8.9 FY -13 FZ -0.2
491 FX 2.1 FY 13 FZ 0.2
494 FX 1.2 FY 2.2
495 FX 1.2 FY -2.2
LOAD 4 LOADTYPE Wind TITLE WL2(-X)
JOINT LOAD
492 FX -4.7 FY 12.3 FZ 0.1
490 FX -4.8 FY -12.3 FZ -0.1
493 FX -8.9 FY 13 FZ -0.2
491 FX -2.1 FY -13 FZ 0.2
494 FX -1.2 FY -2.2
495 FX -1.2 FY 2.2
LOAD 5 LOADTYPE Wind TITLE WL3(+Z)
JOINT LOAD
492 FY 6.1 FZ 2.8
490 FY 7.8 FZ 4.1
493 FY -6.1 FZ 2.6
491 FY -7.8 FZ 3.9
LOAD 6 LOADTYPE Wind TITLE WL4(-Z)
JOINT LOAD
492 FY -6.1 FZ -2.8
490 FY -7.8 FZ -4.1
493 FY 6.1 FZ -2.6
491 FY 7.8 FZ -3.9
***
LOAD COMB 101 1.4 DL
1 1.4
LOAD COMB 102 1.2 (DL) + 1.6 LL
1 1.2 2 1.6
LOAD COMB 103 1.2 (DL) + 1.6 LL
1 1.2 2 1.6
LOAD COMB 104 1.2 DL+ 1.6 WL(+X) + 0.5 LL
1 1.2 3 1.6 2 0.5
LOAD COMB 105 1.2 DL+ 1.6 WL(-X) + 0.5 LL

1 1.2 4 1.6 2 0.5
LOAD COMB 106 1.2 DL+ 1.6 WL(+Z) + 0.5 LL
1 1.2 5 1.6 2 0.5
LOAD COMB 107 1.2 DL+ 1.6 WL(-Z) + 0.5 LL
1 1.2 6 1.6 2 0.5
LOAD COMB 108 0.9 DL + 1.6 WL(+X)
1 0.9 3 1.6
LOAD COMB 109 0.9 DL + 1.6 WL(-X)
1 0.9 4 1.6
LOAD COMB 110 0.9 DL + 1.6 WL(+Z)
1 0.9 5 1.6
LOAD COMB 111 0.9 DL + 1.6 WL(-Z)
1 0.9 6 1.6
LOAD COMB 112 1.2 DL + 0.8 WL(+X)
1 1.2 3 0.8
LOAD COMB 113 1.2 DL + 0.8 WL(-X)
1 1.2 4 0.8
LOAD COMB 114 1.2 DL + 0.8 WL(+Z)
1 1.2 5 0.8
LOAD COMB 115 1.2 DL + 0.8 WL(-Z)
1 1.2 5 0.8
****
****Serviceanbility check
LOAD COMB 201 1. DL
1 1.0
LOAD COMB 202 1 (DL) + 1 LL
1 1.0 2 1.0
LOAD COMB 203 1 (DL) + 1 LL
1 1.0 2 1.0
LOAD COMB 204 1 DL+ 1 WL(+X)
1 1.0 3 1.0
LOAD COMB 205 1 DL+ 1 WL(-X)
1 1.0 4 1.0
LOAD COMB 206 1 DL+ 1 WL(+Z)
1 1.0 5 1.0
LOAD COMB 207 1 DL+ 1 WL(-Z)
1 1.0 6 1.0
LOAD COMB 208 0.6 DL + 1 WL(+X)
1 0.6 3 1.0
LOAD COMB 209 0.6 DL + 1 WL(-X)
1 0.6 4 1.0
LOAD COMB 210 0.6 DL + 1 WL(+Z)
1 0.6 5 1.0
LOAD COMB 211 0.6 DL + 1 WL(-Z)
1 0.6 6 1.0
LOAD COMB 212 1 DL+ 0.75 WL(+X)+ 0.75 LL

```
1 1.0 3 0.75 2 0.75
LOAD COMB 213 1 DL+ 0.75 WL(-X)+ 0.75 LL
1 1.0 4 0.75 2 0.75
LOAD COMB 214 1 DL+ 0.75 WL(+Z)+ 0.75 LL
1 1.0 5 0.75 2 0.75
LOAD COMB 215 1 DL+ 0.75 WL(-Z)+ 0.75 LL
1 1.0 6 0.75 2 0.75
****
PERFORM ANALYSIS
START CONCRETE DESIGN
CODE ACI
UNIT METER KN
CLB 0.04 MEMB 535 540 TO 544 548 TO 553
CLS 0.04 ALL
CLT 0.04 MEMB 535 540 TO 544 548 TO 553
FC 28000 MEMB 535 540 TO 544 548 TO 553
FYMAIN 414000 MEMB 535 540 TO 544 548 TO 553
MINMAIN 16 ALL
MINSEC 10 ALL
DESIGN COLUMN 535 540 TO 544 548 TO 553
END CONCRETE DESIGN
*LOAD LIST 1 TO 10
*PERFORM ANALYSIS
*LOAD LIST 101 TO 123
***
*PARAMETER 1
*CODE LRFD
*FU 41000 MEMB 1 TO 19 24 TO 37 39 41 TO 63 67 TO 107 110 TO 123 125 TO 292
*FYLD 27500 MEMB 1 TO 19 24 TO 37 39 41 TO 63 67 TO 107 110 TO 123 125 TO 292
*KX 1 MEMB 1 TO 19 24 TO 37 39 41 TO 63 67 TO 107 110 TO 123 125 TO 261 268 -
*269 TO 292
*KY 1 MEMB 1 TO 19 24 TO 37 39 41 TO 63 67 TO 107 110 TO 123 125 TO 261 268 -
*269 TO 292
*KZ 0.85 MEMB 1 TO 19 24 TO 37 39 41 TO 63 67 TO 107 110 TO 123 125 TO 292
*LX 1.5 MEMB 81 82 175 TO 184 262 TO 267
*LZ 2 MEMB 262 TO 267
*LY 1.5 MEMB 81 82 175 TO 184 262 TO 267
*LY 5 MEMB 1 TO 4 12 TO 16 18 28 29 31 33 36 37 39 41 47 49 51 53 TO 55 58 -
*123 129 TO 131 201 202 209 TO 211 213 235 236 242 243 256 257 280 TO 282
*LX 3 MEMB 43 45 125 212 214
*LY 3 MEMB 43 45 125 212 214 253
*LZ 3 MEMB 81 82 175 TO 184
*LZ 4 MEMB 8 10 19 27 35 42 48 52 62 113 118 153 TO 160 169 TO 172 194 TO 197 -
*205 TO 208 229 230
*LZ 5 MEMB 1 TO 6 9 11 TO 17 24 25 28 TO 33 36 37 39 41 44 TO 47 49 TO 51 53 -
*54 TO 58 63 67 TO 80 111 116 123 129 TO 131 135 TO 140 144 TO 149 165 167 -
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*185 186 201 202 209 TO 213 258 TO 261 268 TO 279 286
*LZ 5 MEMB 43 45 125 212 214
*LZ 4.5 MEMB 17 25 46 50 111 135 TO 140 270 TO 279 286
*MAIN 200 MEMB 1 TO 19 24 TO 37 39 41 TO 63 67 TO 107 110 TO 123 125 TO 292
*RATIO 1 MEMB 1 TO 19 24 TO 37 39 41 TO 63 67 TO 107 110 TO 123 125 TO 292
*TMAIN 300 MEMB 1 TO 19 24 TO 37 39 41 TO 63 67 TO 107 110 TO 123 125 TO 292
*TRACK 2 MEMB 1 TO 19 24 TO 37 39 41 TO 63 67 TO 107 110 TO 123 125 TO 292
*CHECK CODE MEMB 1 TO 19 24 TO 37 39 41 TO 63 67 TO 107 110 TO 123 125 TO 292
*PARAMETER 2
*CODE LRFD
*DJ2 40 MEMB 69 77 78
*DJ1 41 MEMB 69 77 78
*DJ2 10 MEMB 11 67 70 185
*DJ1 9 MEMB 11 67 70 185
*DJ2 7 MEMB 5 68 71 186
*DJ1 5 MEMB 5 68 71 186
*DJ2 50 MEMB 81 175 TO 179
*DJ1 48 MEMB 81 175 TO 179
*DJ2 51 MEMB 82 180 TO 184
*DJ1 49 MEMB 82 180 TO 184
*DJ2 42 MEMB 72 79 80
*DJ1 43 MEMB 72 79 80
*DFF 500 MEMB 5 6 11 67 TO 74 77 TO 82 175 TO 186 258 259
*DFF 360 MEMB 1 TO 4 12 TO 15 28 29 31 33 36 37 39 41 43 45 47 49 51 -
*53 TO 55 123 125 129 TO 131 201 202 209 TO 214
**FREQUENCY CHECK FOR TOP FLOOR DONE BY DEFLECTION VALUES
*DJ1 4 MEMB 3 15 31 39 130 201 210
*DJ2 8 MEMB 3 15 31 39 130 201 210
*DJ2 6 MEMB 4 13 33 41 202 213
*DJ2 10 MEMB 43 49 53 55 125 131 214
*DJ2 9 MEMB 45 47 51 54 212
*DJ2 5 MEMB 2 12 28 36 211
*DJ1 2 MEMB 4 13 33 41 202 213
*DJ1 29 MEMB 45 47 51 54 212
*DJ1 27 MEMB 43 49 53 55 125 131 214
*DJ1 3 MEMB 1 14 29 37 123 129 209
*DJ1 1 MEMB 2 12 28 36 211
*CHECK CODE MEMB 1 TO 19 24 TO 37 39 41 TO 63 67 TO 107 110 TO 123 125 TO 292
*STEEL TAKE OFF LIST 1 TO 19 24 TO 37 39 41 TO 63 67 TO 107 110 TO 123 125 -
*126 TO 292
FINISH

ATTACHMENT 3 STAAD INPUT FILE FOR F6 DESIGN

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```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 25-May-20
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER MTON
JOINT COORDINATES
167 0 -2.87 0; 464 0 -1.67 0; 490 0 -0.89 0; 491 0 -1.67 -4.145;
492 0 -2.87 -4.145; 493 0 -0.89 -4.145; 494 -6.66 -1.67 -4.145;
495 -6.66 -2.87 -4.145; 496 -6.66 -0.89 -4.145; 497 -6.66 -1.67 0;
498 -6.66 -2.87 0; 499 -6.66 -0.89 0;
MEMBER INCIDENCES
535 464 167; 548 464 490; 549 491 492; 550 491 493; 551 494 495; 552 494 496;
553 497 498; 554 497 499; 555 494 491; 556 494 497; 557 497 464; 558 491 464;
*START GROUP DEFINITION
*FLOOR
*END GROUP DEFINITION
SUPPORTS
167 492 495 498 FIXED
DEFINE MATERIAL START
ISOTROPIC STEEL
E 2.09042e+007
POISSON 0.3
DENSITY 7.83341
ALPHA 1.2e-005
DAMP 0.03
TYPE STEEL
STRENGTH RY 1.5 RT 1.2
ISOTROPIC CONCRETE
E 2.21467e+006
POISSON 0.17
DENSITY 2.40261
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
G 946438
TYPE CONCRETE
STRENGTH FCU 2855.21
STRENGTH FCU 2812.28
END DEFINE MATERIAL
MEMBER PROPERTY EUROPEAN
535 548 TO 554 PRIS YD 0.5 ZD 0.5
UNIT METER KN
MEMBER PROPERTY EUROPEAN
555 TO 558 PRIS YD 0.3 ZD 0.25
UNIT METER MTON
```

```
CONSTANTS
MATERIAL CONCRETE ALL
*****
UNIT METER KN
MEMBER RELEASE
UNIT METER MTON
LOAD 1 LOADTYPE Dead  TITLE DL
UNIT METER KN
SELFWEIGHT Y -1
JOINT LOAD
490 FX 66.44 FY -74.48 FZ 1.17
493 FX 62.9 FY -88.53 FZ -0.08
496 FX -62.21 FY -72.68
499 FX -67.11 FY -74.1
MEMBER LOAD
555 TO 558 UNI GY -5.4
UNIT METER MTON
LOAD 2 LOADTYPE Live  TITLE LL
UNIT METER KN
JOINT LOAD
490 FX 38 FY -51.63 FZ 2.61
499 FX -39.42 FY -50.42 FZ -0.03
496 FX -29.75 FY -47.22
493 FX 31.2 FY -85.43 FZ -0.15
UNIT METER MTON
LOAD 3 LOADTYPE Wind  TITLE WL1(+X)
UNIT METER KN
JOINT LOAD
490 FX 9.35 FY -6.75 FZ 2.18
499 FX 9.35 FY 7.91 FZ -0.09
496 FX 10.46 FY 8.81
493 FX 10.32 FY -9.41 FZ 0.14
UNIT METER MTON
LOAD 4 LOADTYPE Wind  TITLE WL2(-X)
UNIT METER KN
JOINT LOAD
490 FX -9.35 FY 6.72 FZ -2.18
499 FX -9.35 FY -7.91 FZ 0.09
493 FX -10.35 FY 9.42 FZ -0.14
496 FX -10.49 FY -8.84
UNIT METER MTON
LOAD 5 LOADTYPE Wind  TITLE WL3(+Z)
UNIT METER KN
JOINT LOAD
490 FX 1.59 FY -17.99 FZ 31.47
496 FX 5.51 FY 15.18 FZ -0.26
```

493 FX -1.7 FY 17.12 FZ -0.89
499 FX -5.42 FY -15.11 FZ 20.91
UNIT METER MTON
LOAD 6 LOADTYPE Wind TITLE WL4(-Z)
UNIT METER KN
JOINT LOAD
490 FX -1.59 FY 17.99 FZ -31.46
496 FX -5.51 FY -15.21 FZ 0.26
493 FX 1.7 FY -17.12 FZ 0.89
499 FX 5.4 FY 15.14 FZ 20.97
LOAD 7 LOADTYPE None TITLE TL1_TEMP (+25)
JOINT LOAD
490 FX 1.59 FY -0.2 FZ 0.69
499 FX -2.09 FY -0.18 FZ 0.03
496 FX -1.61 FY 0.24
493 FX 2.06 FY -0.06 FZ -0.05
LOAD 8 LOADTYPE None TITLE TL2_TEMP (-25)
JOINT LOAD
490 FX -1.59 FY 0.2 FZ -0.69
496 FX 1.61 FY -0.24
493 FX -2.06 FY 0.06 FZ 0.05
499 FX 2.09 FY 0.03 FZ 0.18
***
UNIT METER MTON
LOAD COMB 101 1.4 DL
1 1.4
LOAD COMB 102 1.2 (DL+TL1) + 1.6 LL
1 1.2 2 1.6 7 1.2
LOAD COMB 103 1.2 (DL+TL2) + 1.6 LL
1 1.2 2 1.6 8 1.2
LOAD COMB 104 1.2 DL+ 1.6 WL(+X) + 0.5 LL
1 1.2 3 1.6 2 0.5
LOAD COMB 105 1.2 DL+ 1.6 WL(-X) + 0.5 LL
1 1.2 4 1.6 2 0.5
LOAD COMB 106 1.2 DL+ 1.6 WL(+Z) + 0.5 LL
1 1.2 5 1.6 2 0.5
LOAD COMB 107 1.2 DL+ 1.6 WL(-Z) + 0.5 LL
1 1.2 6 1.6 2 0.5
LOAD COMB 108 0.9 DL + 1.6 WL(+X)
1 0.9 3 1.6
LOAD COMB 109 0.9 DL + 1.6 WL(-X)
1 0.9 4 1.6
LOAD COMB 110 0.9 DL + 1.6 WL(+Z)
1 0.9 5 1.6
LOAD COMB 111 0.9 DL + 1.6 WL(-Z)
1 0.9 6 1.6

LOAD COMB 112 1.2 DL + 0.8 WL(+X)
1 1.2 3 0.8
LOAD COMB 113 1.2 DL + 0.8 WL(-X)
1 1.2 4 0.8
LOAD COMB 114 1.2 DL + 0.8 WL(+Z)
1 1.2 5 0.8
LOAD COMB 115 1.2 DL + 0.8 WL(-Z)
1 1.2 5 0.8
****
****Serviceanbility check
LOAD COMB 201 1. DL
1 1.0
LOAD COMB 202 1 (DL+TL1) + 1 LL
1 1.0 2 1.0 7 1.0
LOAD COMB 203 1 (DL+TL2) + 1 LL
1 1.0 2 1.0 8 1.0
LOAD COMB 204 1 DL+ 1 WL(+X)
1 1.0 3 1.0
LOAD COMB 205 1 DL+ 1 WL(-X)
1 1.0 4 1.0
LOAD COMB 206 1 DL+ 1 WL(+Z)
1 1.0 5 1.0
LOAD COMB 207 1 DL+ 1 WL(-Z)
1 1.0 6 1.0
LOAD COMB 208 0.6 DL + 1 WL(+X)
1 0.6 3 1.0
LOAD COMB 209 0.6 DL + 1 WL(-X)
1 0.6 4 1.0
LOAD COMB 210 0.6 DL + 1 WL(+Z)
1 0.6 5 1.0
LOAD COMB 211 0.6 DL + 1 WL(-Z)
1 0.6 6 1.0
LOAD COMB 212 1 DL+ 0.75 WL(+X)+ 0.75 LL
1 1.0 3 0.75 2 0.75
LOAD COMB 213 1 DL+ 0.75 WL(-X)+ 0.75 LL
1 1.0 4 0.75 2 0.75
LOAD COMB 214 1 DL+ 0.75 WL(+Z)+ 0.75 LL
1 1.0 5 0.75 2 0.75
LOAD COMB 215 1 DL+ 0.75 WL(-Z)+ 0.75 LL
1 1.0 6 0.75 2 0.75
****
PERFORM ANALYSIS
START CONCRETE DESIGN
CODE ACI
UNIT METER KN
CLB 0.04 ALL

```
CLS 0.04 ALL
CLT 0.04 ALL
FC 28000 ALL
FYMAIN 414000 ALL
MINMAIN 16 ALL
MINSEC 10 ALL
DESIGN BEAM 555 TO 558
DESIGN COLUMN 535 548 TO 554
END CONCRETE DESIGN
*LOAD LIST 1 TO 10
*PERFORM ANALYSIS
*LOAD LIST 101 TO 123
***
*PARAMETER 1
*CODE LRFD
*FU 41000 MEMB 1 TO 19 24 TO 37 39 41 TO 63 67 TO 107 110 TO 123 125 TO 292
*FYLD 27500 MEMB 1 TO 19 24 TO 37 39 41 TO 63 67 TO 107 110 TO 123 125 TO 292
*KX 1 MEMB 1 TO 19 24 TO 37 39 41 TO 63 67 TO 107 110 TO 123 125 TO 261 268 -
*269 TO 292
*KY 1 MEMB 1 TO 19 24 TO 37 39 41 TO 63 67 TO 107 110 TO 123 125 TO 261 268 -
*269 TO 292
*KZ 0.85 MEMB 1 TO 19 24 TO 37 39 41 TO 63 67 TO 107 110 TO 123 125 TO 292
*LX 1.5 MEMB 81 82 175 TO 184 262 TO 267
*LZ 2 MEMB 262 TO 267
*LY 1.5 MEMB 81 82 175 TO 184 262 TO 267
*LY 5 MEMB 1 TO 4 12 TO 16 18 28 29 31 33 36 37 39 41 47 49 51 53 TO 55 58 -
*123 129 TO 131 201 202 209 TO 211 213 235 236 242 243 256 257 280 TO 282
*LX 3 MEMB 43 45 125 212 214
*LY 3 MEMB 43 45 125 212 214 253
*LZ 3 MEMB 81 82 175 TO 184
*LZ 4 MEMB 8 10 19 27 35 42 48 52 62 113 118 153 TO 160 169 TO 172 194 TO 197 -
*205 TO 208 229 230
*LZ 5 MEMB 1 TO 6 9 11 TO 17 24 25 28 TO 33 36 37 39 41 44 TO 47 49 TO 51 53 -
*54 TO 58 63 67 TO 80 111 116 123 129 TO 131 135 TO 140 144 TO 149 165 167 -
*185 186 201 202 209 TO 213 258 TO 261 268 TO 279 286
*LZ 5 MEMB 43 45 125 212 214
*LZ 4.5 MEMB 17 25 46 50 111 135 TO 140 270 TO 279 286
*MAIN 200 MEMB 1 TO 19 24 TO 37 39 41 TO 63 67 TO 107 110 TO 123 125 TO 292
*RATIO 1 MEMB 1 TO 19 24 TO 37 39 41 TO 63 67 TO 107 110 TO 123 125 TO 292
*TMAIN 300 MEMB 1 TO 19 24 TO 37 39 41 TO 63 67 TO 107 110 TO 123 125 TO 292
*TRACK 2 MEMB 1 TO 19 24 TO 37 39 41 TO 63 67 TO 107 110 TO 123 125 TO 292
*CHECK CODE MEMB 1 TO 19 24 TO 37 39 41 TO 63 67 TO 107 110 TO 123 125 TO 292
*PARAMETER 2
*CODE LRFD
*DJ2 40 MEMB 69 77 78
*DJ1 41 MEMB 69 77 78
```

```
*DJ2 10 MEMB 11 67 70 185
*DJ1 9 MEMB 11 67 70 185
*DJ2 7 MEMB 5 68 71 186
*DJ1 5 MEMB 5 68 71 186
*DJ2 50 MEMB 81 175 TO 179
*DJ1 48 MEMB 81 175 TO 179
*DJ2 51 MEMB 82 180 TO 184
*DJ1 49 MEMB 82 180 TO 184
*DJ2 42 MEMB 72 79 80
*DJ1 43 MEMB 72 79 80
*DFF 500 MEMB 5 6 11 67 TO 74 77 TO 82 175 TO 186 258 259
*DFF 360 MEMB 1 TO 4 12 TO 15 28 29 31 33 36 37 39 41 43 45 47 49 51 -
*53 TO 55 123 125 129 TO 131 201 202 209 TO 214
**FREQUENCY CHECK FOR TOP FLOOR DONE BY DEFLECTION VALUES
*DJ1 4 MEMB 3 15 31 39 130 201 210
*DJ2 8 MEMB 3 15 31 39 130 201 210
*DJ2 6 MEMB 4 13 33 41 202 213
*DJ2 10 MEMB 43 49 53 55 125 131 214
*DJ2 9 MEMB 45 47 51 54 212
*DJ2 5 MEMB 2 12 28 36 211
*DJ1 2 MEMB 4 13 33 41 202 213
*DJ1 29 MEMB 45 47 51 54 212
*DJ1 27 MEMB 43 49 53 55 125 131 214
*DJ1 3 MEMB 1 14 29 37 123 129 209
*DJ1 1 MEMB 2 12 28 36 211
*CHECK CODE MEMB 1 TO 19 24 TO 37 39 41 TO 63 67 TO 107 110 TO 123 125 TO 292
*STEEL TAKE OFF LIST 1 TO 19 24 TO 37 39 41 TO 63 67 TO 107 110 TO 123 125 -
*126 TO 292
FINISH
```

ATTACHMENT 4 STAAD INPUT FILE FOR F7 DESIGN

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STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 25-May-20
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER MTON
JOINT COORDINATES
167 0 -2.57 0; 464 0 -2.17 0; 490 0 -0.72 0; 491 1.8 -2.17 0; 492 1.8 -2.57 0;
493 1.8 -0.72 0; 494 2.3 -2.57 0.75; 495 2.3 -2.57 -0.75; 496 -0.5 -2.57 0.75;
497 -0.5 -2.57 -0.75; 498 -0.45 -2.57 -0.75; 499 -0.45 -2.57 -0.7;
500 -0.5 -2.57 -0.7; 501 -0.4 -2.57 -0.75; 502 -0.4 -2.57 -0.7;
503 -0.35 -2.57 -0.75; 504 -0.35 -2.57 -0.7; 505 -0.3 -2.57 -0.75;
506 -0.3 -2.57 -0.7; 507 -0.25 -2.57 -0.75; 508 -0.25 -2.57 -0.7;
509 -0.2 -2.57 -0.75; 510 -0.2 -2.57 -0.7; 511 -0.15 -2.57 -0.75;
512 -0.15 -2.57 -0.7; 513 -0.1 -2.57 -0.75; 514 -0.1 -2.57 -0.7;
515 -0.05 -2.57 -0.75; 516 -0.05 -2.57 -0.7; 517 3.40598e-009 -2.57 -0.75;
518 6.18753e-009 -2.57 -0.7; 519 0.05 -2.57 -0.75; 520 0.05 -2.57 -0.7;
521 0.1 -2.57 -0.75; 522 0.1 -2.57 -0.7; 523 0.15 -2.57 -0.75;
524 0.15 -2.57 -0.7; 525 0.2 -2.57 -0.75; 526 0.2 -2.57 -0.7;
527 0.25 -2.57 -0.75; 528 0.25 -2.57 -0.7; 529 0.3 -2.57 -0.75;
530 0.3 -2.57 -0.7; 531 0.35 -2.57 -0.75; 532 0.35 -2.57 -0.7;
533 0.4 -2.57 -0.75; 534 0.4 -2.57 -0.7; 535 0.45 -2.57 -0.75;
536 0.45 -2.57 -0.7; 537 0.5 -2.57 -0.75; 538 0.5 -2.57 -0.7;
539 0.55 -2.57 -0.75; 540 0.55 -2.57 -0.7; 541 0.6 -2.57 -0.75;
542 0.6 -2.57 -0.7; 543 0.65 -2.57 -0.75; 544 0.65 -2.57 -0.7;
545 0.7 -2.57 -0.75; 546 0.7 -2.57 -0.7; 547 0.75 -2.57 -0.75;
548 0.75 -2.57 -0.7; 549 0.8 -2.57 -0.75; 550 0.8 -2.57 -0.7;
551 0.85 -2.57 -0.75; 552 0.85 -2.57 -0.7; 553 0.9 -2.57 -0.75;
554 0.9 -2.57 -0.7; 555 0.95 -2.57 -0.75; 556 0.95 -2.57 -0.7;
557 1 -2.57 -0.75; 558 1 -2.57 -0.7; 559 1.05 -2.57 -0.75; 560 1.05 -2.57 -0.7;
561 1.1 -2.57 -0.75; 562 1.1 -2.57 -0.7; 563 1.15 -2.57 -0.75;
564 1.15 -2.57 -0.7; 565 1.2 -2.57 -0.75; 566 1.2 -2.57 -0.7;
567 1.25 -2.57 -0.75; 568 1.25 -2.57 -0.7; 569 1.3 -2.57 -0.75;
570 1.3 -2.57 -0.7; 571 1.35 -2.57 -0.75; 572 1.35 -2.57 -0.7;
573 1.4 -2.57 -0.75; 574 1.4 -2.57 -0.7; 575 1.45 -2.57 -0.75;
576 1.45 -2.57 -0.7; 577 1.5 -2.57 -0.75; 578 1.5 -2.57 -0.7;
579 1.55 -2.57 -0.75; 580 1.55 -2.57 -0.7; 581 1.6 -2.57 -0.75;
582 1.6 -2.57 -0.7; 583 1.65 -2.57 -0.75; 584 1.65 -2.57 -0.7;
585 1.7 -2.57 -0.75; 586 1.7 -2.57 -0.7; 587 1.75 -2.57 -0.75;
588 1.75 -2.57 -0.7; 589 1.8 -2.57 -0.75; 590 1.8 -2.57 -0.7;
591 1.85 -2.57 -0.75; 592 1.85 -2.57 -0.7; 593 1.9 -2.57 -0.75;
594 1.9 -2.57 -0.7; 595 1.95 -2.57 -0.75; 596 1.95 -2.57 -0.7;
597 2 -2.57 -0.75; 598 2 -2.57 -0.7; 599 2.05 -2.57 -0.75; 600 2.05 -2.57 -0.7;
601 2.1 -2.57 -0.75; 602 2.1 -2.57 -0.7; 603 2.15 -2.57 -0.75;
604 2.15 -2.57 -0.7; 605 2.2 -2.57 -0.75; 606 2.2 -2.57 -0.7;
607 2.25 -2.57 -0.75; 608 2.25 -2.57 -0.7; 609 2.3 -2.57 -0.7;

610 -0.45 -2.57 -0.65; 611 -0.5 -2.57 -0.65; 612 -0.4 -2.57 -0.65;
613 -0.35 -2.57 -0.65; 614 -0.3 -2.57 -0.65; 615 -0.25 -2.57 -0.65;
616 -0.2 -2.57 -0.65; 617 -0.15 -2.57 -0.65; 618 -0.1 -2.57 -0.65;
619 -0.05 -2.57 -0.65; 620 8.96908e-009 -2.57 -0.65; 621 0.05 -2.57 -0.65;
622 0.1 -2.57 -0.65; 623 0.15 -2.57 -0.65; 624 0.2 -2.57 -0.65;
625 0.25 -2.57 -0.65; 626 0.3 -2.57 -0.65; 627 0.35 -2.57 -0.65;
628 0.4 -2.57 -0.65; 629 0.45 -2.57 -0.65; 630 0.5 -2.57 -0.65;
631 0.55 -2.57 -0.65; 632 0.6 -2.57 -0.65; 633 0.65 -2.57 -0.65;
634 0.7 -2.57 -0.65; 635 0.75 -2.57 -0.65; 636 0.8 -2.57 -0.65;
637 0.85 -2.57 -0.65; 638 0.9 -2.57 -0.65; 639 0.95 -2.57 -0.65;
640 1 -2.57 -0.65; 641 1.05 -2.57 -0.65; 642 1.1 -2.57 -0.65;
643 1.15 -2.57 -0.65; 644 1.2 -2.57 -0.65; 645 1.25 -2.57 -0.65;
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649 1.45 -2.57 -0.65; 650 1.5 -2.57 -0.65; 651 1.55 -2.57 -0.65;
652 1.6 -2.57 -0.65; 653 1.65 -2.57 -0.65; 654 1.7 -2.57 -0.65;
655 1.75 -2.57 -0.65; 656 1.8 -2.57 -0.65; 657 1.85 -2.57 -0.65;
658 1.9 -2.57 -0.65; 659 1.95 -2.57 -0.65; 660 2 -2.57 -0.65;
661 2.05 -2.57 -0.65; 662 2.1 -2.57 -0.65; 663 2.15 -2.57 -0.65;
664 2.2 -2.57 -0.65; 665 2.25 -2.57 -0.65; 666 2.3 -2.57 -0.65;
667 -0.45 -2.57 -0.6; 668 -0.5 -2.57 -0.6; 669 -0.4 -2.57 -0.6;
670 -0.35 -2.57 -0.6; 671 -0.3 -2.57 -0.6; 672 -0.25 -2.57 -0.6;
673 -0.2 -2.57 -0.6; 674 -0.15 -2.57 -0.6; 675 -0.1 -2.57 -0.6;
676 -0.05 -2.57 -0.6; 677 1.17506e-008 -2.57 -0.6; 678 0.05 -2.57 -0.6;
679 0.1 -2.57 -0.6; 680 0.15 -2.57 -0.6; 681 0.2 -2.57 -0.6;
682 0.25 -2.57 -0.6; 683 0.3 -2.57 -0.6; 684 0.35 -2.57 -0.6;
685 0.4 -2.57 -0.6; 686 0.45 -2.57 -0.6; 687 0.5 -2.57 -0.6;
688 0.55 -2.57 -0.6; 689 0.6 -2.57 -0.6; 690 0.65 -2.57 -0.6;
691 0.7 -2.57 -0.6; 692 0.75 -2.57 -0.6; 693 0.8 -2.57 -0.6;
694 0.85 -2.57 -0.6; 695 0.9 -2.57 -0.6; 696 0.95 -2.57 -0.6; 697 1 -2.57 -0.6;
698 1.05 -2.57 -0.6; 699 1.1 -2.57 -0.6; 700 1.15 -2.57 -0.6;
701 1.2 -2.57 -0.6; 702 1.25 -2.57 -0.6; 703 1.3 -2.57 -0.6;
704 1.35 -2.57 -0.6; 705 1.4 -2.57 -0.6; 706 1.45 -2.57 -0.6;
707 1.5 -2.57 -0.6; 708 1.55 -2.57 -0.6; 709 1.6 -2.57 -0.6;
710 1.65 -2.57 -0.6; 711 1.7 -2.57 -0.6; 712 1.75 -2.57 -0.6;
713 1.8 -2.57 -0.6; 714 1.85 -2.57 -0.6; 715 1.9 -2.57 -0.6;
716 1.95 -2.57 -0.6; 717 2 -2.57 -0.6; 718 2.05 -2.57 -0.6; 719 2.1 -2.57 -0.6;
720 2.15 -2.57 -0.6; 721 2.2 -2.57 -0.6; 722 2.25 -2.57 -0.6;
723 2.3 -2.57 -0.6; 724 -0.45 -2.57 -0.55; 725 -0.5 -2.57 -0.55;
726 -0.4 -2.57 -0.55; 727 -0.35 -2.57 -0.55; 728 -0.3 -2.57 -0.55;
729 -0.25 -2.57 -0.55; 730 -0.2 -2.57 -0.55; 731 -0.15 -2.57 -0.55;
732 -0.1 -2.57 -0.55; 733 -0.05 -2.57 -0.55; 734 1.45322e-008 -2.57 -0.55;
735 0.05 -2.57 -0.55; 736 0.1 -2.57 -0.55; 737 0.15 -2.57 -0.55;
738 0.2 -2.57 -0.55; 739 0.25 -2.57 -0.55; 740 0.3 -2.57 -0.55;
741 0.35 -2.57 -0.55; 742 0.4 -2.57 -0.55; 743 0.45 -2.57 -0.55;
744 0.5 -2.57 -0.55; 745 0.55 -2.57 -0.55; 746 0.6 -2.57 -0.55;
747 0.65 -2.57 -0.55; 748 0.7 -2.57 -0.55; 749 0.75 -2.57 -0.55;

750 0.8 -2.57 -0.55; 751 0.85 -2.57 -0.55; 752 0.9 -2.57 -0.55;
753 0.95 -2.57 -0.55; 754 1 -2.57 -0.55; 755 1.05 -2.57 -0.55;
756 1.1 -2.57 -0.55; 757 1.15 -2.57 -0.55; 758 1.2 -2.57 -0.55;
759 1.25 -2.57 -0.55; 760 1.3 -2.57 -0.55; 761 1.35 -2.57 -0.55;
762 1.4 -2.57 -0.55; 763 1.45 -2.57 -0.55; 764 1.5 -2.57 -0.55;
765 1.55 -2.57 -0.55; 766 1.6 -2.57 -0.55; 767 1.65 -2.57 -0.55;
768 1.7 -2.57 -0.55; 769 1.75 -2.57 -0.55; 770 1.8 -2.57 -0.55;
771 1.85 -2.57 -0.55; 772 1.9 -2.57 -0.55; 773 1.95 -2.57 -0.55;
774 2 -2.57 -0.55; 775 2.05 -2.57 -0.55; 776 2.1 -2.57 -0.55;
777 2.15 -2.57 -0.55; 778 2.2 -2.57 -0.55; 779 2.25 -2.57 -0.55;
780 2.3 -2.57 -0.55; 781 -0.45 -2.57 -0.5; 782 -0.5 -2.57 -0.5;
783 -0.4 -2.57 -0.5; 784 -0.35 -2.57 -0.5; 785 -0.3 -2.57 -0.5;
786 -0.25 -2.57 -0.5; 787 -0.2 -2.57 -0.5; 788 -0.15 -2.57 -0.5;
789 -0.1 -2.57 -0.5; 790 -0.05 -2.57 -0.5; 791 1.73137e-008 -2.57 -0.5;
792 0.05 -2.57 -0.5; 793 0.1 -2.57 -0.5; 794 0.15 -2.57 -0.5;
795 0.2 -2.57 -0.5; 796 0.25 -2.57 -0.5; 797 0.3 -2.57 -0.5;
798 0.35 -2.57 -0.5; 799 0.4 -2.57 -0.5; 800 0.45 -2.57 -0.5;
801 0.5 -2.57 -0.5; 802 0.55 -2.57 -0.5; 803 0.6 -2.57 -0.5;
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*FLOOR
*END GROUP DEFINITION
ELEMENT PROPERTY
552 TO 2231 2233 TO 2952 2955 TO 3754 THICKNESS 0.5
SUPPORTS
552 TO 2231 2233 TO 2952 2955 TO 3754 PLATE MAT DIRECT Y SUBGRADE 4129.85
PRINT
DEFINE MATERIAL START
ISOTROPIC STEEL
E 2.09042e+007
POISSON 0.3
DENSITY 7.83341
ALPHA 1.2e-005
DAMP 0.03
TYPE STEEL
STRENGTH RY 1.5 RT 1.2
ISOTROPIC CONCRETE
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POISSON 0.17
DENSITY 2.40261
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DAMP 0.05
TYPE CONCRETE
G 946438
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STRENGTH FCU 2855.21
STRENGTH FCU 2812.28
END DEFINE MATERIAL
MEMBER PROPERTY EUROPEAN
535 548 TO 550 PRIS YD 0.5 ZD 0.5
CONSTANTS
MATERIAL CONCRETE ALL
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LOAD 1 LOADTYPE Dead TITLE DL
UNIT METER KN
ELEMENT LOAD
552 TO 2231 2233 TO 2952 2955 TO 3754 PR GY -14.4
JOINT LOAD
490 FX 4.1 FY -16.67 FZ -8.63
493 FX -4.31 FY -64.58 FZ 0.05

SELFWEIGHT Y -1
UNIT METER MTON
LOAD 2 LOADTYPE Live TITLE LL
UNIT METER KN
JOINT LOAD
490 FX 10.26 FY 4.47 FZ -15.3
493 FX -11.66 FY -154.65 FZ 0.23
UNIT METER MTON
LOAD 3 LOADTYPE Wind TITLE WL1(+X)
UNIT METER KN
JOINT LOAD
490 FX 41.94 FY 115.08 FZ 2.33
493 FX -0.89 FY -105.51 FZ 0.24
UNIT METER MTON
LOAD 4 LOADTYPE Wind TITLE WL2(-X)
UNIT METER KN
JOINT LOAD
490 FX -41.96 FY -115.13 FZ -2.33
493 FX 0.89 FY 105.56 FZ -0.24
UNIT METER MTON
LOAD 5 LOADTYPE Wind TITLE WL3(+Z)
UNIT METER KN
JOINT LOAD
490 FX 12.2 FY 32.81 FZ 9.62
493 FX 1.5 FY -61.28 FZ 2.57
UNIT METER MTON
LOAD 6 LOADTYPE Wind TITLE WL4(-Z)
UNIT METER KN
JOINT LOAD
490 FX -12.02 FY -32.49 FZ -8.75
493 FX -1.52 FY 60.03 FZ -2.55
LOAD 7 LOADTYPE None TITLE TL1_TEMP (+25)
JOINT LOAD
490 FX 3.11 FY 11.87 FZ 2.55
493 FX 0.5 FY -17.84 FZ -0.36
LOAD 8 LOADTYPE None TITLE TL2_TEMP (-25)
JOINT LOAD
490 FX -3.11 FY -11.87 FZ -2.55
493 FX -0.5 FY 17.84 FZ 0.36
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UNIT METER MTON
LOAD COMB 101 1.4 DL
1 1.4
LOAD COMB 102 1.2 (DL+TL1) + 1.6 LL
1 1.2 2 1.6 7 1.2
LOAD COMB 103 1.2 (DL+TL2) + 1.6 LL

1 1.2 2 1.6 8 1.2
LOAD COMB 104 1.2 DL+ 1.6 WL(+X) + 0.5 LL
1 1.2 3 1.6 2 0.5
LOAD COMB 105 1.2 DL+ 1.6 WL(-X) + 0.5 LL
1 1.2 4 1.6 2 0.5
LOAD COMB 106 1.2 DL+ 1.6 WL(+Z) + 0.5 LL
1 1.2 5 1.6 2 0.5
LOAD COMB 107 1.2 DL+ 1.6 WL(-Z) + 0.5 LL
1 1.2 6 1.6 2 0.5
LOAD COMB 108 0.9 DL + 1.6 WL(+X)
1 0.9 3 1.6
LOAD COMB 109 0.9 DL + 1.6 WL(-X)
1 0.9 4 1.6
LOAD COMB 110 0.9 DL + 1.6 WL(+Z)
1 0.9 5 1.6
LOAD COMB 111 0.9 DL + 1.6 WL(-Z)
1 0.9 6 1.6
LOAD COMB 112 1.2 DL + 0.8 WL(+X)
1 1.2 3 0.8
LOAD COMB 113 1.2 DL + 0.8 WL(-X)
1 1.2 4 0.8
LOAD COMB 114 1.2 DL + 0.8 WL(+Z)
1 1.2 5 0.8
LOAD COMB 115 1.2 DL + 0.8 WL(-Z)
1 1.2 5 0.8
****
****Serviceanbility check
LOAD COMB 201 1. DL
1 1.0
LOAD COMB 202 1 (DL+TL1) + 1 LL
1 1.0 2 1.0 7 1.0
LOAD COMB 203 1 (DL+TL2) + 1 LL
1 1.0 2 1.0 8 1.0
LOAD COMB 204 1 DL+ 1 WL(+X)
1 1.0 3 1.0
LOAD COMB 205 1 DL+ 1 WL(-X)
1 1.0 4 1.0
LOAD COMB 206 1 DL+ 1 WL(+Z)
1 1.0 5 1.0
LOAD COMB 207 1 DL+ 1 WL(-Z)
1 1.0 6 1.0
LOAD COMB 208 0.6 DL + 1 WL(+X)
1 0.6 3 1.0
LOAD COMB 209 0.6 DL + 1 WL(-X)
1 0.6 4 1.0
LOAD COMB 210 0.6 DL + 1 WL(+Z)

1 0.6 5 1.0
LOAD COMB 211 0.6 DL + 1 WL(-Z)
1 0.6 6 1.0
LOAD COMB 212 1 DL+ 0.75 WL(+X)+ 0.75 LL
1 1.0 3 0.75 2 0.75
LOAD COMB 213 1 DL+ 0.75 WL(-X)+ 0.75 LL
1 1.0 4 0.75 2 0.75
LOAD COMB 214 1 DL+ 0.75 WL(+Z)+ 0.75 LL
1 1.0 5 0.75 2 0.75
LOAD COMB 215 1 DL+ 0.75 WL(-Z)+ 0.75 LL
1 1.0 6 0.75 2 0.75
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PERFORM ANALYSIS
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CODE ACI
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CLB 0.04 MEMB 548 550
CLS 0.04 MEMB 548 550
CLT 0.04 MEMB 548 550
FC 28000 MEMB 548 550
FYMAIN 414000 MEMB 548 550
MINMAIN 16 MEMB 548 550
MINSEC 10 MEMB 548 550
DESIGN COLUMN 548 550
END CONCRETE DESIGN
*LOAD LIST 1 TO 10
*PERFORM ANALYSIS
*LOAD LIST 101 TO 123
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*PARAMETER 1
*CODE LRFD
*FU 41000 MEMB 1 TO 19 24 TO 37 39 41 TO 63 67 TO 107 110 TO 123 125 TO 292
*FYLD 27500 MEMB 1 TO 19 24 TO 37 39 41 TO 63 67 TO 107 110 TO 123 125 TO 292
*KX 1 MEMB 1 TO 19 24 TO 37 39 41 TO 63 67 TO 107 110 TO 123 125 TO 261 268 -
*269 TO 292
*KY 1 MEMB 1 TO 19 24 TO 37 39 41 TO 63 67 TO 107 110 TO 123 125 TO 261 268 -
*269 TO 292
*KZ 0.85 MEMB 1 TO 19 24 TO 37 39 41 TO 63 67 TO 107 110 TO 123 125 TO 292
*LX 1.5 MEMB 81 82 175 TO 184 262 TO 267
*LZ 2 MEMB 262 TO 267
*LY 1.5 MEMB 81 82 175 TO 184 262 TO 267
*LY 5 MEMB 1 TO 4 12 TO 16 18 28 29 31 33 36 37 39 41 47 49 51 53 TO 55 58 -
*123 129 TO 131 201 202 209 TO 211 213 235 236 242 243 256 257 280 TO 282
*LX 3 MEMB 43 45 125 212 214
*LY 3 MEMB 43 45 125 212 214 253
*LZ 3 MEMB 81 82 175 TO 184

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*LZ 4 MEMB 8 10 19 27 35 42 48 52 62 113 118 153 TO 160 169 TO 172 194 TO 197 -
*205 TO 208 229 230
*LZ 5 MEMB 1 TO 6 9 11 TO 17 24 25 28 TO 33 36 37 39 41 44 TO 47 49 TO 51 53 -
*54 TO 58 63 67 TO 80 111 116 123 129 TO 131 135 TO 140 144 TO 149 165 167 -
*185 186 201 202 209 TO 213 258 TO 261 268 TO 279 286
*LZ 5 MEMB 43 45 125 212 214
*LZ 4.5 MEMB 17 25 46 50 111 135 TO 140 270 TO 279 286
*MAIN 200 MEMB 1 TO 19 24 TO 37 39 41 TO 63 67 TO 107 110 TO 123 125 TO 292
*RATIO 1 MEMB 1 TO 19 24 TO 37 39 41 TO 63 67 TO 107 110 TO 123 125 TO 292
*TMAIN 300 MEMB 1 TO 19 24 TO 37 39 41 TO 63 67 TO 107 110 TO 123 125 TO 292
*TRACK 2 MEMB 1 TO 19 24 TO 37 39 41 TO 63 67 TO 107 110 TO 123 125 TO 292
*CHECK CODE MEMB 1 TO 19 24 TO 37 39 41 TO 63 67 TO 107 110 TO 123 125 TO 292
*PARAMETER 2
*CODE LRFD
*DJ2 40 MEMB 69 77 78
*DJ1 41 MEMB 69 77 78
*DJ2 10 MEMB 11 67 70 185
*DJ1 9 MEMB 11 67 70 185
*DJ2 7 MEMB 5 68 71 186
*DJ1 5 MEMB 5 68 71 186
*DJ2 50 MEMB 81 175 TO 179
*DJ1 48 MEMB 81 175 TO 179
*DJ2 51 MEMB 82 180 TO 184
*DJ1 49 MEMB 82 180 TO 184
*DJ2 42 MEMB 72 79 80
*DJ1 43 MEMB 72 79 80
*DFF 500 MEMB 5 6 11 67 TO 74 77 TO 82 175 TO 186 258 259
*DFF 360 MEMB 1 TO 4 12 TO 15 28 29 31 33 36 37 39 41 43 45 47 49 51 -
*53 TO 55 123 125 129 TO 131 201 202 209 TO 214
**FREQUENCY CHECK FOR TOP FLOOR DONE BY DEFLECTION VALUES
*DJ1 4 MEMB 3 15 31 39 130 201 210
*DJ2 8 MEMB 3 15 31 39 130 201 210
*DJ2 6 MEMB 4 13 33 41 202 213
*DJ2 10 MEMB 43 49 53 55 125 131 214
*DJ2 9 MEMB 45 47 51 54 212
*DJ2 5 MEMB 2 12 28 36 211
*DJ1 2 MEMB 4 13 33 41 202 213
*DJ1 29 MEMB 45 47 51 54 212
*DJ1 27 MEMB 43 49 53 55 125 131 214
*DJ1 3 MEMB 1 14 29 37 123 129 209
*DJ1 1 MEMB 2 12 28 36 211
*CHECK CODE MEMB 1 TO 19 24 TO 37 39 41 TO 63 67 TO 107 110 TO 123 125 TO 292
*STEEL TAKE OFF LIST 1 TO 19 24 TO 37 39 41 TO 63 67 TO 107 110 TO 123 125 -
*126 TO 292
FINISH
```

ATTACHMENT 5 STAAD INPUT FILE FOR F8 & F9 DESIGN

PROPRIETARY & CONFIDENTIAL

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STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 25-May-20
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER MTON
JOINT COORDINATES
167 0 -2.57 0; 464 0 -2.17 0; 490 0 -1.22 0; 491 1.4 -2.17 0; 492 1.4 -2.57 0;
493 1.4 -1.22 0; 494 1.7 -2.57 0.75; 495 1.7 -2.57 -0.75; 496 -1.1 -2.57 0.75;
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1938 1904 1905 1962 1961; 1939 1905 1906 1963 1962; 1940 1906 1907 1964 1963;
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1947 1913 1914 1971 1970; 1948 1914 1915 1972 1971; 1949 1915 1916 1973 1972;
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2127 2097 2098 2155 2154; 2128 2098 2099 2156 2155; 2129 2099 2100 2157 2156;
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2139 2109 2110 2167 2166; 2140 2110 2111 2168 2167; 2141 2111 2112 2169 2168;
2142 2112 2113 2170 2169; 2143 2113 2114 2171 2170; 2144 2114 2115 2172 2171;
2145 2115 2116 2173 2172; 2146 2116 2117 2174 2173; 2147 2117 2118 2175 2174;
2148 2118 2119 2176 2175; 2149 2119 2120 2177 2176; 2150 2120 2121 2178 2177;
2151 2121 2122 2179 2178; 2152 2122 2123 2180 2179; 2153 2123 2124 2181 2180;
2154 2124 2125 2182 2181; 2155 2125 2126 2183 2182; 2156 2126 2127 2184 2183;
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2220 2191 2192 2248 2247; 2221 2192 2193 2249 2248; 2222 2193 2194 2250 2249;
2223 2194 2195 2251 2250; 2224 2195 2196 2252 2251; 2225 2196 2197 2253 2252;
2226 2197 2198 2254 2253; 2227 2198 2199 2255 2254; 2228 2199 2200 2256 2255;
2229 2200 2201 2257 2256; 2230 2201 2202 2258 2257; 2231 2202 2203 494 2258;
*START GROUP DEFINITION
*FLOOR
*END GROUP DEFINITION
ELEMENT PROPERTY
552 TO 2231 THICKNESS 0.4
SUPPORTS
552 TO 2231 PLATE MAT DIRECT Y SUBGRADE 4129.85 PRINT
DEFINE MATERIAL START
ISOTROPIC STEEL
E 2.09042e+007
POISSON 0.3
DENSITY 7.83341
ALPHA 1.2e-005
DAMP 0.03
TYPE STEEL
STRENGTH RY 1.5 RT 1.2
ISOTROPIC CONCRETE
E 2.21467e+006
POISSON 0.17
DENSITY 2.40261
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
G 946438
TYPE CONCRETE
STRENGTH FCU 2855.21
STRENGTH FCU 2812.28
END DEFINE MATERIAL
MEMBER PROPERTY EUROPEAN
535 548 TO 550 PRIS YD 0.45 ZD 0.45
CONSTANTS
MATERIAL CONCRETE ALL
*****
LOAD 1 LOADTYPE Dead TITLE DL

UNIT METER KN
SELFWEIGHT Y -1 LIST ALL
ELEMENT LOAD
552 TO 2231 PR GY -14.4
JOINT LOAD
490 FX -0.9 FY -25.64 FZ 0.57
493 FX 0.78 FY -14.7 FZ -0.62
UNIT METER MTON
LOAD 2 LOADTYPE Live TITLE LL
UNIT METER KN
JOINT LOAD
490 FX -1.91 FY -55.95 FZ 1.67
493 FX 1.38 FY -36.08 FZ -1.55
UNIT METER MTON
LOAD 3 LOADTYPE Wind TITLE WL1(+X)
UNIT METER KN
JOINT LOAD
490 FX 14.81 FY 16.05 FZ 1.4
493 FX 12.81 FY -15.68 FZ 0.06
UNIT METER MTON
LOAD 4 LOADTYPE Wind TITLE WL2(-X)
UNIT METER KN
JOINT LOAD
490 FX -14.61 FY -16.05 FZ -1.41
493 FX -13.01 FY 15.68 FZ -0.05
UNIT METER MTON
LOAD 5 LOADTYPE Wind TITLE WL3(+Z)
UNIT METER KN
JOINT LOAD
490 FY 0.02 FZ 12.63
493 FX -0.03 FY -0.03 FZ 11.22
UNIT METER MTON
LOAD 6 LOADTYPE Wind TITLE WL4(-Z)
UNIT METER KN
JOINT LOAD
490 FY -0.02 FZ -12.63
493 FX 0.03 FY 0.03 FZ -11.27
LOAD 7 LOADTYPE None TITLE TL1_Temp (+25)
JOINT LOAD
490 FX -38.21 FY -2.52 FZ -2.72
493 FX 36.44 FY 2.45 FZ 0.23
LOAD 8 LOADTYPE None TITLE TL2_Temp (-25)
JOINT LOAD
490 FX 38.21 FY 2.52 FZ 2.72
493 FX -36.44 FY -2.45 FZ -0.23
***

UNIT METER MTON
LOAD COMB 101 1.4 DL
1 1.4
LOAD COMB 102 1.2 (DL+TL1) + 1.6 LL
1 1.2 2 1.6 7 1.2
LOAD COMB 103 1.2 (DL+TL2) + 1.6 LL
1 1.2 2 1.6 8 1.2
LOAD COMB 104 1.2 DL+ 1.6 WL(+X) + 0.5 LL
1 1.2 3 1.6 2 0.5
LOAD COMB 105 1.2 DL+ 1.6 WL(-X) + 0.5 LL
1 1.2 4 1.6 2 0.5
LOAD COMB 106 1.2 DL+ 1.6 WL(+Z) + 0.5 LL
1 1.2 5 1.6 2 0.5
LOAD COMB 107 1.2 DL+ 1.6 WL(-Z) + 0.5 LL
1 1.2 6 1.6 2 0.5
LOAD COMB 108 0.9 DL + 1.6 WL(+X)
1 0.9 3 1.6
LOAD COMB 109 0.9 DL + 1.6 WL(-X)
1 0.9 4 1.6
LOAD COMB 110 0.9 DL + 1.6 WL(+Z)
1 0.9 5 1.6
LOAD COMB 111 0.9 DL + 1.6 WL(-Z)
1 0.9 6 1.6
LOAD COMB 112 1.2 DL + 0.8 WL(+X)
1 1.2 3 0.8
LOAD COMB 113 1.2 DL + 0.8 WL(-X)
1 1.2 4 0.8
LOAD COMB 114 1.2 DL + 0.8 WL(+Z)
1 1.2 5 0.8
LOAD COMB 115 1.2 DL + 0.8 WL(-Z)
1 1.2 5 0.8
****
****Serviceanbility check
LOAD COMB 201 1. DL
1 1.0
LOAD COMB 202 1 (DL+TL1) + 1 LL
1 1.0 2 1.0 7 1.0
LOAD COMB 203 1 (DL+TL2) + 1 LL
1 1.0 2 1.0 8 1.0
LOAD COMB 204 1 DL+ 1 WL(+X)
1 1.0 3 1.0
LOAD COMB 205 1 DL+ 1 WL(-X)
1 1.0 4 1.0
LOAD COMB 206 1 DL+ 1 WL(+Z)
1 1.0 5 1.0
LOAD COMB 207 1 DL+ 1 WL(-Z)

```
1 1.0 6 1.0
LOAD COMB 208 0.6 DL + 1 WL(+X)
1 0.6 3 1.0
LOAD COMB 209 0.6 DL + 1 WL(-X)
1 0.6 4 1.0
LOAD COMB 210 0.6 DL + 1 WL(+Z)
1 0.6 5 1.0
LOAD COMB 211 0.6 DL + 1 WL(-Z)
1 0.6 6 1.0
LOAD COMB 212 1 DL+ 0.75 WL(+X)+ 0.75 LL
1 1.0 3 0.75 2 0.75
LOAD COMB 213 1 DL+ 0.75 WL(-X)+ 0.75 LL
1 1.0 4 0.75 2 0.75
LOAD COMB 214 1 DL+ 0.75 WL(+Z)+ 0.75 LL
1 1.0 5 0.75 2 0.75
LOAD COMB 215 1 DL+ 0.75 WL(-Z)+ 0.75 LL
1 1.0 6 0.75 2 0.75
****
PERFORM ANALYSIS
START CONCRETE DESIGN
UNIT METER KN
CODE ACI
CLB 0.04 MEMB 548 550
CLS 0.04 MEMB 548 550
CLT 0.04 MEMB 548 550
FC 28000 MEMB 548 550
FYMAIN 414000 MEMB 548 550
MINMAIN 16 MEMB 548 550
MINSEC 10 MEMB 548 550
DESIGN COLUMN 548 550
END CONCRETE DESIGN
*LOAD LIST 1 TO 10
*PERFORM ANALYSIS
*LOAD LIST 101 TO 123
***
*PARAMETER 1
*CODE LRFD
*FU 41000 MEMB 1 TO 19 24 TO 37 39 41 TO 63 67 TO 107 110 TO 123 125 TO 292
*FYLD 27500 MEMB 1 TO 19 24 TO 37 39 41 TO 63 67 TO 107 110 TO 123 125 TO 292
*KX 1 MEMB 1 TO 19 24 TO 37 39 41 TO 63 67 TO 107 110 TO 123 125 TO 261 268 -
*269 TO 292
*KY 1 MEMB 1 TO 19 24 TO 37 39 41 TO 63 67 TO 107 110 TO 123 125 TO 261 268 -
*269 TO 292
*KZ 0.85 MEMB 1 TO 19 24 TO 37 39 41 TO 63 67 TO 107 110 TO 123 125 TO 292
*LX 1.5 MEMB 81 82 175 TO 184 262 TO 267
*LZ 2 MEMB 262 TO 267
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*LY 1.5 MEMB 81 82 175 TO 184 262 TO 267
*LY 5 MEMB 1 TO 4 12 TO 16 18 28 29 31 33 36 37 39 41 47 49 51 53 TO 55 58 -
*123 129 TO 131 201 202 209 TO 211 213 235 236 242 243 256 257 280 TO 282
*LX 3 MEMB 43 45 125 212 214
*LY 3 MEMB 43 45 125 212 214 253
*LZ 3 MEMB 81 82 175 TO 184
*LZ 4 MEMB 8 10 19 27 35 42 48 52 62 113 118 153 TO 160 169 TO 172 194 TO 197 -
*205 TO 208 229 230
*LZ 5 MEMB 1 TO 6 9 11 TO 17 24 25 28 TO 33 36 37 39 41 44 TO 47 49 TO 51 53 -
*54 TO 58 63 67 TO 80 111 116 123 129 TO 131 135 TO 140 144 TO 149 165 167 -
*185 186 201 202 209 TO 213 258 TO 261 268 TO 279 286
*LZ 5 MEMB 43 45 125 212 214
*LZ 4.5 MEMB 17 25 46 50 111 135 TO 140 270 TO 279 286
*MAIN 200 MEMB 1 TO 19 24 TO 37 39 41 TO 63 67 TO 107 110 TO 123 125 TO 292
*RATIO 1 MEMB 1 TO 19 24 TO 37 39 41 TO 63 67 TO 107 110 TO 123 125 TO 292
*TMAIN 300 MEMB 1 TO 19 24 TO 37 39 41 TO 63 67 TO 107 110 TO 123 125 TO 292
*TRACK 2 MEMB 1 TO 19 24 TO 37 39 41 TO 63 67 TO 107 110 TO 123 125 TO 292
*CHECK CODE MEMB 1 TO 19 24 TO 37 39 41 TO 63 67 TO 107 110 TO 123 125 TO 292
*PARAMETER 2
*CODE LRFD
*DJ2 40 MEMB 69 77 78
*DJ1 41 MEMB 69 77 78
*DJ2 10 MEMB 11 67 70 185
*DJ1 9 MEMB 11 67 70 185
*DJ2 7 MEMB 5 68 71 186
*DJ1 5 MEMB 5 68 71 186
*DJ2 50 MEMB 81 175 TO 179
*DJ1 48 MEMB 81 175 TO 179
*DJ2 51 MEMB 82 180 TO 184
*DJ1 49 MEMB 82 180 TO 184
*DJ2 42 MEMB 72 79 80
*DJ1 43 MEMB 72 79 80
*DFF 500 MEMB 5 6 11 67 TO 74 77 TO 82 175 TO 186 258 259
*DFF 360 MEMB 1 TO 4 12 TO 15 28 29 31 33 36 37 39 41 43 45 47 49 51 -
*53 TO 55 123 125 129 TO 131 201 202 209 TO 214
**FREQUENCY CHECK FOR TOP FLOOR DONE BY DEFLECTION VALUES
*DJ1 4 MEMB 3 15 31 39 130 201 210
*DJ2 8 MEMB 3 15 31 39 130 201 210
*DJ2 6 MEMB 4 13 33 41 202 213
*DJ2 10 MEMB 43 49 53 55 125 131 214
*DJ2 9 MEMB 45 47 51 54 212
*DJ2 5 MEMB 2 12 28 36 211
*DJ1 2 MEMB 4 13 33 41 202 213
*DJ1 29 MEMB 45 47 51 54 212
*DJ1 27 MEMB 43 49 53 55 125 131 214
*DJ1 3 MEMB 1 14 29 37 123 129 209
```

*DJ1 1 MEMB 2 12 28 36 211
*CHECK CODE MEMB 1 TO 19 24 TO 37 39 41 TO 63 67 TO 107 110 TO 123 125 TO 292
*STEEL TAKE OFF LIST 1 TO 19 24 TO 37 39 41 TO 63 67 TO 107 110 TO 123 125 -
*126 TO 292

ATTACHMENT 6 STAAD INPUT FILE FOR F16 DESIGN

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```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 25-May-20
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER MTON
JOINT COORDINATES
494 -6.66 -1.67 -4.145; 495 -6.66 -2.87 -4.145; 496 -6.66 -0.67 -4.145;
497 -5.28 -1.67 -4.145; 498 -5.28 -2.87 -4.145; 499 -5.28 -0.67 -4.145;
MEMBER INCIDENCES
551 494 495; 552 494 496; 553 497 498; 554 497 499;
*START GROUP DEFINITION
*FLOOR
*END GROUP DEFINITION
SUPPORTS
495 498 FIXED
DEFINE MATERIAL START
ISOTROPIC STEEL
E 2.09042e+007
POISSON 0.3
DENSITY 7.83341
ALPHA 1.2e-005
DAMP 0.03
TYPE STEEL
STRENGTH RY 1.5 RT 1.2
ISOTROPIC CONCRETE
E 2.21467e+006
POISSON 0.17
DENSITY 2.40261
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
G 946438
TYPE CONCRETE
STRENGTH FCU 2855.21
STRENGTH FCU 2812.28
END DEFINE MATERIAL
MEMBER PROPERTY EUROPEAN
551 TO 554 PRIS YD 0.5 ZD 0.5
CONSTANTS
MATERIAL CONCRETE ALL
*****
LOAD 1 LOADTYPE Dead  TITLE DL
UNIT METER KN
SELFWEIGHT Y -1
JOINT LOAD
```

496 FX 0.26 FY -7.89 FZ 0.08
499 FX -0.42 FY -4.38
UNIT METER MTON
LOAD 2 LOADTYPE Live TITLE LL
UNIT METER KN
JOINT LOAD
496 FX 0.69 FY -15.8 FZ 0.2
499 FX -1.02 FY -1.47 FZ 0.02
UNIT METER MTON
LOAD 3 LOADTYPE Wind TITLE WL1(+X)
UNIT METER KN
JOINT LOAD
496 FX -1.46 FY 23.91 FZ -0.05
499 FX 11.22 FY -23.82 FZ -0.08
UNIT METER MTON
LOAD 4 LOADTYPE Wind TITLE WL2(-X)
UNIT METER KN
JOINT LOAD
496 FX 1.46 FY -23.9 FZ 0.05
499 FX -11.22 FY 23.82 FZ 0.08
UNIT METER MTON
LOAD 5 LOADTYPE Wind TITLE WL3(+Z)
UNIT METER KN
JOINT LOAD
496 FX -0.03 FY 1.07 FZ 0.36
499 FX 0.83 FY -1.86 FZ 0.51
UNIT METER MTON
LOAD 6 LOADTYPE Wind TITLE WL4(-Z)
UNIT METER KN
JOINT LOAD
496 FX 0.03 FY -1.08 FZ -0.33
499 FX -0.84 FY 1.95 FZ -0.48
LOAD 7 LOADTYPE None TITLE TL1_TEMP (+25)
JOINT LOAD
496 FX -0.2 FY -0.14 FZ 0.41
499 FX 0.12 FY -0.2 FZ 0.41
LOAD 8 LOADTYPE None TITLE TL2_TEMP (-25)
JOINT LOAD
496 FX 0.2 FY 0.14 FZ -0.41
499 FX -0.12 FY 0.2 FZ -0.41
***
UNIT METER MTON
LOAD COMB 101 1.4 DL
1 1.4
LOAD COMB 102 1.2 (DL+TL1) + 1.6 LL
1 1.2 2 1.6 7 1.2

LOAD COMB 103 1.2 (DL+TL2) + 1.6 LL
1 1.2 2 1.6 8 1.2
LOAD COMB 104 1.2 DL+ 1.6 WL(+X) + 0.5 LL
1 1.2 3 1.6 2 0.5
LOAD COMB 105 1.2 DL+ 1.6 WL(-X) + 0.5 LL
1 1.2 4 1.6 2 0.5
LOAD COMB 106 1.2 DL+ 1.6 WL(+Z) + 0.5 LL
1 1.2 5 1.6 2 0.5
LOAD COMB 107 1.2 DL+ 1.6 WL(-Z) + 0.5 LL
1 1.2 6 1.6 2 0.5
LOAD COMB 108 0.9 DL + 1.6 WL(+X)
1 0.9 3 1.6
LOAD COMB 109 0.9 DL + 1.6 WL(-X)
1 0.9 4 1.6
LOAD COMB 110 0.9 DL + 1.6 WL(+Z)
1 0.9 5 1.6
LOAD COMB 111 0.9 DL + 1.6 WL(-Z)
1 0.9 6 1.6
LOAD COMB 112 1.2 DL + 0.8 WL(+X)
1 1.2 3 0.8
LOAD COMB 113 1.2 DL + 0.8 WL(-X)
1 1.2 4 0.8
LOAD COMB 114 1.2 DL + 0.8 WL(+Z)
1 1.2 5 0.8
LOAD COMB 115 1.2 DL + 0.8 WL(-Z)
1 1.2 5 0.8
****
****Serviceanbility check
LOAD COMB 201 1. DL
1 1.0
LOAD COMB 202 1 (DL+TL1) + 1 LL
1 1.0 2 1.0 7 1.0
LOAD COMB 203 1 (DL+TL2) + 1 LL
1 1.0 2 1.0 8 1.0
LOAD COMB 204 1 DL+ 1 WL(+X)
1 1.0 3 1.0
LOAD COMB 205 1 DL+ 1 WL(-X)
1 1.0 4 1.0
LOAD COMB 206 1 DL+ 1 WL(+Z)
1 1.0 5 1.0
LOAD COMB 207 1 DL+ 1 WL(-Z)
1 1.0 6 1.0
LOAD COMB 208 0.6 DL + 1 WL(+X)
1 0.6 3 1.0
LOAD COMB 209 0.6 DL + 1 WL(-X)
1 0.6 4 1.0

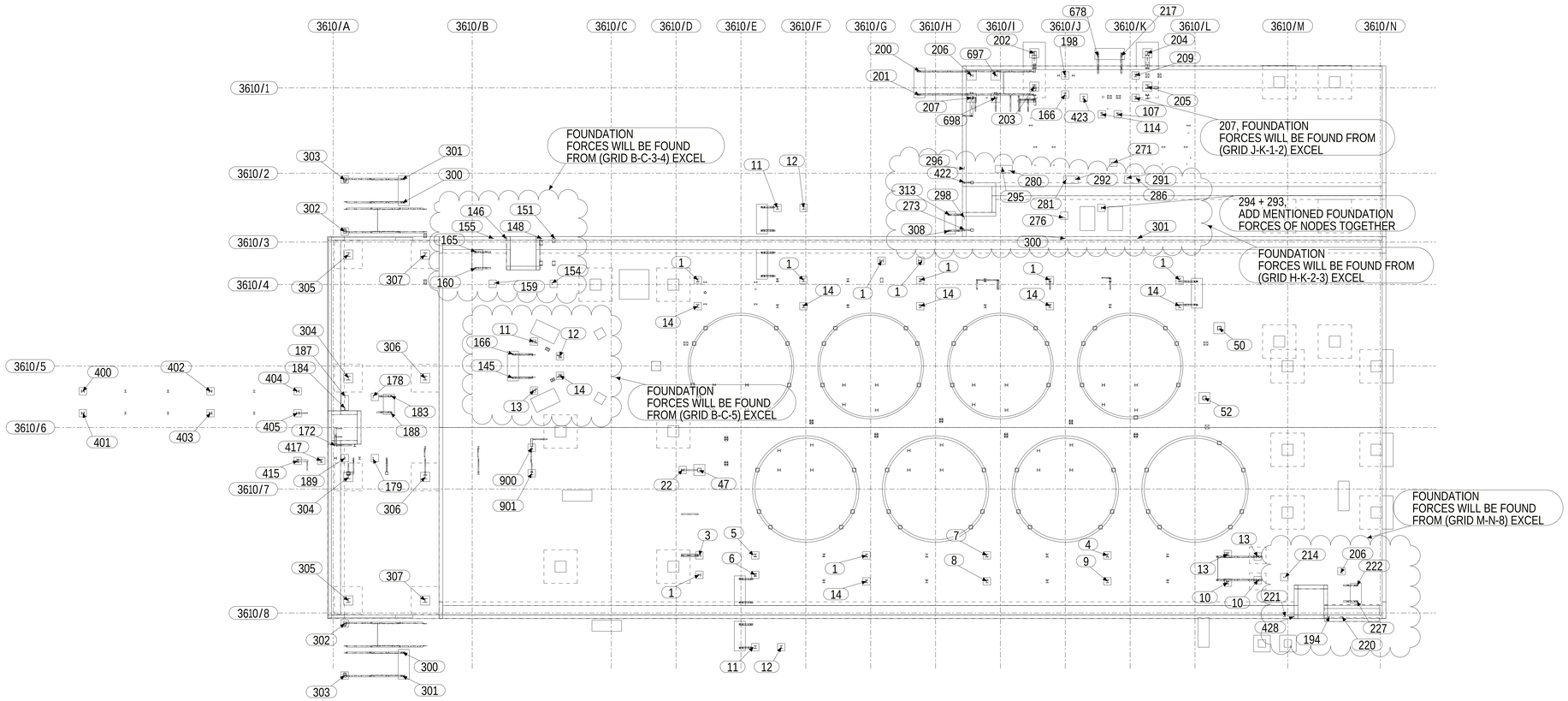
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LOAD COMB 210 0.6 DL + 1 WL(+Z)
1 0.6 5 1.0
LOAD COMB 211 0.6 DL + 1 WL(-Z)
1 0.6 6 1.0
LOAD COMB 212 1 DL+ 0.75 WL(+X)+ 0.75 LL
1 1.0 3 0.75 2 0.75
LOAD COMB 213 1 DL+ 0.75 WL(-X)+ 0.75 LL
1 1.0 4 0.75 2 0.75
LOAD COMB 214 1 DL+ 0.75 WL(+Z)+ 0.75 LL
1 1.0 5 0.75 2 0.75
LOAD COMB 215 1 DL+ 0.75 WL(-Z)+ 0.75 LL
1 1.0 6 0.75 2 0.75
****
PERFORM ANALYSIS
START CONCRETE DESIGN
CODE ACI
UNIT METER KN
CLB 0.04 ALL
CLS 0.04 ALL
CLT 0.04 ALL
FC 28000 ALL
FYMAIN 414000 ALL
MINMAIN 16 ALL
MINSEC 10 ALL
DESIGN COLUMN 551 TO 554
END CONCRETE DESIGN
*LOAD LIST 1 TO 10
*PERFORM ANALYSIS
*LOAD LIST 101 TO 123
***
*PARAMETER 1
*CODE LRFD
*FU 41000 MEMB 1 TO 19 24 TO 37 39 41 TO 63 67 TO 107 110 TO 123 125 TO 292
*FYLD 27500 MEMB 1 TO 19 24 TO 37 39 41 TO 63 67 TO 107 110 TO 123 125 TO 292
*KX 1 MEMB 1 TO 19 24 TO 37 39 41 TO 63 67 TO 107 110 TO 123 125 TO 261 268 -
*269 TO 292
*KY 1 MEMB 1 TO 19 24 TO 37 39 41 TO 63 67 TO 107 110 TO 123 125 TO 261 268 -
*269 TO 292
*KZ 0.85 MEMB 1 TO 19 24 TO 37 39 41 TO 63 67 TO 107 110 TO 123 125 TO 292
*LX 1.5 MEMB 81 82 175 TO 184 262 TO 267
*LZ 2 MEMB 262 TO 267
*LY 1.5 MEMB 81 82 175 TO 184 262 TO 267
*LY 5 MEMB 1 TO 4 12 TO 16 18 28 29 31 33 36 37 39 41 47 49 51 53 TO 55 58 -
*123 129 TO 131 201 202 209 TO 211 213 235 236 242 243 256 257 280 TO 282
*LX 3 MEMB 43 45 125 212 214
*LY 3 MEMB 43 45 125 212 214 253
```

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*LZ 3 MEMB 81 82 175 TO 184
*LZ 4 MEMB 8 10 19 27 35 42 48 52 62 113 118 153 TO 160 169 TO 172 194 TO 197 -
*205 TO 208 229 230
*LZ 5 MEMB 1 TO 6 9 11 TO 17 24 25 28 TO 33 36 37 39 41 44 TO 47 49 TO 51 53 -
*54 TO 58 63 67 TO 80 111 116 123 129 TO 131 135 TO 140 144 TO 149 165 167 -
*185 186 201 202 209 TO 213 258 TO 261 268 TO 279 286
*LZ 5 MEMB 43 45 125 212 214
*LZ 4.5 MEMB 17 25 46 50 111 135 TO 140 270 TO 279 286
*MAIN 200 MEMB 1 TO 19 24 TO 37 39 41 TO 63 67 TO 107 110 TO 123 125 TO 292
*RATIO 1 MEMB 1 TO 19 24 TO 37 39 41 TO 63 67 TO 107 110 TO 123 125 TO 292
*TMAIN 300 MEMB 1 TO 19 24 TO 37 39 41 TO 63 67 TO 107 110 TO 123 125 TO 292
*TRACK 2 MEMB 1 TO 19 24 TO 37 39 41 TO 63 67 TO 107 110 TO 123 125 TO 292
*CHECK CODE MEMB 1 TO 19 24 TO 37 39 41 TO 63 67 TO 107 110 TO 123 125 TO 292
*PARAMETER 2
*CODE LRFD
*DJ2 40 MEMB 69 77 78
*DJ1 41 MEMB 69 77 78
*DJ2 10 MEMB 11 67 70 185
*DJ1 9 MEMB 11 67 70 185
*DJ2 7 MEMB 5 68 71 186
*DJ1 5 MEMB 5 68 71 186
*DJ2 50 MEMB 81 175 TO 179
*DJ1 48 MEMB 81 175 TO 179
*DJ2 51 MEMB 82 180 TO 184
*DJ1 49 MEMB 82 180 TO 184
*DJ2 42 MEMB 72 79 80
*DJ1 43 MEMB 72 79 80
*DFF 500 MEMB 5 6 11 67 TO 74 77 TO 82 175 TO 186 258 259
*DFF 360 MEMB 1 TO 4 12 TO 15 28 29 31 33 36 37 39 41 43 45 47 49 51 -
*53 TO 55 123 125 129 TO 131 201 202 209 TO 214
**FREQUENCY CHECK FOR TOP FLOOR DONE BY DEFLECTION VALUES
*DJ1 4 MEMB 3 15 31 39 130 201 210
*DJ2 8 MEMB 3 15 31 39 130 201 210
*DJ2 6 MEMB 4 13 33 41 202 213
*DJ2 10 MEMB 43 49 53 55 125 131 214
*DJ2 9 MEMB 45 47 51 54 212
*DJ2 5 MEMB 2 12 28 36 211
*DJ1 2 MEMB 4 13 33 41 202 213
*DJ1 29 MEMB 45 47 51 54 212
*DJ1 27 MEMB 43 49 53 55 125 131 214
*DJ1 3 MEMB 1 14 29 37 123 129 209
*DJ1 1 MEMB 2 12 28 36 211
*CHECK CODE MEMB 1 TO 19 24 TO 37 39 41 TO 63 67 TO 107 110 TO 123 125 TO 292
*STEEL TAKE OFF LIST 1 TO 19 24 TO 37 39 41 TO 63 67 TO 107 110 TO 123 125 -
*126 TO 292
FINISH
```

ATTACHMENT 7 FOUNDATION DESIGN LOAD DATA

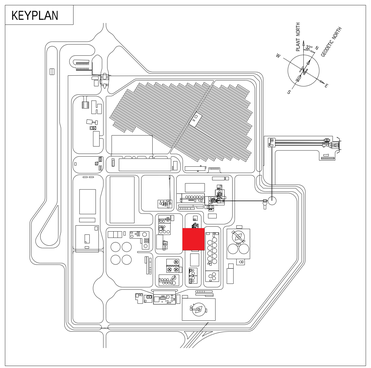
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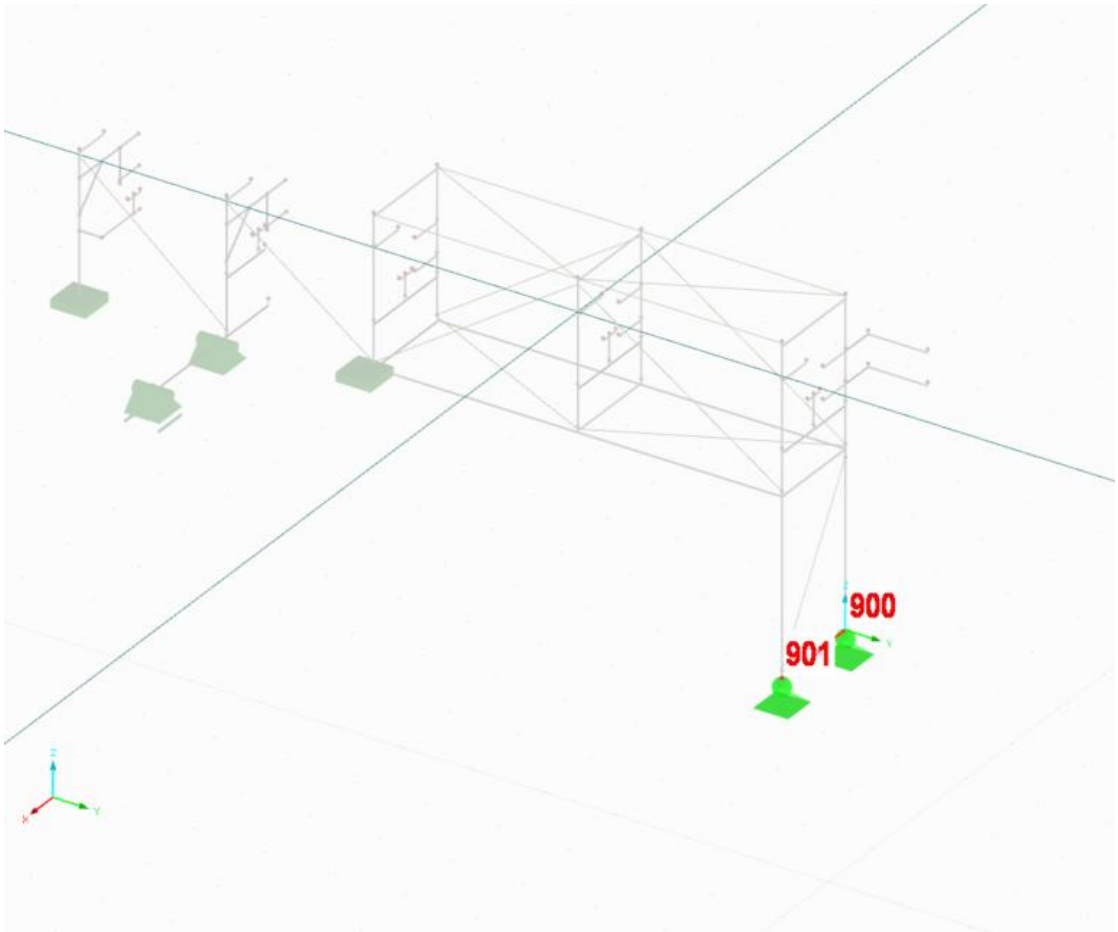
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KEYPLAN



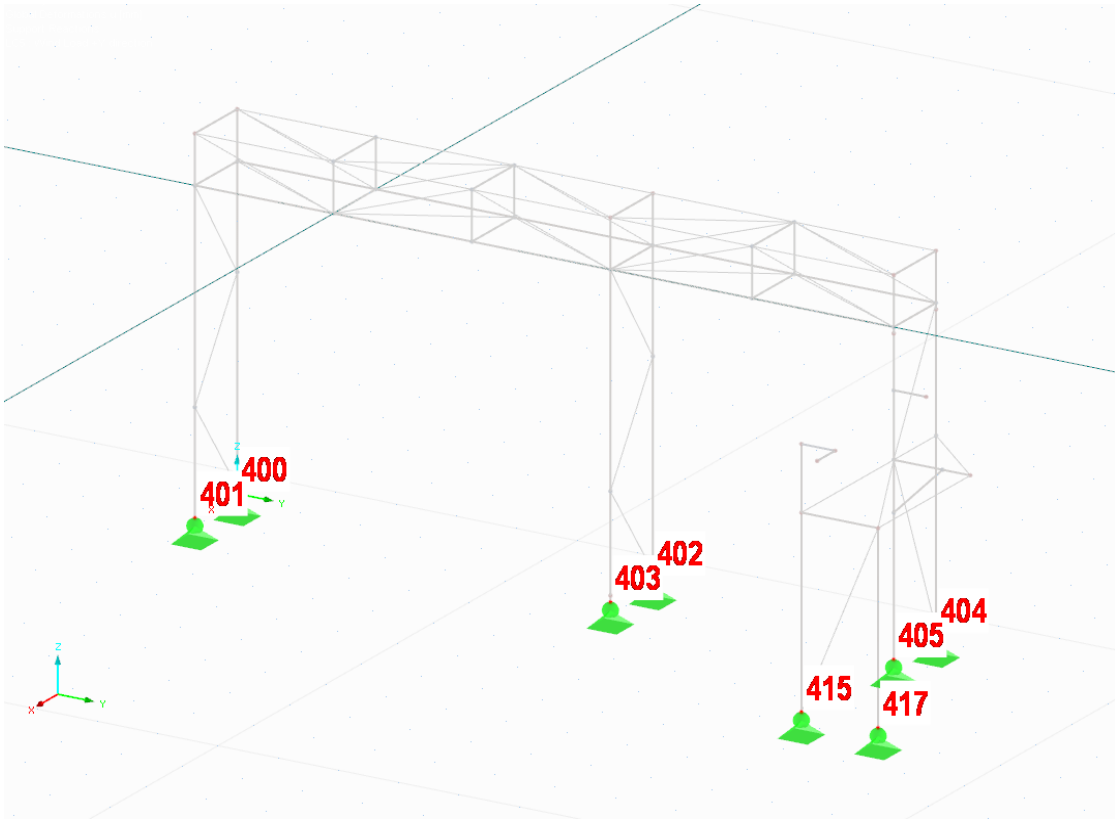
| | | |

Node No.	LC/CO	Support Forces [kN]		
		$P_{X'}$	$P_{Y'}$	$P_{Z'}$
900	LC1 - Self Weight + Secondary Structures	0,26	0,08	-7,89
	LC2 - Live Load + Technical load	0,69	0,2	-15,8
	LC3 - Wind load +X direction	-1,46	-0,05	23,91
	LC4 - Wind load -X direction	1,46	0,05	-23,9
	LC5 - Wind load +Y direction	-0,03	0,36	1,07
	LC6 - Wind load -Y direction	0,03	-0,33	-1,08
	LC7 - Temperature +25	-0,2	0,41	-0,14
	LC8 - Temperature -25	0,2	-0,41	0,14
901	LC1 - Self Weight + Secondary Structures	-0,42	0	-4,38
	LC2 - Live Load + Technical load	-1,02	0,02	-1,47
	LC3 - Wind load +X direction	11,22	-0,08	-23,82
	LC4 - Wind load -X direction	-11,22	0,08	23,82
	LC5 - Wind load +Y direction	0,83	0,51	-1,86
	LC6 - Wind load -Y direction	-0,84	-0,48	1,95
	LC7 - Temperature +25	0,12	0,41	-0,2
	LC8 - Temperature -25	-0,12	-0,41	0,2

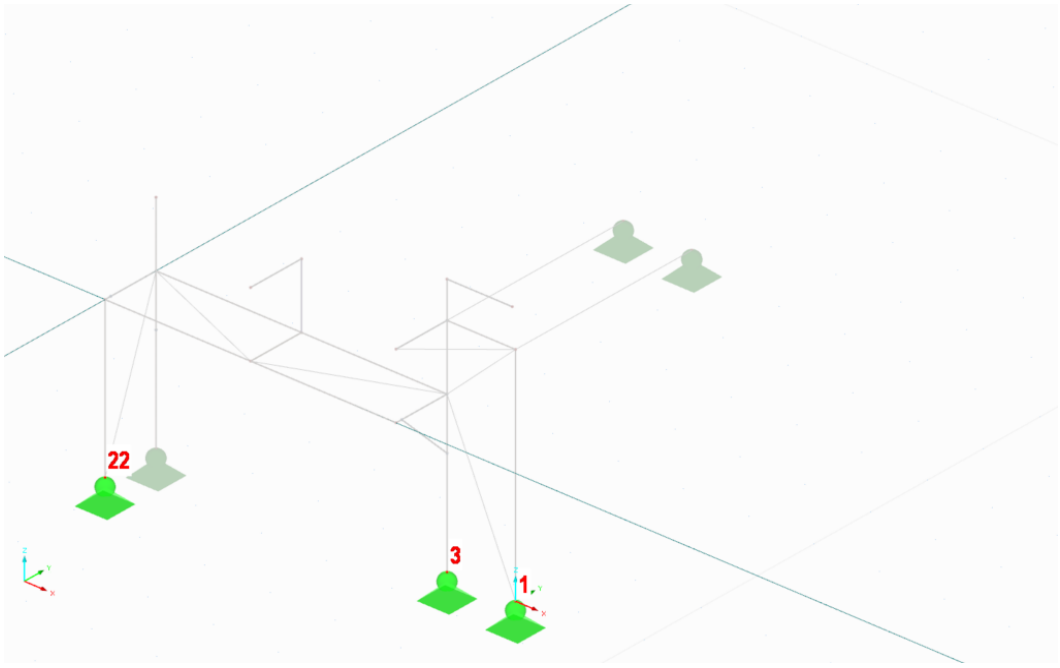

900
901

Node		Support Forces [kN]		
No.	LC/CO	$P_{X'}$	$P_{Y'}$	$P_{Z'}$
400	LC1 - Self Weight + Secondary Structures	-0,02	-0,03	-6,32
	LC2 - Live Load + Technical load	-0,03	-0,02	-5,63
	LC3 - Wind load +X direction	8,27	0,05	30,45
	LC4 - Wind load -X direction	-8,27	-0,05	-30,44
	LC5 - Wind load +Y direction	-0,56	2,27	-0,69
	LC6 - Wind load -Y direction	0,56	-2,27	0,69
	LC7 - Temperature +25	-0,2	-0,2	-0,35
	LC8 - Temperature -25	0,2	0,2	0,35
401	LC1 - Self Weight + Secondary Structures	0,02	-0,03	-6,29
	LC2 - Live Load + Technical load	0,02	-0,02	-5,6
	LC3 - Wind load +X direction	0,21	0,05	-30,36
	LC4 - Wind load -X direction	-0,21	-0,05	30,36
	LC5 - Wind load +Y direction	0	2,24	3,75
	LC6 - Wind load -Y direction	0	-2,25	-3,72
	LC7 - Temperature +25	0,15	-0,2	0,06
	LC8 - Temperature -25	-0,15	0,2	-0,06
402	LC1 - Self Weight + Secondary Structures	0	0,05	-8,87
	LC2 - Live Load + Technical load	0,06	0,11	-10,44
	LC3 - Wind load +X direction	10,89	0,05	41,27
	LC4 - Wind load -X direction	-10,89	-0,05	-41,27
	LC5 - Wind load +Y direction	-0,08	2,3	0,06
	LC6 - Wind load -Y direction	0,08	-2,3	-0,11
	LC7 - Temperature +25	-0,15	-0,02	0,29
	LC8 - Temperature -25	0,15	0,02	-0,29
403	LC1 - Self Weight + Secondary Structures	0	0,05	-8,84
	LC2 - Live Load + Technical load	0	0,11	-10,8
	LC3 - Wind load +X direction	0,21	0,05	-41,25
	LC4 - Wind load -X direction	-0,21	-0,05	41,25
	LC5 - Wind load +Y direction	0,05	2,25	0,77
	LC6 - Wind load -Y direction	-0,05	-2,25	-0,8
	LC7 - Temperature +25	0,15	-0,02	0,27
	LC8 - Temperature -25	-0,15	0,02	-0,27
404	LC1 - Self Weight + Secondary Structures	0	0,08	-5,27
	LC2 - Live Load + Technical load	0,06	0,15	-3,54
	LC3 - Wind load +X direction	8,84	0,15	30,23
	LC4 - Wind load -X direction	-8,84	-0,15	-30,23
	LC5 - Wind load +Y direction	0,59	3,87	0,63
	LC6 - Wind load -Y direction	-0,59	-3,87	-0,59
	LC7 - Temperature +25	0	0,26	0,06
	LC8 - Temperature -25	0	-0,26	-0,06
405	LC1 - Self Weight + Secondary Structures	0	-0,17	-6,84
	LC2 - Live Load + Technical load	0	-0,44	-8,51
	LC3 - Wind load +X direction	0,56	0,02	-30,33

	LC4 - Wind load -X direction	-0,56	-0,02	30,33
	LC5 - Wind load +Y direction	-0,02	0,05	-4,52
	LC6 - Wind load -Y direction	0,02	-0,06	4,52
	LC7 - Temperature +25	0,05	-0,06	-0,33
	LC8 - Temperature -25	-0,05	0,06	0,33
415	LC1 - Self Weight + Secondary Structures	0	0,05	-2,42
	LC2 - Live Load + Technical load	-0,12	0,09	-3,39
	LC3 - Wind load +X direction	0,69	-0,35	-0,84
	LC4 - Wind load -X direction	-0,69	0,35	0,84
	LC5 - Wind load +Y direction	0	5,28	11,57
	LC6 - Wind load -Y direction	0	-5,63	-12,54
	LC7 - Temperature +25	0	0,24	0,57
	LC8 - Temperature -25	0	-0,24	-0,57
417	LC1 - Self Weight + Secondary Structures	0	0	-2,21
	LC2 - Live Load + Technical load	0	0	-2,76
	LC3 - Wind load +X direction	0,53	0	0,84
	LC4 - Wind load -X direction	-0,53	0	-0,84
	LC5 - Wind load +Y direction	0	0,51	-11,57
	LC6 - Wind load -Y direction	0	-0,51	12,54
	LC7 - Temperature +25	0	0	-0,57
	LC8 - Temperature -25	0	0	0,57

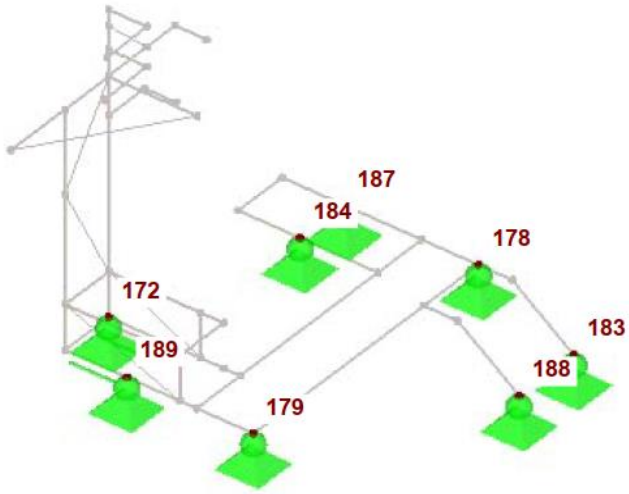

400
401
402
403
404
405
415
417
Z
Y
X

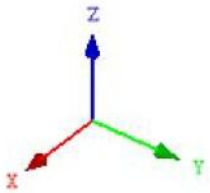
Node No.	LC/CO	Support Forces [kN]		
		$P_{X'}$	$P_{Y'}$	$P_{Z'}$
1	LC1 - Self Weight + Secondary Structures	0,44	0	-3,28
	LC2 - Live Load + Technical load	1	0	-4,28
	LC3 - Wind load +X direction	5,9	0	-13,5
	LC4 - Wind load -X direction	-6,64	0	15,54
	LC5 - Wind load +Y direction	-0,08	1,06	0,46
	LC6 - Wind load -Y direction	0,08	-1,06	-0,46
	LC7 - Temperature +25	1,32	0	-3,18
	LC8 - Temperature -25	-1,32	0	3,18
3	LC1 - Self Weight + Secondary Structures	0,18	0,56	-7,54
	LC2 - Live Load + Technical load	0,38	1,46	-14,64
	LC3 - Wind load +X direction	0,6	0,04	13,38
	LC4 - Wind load -X direction	-0,66	-0,06	-15,38
	LC5 - Wind load +Y direction	0,06	1,22	-0,56
	LC6 - Wind load -Y direction	-0,06	-1,22	0,56
	LC7 - Temperature +25	-0,16	-0,06	3,92
	LC8 - Temperature -25	0,16	0,06	-3,92
16	LC1 - Self Weight + Secondary Structures	-0,26	0	-3,86
	LC2 - Live Load + Technical load	-0,58	0	-3,56
	LC3 - Wind load +X direction	0,6	0	0,52
	LC4 - Wind load -X direction	-0,62	0	-0,64
	LC5 - Wind load +Y direction	-0,08	0,32	-10,56
	LC6 - Wind load -Y direction	0,08	-0,32	10,56
	LC7 - Temperature +25	-0,56	0	-0,08
	LC8 - Temperature -25	0,56	0	0,08
22	LC1 - Self Weight + Secondary Structures	-0,36	-0,08	-3,9
	LC2 - Live Load + Technical load	-0,82	-0,16	-5,2
	LC3 - Wind load +X direction	0,84	-0,14	-0,38
	LC4 - Wind load -X direction	-0,86	0,18	0,48
	LC5 - Wind load +Y direction	0,12	14,38	10,68
	LC6 - Wind load -Y direction	-0,12	-14,38	-10,68
	LC7 - Temperature +25	-0,62	-0,12	-0,68
	LC8 - Temperature -25	0,62	0,12	0,68


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Node No.	LC/CO	Support Forces [kN] $P_{X'}$	$P_{Y'}$	$P_{Z'}$
172	LC1 - Self Weight + Secondary Structures	0	0	-5,54
	LC2 - Live Load + Technical load	-0,02	0	-6,3
	LC3 - Wind load +X direction	4,04	0	12,86
	LC4 - Wind load -X direction	-4,38	0	-12,86
	LC5 - Wind load +Y direction	-2,4	0	5,02
	LC6 - Wind load -Y direction	1,72	0	-5,02
	LC7 - Temperature +25	-1,94	0	-0,02
	LC8 - Temperature -25	1,94	0	0,02
178	LC1 - Self Weight + Secondary Structures	0,04	0	-8,6
	LC2 - Live Load + Technical load	0,06	0	-17,3
	LC3 - Wind load +X direction	2,28	0	-0,06
	LC4 - Wind load -X direction	-0,18	0,28	-0,14
	LC5 - Wind load +Y direction	-0,46	0	-0,06
	LC6 - Wind load -Y direction	0,02	-1,28	0,06
	LC7 - Temperature +25	-5,62	0	1,84
	LC8 - Temperature -25	5,62	0	-1,84
179	LC1 - Self Weight + Secondary Structures	0	0	-5,06
	LC2 - Live Load + Technical load	0,02	0	-4,74
	LC3 - Wind load +X direction	0,32	0	6,32
	LC4 - Wind load -X direction	-1,36	-0,28	-6,14
	LC5 - Wind load +Y direction	0,76	0	-7,28
	LC6 - Wind load -Y direction	0	-2,28	7,28
	LC7 - Temperature +25	0,66	0	0
	LC8 - Temperature -25	-0,66	0	0
183	LC1 - Self Weight + Secondary Structures	-0,04	0	-0,4
	LC2 - Live Load + Technical load	-0,08	0	-1,68
	LC3 - Wind load +X direction	0,64	0	0
	LC4 - Wind load -X direction	0	0	-0,02
	LC5 - Wind load +Y direction	0	1,5	-0,28
	LC6 - Wind load -Y direction	0	-1,5	0,28
	LC7 - Temperature +25	0	0	-1,08
	LC8 - Temperature -25	0	0	1,08
184	LC1 - Self Weight + Secondary Structures	0	0	-3,08
	LC2 - Live Load + Technical load	-0,02	0	-4,62
	LC3 - Wind load +X direction	0,36	0	0
	LC4 - Wind load -X direction	0	0	0
	LC5 - Wind load +Y direction	1,44	0,46	-0,02
	LC6 - Wind load -Y direction	-0,02	0	0,02
	LC7 - Temperature +25	6,44	0	0
	LC8 - Temperature -25	-6,44	0	0
187	LC1 - Self Weight + Secondary Structures	0	0	-4,14
	LC2 - Live Load + Technical load	0	0	-6,66
	LC3 - Wind load +X direction	1,9	0,02	-0,02

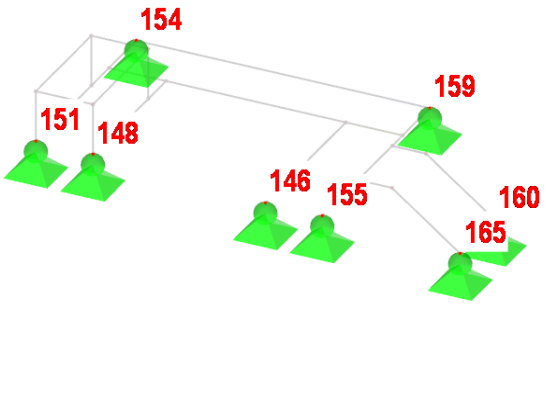
	LC4 - Wind load -X direction	-0,12	0,02	0,02
	LC5 - Wind load +Y direction	-0,3	1,74	-0,34
	LC6 - Wind load -Y direction	0	0	0,34
	LC7 - Temperature +25	-5,62	1,28	-0,76
	LC8 - Temperature -25	5,62	-1,28	0,76
188	LC1 - Self Weight + Secondary Structures	0	0	-1,14
	LC2 - Live Load + Technical load	0	0	-3,2
	LC3 - Wind load +X direction	0	0	0
	LC4 - Wind load -X direction	-1,8	0	0
	LC5 - Wind load +Y direction	0	1,5	-0,38
	LC6 - Wind load -Y direction	0	-1,5	0,38
	LC7 - Temperature +25	0	0	0
	LC8 - Temperature -25	0	0	0
189	LC1 - Self Weight + Secondary Structures	0	0	-8,38
	LC2 - Live Load + Technical load	0,02	0	-30,24
	LC3 - Wind load +X direction	0,92	-0,02	-19,1
	LC4 - Wind load -X direction	-2,52	-0,02	19,12
	LC5 - Wind load +Y direction	0,96	8,7	3,32
	LC6 - Wind load -Y direction	-1,74	-7,1	-3,32
	LC7 - Temperature +25	6,08	-1,28	0,02
	LC8 - Temperature -25	-6,08	1,28	-0,02


187
184
178
172
183
189
188
179


Z
X
Y

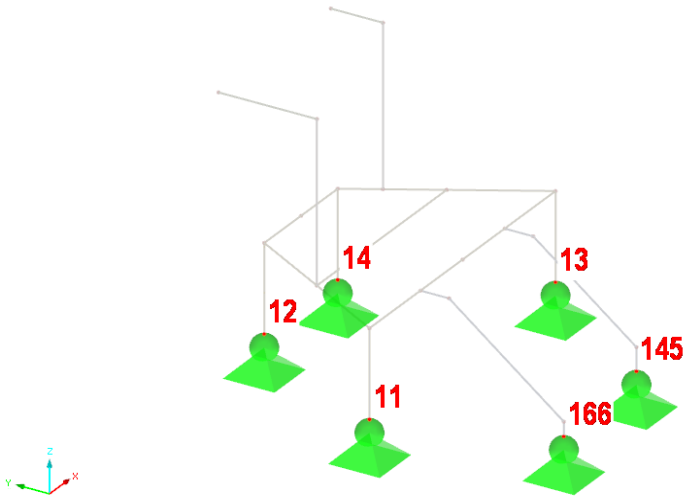
Node No.	LC/CO	Support Forces [kN] $P_{X'}$	$P_{Y'}$	$P_{Z'}$
146	LC1 - Self Weight + Secondary Structures	0	0	-0,89
	LC2 - Live Load + Technical load	0	-0,02	-2,27
	LC3 - Wind load +X direction	0	0,45	0
	LC4 - Wind load -X direction	0	0	0
	LC5 - Wind load +Y direction	0	0,08	0
	LC6 - Wind load -Y direction	0	0,05	0
	LC7 - Temperature +25	0	0,42	0
	LC8 - Temperature -25	0	-0,42	0
148	LC1 - Self Weight + Secondary Structures	-1,02	-0,21	-2,12
	LC2 - Live Load + Technical load	-2,4	-0,48	-3,8
	LC3 - Wind load +X direction	0,23	-0,29	0,05
	LC4 - Wind load -X direction	-0,32	0	-0,11
	LC5 - Wind load +Y direction	0,23	0,48	0,95
	LC6 - Wind load -Y direction	-0,29	-0,54	-0,99
	LC7 - Temperature +25	0,98	-0,53	0,68
	LC8 - Temperature -25	-0,98	0,53	-0,68
151	LC1 - Self Weight + Secondary Structures	1,04	-0,18	-1,79
	LC2 - Live Load + Technical load	2,45	-0,47	-2,7
	LC3 - Wind load +X direction	1,68	-0,2	0,14
	LC4 - Wind load -X direction	-0,36	0	-0,08
	LC5 - Wind load +Y direction	-0,24	0,48	-0,96
	LC6 - Wind load -Y direction	0,21	-1,32	1,01
	LC7 - Temperature +25	-1,35	1,65	-0,63
	LC8 - Temperature -25	1,35	-1,65	0,63
154	LC1 - Self Weight + Secondary Structures	0	0,2	-4,68
	LC2 - Live Load + Technical load	0	0,44	-10,14
	LC3 - Wind load +X direction	0	0,21	-0,17
	LC4 - Wind load -X direction	-1,76	0	0,15
	LC5 - Wind load +Y direction	0	0,29	-0,05
	LC6 - Wind load -Y direction	0	-1,64	0,05
	LC7 - Temperature +25	0	2,39	0,09
	LC8 - Temperature -25	0	-2,39	-0,09
155	LC1 - Self Weight + Secondary Structures	-0,02	0,03	-2,37
	LC2 - Live Load + Technical load	-0,03	0,08	-4,31
	LC3 - Wind load +X direction	1,86	-0,17	0
	LC4 - Wind load -X direction	-0,92	0,14	0
	LC5 - Wind load +Y direction	0,02	1,35	-0,3
	LC6 - Wind load -Y direction	0,08	-0,08	0,32
	LC7 - Temperature +25	0,32	-0,6	-0,02
	LC8 - Temperature -25	-0,32	0,6	0,02
159	LC1 - Self Weight + Secondary Structures	0	0,18	-4,83
	LC2 - Live Load + Technical load	0	0,44	-10,02
	LC3 - Wind load +X direction	0	0	0

	LC4 - Wind load -X direction	-1,76	-0,17	0,02
	LC5 - Wind load +Y direction	0	1,4	-0,27
	LC6 - Wind load -Y direction	0	-0,38	0,27
	LC7 - Temperature +25	0	-3,35	-0,11
	LC8 - Temperature -25	0	3,35	0,11
160	LC1 - Self Weight + Secondary Structures	0	0	-0,69
	LC2 - Live Load + Technical load	0,02	0	0
	LC3 - Wind load +X direction	0	0	0
	LC4 - Wind load -X direction	-0,53	0	0
	LC5 - Wind load +Y direction	0	1,23	0,32
	LC6 - Wind load -Y direction	0	-1,23	-0,32
	LC7 - Temperature +25	0,05	0	0
	LC8 - Temperature -25	-0,05	0	0
165	LC1 - Self Weight + Secondary Structures	-0,02	0	-0,69
	LC2 - Live Load + Technical load	-0,02	0	0
	LC3 - Wind load +X direction	0,53	0	0
	LC4 - Wind load -X direction	0	0	0
	LC5 - Wind load +Y direction	0	1,23	0,32
	LC6 - Wind load -Y direction	0	-1,23	-0,32
	LC7 - Temperature +25	0,02	0	0
	LC8 - Temperature -25	-0,02	0	0


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Node No.	LC/CO	Support Forces [kN]		
		$P_{X'}$	$P_{Y'}$	$P_{Z'}$
11	LC1 - Self Weight + Secondary Structures	-0,77	0	-4,85
	LC2 - Live Load + Technical load	-1,43	0	-8,16
	LC3 - Wind load +X direction	1,25	0	1,07
	LC4 - Wind load -X direction	-1,29	0	-1,07
	LC5 - Wind load +Y direction	-1,62	0,14	-2,03
	LC6 - Wind load -Y direction	1,38	-0,14	1,38
	LC7 - Temperature +25	-2,19	0	0,03
	LC8 - Temperature -25	2,19	0	-0,03
12	LC1 - Self Weight + Secondary Structures	-0,03	0	-3,09
	LC2 - Live Load + Technical load	0,11	0	-8,3
	LC3 - Wind load +X direction	1,59	0	2,88
	LC4 - Wind load -X direction	-1,59	0	-2,88
	LC5 - Wind load +Y direction	0,12	0,14	-0,77
	LC6 - Wind load -Y direction	-0,15	-0,14	0,9
	LC7 - Temperature +25	-1,05	0	0,09
	LC8 - Temperature -25	1,05	0	-0,09
13	LC1 - Self Weight + Secondary Structures	0,77	0	-4,68
	LC2 - Live Load + Technical load	1,44	0	-7,52
	LC3 - Wind load +X direction	1,44	0	-1,52
	LC4 - Wind load -X direction	-1,37	0	1,5
	LC5 - Wind load +Y direction	1,58	0,14	-1,94
	LC6 - Wind load -Y direction	-1,29	-0,14	1,26
	LC7 - Temperature +25	2,06	0	0
	LC8 - Temperature -25	-2,06	0	0
14	LC1 - Self Weight + Secondary Structures	0,02	0	-3,12
	LC2 - Live Load + Technical load	-0,12	-0,32	-8,16
	LC3 - Wind load +X direction	1,34	-0,35	-2,69
	LC4 - Wind load -X direction	-1,34	0,35	2,69
	LC5 - Wind load +Y direction	-0,08	-0,14	-1,11
	LC6 - Wind load -Y direction	0,08	0,18	1,37
	LC7 - Temperature +25	1,19	0	-0,11
	LC8 - Temperature -25	-1,19	0	0,11
145	LC1 - Self Weight + Secondary Structures	0	0	-1,01
	LC2 - Live Load + Technical load	0	0,27	-2,36
	LC3 - Wind load +X direction	0	0,06	0,05
	LC4 - Wind load -X direction	-0,83	-0,05	-0,05
	LC5 - Wind load +Y direction	0	4,34	2,99
	LC6 - Wind load -Y direction	0	-3,74	-2,51
	LC7 - Temperature +25	0	0,15	0,12
	LC8 - Temperature -25	0	-0,15	-0,12
166	LC1 - Self Weight + Secondary Structures	0	0	-1,04
	LC2 - Live Load + Technical load	0	0,05	-2,78
	LC3 - Wind load +X direction	0,86	0,3	0,21

	LC4 - Wind load -X direction	0	-0,3	-0,21
	LC5 - Wind load +Y direction	0	4,62	2,85
	LC6 - Wind load -Y direction	0	-3,98	-2,39
	LC7 - Temperature +25	0	-0,17	-0,12
	LC8 - Temperature -25	0	0,17	0,12

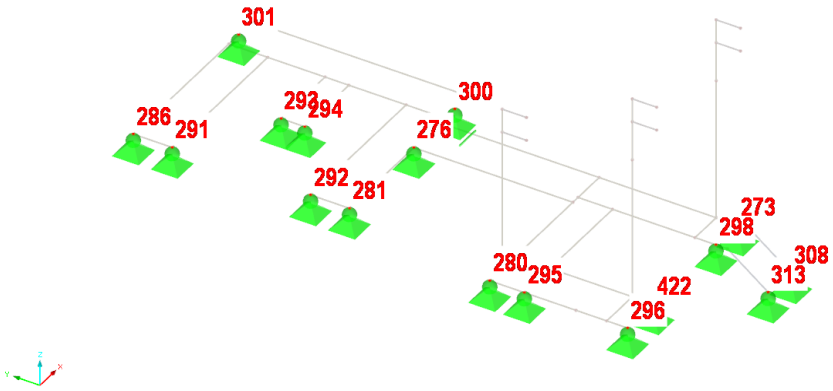

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Node No.	LC/CO	Support Forces [kN] $P_{X'}$	$P_{Y'}$	$P_{Z'}$
273	LC1 - Self Weight + Secondary Structures	0	0	-20,25
	LC2 - Live Load + Technical load	0,02	0	-41,67
	LC3 - Wind load +X direction	1,58	-0,24	-7,8
	LC4 - Wind load -X direction	-5,63	4,82	7,8
	LC5 - Wind load +Y direction	0	2,76	-4,77
	LC6 - Wind load -Y direction	-0,02	-2,19	4,77
	LC7 - Temperature +25	0,11	-1,1	0,27
	LC8 - Temperature -25	-0,11	1,1	-0,27
276	LC1 - Self Weight + Secondary Structures	0	0	-8,54
	LC2 - Live Load + Technical load	0	0	-15,12
	LC3 - Wind load +X direction	2,84	0,03	-0,11
	LC4 - Wind load -X direction	-5,1	2,22	0,11
	LC5 - Wind load +Y direction	0	0	-0,21
	LC6 - Wind load -Y direction	0,08	-0,02	0,21
	LC7 - Temperature +25	1,08	-0,81	-0,06
	LC8 - Temperature -25	-1,08	0,81	0,06
280	LC1 - Self Weight + Secondary Structures	0	0	-3,96
	LC2 - Live Load + Technical load	0	0	-5,45
	LC3 - Wind load +X direction	2,73	0	-0,03
	LC4 - Wind load -X direction	-1,13	0,03	0,03
	LC5 - Wind load +Y direction	-0,87	2	-6,24
	LC6 - Wind load -Y direction	0,38	-1,98	6,24
	LC7 - Temperature +25	-4,62	0,86	-0,03
	LC8 - Temperature -25	4,62	-0,86	0,03
281	LC1 - Self Weight + Secondary Structures	0	0	-1,16
	LC2 - Live Load + Technical load	0	0	-1,76
	LC3 - Wind load +X direction	0,39	0	0
	LC4 - Wind load -X direction	0	0	0
	LC5 - Wind load +Y direction	0	0	0,02
	LC6 - Wind load -Y direction	0	0	-0,02
	LC7 - Temperature +25	0	0	0
	LC8 - Temperature -25	0	0	0
286	LC1 - Self Weight + Secondary Structures	0	0	-2,76
	LC2 - Live Load + Technical load	0	0	-4,55
	LC3 - Wind load +X direction	0,39	-0,12	0
	LC4 - Wind load -X direction	0	0	0
	LC5 - Wind load +Y direction	0	0	0
	LC6 - Wind load -Y direction	0	-1,7	0
	LC7 - Temperature +25	0	0	0
	LC8 - Temperature -25	0	0	0
291	LC1 - Self Weight + Secondary Structures	0	0	-2,43
	LC2 - Live Load + Technical load	0	0	-3,99
	LC3 - Wind load +X direction	0,39	-0,26	0

	LC4 - Wind load -X direction	0	0	0
	LC5 - Wind load +Y direction	0	0	0
	LC6 - Wind load -Y direction	0	-0,03	0
	LC7 - Temperature +25	0	0,66	0
	LC8 - Temperature -25	0	-0,66	0
292	LC1 - Self Weight + Secondary Structures	0	0	-2,42
	LC2 - Live Load + Technical load	0	0	-3,95
	LC3 - Wind load +X direction	0,39	0,27	0
	LC4 - Wind load -X direction	0	0	0
	LC5 - Wind load +Y direction	0	0	0
	LC6 - Wind load -Y direction	0	0,03	0
	LC7 - Temperature +25	0	0,3	0
	LC8 - Temperature -25	0	-0,3	0
293	LC1 - Self Weight + Secondary Structures	0	0	-0,87
	LC2 - Live Load + Technical load	0	0	-1,14
	LC3 - Wind load +X direction	0,24	-0,38	0
	LC4 - Wind load -X direction	0	0,02	0
	LC5 - Wind load +Y direction	0	0	0
	LC6 - Wind load -Y direction	0	0,03	0
	LC7 - Temperature +25	0	3,24	0
	LC8 - Temperature -25	0	-3,24	0
294	LC1 - Self Weight + Secondary Structures	0	0	-0,87
	LC2 - Live Load + Technical load	0	0	-1,14
	LC3 - Wind load +X direction	0,24	0,05	0
	LC4 - Wind load -X direction	0	0,02	0
	LC5 - Wind load +Y direction	0	0	0
	LC6 - Wind load -Y direction	0	0,06	0
	LC7 - Temperature +25	0	2,52	0
	LC8 - Temperature -25	0	-2,52	0
295	LC1 - Self Weight + Secondary Structures	0	0	-6,3
	LC2 - Live Load + Technical load	0	0	-12,9
	LC3 - Wind load +X direction	1,44	0	1,44
	LC4 - Wind load -X direction	-0,15	-0,03	-1,44
	LC5 - Wind load +Y direction	0,87	1,74	3,81
	LC6 - Wind load -Y direction	-0,36	-0,02	-3,81
	LC7 - Temperature +25	1,08	1,98	0
	LC8 - Temperature -25	-1,08	-1,98	0
296	LC1 - Self Weight + Secondary Structures	0	0	-1,43
	LC2 - Live Load + Technical load	0	0	-2,34
	LC3 - Wind load +X direction	1,11	0,38	3,42
	LC4 - Wind load -X direction	-0,02	-0,15	-3,42
	LC5 - Wind load +Y direction	0,02	0,12	0
	LC6 - Wind load -Y direction	0	-0,02	0
	LC7 - Temperature +25	0,11	1,22	0
	LC8 - Temperature -25	-0,11	-1,22	0

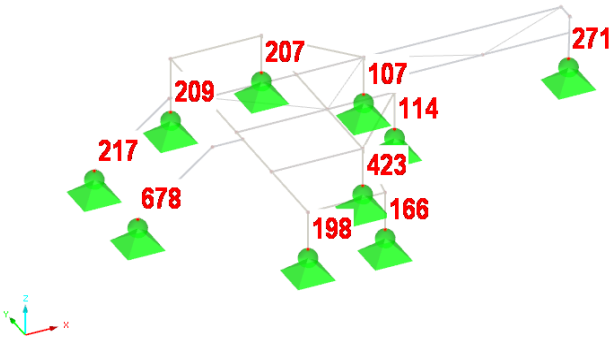
298	LC1 - Self Weight + Secondary Structures	0,02	0	-13,16
	LC2 - Live Load + Technical load	0,02	0	-22,4
	LC3 - Wind load +X direction	0,75	0,24	5,3
	LC4 - Wind load -X direction	-0,11	-4,83	-5,3
	LC5 - Wind load +Y direction	-0,21	1,4	-2,37
	LC6 - Wind load -Y direction	-0,02	-0,15	2,37
	LC7 - Temperature +25	1,71	-1,74	0,68
	LC8 - Temperature -25	-1,71	1,74	-0,68
300	LC1 - Self Weight + Secondary Structures	0	0	-12,08
	LC2 - Live Load + Technical load	0	0	-20,37
	LC3 - Wind load +X direction	0	0,29	-0,14
	LC4 - Wind load -X direction	0	-2,25	0,14
	LC5 - Wind load +Y direction	0	0	-0,29
	LC6 - Wind load -Y direction	0	-0,06	0,29
	LC7 - Temperature +25	0	-5,91	-0,08
	LC8 - Temperature -25	0	5,91	0,08
301	LC1 - Self Weight + Secondary Structures	0	0	-11,34
	LC2 - Live Load + Technical load	0	0	-19,16
	LC3 - Wind load +X direction	1,4	0,12	0
	LC4 - Wind load -X direction	-1,74	0	0
	LC5 - Wind load +Y direction	0	0	0
	LC6 - Wind load -Y direction	-0,06	-2,19	0
	LC7 - Temperature +25	-1,01	0	0
	LC8 - Temperature -25	1,01	0	0
308	LC1 - Self Weight + Secondary Structures	0	0	8,57
	LC2 - Live Load + Technical load	-0,02	0	12,44
	LC3 - Wind load +X direction	-0,02	0	2,04
	LC4 - Wind load -X direction	-0,5	0	-2,04
	LC5 - Wind load +Y direction	-0,02	1,22	4,92
	LC6 - Wind load -Y direction	0,02	-1,22	-4,92
	LC7 - Temperature +25	0	0	-0,3
	LC8 - Temperature -25	0	0	0,3
313	LC1 - Self Weight + Secondary Structures	-0,02	0	6,26
	LC2 - Live Load + Technical load	-0,02	0	9,23
	LC3 - Wind load +X direction	0,5	0	-0,45
	LC4 - Wind load -X direction	0	0	0,45
	LC5 - Wind load +Y direction	-0,02	1,22	1,82
	LC6 - Wind load -Y direction	0,02	-1,22	-1,82
	LC7 - Temperature +25	0	0	-0,5
	LC8 - Temperature -25	0	0	0,5
422	LC1 - Self Weight + Secondary Structures	0	0	-3,35
	LC2 - Live Load + Technical load	0	0	-14,48
	LC3 - Wind load +X direction	1,52	-0,38	-3,71
	LC4 - Wind load -X direction	-1,52	0,15	3,71
	LC5 - Wind load +Y direction	0,2	2,16	3,29

	LC6 - Wind load -Y direction	0	-2,27	-3,29
	LC7 - Temperature +25	1,56	-1,22	0
	LC8 - Temperature -25	-1,56	1,22	0


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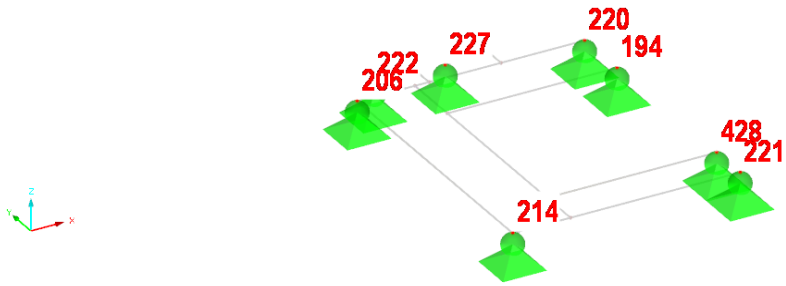
Node No.	LC/CO	Support Forces [kN]		
		$P_{X'}$	$P_{Y'}$	$P_{Z'}$
107	LC1 - Self Weight + Secondary Structures	0,02	0	-4,05
	LC2 - Live Load + Technical load	0,03	0	-5,88
	LC3 - Wind load +X direction	0,09	0	0
	LC4 - Wind load -X direction	-0,09	0	-0,02
	LC5 - Wind load +Y direction	0	0,09	-0,02
	LC6 - Wind load -Y direction	0	-0,09	0,02
	LC7 - Temperature +25	0	0	0,08
	LC8 - Temperature -25	0	0	-0,08
114	LC1 - Self Weight + Secondary Structures	3,9	0	-5,18
	LC2 - Live Load + Technical load	8,01	0	-8,9
	LC3 - Wind load +X direction	1,25	0	-0,71
	LC4 - Wind load -X direction	-1,4	0	0,78
	LC5 - Wind load +Y direction	0,99	0,11	-0,59
	LC6 - Wind load -Y direction	-0,92	-0,11	0,54
	LC7 - Temperature +25	4,37	0	-2,63
	LC8 - Temperature -25	-4,37	0	2,63
166	LC1 - Self Weight + Secondary Structures	0,77	0,02	-1,13
	LC2 - Live Load + Technical load	1,62	0,03	-0,87
	LC3 - Wind load +X direction	1,28	0	-0,63
	LC4 - Wind load -X direction	-1,4	0	0,69
	LC5 - Wind load +Y direction	0,53	0,09	-0,29
	LC6 - Wind load -Y direction	-0,57	-0,09	0,3
	LC7 - Temperature +25	0,18	0	-0,11
	LC8 - Temperature -25	-0,18	0	0,11
198	LC1 - Self Weight + Secondary Structures	0	-0,02	-2,01
	LC2 - Live Load + Technical load	0	-0,02	-2,46
	LC3 - Wind load +X direction	0,09	0	-0,02
	LC4 - Wind load -X direction	-0,09	0	0
	LC5 - Wind load +Y direction	0	0,09	0,18
	LC6 - Wind load -Y direction	0	-0,09	-0,21
	LC7 - Temperature +25	0	0	1,47
	LC8 - Temperature -25	0	0	-1,47
207	LC1 - Self Weight + Secondary Structures	0	0	-5,1
	LC2 - Live Load + Technical load	-0,02	0,02	-9,14
	LC3 - Wind load +X direction	0,09	0	0,35
	LC4 - Wind load -X direction	-0,09	0	-0,39
	LC5 - Wind load +Y direction	-0,02	0,11	-0,87
	LC6 - Wind load -Y direction	0,02	-0,11	0,92
	LC7 - Temperature +25	0,02	0,02	1,23
	LC8 - Temperature -25	-0,02	-0,02	-1,23
209	LC1 - Self Weight + Secondary Structures	0	2,07	-5,24
	LC2 - Live Load + Technical load	0	4,19	-9,54
	LC3 - Wind load +X direction	0,14	0,54	-1,59

	LC4 - Wind load -X direction	-0,14	-0,59	1,82
	LC5 - Wind load +Y direction	0	1,38	0,93
	LC6 - Wind load -Y direction	0	-1,41	-0,9
	LC7 - Temperature +25	0	1,61	1,4
	LC8 - Temperature -25	0	-1,61	-1,4
217	LC1 - Self Weight + Secondary Structures	-0,66	0	-1,47
	LC2 - Live Load + Technical load	-1,65	0	-4,04
	LC3 - Wind load +X direction	3,2	0	1,34
	LC4 - Wind load -X direction	-3,65	0	-1,59
	LC5 - Wind load +Y direction	-4,74	0	-2,64
	LC6 - Wind load -Y direction	4,67	-0,87	2,6
	LC7 - Temperature +25	-2,3	0	-1,28
	LC8 - Temperature -25	2,3	0	1,28
271	LC1 - Self Weight + Secondary Structures	-0,02	0,27	-3,84
	LC2 - Live Load + Technical load	0,03	0,42	-5,52
	LC3 - Wind load +X direction	0,14	0	-0,02
	LC4 - Wind load -X direction	-0,15	0	0,02
	LC5 - Wind load +Y direction	-0,03	0,14	0
	LC6 - Wind load -Y direction	0,03	-0,14	-0,02
	LC7 - Temperature +25	0,41	-0,02	-0,09
	LC8 - Temperature -25	-0,41	0,02	0,09
423	LC1 - Self Weight + Secondary Structures	0,02	-2,36	-4,65
	LC2 - Live Load + Technical load	0,05	-4,62	-7,59
	LC3 - Wind load +X direction	0,11	-0,54	0,36
	LC4 - Wind load -X direction	-0,11	0,6	-0,41
	LC5 - Wind load +Y direction	0,02	5,72	1,47
	LC6 - Wind load -Y direction	-0,02	-5,88	-1,47
	LC7 - Temperature +25	0	-1,61	1,41
	LC8 - Temperature -25	0	1,61	-1,41
678	LC1 - Self Weight + Secondary Structures	-4,05	0	-3,36
	LC2 - Live Load + Technical load	-8,07	0	-7,61
	LC3 - Wind load +X direction	2,45	0	0,92
	LC4 - Wind load -X direction	-2,46	0	-0,92
	LC5 - Wind load +Y direction	3,26	0,89	1,82
	LC6 - Wind load -Y direction	-3,21	0	-1,79
	LC7 - Temperature +25	-2,67	0	-1,49
	LC8 - Temperature -25	2,67	0	1,49


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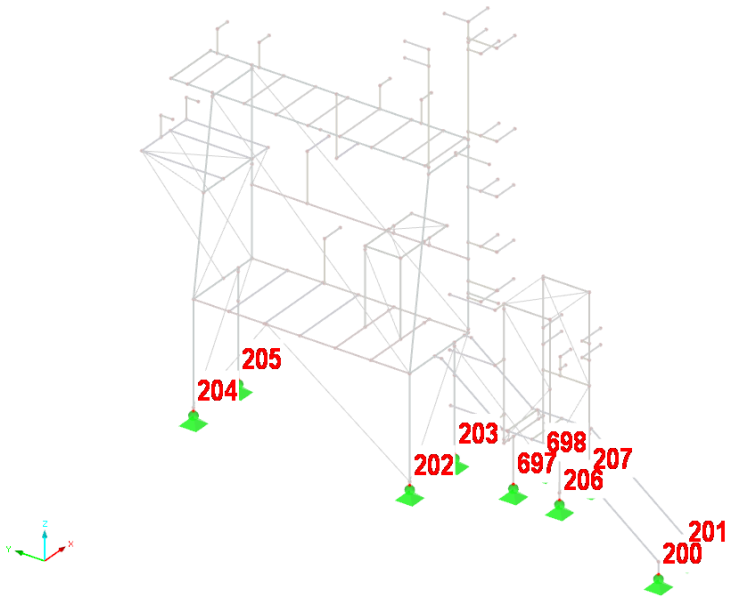
Node No.	LC/CO	Support Forces [kN] $P_{X'}$	$P_{Y'}$	$P_{Z'}$
194	LC1 - Self Weight + Secondary Structures	0	0	-1,35
	LC2 - Live Load + Technical load	0	0	-2,28
	LC3 - Wind load +X direction	0	-0,02	0
	LC4 - Wind load -X direction	0	-0,39	0
	LC5 - Wind load +Y direction	0	0,09	0
	LC6 - Wind load -Y direction	0	-0,02	0
	LC7 - Temperature +25	0	1,7	0
	LC8 - Temperature -25	0	-1,7	0
206	LC1 - Self Weight + Secondary Structures	0	0	-6,83
	LC2 - Live Load + Technical load	0	0	-12,03
	LC3 - Wind load +X direction	1,64	0,12	0
	LC4 - Wind load -X direction	0	0,14	0
	LC5 - Wind load +Y direction	0	0	0
	LC6 - Wind load -Y direction	0	-0,92	-0,47
	LC7 - Temperature +25	0	0,18	0
	LC8 - Temperature -25	0	-0,18	0
214	LC1 - Self Weight + Secondary Structures	0	0	-5,78
	LC2 - Live Load + Technical load	0	0	-9,08
	LC3 - Wind load +X direction	1,64	0	0
	LC4 - Wind load -X direction	0	-0,06	0
	LC5 - Wind load +Y direction	0	1,2	0
	LC6 - Wind load -Y direction	0	0	0
	LC7 - Temperature +25	0	-1,83	0
	LC8 - Temperature -25	0	1,83	0
220	LC1 - Self Weight + Secondary Structures	0	0	-3,32
	LC2 - Live Load + Technical load	0	0	-6,39
	LC3 - Wind load +X direction	0,9	-0,12	0
	LC4 - Wind load -X direction	-2,7	-0,14	0
	LC5 - Wind load +Y direction	0,05	0	0
	LC6 - Wind load -Y direction	0,02	-0,93	-0,42
	LC7 - Temperature +25	0,6	-0,18	0,03
	LC8 - Temperature -25	-0,6	0,18	-0,03
221	LC1 - Self Weight + Secondary Structures	0	0	-2,25
	LC2 - Live Load + Technical load	0	0	-3,47
	LC3 - Wind load +X direction	-0,02	0	0
	LC4 - Wind load -X direction	-1,47	-0,02	0
	LC5 - Wind load +Y direction	-0,05	1,32	0
	LC6 - Wind load -Y direction	-0,02	0	0
	LC7 - Temperature +25	-0,51	-0,36	0
	LC8 - Temperature -25	0,51	0,36	0
222	LC1 - Self Weight + Secondary Structures	-0,02	0	-0,86
	LC2 - Live Load + Technical load	-0,03	0	-2,43
	LC3 - Wind load +X direction	0,5	0	0

	LC4 - Wind load -X direction	0	0	0
	LC5 - Wind load +Y direction	0	0	0
	LC6 - Wind load -Y direction	0	-1,38	0,45
	LC7 - Temperature +25	-0,05	0	-0,02
	LC8 - Temperature -25	0,05	0	0,02
227	LC1 - Self Weight + Secondary Structures	0,02	0	-0,87
	LC2 - Live Load + Technical load	0,02	0	-2,45
	LC3 - Wind load +X direction	0	0	0
	LC4 - Wind load -X direction	-0,5	0	0
	LC5 - Wind load +Y direction	0	0	0
	LC6 - Wind load -Y direction	0	-1,38	0,45
	LC7 - Temperature +25	-0,03	0	0
	LC8 - Temperature -25	0,03	0	0
428	LC1 - Self Weight + Secondary Structures	0	0	-1,1
	LC2 - Live Load + Technical load	0	0	-1,65
	LC3 - Wind load +X direction	0	0,02	0
	LC4 - Wind load -X direction	0	0,45	0
	LC5 - Wind load +Y direction	0	0,09	0
	LC6 - Wind load -Y direction	0	0,02	0
	LC7 - Temperature +25	0	0,48	0
	LC8 - Temperature -25	0	-0,48	0


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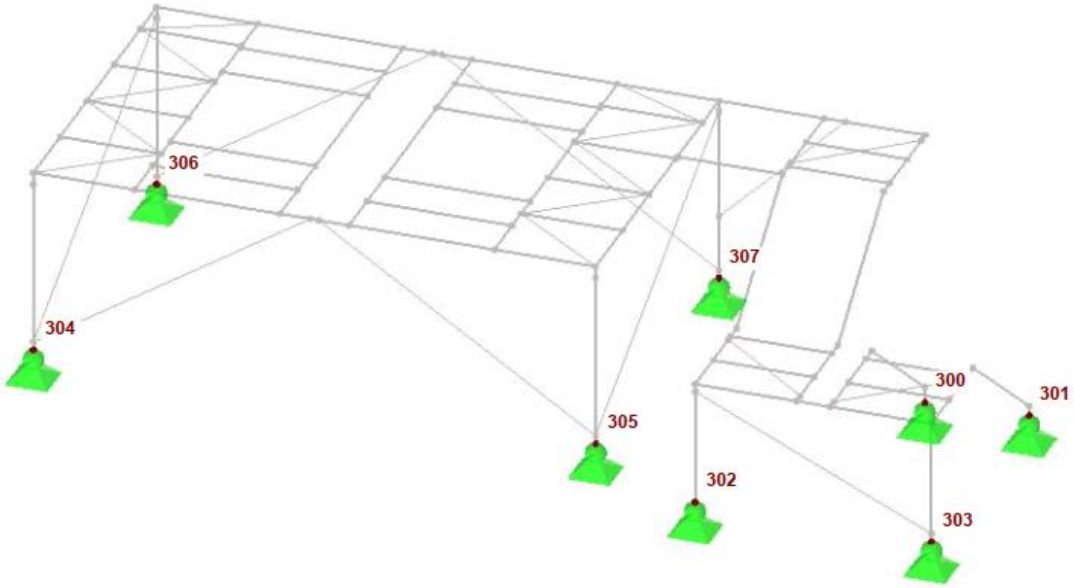
Node No.	LC/CO	Support Forces [kN]		
		$P_{X'}$	$P_{Y'}$	$P_{Z'}$
200	LC1 - Self Weight + Secondary Structures	0	0,09	-3,18
	LC2 - Live Load + Technical load	0	2,88	-2,36
	LC3 - Wind load +X direction	1,76	2,91	2,55
	LC4 - Wind load -X direction	-1,76	-2,91	-2,55
	LC5 - Wind load +Y direction	0	10,44	5,69
	LC6 - Wind load -Y direction	0	-10,37	-5,63
	LC7 - Temperature +25	-0,02	-12,62	-10,5
	LC8 - Temperature -25	0,02	12,62	10,5
201	LC1 - Self Weight + Secondary Structures	0	-0,74	-3,83
	LC2 - Live Load + Technical load	0	0,11	-4,49
	LC3 - Wind load +X direction	1,68	0,62	0,47
	LC4 - Wind load -X direction	-1,68	-0,63	-0,48
	LC5 - Wind load +Y direction	0	13,31	7,7
	LC6 - Wind load -Y direction	0	-13,23	-7,64
	LC7 - Temperature +25	0,02	-7,04	-6,27
	LC8 - Temperature -25	-0,02	7,04	6,27
202	LC1 - Self Weight + Secondary Structures	4,1	-8,63	-16,67
	LC2 - Live Load + Technical load	10,26	-15,3	4,47
	LC3 - Wind load +X direction	41,94	2,33	115,08
	LC4 - Wind load -X direction	-41,96	-2,33	-115,13
	LC5 - Wind load +Y direction	12,2	9,62	32,81
	LC6 - Wind load -Y direction	-12,02	-8,75	-32,49
	LC7 - Temperature +25	3,11	2,55	11,87
	LC8 - Temperature -25	-3,11	-2,55	-11,87
203	LC1 - Self Weight + Secondary Structures	-4,31	0,05	-64,58
	LC2 - Live Load + Technical load	-11,66	0,23	-154,65
	LC3 - Wind load +X direction	-0,89	0,24	-105,51
	LC4 - Wind load -X direction	0,89	-0,24	105,56
	LC5 - Wind load +Y direction	1,5	2,57	-61,28
	LC6 - Wind load -Y direction	-1,52	-2,55	60,03
	LC7 - Temperature +25	0,5	-0,36	-17,84
	LC8 - Temperature -25	-0,5	0,36	17,84
204	LC1 - Self Weight + Secondary Structures	5,42	9,44	-18,29
	LC2 - Live Load + Technical load	11,85	16,82	-10,47
	LC3 - Wind load +X direction	38,39	0,44	105,92
	LC4 - Wind load -X direction	-38,39	-0,44	-105,95
	LC5 - Wind load +Y direction	0,33	6,45	-9,65
	LC6 - Wind load -Y direction	-0,33	-6,44	9,63
	LC7 - Temperature +25	-0,99	3,21	-4,29
	LC8 - Temperature -25	0,99	-3,21	4,29
205	LC1 - Self Weight + Secondary Structures	-5,6	0	-89,07
	LC2 - Live Load + Technical load	-12,53	0	-187,65
	LC3 - Wind load +X direction	-2,84	0	-107,28

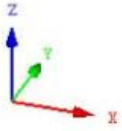
	LC4 - Wind load -X direction	2,84	0	107,31
	LC5 - Wind load +Y direction	-1,73	0,81	-21,98
	LC6 - Wind load -Y direction	1,73	-0,81	21,98
	LC7 - Temperature +25	0,74	0	1,44
	LC8 - Temperature -25	-0,74	0	-1,44
206	LC1 - Self Weight + Secondary Structures	-0,17	0	-14,67
	LC2 - Live Load + Technical load	-0,32	0,02	-15
	LC3 - Wind load +X direction	16,38	0,02	14,81
	LC4 - Wind load -X direction	-16,38	-0,02	-14,81
	LC5 - Wind load +Y direction	0,06	0,02	-10,35
	LC6 - Wind load -Y direction	-0,06	-0,02	10,31
	LC7 - Temperature +25	-0,27	0	6,56
	LC8 - Temperature -25	0,27	0	-6,56
207	LC1 - Self Weight + Secondary Structures	0,15	0,44	-6,44
	LC2 - Live Load + Technical load	-0,17	0,71	-13,76
	LC3 - Wind load +X direction	-1,46	0,36	-14,88
	LC4 - Wind load -X direction	1,46	-0,38	14,88
	LC5 - Wind load +Y direction	0,05	5,36	14,06
	LC6 - Wind load -Y direction	-0,05	-5,33	-13,95
	LC7 - Temperature +25	0,26	1,01	6,15
	LC8 - Temperature -25	-0,26	-1,01	-6,15
697	LC1 - Self Weight + Secondary Structures	0,14	0	-5,51
	LC2 - Live Load + Technical load	0,5	0	-12,75
	LC3 - Wind load +X direction	0,95	0	4,23
	LC4 - Wind load -X direction	-0,95	0	-4,23
	LC5 - Wind load +Y direction	-0,02	0,02	-0,59
	LC6 - Wind load -Y direction	0,02	-0,02	0,59
	LC7 - Temperature +25	0,02	0	0,27
	LC8 - Temperature -25	-0,02	0	-0,27
698	LC1 - Self Weight + Secondary Structures	0,3	-0,68	-6,26
	LC2 - Live Load + Technical load	2,07	-5,45	-19,26
	LC3 - Wind load +X direction	3,42	-6,92	-15,38
	LC4 - Wind load -X direction	-3,42	6,92	15,39
	LC5 - Wind load +Y direction	-12,39	51,06	43,59
	LC6 - Wind load -Y direction	12,21	-50,39	-42,86
	LC7 - Temperature +25	-3,35	13,28	12,65
	LC8 - Temperature -25	3,35	-13,28	-12,65


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Node No.	LC/CO	Support Forces [kN]		
		$P_{X'}$	$P_{Y'}$	$P_{Z'}$
300	LC1 - Self Weight + Secondary Structures	0	1,32	-3,33
	LC2 - Live Load + Technical load	0	2,94	-8,67
	LC3 - Wind load +X direction	1,26	2,39	-1,2
	LC4 - Wind load -X direction	-1,26	-2,39	1,2
	LC5 - Wind load +Y direction	0,02	12	-4,77
	LC6 - Wind load -Y direction	-0,02	-12	4,77
	LC7 - Temperature +25	0	0,83	-0,42
	LC8 - Temperature -25	0	-0,83	0,42
301	LC1 - Self Weight + Secondary Structures	0	-0,23	-2,64
	LC2 - Live Load + Technical load	0	-0,53	-7,19
	LC3 - Wind load +X direction	1,31	-4,52	2,19
	LC4 - Wind load -X direction	-1,31	4,52	-2,19
	LC5 - Wind load +Y direction	0,02	6,51	-1,8
	LC6 - Wind load -Y direction	-0,02	-6,51	1,8
	LC7 - Temperature +25	0	-0,15	0,08
	LC8 - Temperature -25	0	0,15	-0,08
302	LC1 - Self Weight + Secondary Structures	0	0	-7,65
	LC2 - Live Load + Technical load	0,02	0	-17,34
	LC3 - Wind load +X direction	-0,54	0	5,88
	LC4 - Wind load -X direction	0,54	0	-5,88
	LC5 - Wind load +Y direction	0	0	1,08
	LC6 - Wind load -Y direction	0	0	-1,08
	LC7 - Temperature +25	-0,02	0	0,3
	LC8 - Temperature -25	0,02	0	-0,3
303	LC1 - Self Weight + Secondary Structures	-0,02	0	-6,45
	LC2 - Live Load + Technical load	-0,05	0	-14,03
	LC3 - Wind load +X direction	12,38	0	-7,46
	LC4 - Wind load -X direction	-12,38	0	7,46
	LC5 - Wind load +Y direction	-0,02	0	4,7
	LC6 - Wind load -Y direction	0,02	0	-4,7
	LC7 - Temperature +25	0,05	0	0,23
	LC8 - Temperature -25	-0,05	0	-0,23
304	LC1 - Self Weight + Secondary Structures	-67,11	0	-74,1
	LC2 - Live Load + Technical load	-39,42	0,03	-50,42
	LC3 - Wind load +X direction	9,35	0,09	7,91
	LC4 - Wind load -X direction	-9,35	-0,09	-7,91
	LC5 - Wind load +Y direction	5,4	20,97	15,14
	LC6 - Wind load -Y direction	-5,42	-20,91	-15,11
	LC7 - Temperature +25	-2,09	-0,03	-0,18
	LC8 - Temperature -25	2,09	0,03	0,18
305	LC1 - Self Weight + Secondary Structures	66,44	-1,17	-74,48
	LC2 - Live Load + Technical load	38	-2,61	-51,63
	LC3 - Wind load +X direction	9,35	2,18	-6,72

	LC4 - Wind load -X direction	-9,35	-2,18	6,72
	LC5 - Wind load +Y direction	-1,59	31,46	17,99
	LC6 - Wind load -Y direction	1,59	-31,47	-17,99
	LC7 - Temperature +25	1,59	-0,69	-0,2
	LC8 - Temperature -25	-1,59	0,69	0,2
306	LC1 - Self Weight + Secondary Structures	-62,21	0	-72,68
	LC2 - Live Load + Technical load	-29,75	0	-47,22
	LC3 - Wind load +X direction	10,46	0	8,81
	LC4 - Wind load -X direction	-10,49	0	-8,84
	LC5 - Wind load +Y direction	-5,51	-0,26	-15,21
	LC6 - Wind load -Y direction	5,51	0,26	15,18
	LC7 - Temperature +25	-1,61	0	0,24
	LC8 - Temperature -25	1,61	0	-0,24
307	LC1 - Self Weight + Secondary Structures	62,9	0,08	-88,53
	LC2 - Live Load + Technical load	31,2	0,15	-85,43
	LC3 - Wind load +X direction	10,32	-0,14	-9,41
	LC4 - Wind load -X direction	-10,35	0,14	9,42
	LC5 - Wind load +Y direction	1,7	-0,89	-17,12
	LC6 - Wind load -Y direction	-1,7	0,89	17,12
	LC7 - Temperature +25	2,06	0,05	-0,06
	LC8 - Temperature -25	-2,06	-0,05	0,06

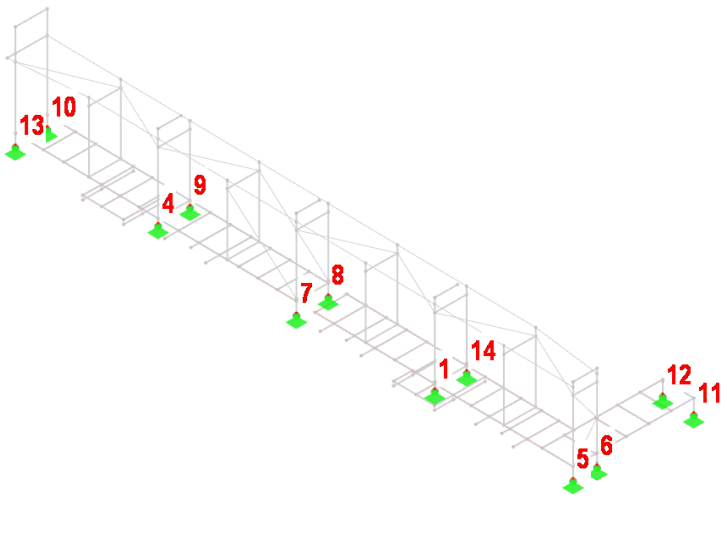

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Z
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Node No.	LC/CO	Support Forces [kN] $P_{X'}$	$P_{Y'}$	$P_{Z'}$
1	LC1 - Self Weight + Secondary Structures	-0,9	0,57	-25,64
	LC2 - Live Load + Technical load	-1,91	1,67	-55,95
	LC3 - Wind load +X direction	14,81	1,4	16,05
	LC4 - Wind load -X direction	-14,61	-1,41	-16,05
	LC5 - Wind load +Y direction	0	12,63	0,02
	LC6 - Wind load -Y direction	0	-12,63	-0,02
	LC7 - Temperature +25	-38,21	-2,72	-2,52
	LC8 - Temperature -25	38,21	2,72	2,52
4	LC1 - Self Weight + Secondary Structures	-0,57	6,75	-23,43
	LC2 - Live Load + Technical load	-1,14	14,75	-48,9
	LC3 - Wind load +X direction	9,42	0,09	16,97
	LC4 - Wind load -X direction	-9,23	-0,09	-16,97
	LC5 - Wind load +Y direction	0,15	6,27	-1,64
	LC6 - Wind load -Y direction	-0,15	-6,27	1,65
	LC7 - Temperature +25	-37,32	-17,52	0,86
	LC8 - Temperature -25	37,32	17,52	-0,86
5	LC1 - Self Weight + Secondary Structures	-0,15	-1,01	-17,22
	LC2 - Live Load + Technical load	-0,23	-2,43	-31,83
	LC3 - Wind load +X direction	3,81	-1,11	14,48
	LC4 - Wind load -X direction	-3,8	1,11	-14,48
	LC5 - Wind load +Y direction	-0,06	0,45	0,2
	LC6 - Wind load -Y direction	0,06	-0,45	-0,21
	LC7 - Temperature +25	-10,47	-13,88	3,86
	LC8 - Temperature -25	10,47	13,88	-3,86
6	LC1 - Self Weight + Secondary Structures	0,02	-1,19	-6,8
	LC2 - Live Load + Technical load	0,09	-2,72	-15,6
	LC3 - Wind load +X direction	3,92	-0,05	-14,51
	LC4 - Wind load -X direction	-3,93	0,06	14,52
	LC5 - Wind load +Y direction	-0,06	0,26	0,14
	LC6 - Wind load -Y direction	0,05	-0,24	-0,14
	LC7 - Temperature +25	11,66	-11,36	-4,35
	LC8 - Temperature -25	-11,66	11,36	4,35
7	LC1 - Self Weight + Secondary Structures	0,03	-0,99	-18,2
	LC2 - Live Load + Technical load	0,23	-2,13	-40,05
	LC3 - Wind load +X direction	6,96	0,05	19,71
	LC4 - Wind load -X direction	-6,89	-0,05	-19,71
	LC5 - Wind load +Y direction	0	0,12	2,28
	LC6 - Wind load -Y direction	0	-0,12	-2,28
	LC7 - Temperature +25	-7,77	14	-0,71
	LC8 - Temperature -25	7,77	-14	0,71
8	LC1 - Self Weight + Secondary Structures	0,6	-0,66	-20,27
	LC2 - Live Load + Technical load	1,29	-1,49	-47,52
	LC3 - Wind load +X direction	6,9	0,26	-19,86

	LC4 - Wind load -X direction	-6,98	-0,26	19,86
	LC5 - Wind load +Y direction	0	-0,38	2,73
	LC6 - Wind load -Y direction	0	0,36	-2,73
	LC7 - Temperature +25	7,34	13,53	0,98
	LC8 - Temperature -25	-7,34	-13,53	-0,98
9	LC1 - Self Weight + Secondary Structures	0,45	4,55	-14,64
	LC2 - Live Load + Technical load	0,83	10,37	-32,58
	LC3 - Wind load +X direction	9,17	-1,82	-17,01
	LC4 - Wind load -X direction	-9,35	1,82	17,01
	LC5 - Wind load +Y direction	0,15	6,45	-3,36
	LC6 - Wind load -Y direction	-0,15	-6,45	3,35
	LC7 - Temperature +25	36,59	11,75	-0,36
	LC8 - Temperature -25	-36,59	-11,75	0,36
10	LC1 - Self Weight + Secondary Structures	0,11	-3,51	-10,47
	LC2 - Live Load + Technical load	0,24	-7,86	-18,42
	LC3 - Wind load +X direction	4,01	-0,21	-11,76
	LC4 - Wind load -X direction	-4,02	0,21	11,76
	LC5 - Wind load +Y direction	-0,09	-3,02	0,24
	LC6 - Wind load -Y direction	0,09	3,02	-0,24
	LC7 - Temperature +25	8,39	-4,11	-1,88
	LC8 - Temperature -25	-8,39	4,11	1,88
11	LC1 - Self Weight + Secondary Structures	0	-0,24	-4,29
	LC2 - Live Load + Technical load	-0,02	-0,6	-8,76
	LC3 - Wind load +X direction	0,08	-0,02	-0,02
	LC4 - Wind load -X direction	-0,08	0,02	0,02
	LC5 - Wind load +Y direction	0	0,93	0,78
	LC6 - Wind load -Y direction	0	-0,92	-0,77
	LC7 - Temperature +25	0	-3,68	-0,32
	LC8 - Temperature -25	0	3,68	0,32
12	LC1 - Self Weight + Secondary Structures	0	0,23	-6,8
	LC2 - Live Load + Technical load	0,02	0,56	-14,3
	LC3 - Wind load +X direction	0,12	-0,02	-0,15
	LC4 - Wind load -X direction	-0,12	0,02	0,15
	LC5 - Wind load +Y direction	0	0,93	-0,68
	LC6 - Wind load -Y direction	0	-0,93	0,68
	LC7 - Temperature +25	0	2,96	0,74
	LC8 - Temperature -25	0	-2,96	-0,74
13	LC1 - Self Weight + Secondary Structures	-0,36	-3,89	-12,81
	LC2 - Live Load + Technical load	-0,78	-8,58	-22,88
	LC3 - Wind load +X direction	3,89	1,37	11,78
	LC4 - Wind load -X direction	-3,87	-1,37	-11,78
	LC5 - Wind load +Y direction	-0,08	-3,06	-0,68
	LC6 - Wind load -Y direction	0,08	3,06	0,68
	LC7 - Temperature +25	-6,63	10,8	1,28
	LC8 - Temperature -25	6,63	-10,8	-1,28

14	LC1 - Self Weight + Secondary Structures	0,78	-0,62	-14,7
	LC2 - Live Load + Technical load	1,38	-1,55	-36,08
	LC3 - Wind load +X direction	12,81	0,06	-15,68
	LC4 - Wind load -X direction	-13,01	-0,05	15,68
	LC5 - Wind load +Y direction	-0,03	11,22	-0,03
	LC6 - Wind load -Y direction	0,03	-11,27	0,03
	LC7 - Temperature +25	36,44	0,23	2,45
	LC8 - Temperature -25	-36,44	-0,23	-2,45


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LC1 - Self-weight

Node	Support Forces [kN]			Support Moments [kNm]		
No.	P_X	P_Y	P_Z	M_X	M_Y	M_Z
47	0,06	0,00	-19,12	0,00	0,00	0,00
50	0,00	0,00	-26,23	0,00	0,00	0,00
52	0,00	0,00	-17,55	0,00	0,00	0,00

LC2 - Live load

Node	Support Forces [kN]			Support Moments [kNm]		
No.	P_X	P_Y	P_Z	M_X	M_Y	M_Z
47	0,07	0,00	-20,51	0,00	0,00	0,00
50	0,00	0,00	-27,08	0,00	0,00	0,00
52	0,00	0,00	-35,25	0,00	0,00	0,00

LC3 - Wind +X

Node	Support Forces [kN]			Support Moments [kNm]		
No.	P_X	P_Y	P_Z	M_X	M_Y	M_Z
47	0,00	0,00	0,00	0,00	0,00	0,00
50	0,00	0,00	1,64	0,00	0,00	0,00
52	0,00	0,00	-0,56	0,00	0,00	-0,04

LC4 - Wind -X

Node	Support Forces [kN]			Support Moments [kNm]		
No.	P_X	P_Y	P_Z	M_X	M_Y	M_Z
47	0,00	0,00	0,00	0,00	0,00	0,00
50	0,00	0,00	-1,64	0,00	0,00	0,00
52	0,00	0,00	0,56	0,00	0,00	0,04

LC5 - Wind +Y

Node	Support Forces [kN]			Support Moments [kNm]		
No.	P_X	P_Y	P_Z	M_X	M_Y	M_Z
47	0,00	0,00	0,00	0,00	0,00	0,00
50	0,00	0,00	-0,17	0,00	0,00	0,01
52	0,00	0,00	-0,15	0,00	0,00	0,00

LC6 - Wind -Y

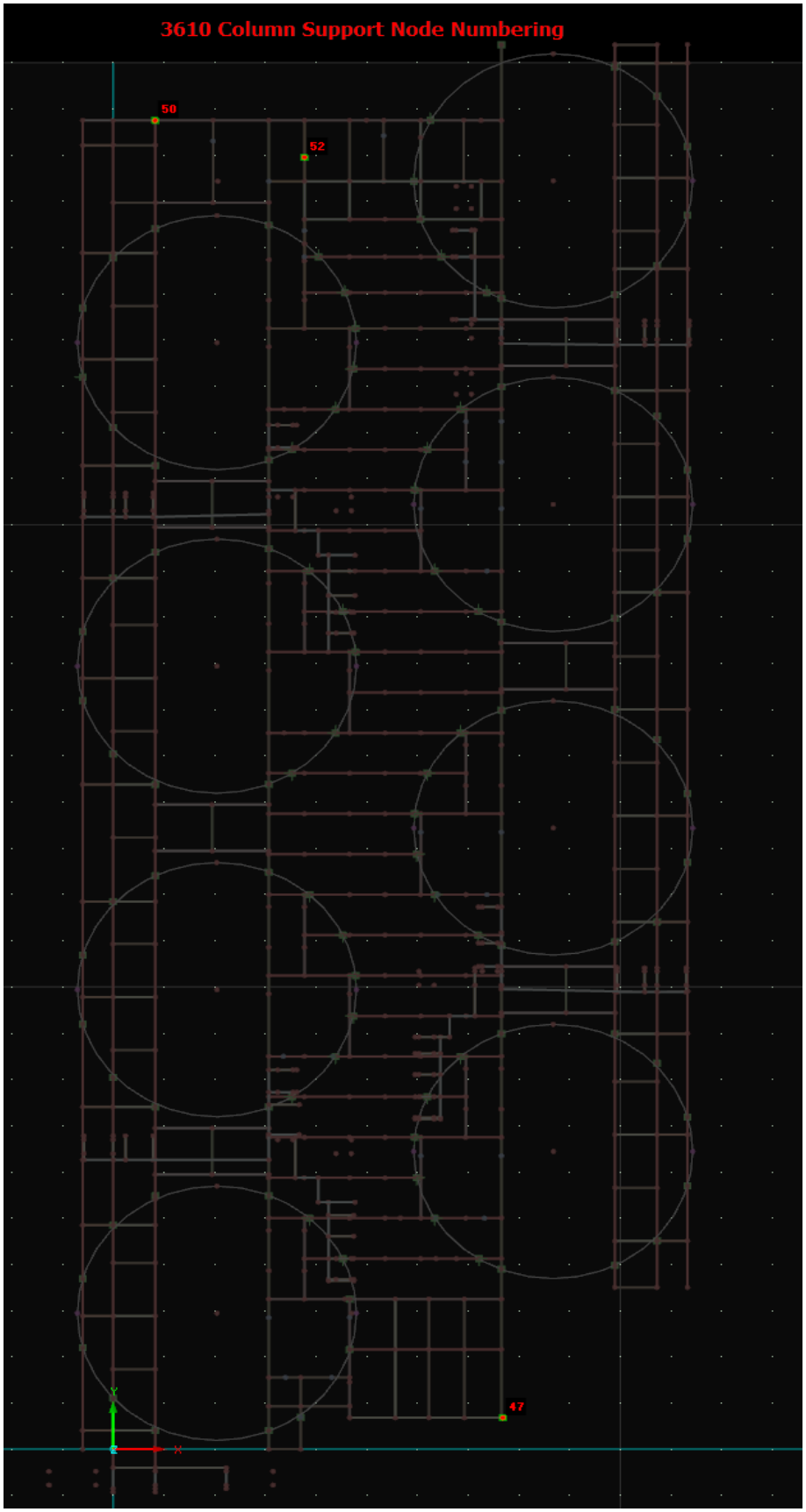
Node	Support Forces [kN]			Support Moments [kNm]		
No.	P_X	P_Y	P_Z	M_X	M_Y	M_Z
47	0,00	0,00	0,00	0,00	0,00	0,00
50	0,00	0,00	0,17	0,00	0,00	-0,02
52	0,00	0,00	0,11	0,00	0,00	0,00

LC7 - Temperature +25

Node	Support Forces [kN]			Support Moments [kNm]		
No.	P_X	P_Y	P_Z	M_X	M_Y	M_Z
47	0,00	0,00	0,00	0,00	0,00	0,00
50	0,00	0,00	0,02	0,00	0,00	0,46
52	0,00	0,00	0,16	0,00	0,00	0,71

LC8 - Temperature -25

Node	Support Forces [kN]			Support Moments [kNm]		
No.	P_X	P_Y	P_Z	M_X	M_Y	M_Z
47	0,00	0,00	0,00	0,00	0,00	0,00
50	0,00	0,00	-0,02	0,00	0,00	-0,46
52	0,00	0,00	-0,16	0,00	0,00	-0,71


3610 Column Support Node Numbering
50
52
47
Y
X